华沿机器人
HUAYAN ROBOTICS



Preface

How to use this manual

This manual is intended for the operators of Elfin series robots. If you need to carry out large-scale script development, you should have some knowledge of Python programming. This manual will guide the operators to use Elfin from the following aspects:

- Software control: Guide the operator to install and run the software.
- Parameter configuration: Introduce the basic configuration and related functions of Elfin software robot.
- Robot Programming: Detailed introduction to Elfin programming instructions.

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technical support

Shenzhen Hua Yan Robot Co., Ltd. provides you with long-term technical services. When you encounter technical problems or have other needs during the use, you are welcome to visit our company website: www.huayan-robotics.com, or directly contact our engineers.

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Glossary of Terms:

Terms	Explanations
Robot Arm	Usually refers to the joints, links, and electrical control boxes of the robot.
Robot	A complete equipment consisting of robot arm and electrical control box.
Controller Software Operation Interface	The graphical interfaced for operating the robot.
System Software	A collection of operation interfaces and back-end services.
Program	A collection of commands written by a programming module.
Program Files	A file that stores program commands.
User Coordinate System (UCS)	User-defined reference coordinate system.
Tool Center Point (TCP)	The coordinate system centered on the tool at the robot arm's end-effector.
I/O	Short for input/output, including CI/CO, DI/DO, AI/AO, EndDI/EndDI, EndAI/EndAO.
Script	The Script module inserted in programming.
Waypoint	The robot arm target position used in the program.
Point	The specific position information of the robot arm recorded and stored.

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Chapter 1 Software Introduction

Huayan S.p.A Teach Pendent has an operating software that enables manual operation, programming, parameter configuration and user monitoring of robots. The user can control the robot through the function buttons on the operation interface, and enable the robot to move along the user-specified path and complete the expected actions. The teach pendant serves as the steering wheel that drives the robot.

There are two ways to access the interface of the display controller software to operate the robot:

◆ Through Huayan S.p.A Teach Pendent.

The user can directly operate the robot through the teach pendant; in this case, the teach pendant only serves as the display of the robot.

◆ Through different end-effectors.

End-effector refers to personal computers and tablets. Refer to the Appendix for detailed connections.

Supported end-effector devices are personal computers, tablets and etc., which are connected to the robot through TCP protocol in B/S mode. Refer to the Appendix for specific connection methods.

1.1. User Login

After starting the software and accessing the login page, users with login permission can enter their usernames and passwords to log in.

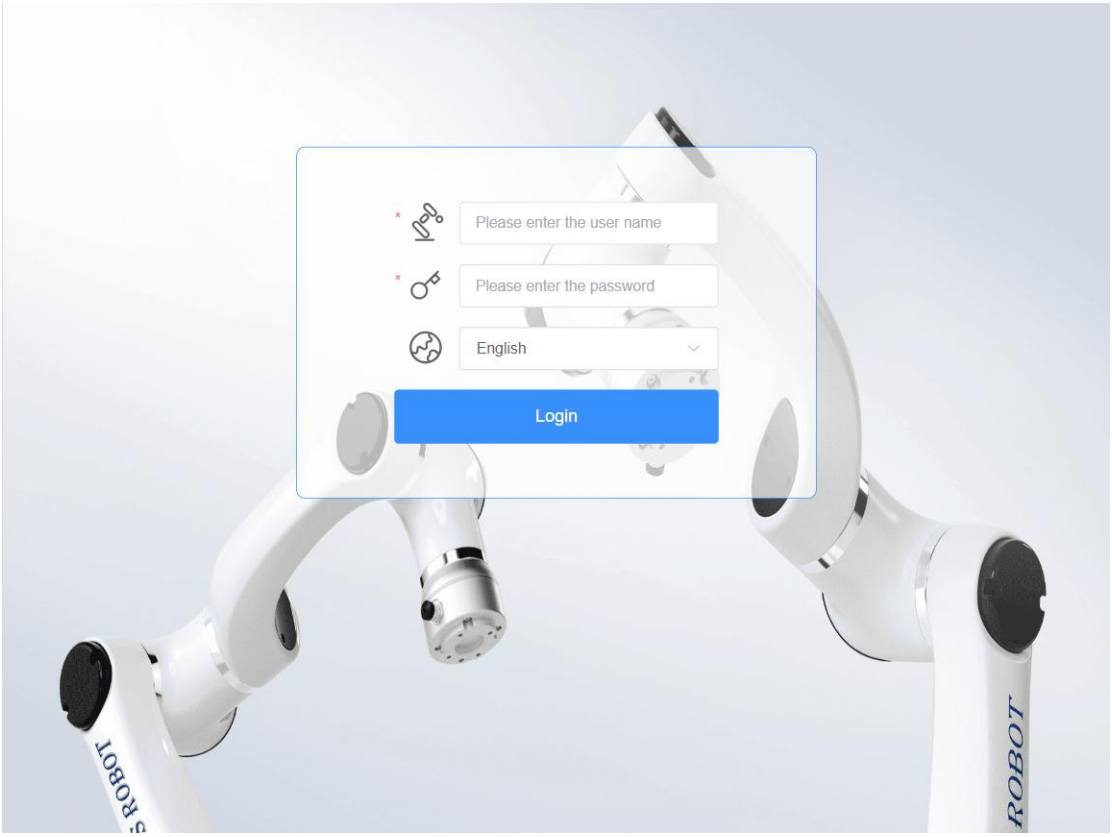


Fig. 1.3.1 User Login

Chapter 2 Operation Interface

2.1. Operation Interface

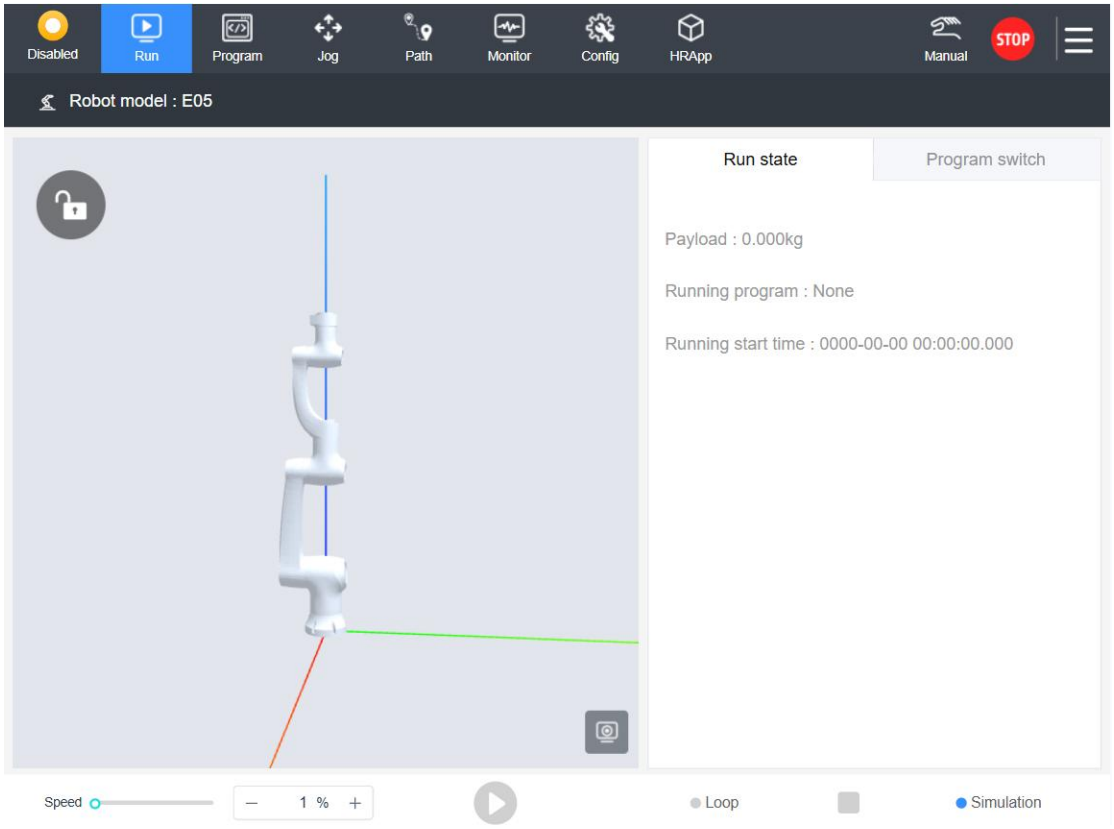


Fig. 2.1.1 Robot Operation Interface

Main components on the robot control interface:

1. Menu Bar
2. Robot ON/OFF
3. Action Bar

2.1.1. Menu Bar

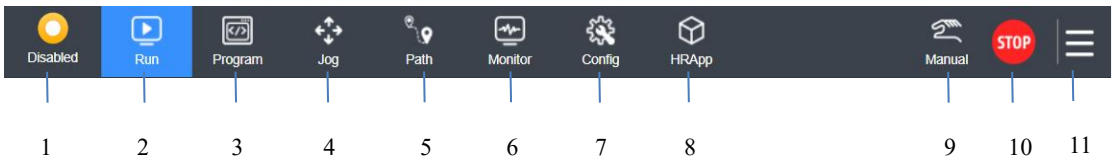


Table 2.1.2 Operation Function Interface

SN	Name	Function	Reference
1	Robot Status	Viewing the robot running status	
2	Run	Controller software operation homepage	7 Run
3	Program	Control of robot arm motion through programming	6 Program
4	Jog	Controlling the pose and motion of the robot arm	4 Jog
5	Path	Establishing the continuous path motion of robot arm	8 Path
6	Monitor	Monitoring the robot status	9 Monitor
7	Config	Configuring the robot	5 Config
8	HRApp	Huayan S.p.A plug-in platform	11 HRApp
9	Mode	Automatic/manual mode switching	3 Manual/Auto Mode
10	Stop	Emergency stop	Emergency stop button
11	Stack Menu	System Language About	2.3 Stack Menu

		Log Out Restart Shutdown	

2.1.2. Robot ON/OFF

SN	Name	Function	Reference
1	Robot ON/OFF	Viewing the startup status of the robot Turning the robotic system ON or OFF	2.2 Status Indicator

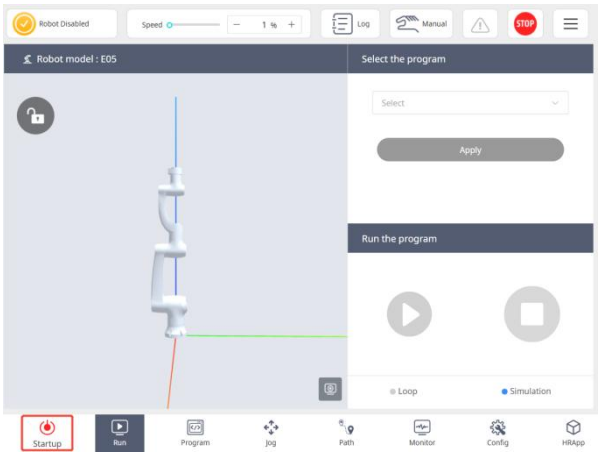
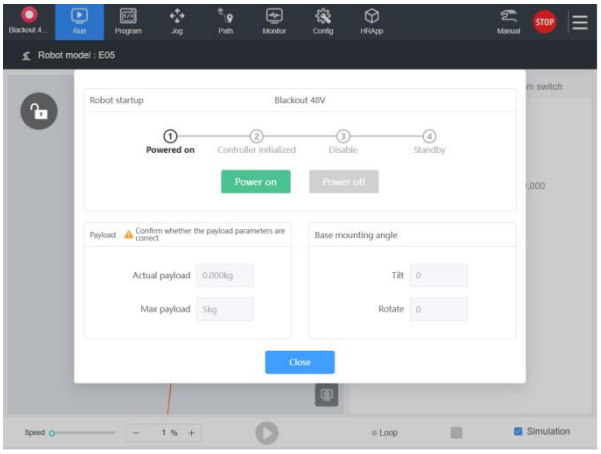
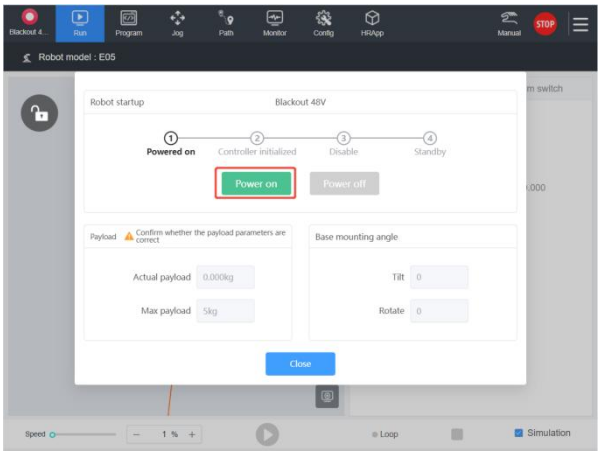
2.2. Robot ON/OFF

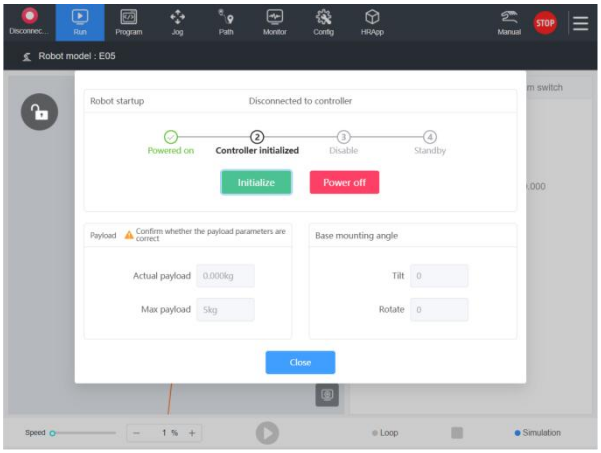
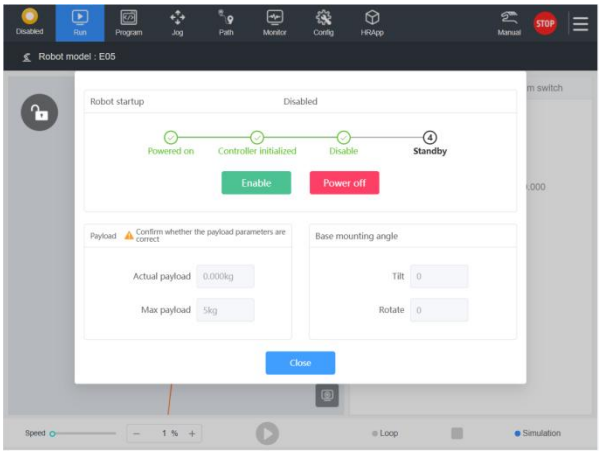
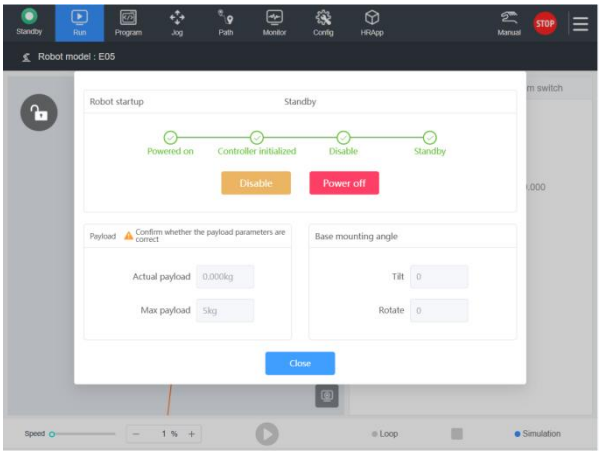
The status indicator indicates the current status of the robot arm: the color yellow indicates the disabled state, green indicates the enabled state, purple indicates the running state, blue indicates the free drive state, and red indicates the uninitialized state or abnormal state.

2.2.1. Robotic System ON/OFF

The steps to turn on the robot are as follows:

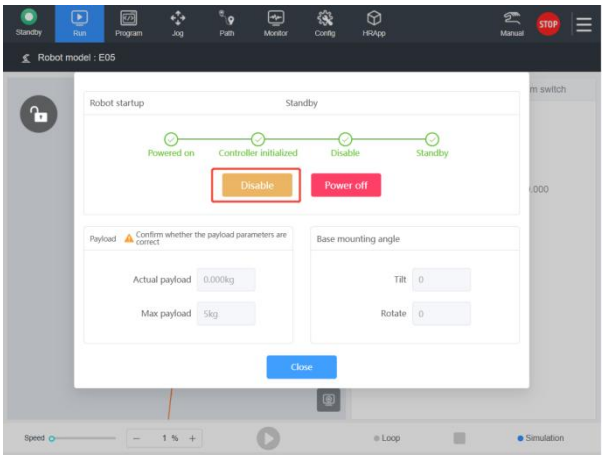
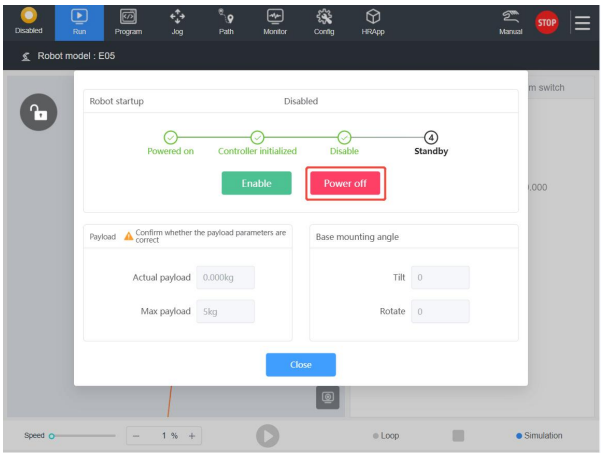
SN	Operation	Examples
1	Enter the controller software operation interface, connect the robot, and log in to the controller;	

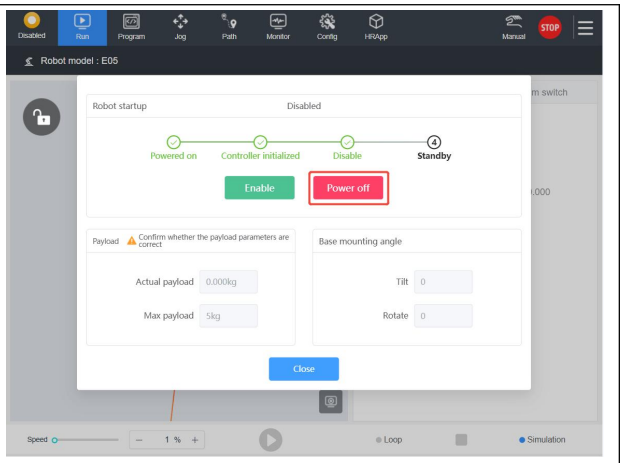
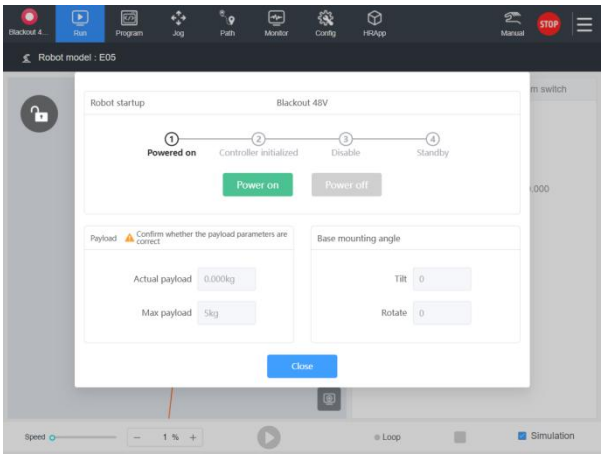
2	After logging in, the robot is still not electrified; power on and enable the robot motion;	
3	Click the 【Robot Blackout】 indicator button to access the Power on popup;	
4	After clicking 【Power on】, the robot is electrified, but the joint brakes are not released, and the robot arm motion is not enabled;	

5	Click 【Initialize】 to initialize the controller;	
6	After successful initialization, the robot will be stably in the disabled state. Before clicking the 【Enable】 button, be sure to confirm whether the robot settings information (current payload, base mounting angle position, TCP and other parameters) is correct. **The current payload and mounting angle are displayed below the 【Enable】 button for verification;**	
7	Click the 【Enable】 button, the robot arm is powered on, and the status indicator turns green, showing "Standby", and the indicating enabling successful;	

Note: Before enabling the robot, please carefully check the robot configurations. If configured incorrectly, the robot arm or control box may not work properly, and personnel or equipment hazards may even occur. Also, please confirm whether the pose of the robot arm is consistent with the actual pose; otherwise, it is necessary to check the match between the robot arm and the electrical control box.

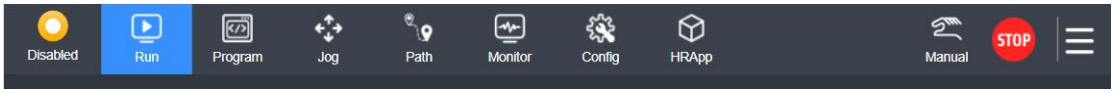
The steps to turn off the robot are as follows:

SN	Operation	Examples
1	Click the 【Disabled】 button; and when the status indicator turns yellow or gray, it indicates that the robot is disabled;	
2	Click【Power off】until the status indicator turns red, and the robot is powered off;	

		
3	The current status indicates successful blackout;	

2.2.1.1. Robot State Description

During the entire process of the robot from startup to normal operation, the page will display the status prompts in real time:



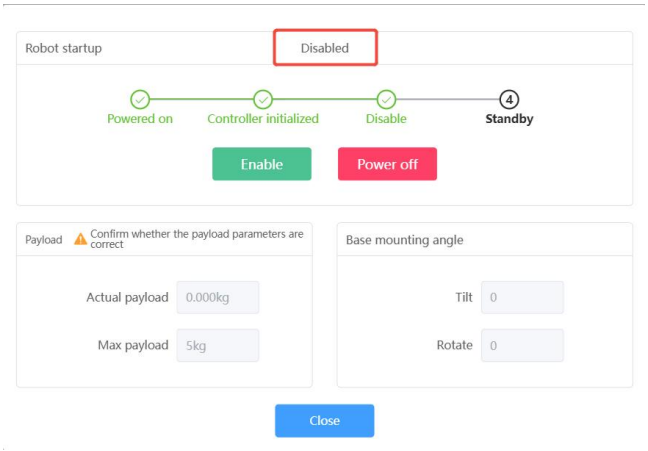


Fig. 2.2.1: Page Prompt

- After Successfully Starting the Teach Pendant: it prompts "The robotic arm is powered off". If the robot fails to power on, the corresponding error and cause will be prompted. Please check whether there are hardware failures according to the prompts;
- If Initialization Fails After Power-on: it prompts "The controller is not initialized", indicating that the current robot has been powered on but not initialized;
- If Not Enabled After Initialization: it prompts "Disabled", indicating that it is currently initialized;
- After Enabling: it prompts "Standby", indicating that the robot arm is ready for the next operation;
- Free Drive: "Free drive" means that the robot arm is in free drive mode, and within a safe range, the robot arm can be freely dragged and moved to the target position;
- Running: "Program Running" indicates that the robot arm is executing the program. Meanwhile, no settings or programming is allowed. Click 【Pause】 or 【Stop】, and the robot will pause or stop the running program;

2.3. Stack Menu

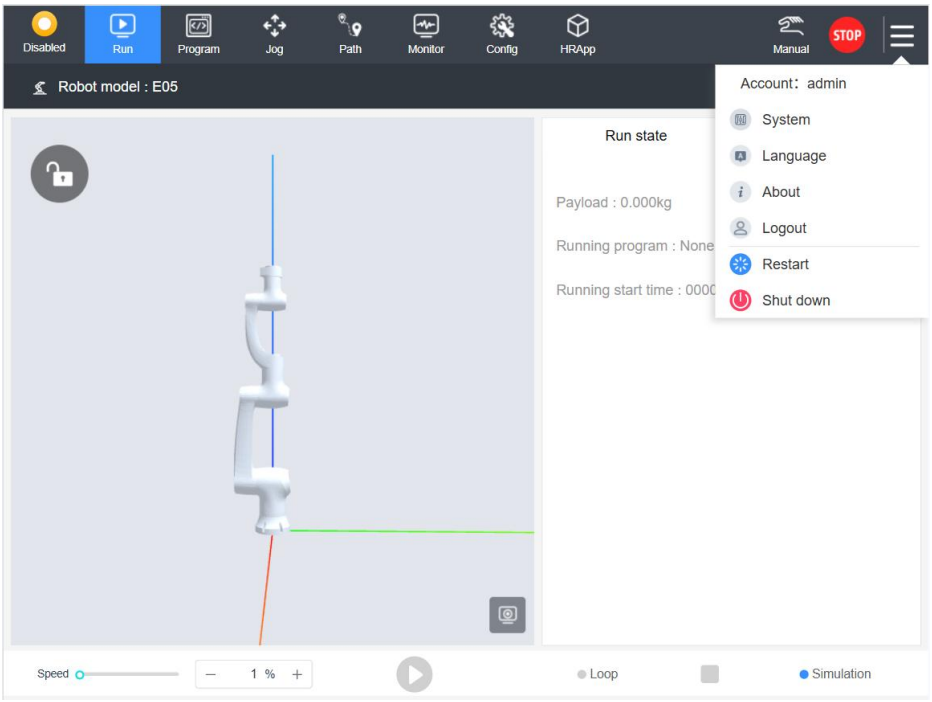


Fig. 2.3.1 Stack Menu

- **Management: Click【Admin】in the Stack Menu to access the System page**, where authorization management, network configuration, system update and user management operations are available. Refer to Chapter 9 Management for details;

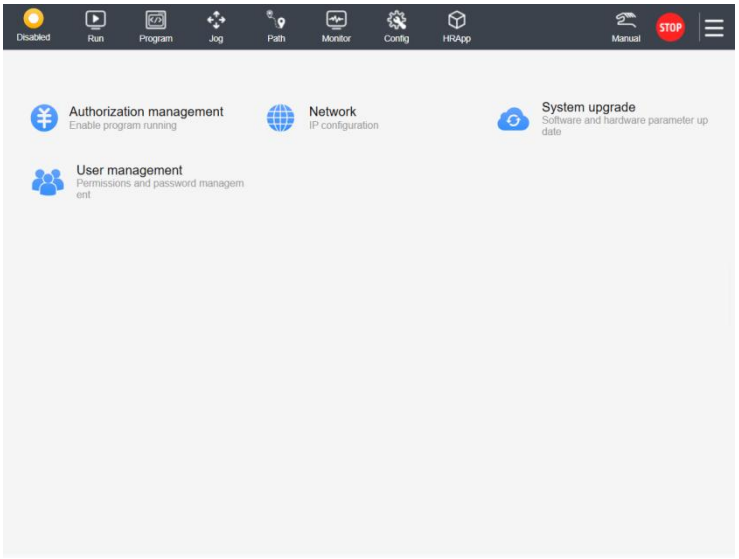


Fig. 2.3.2 System

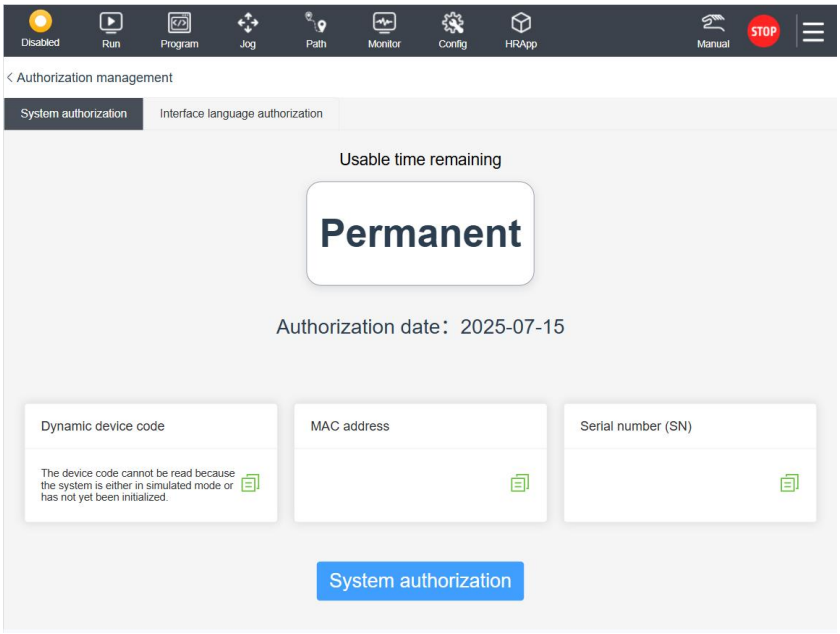


Fig. 2.3.3 Administration

- **Language:** Users can switch the language of the software system. The following languages are available: Chinese, English, Korean, German, Japanese, Spanish, Russian, Polish, French, Italian and Czech;
- **About:** Viewing the version information of the controller software system;
- **User Logout:** If the account is currently logged in, the user can click 【Log out】 to log out and further log in in the menu;
- **Restart:** The software system will perform a restart operation. If there are currently unsaved files, a prompt will appear to remind the user to save the files. Users can choose to restart after saving the files, or restart directly without saving the files, or cancel the restart operation;
- **Shutdown:** To turn off the power supply of the teach pendant, electrical control box and robot arm. If there are currently unsaved files, a prompt will appear to remind the user to save the files. Users can choose to restart after saving the files, or restart directly without saving the files, or cancel the restart operation;

Chapter 3 Manual/Auto Mode

3.1. Auto Mode

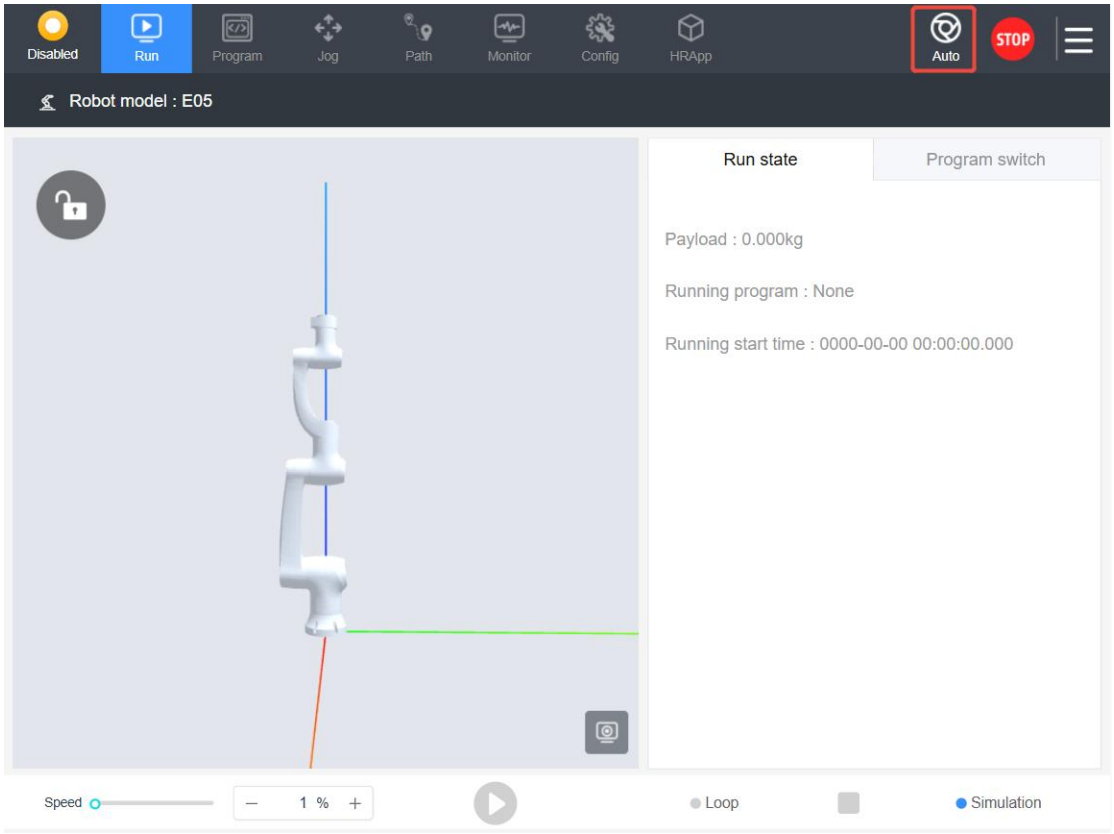


Fig. 3.1.1 Auto Mode Interface

The auto mode refers to an user operation mode. When enabled, only the saved runnable program, robot arm jog functions and robot monitoring functions can be executed. Editing the configuration and the run-able programs of the robot is not allowed.

3.2. Manual Mode



Fig. 3.2.1 Manual Mode Interface

When the manual mode is optioned the user can carry out all system operations. To enhance operational safety, it is recommended to operate in conjunction with the

three-position button.

3.3. Mode Switching

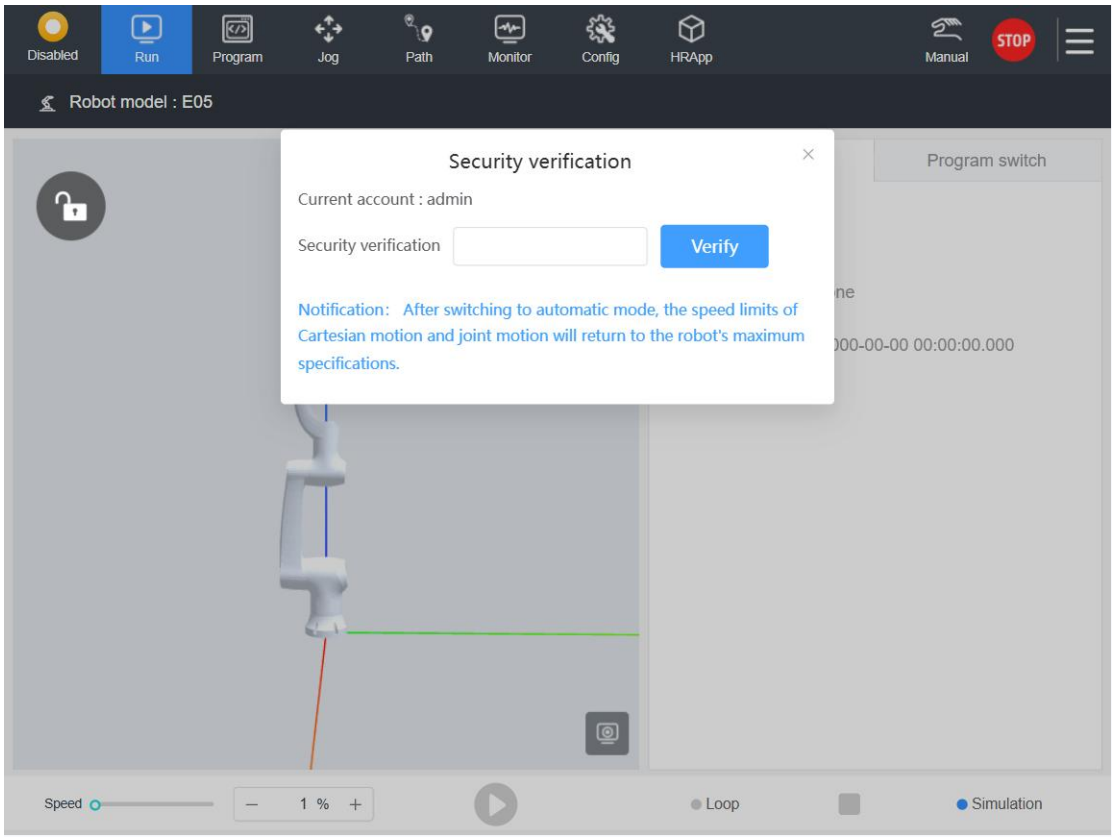


Fig. 3.3.1 Mode Switching

- When switching from auto mode to manual mode, password verification is required;
- When switching from manual mode to auto mode, password verification is not required;

Chapter 4 Jog

This section mainly introduces how to control the robot arm motion to the specified target position. This contents below can be found on the jog interface. The jog teaching interface is intended for operating the basic motions of the robot arm, including joint motion, space motion and basic free drive.

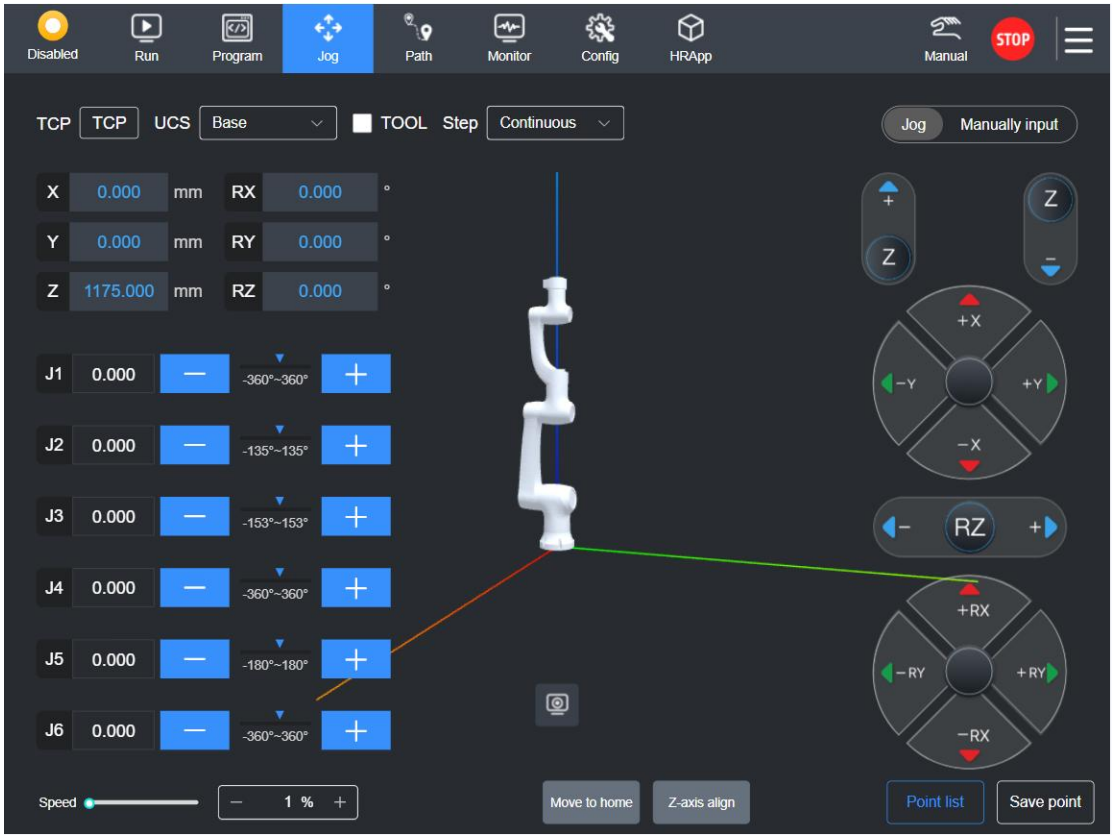


Fig. 4.1 Jog Interface

4.1. Coordinate System



Fig. 4.1.1 Coordinate System Settings

The user coordinate system currently used to control the robot arm can be defined, and different user coordinate systems are optional in the drop-down menu. The robot moves based on the selected coordinate system. If the coordinate system is selected incorrectly, the robot may perform unexpected actions, resulting in collision.

4.2. TOOL Mode

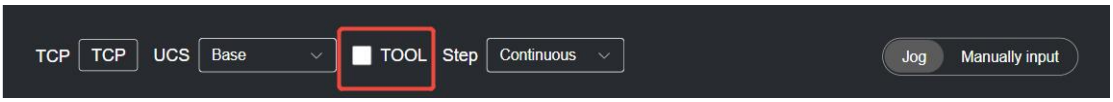


Fig. 4.2.1 TOOL Mode

When this checkbox is ticked, it enters the TOOL mode. In TOOL mode, the robot arm moves based on the currently selected TOOL coordinate system.

4.3. Step

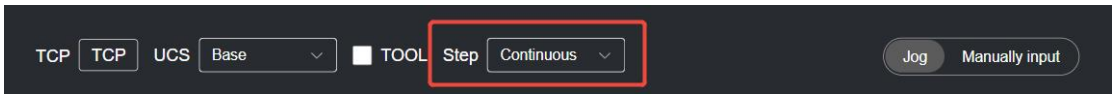


Fig. 4.3.1 Step Button



Fig. 4.3.2 drop-down menu Display

The motion step value is used to control the single-step adjustment of the robot arm. If any option other than Continuous is selected, the pose can only be adjusted in a single step. For example, if the step is 0.1, click J1- once, the value change is -0.100; and click twice, the value changes by -0.200.

The step range is 0.01 ~ 50.00.

Note:

◆ **In joint motion mode, the unit of step size is angle (°);**

◆ **In Cartesian space motion mode, the unit of step size is millimeter (mm), and the unit of pose step is angle (°).**

4.4. Speed Bar

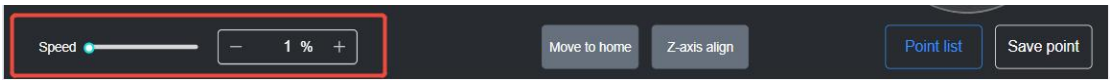


Fig. 4.4.1 Speed Bar

The speed bar shows the current speed ratio of the robot, which is restricted by the safety limits. Users can adjust the speed ratio by dragging the slider bar, clicking the +/- buttons or directly enter the parameters. The adjustment range is 1% ~ 100%. The configured speed multiplied by the speed ratio is the maximum motion speed in the current settings.

When the robot is in motion, the speed ratio cannot be set.

In the reduced mode under safety limits, the maximum reduced speed ratio is 30% by default, and the speed bar is in green.

4.5. Cartesian Space Motion

The rotation of the current robot arm pose is based on XYZ fixed-angle rotation mode.

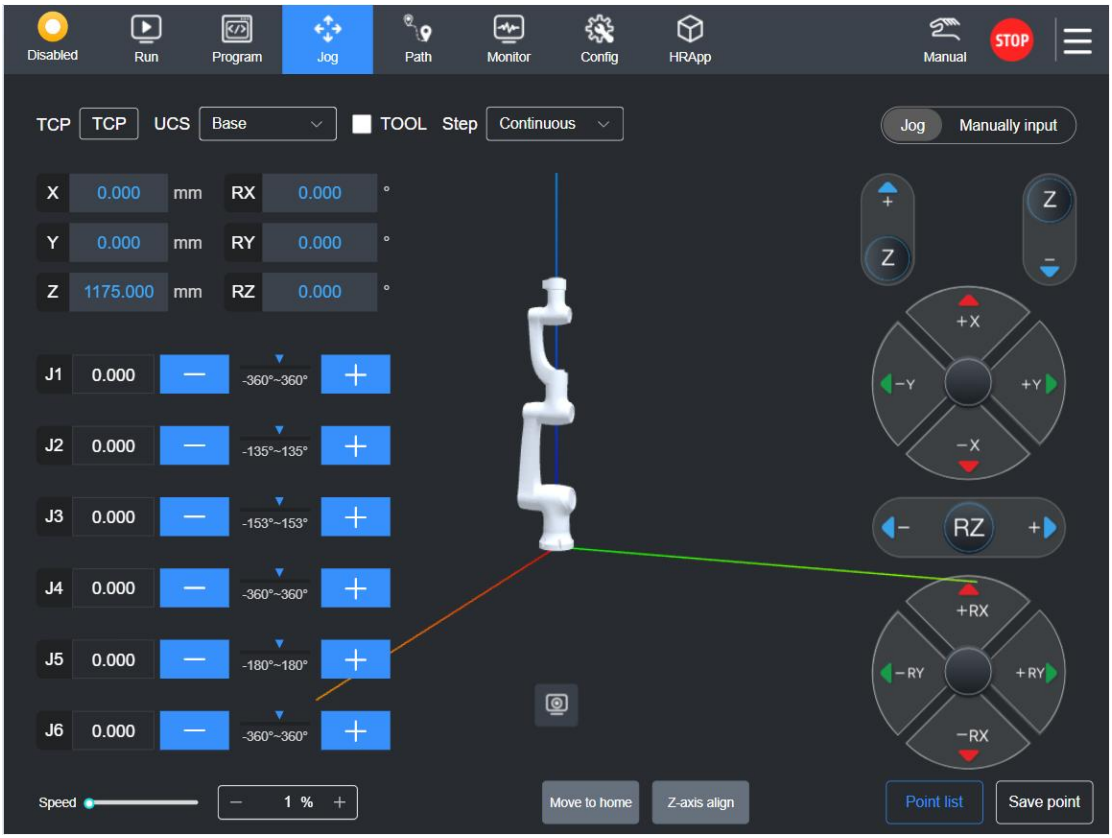


Fig. 4.5.1 Cartesian Space Interface

◆ **Definition of +/- Direction Motion Along X, Y, and Z Axes**



Fig. 4.5.2 Linear Motion Control

Within the safe space, the robot end TCP can move forward or reverse in the corresponding axis direction of the currently selected coordinate system:

- Press and Hold the Button: the robot TCP at the end-effector will continue to move in the +/- direction along the corresponding axis, and stop moving after releasing the button;
- Single Click on Button (Step Mode): If the step mode is not "Continuous",

click the button once, and the robot TCP at the end-effector will move by a step value along the +/- direction of the corresponding axial direction;

◆ **Definition of +/- Direction Rotation Along RX, RY and RZ Axes**



Fig. 4.5.3 Rotary Motion Control

RX, RY, RZ referred to here vary from that of TCP (tool center point) displayed at the top left of the page (which indicates the rotation of the TCP relative to its own tool coordinate system). Instead, RX, RY, RZ here refer to the physical rotation of the TCP around its user coordinate system, which may cause multiple angle values to change simultaneously due to rotational coupling.

RX indicates that the robot TCP at the end-effector rotates around the corresponding direction of the X axis of the current coordinate system; while RY and RZ represent rotation about the corresponding directions of the Y axis and Z axis separately.

- Press and Hold the Button: the robot TCP at the end-effector will continue to rotate along the corresponding direction of the corresponding axis, and stop moving after releasing the button;
- Single Click on Button (Step Mode): If the step mode is not "Continuous", click the button once, and the robot TCP at the end-effector will rotate by a step value along the corresponding direction of the corresponding axis;

4.6. Joint Motion



Fig. 4.6.1 Robot Arm Joint Motion Display

J1-J6 sequentially represent the 1st axis to the 6th axis of the robot arm. The text box displays the angular coordinate values of each corresponding axis position of the current robot arm, in ° (angle).

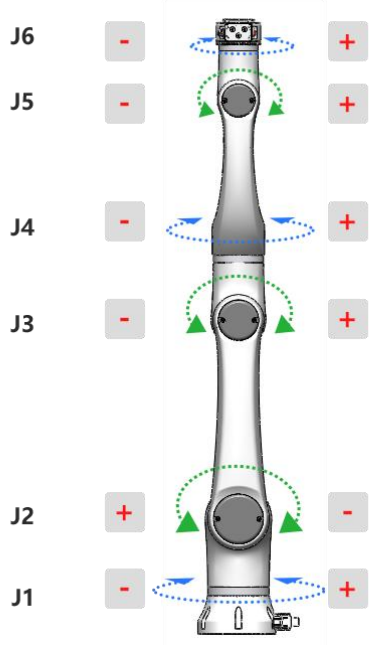


Fig. 4.6.2 Motion Directions of Each Joint

◆ **+/- Direction of Joints**

Within the safe space, the corresponding axis of the robot arm can move in the +/- direction, and the text box will display the current position of the axis in real time.

- Single Click on Button: A single click on the button, the corresponding axis moves by 2° in the +/- direction;
- Press and Hold the Button: A long press on the button, the corresponding axis continuously moves in the +/- direction, and the motion stops after releasing the button;
- Fine-Tuning Function: If "Fine-tuning" is selected, the corresponding axis will move in the +/- direction according to the fine-tuning value input by the user;

4.7. Manually Input

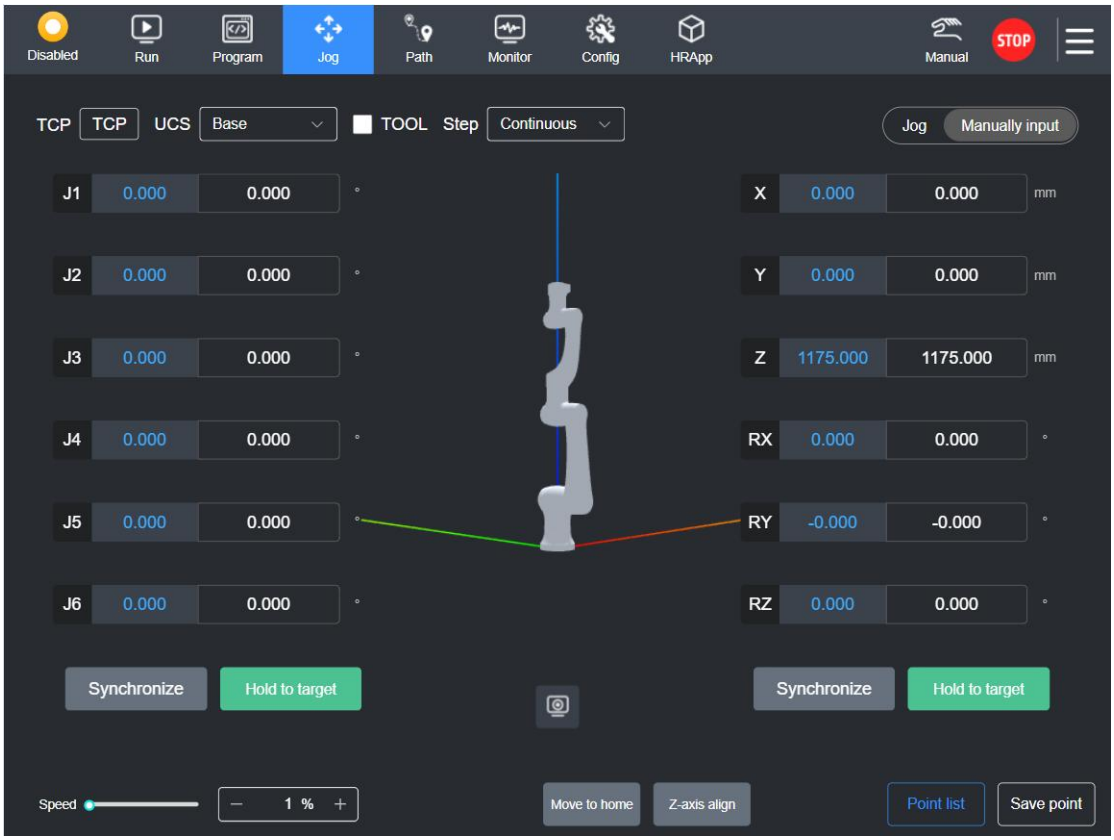


Fig. 4.7.1 Manually Input Interface

Click the 【Manually input】 button, and the user can manually enter the coordinate value of known spatial coordinate values of the point or joint angle.

4.8. Hold to Target

After entering the value, press and hold the 【Hold to Target】 button, the robot arm moves towards the target position. After reaching the target position, it prompts to notify and the robot arm stops moving.

Joint angle input: If the input joint angle is beyond the valid range, the system will prompt the "Available range" of the axis.

Space value input: Press and hold the 【Hold to Target】 button, the system starts to calculate the target point. If the robot arm cannot reach the target point, the system will prompt the error "Target point unreachable", and the user needs to re-enter the value.

If the result of the current input value is normal, and the safety range is exceeded during 【Hold to Target】, it will prompt "Target pose exceeds Cartesian safety space" and a reset prompt appears.

◆ **TOOL Mode:** After entering TOOL mode, press and hold the 【Hold to Target】 button, and the robot arm will move to the target point in the MoveL motion mode of the TOOL coordinate system;
◆ **Joint Angle:** Enter a numerical value in the joint angle text box, and then press and hold the 【Hold to Target】 button to enable the robot arm's end-effector to reach the specified target point in the MoveL motion mode;
◆ **Spatial Coordinates:** Enter a numerical value in the spatial coordinate text box, and then press and hold the 【Hold to Target】 button to enable the robot arm's end-effector to reach the specified target point through MoveL motion mode;
◆ **Hold to Target:** Press and hold the button to enable the robot arm to move to the target point. If the target point is reached, the robot arm stops moving and prompts "Target reached"; if the button is released during motion, the robot arm stops as well;
◆ **Synchronization:** To synchronize the text box data with the actual position value of the current robot arm;

4.9. Point List

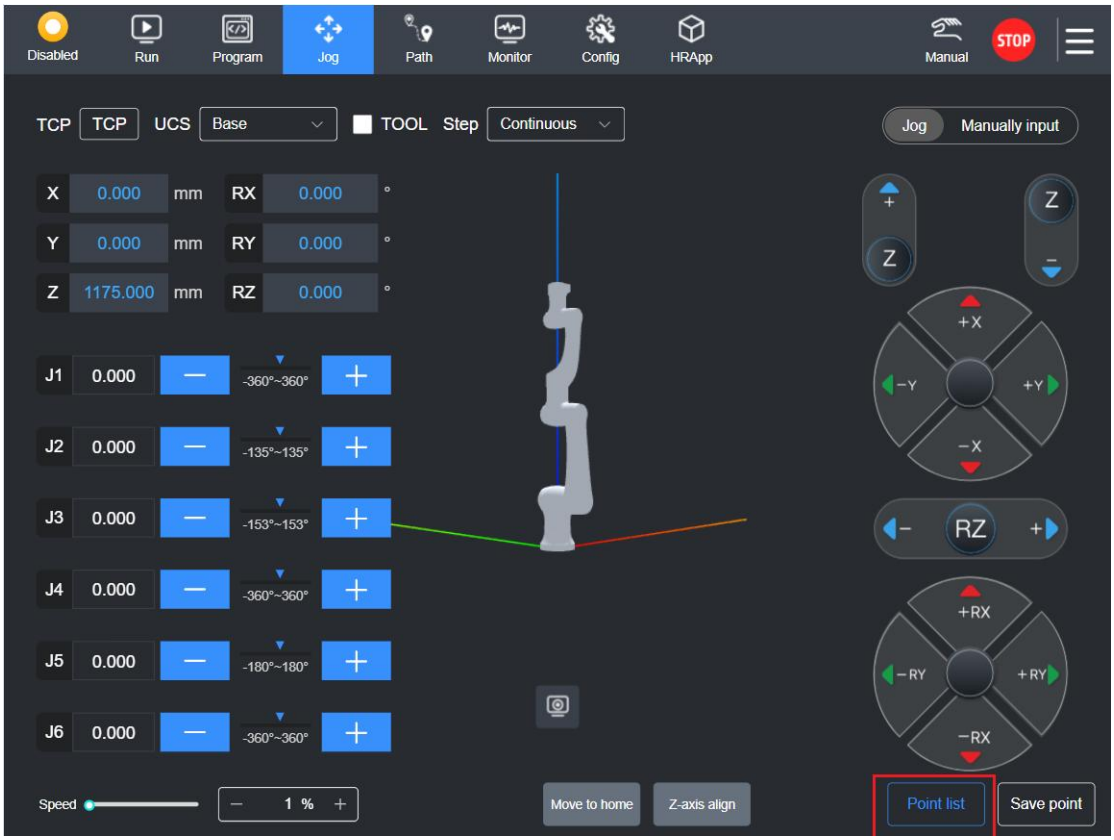


Fig. 4.9.1 Jog Interface

4.9.1. Point List Application

There may be multiple waypoints in the program that use the same point. The point can be saved to the point list in advance to facilitate subsequent direct call. To modify a point, just update the corresponding point in the point list, and the point data in the script will be updated synchronously.

4.9.2. Save Point

After clicking 【Save Point】, the current point data of the robot arm will be saved in the point list, and the point name will be saved according to the system preset. Any further modification can be done in the point list.

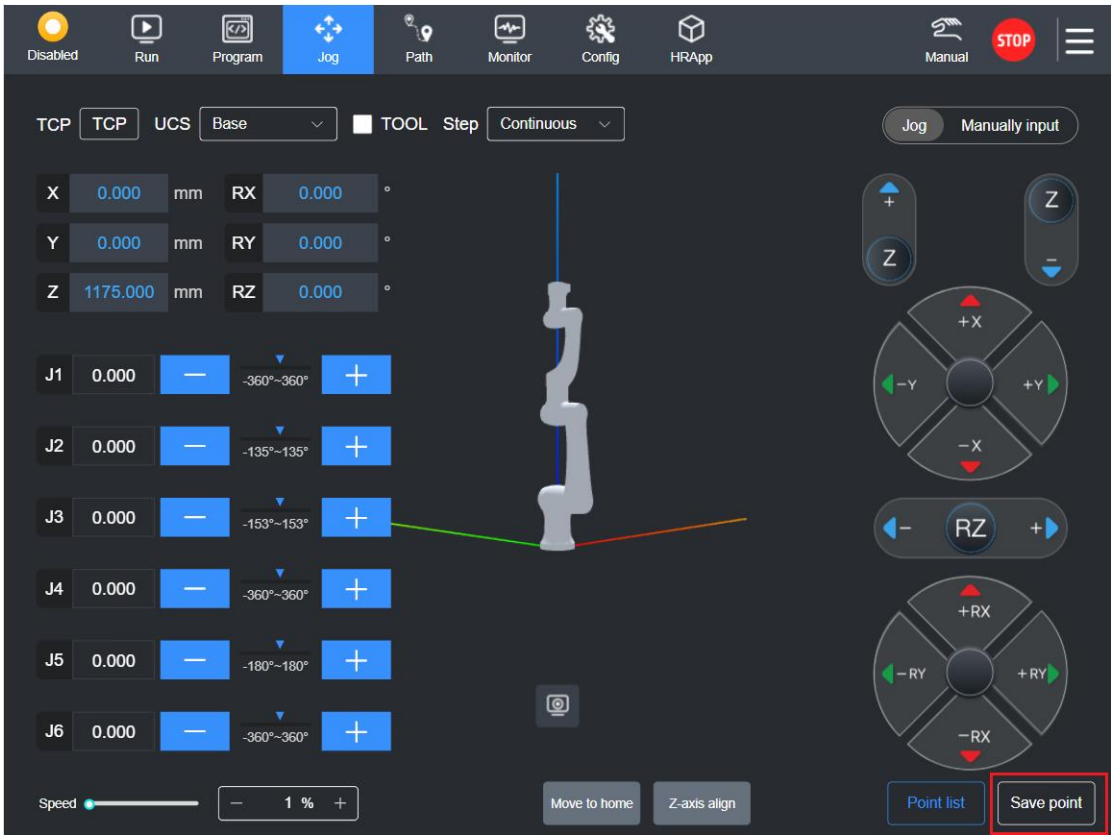


Fig. 4.9.2 Save Point

4.9.3. Point List

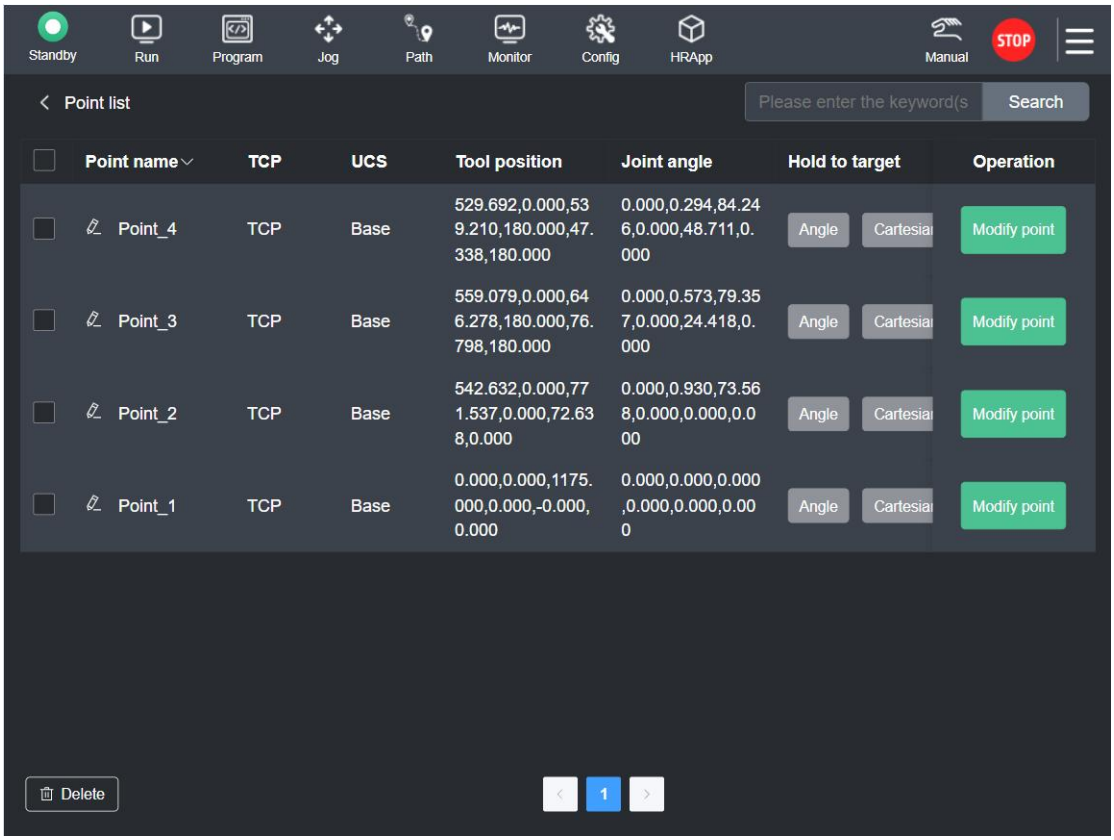


Fig. 4.9.3 Point List

The stored points are included in the point list;

- Point Name: The name of the currently saved point in the list;
  - Modify Point: Click before the point name to modify the point name;
  - Sort: Click the drop-down menu of point name to select the name sorting method;
    - Default Sort: sort according to the order of point generation time, from the latest to previous;
    - Names in Order: sort according to the order of characters in the ASCII code table;
    - Names in Reverse Order: sort according to the reverse order of characters in the ASCII code table;
- TCP: The TCP name used by the currently saved point in the list;
- UCS (Coordinate System): The name of the user coordinate system used for the saved points in the list;
- Tool Position: The saved point of the robot arm and the tool position of the robot arm's end-effector;
- Joint Angle: The joint angle parameters of the saved points in the list;

◆ Hold to Target:

■ Angle: Press and hold 【Angle】, the robot arm will move to the joint angle position of the point in the MoveJ motion mode;

■ Space: Press and hold 【Space】, the robot arm will move to the spatial position of the point in the MoveJ motion mode;

■ Modify Point: Jump to the jog interface to modify the point, and the modification is successful after saving; after modification, the waypoints using this point will be modified synchronously;

■ Delete: Tick the box in front of the point and click to delete the currently selected point;

◆ Modify Point: To modify the point information of the corresponding point; only displayed when entered from the "Jog" page.

Click the point name, and sort by default order, name order or name reverse order are available in the drop-down menu;

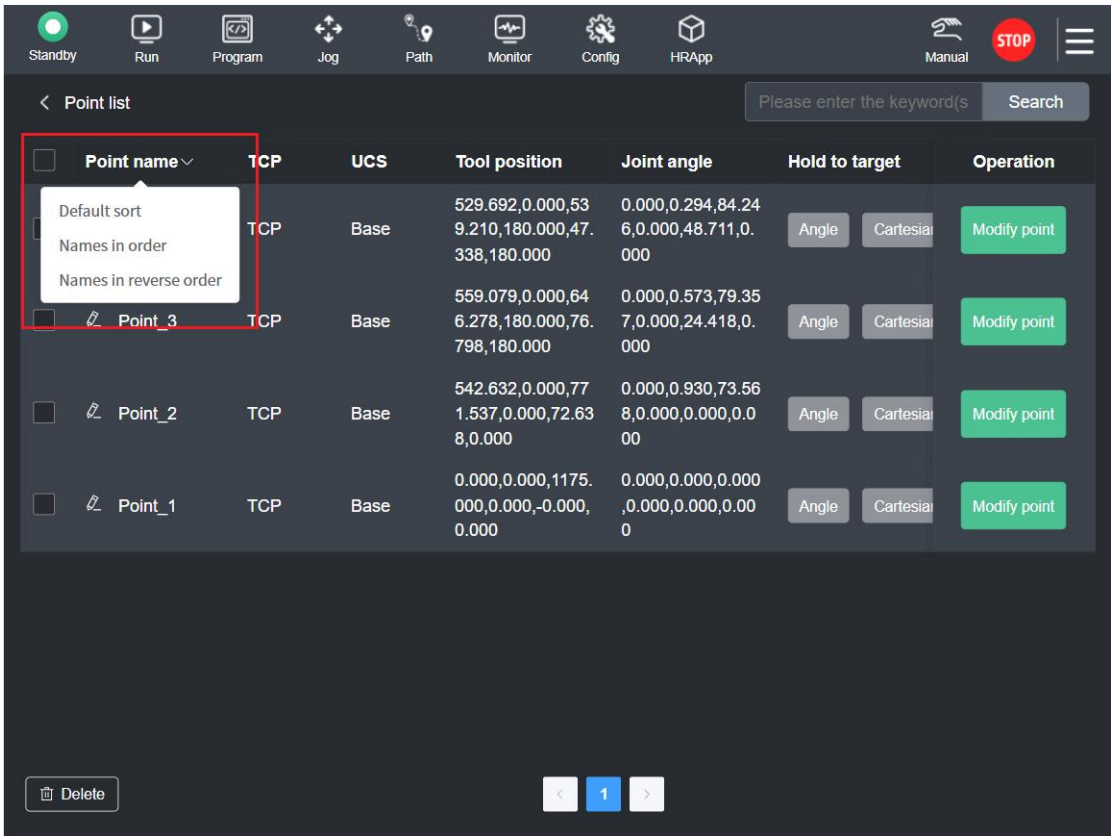


Fig. 4.9.3 Point Sort

Enter keywords in the upper search bar to conduct fuzzy search for point names. When there are a large number of points, the point search function can be of great help.

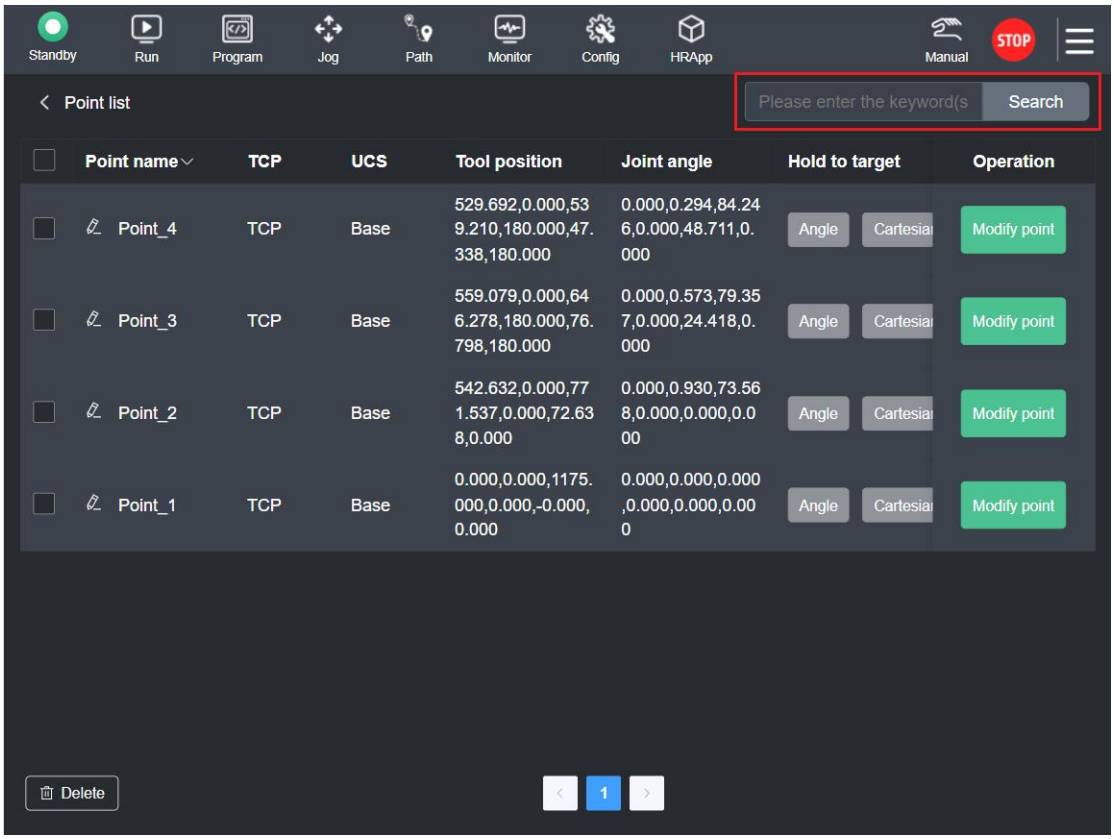


Fig. 4.9.4 Point Search

Note: If a point used by a waypoint is deleted from the point list, an error will be reported for the relevant waypoint. Therefore, before deleting a point, firstly check if it is used by any waypoint.

4.10. Move to Home

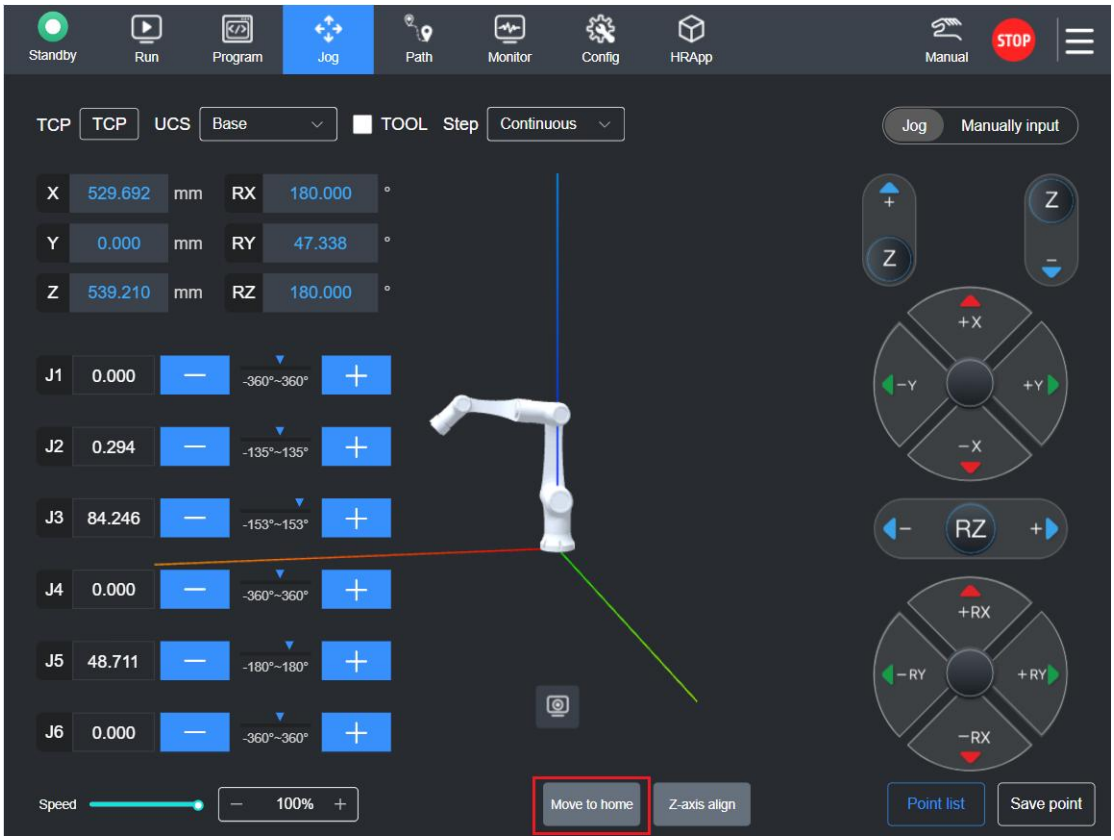


Fig. 4.10.1 Move to Home Button

Press and hold the 【Move to home】 button, the robot arm moves from the current pose to the home pose in MoveJ motion mode.

The default home pose is the mechanical zero point. For E-series robots, it is an angle of 0° for all joints; for S-Series robots, it is an angle of -90° for the 2nd and 4th joints, and 0° for the remaining joints. The home position can be modified by **Configuration-> Home**.

Note: Please pay close attention to any obstacles around the robot arm, as the robot arm may collide with obstacles during the auto homing. It is recommended to move the robot arm to a safe position before moving to home.

4.11. Z-axis Align

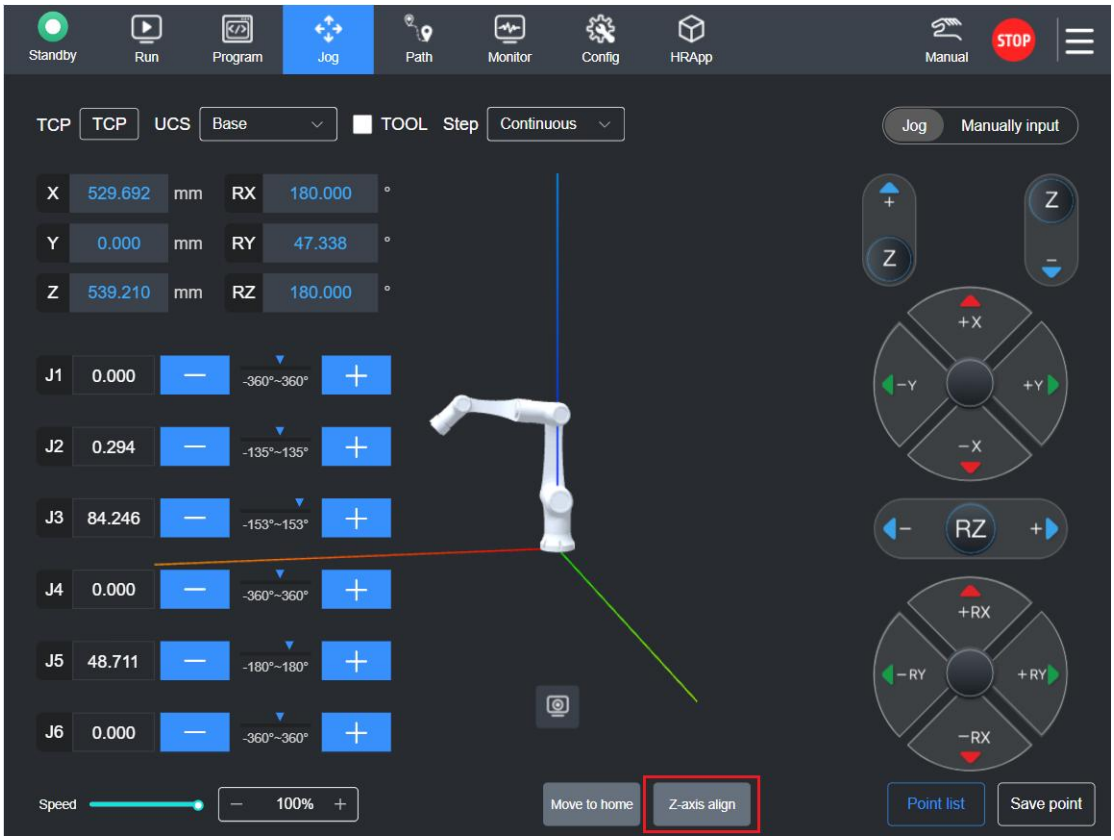


Fig. 4.11.1 Z-axis Align Button

In any pose of the robot arm (except when it is at a singularity point), press and hold the 【Z-axis align】 button, the TCP Z direction of the end-effector tool of the robot arm will move to an pose parallel to the Z direction of the user coordinates in the MoveJ motion mode. During the motion, the spatial position parameters (X, Y, Z) and orientation parameters (RX, RY, RZ) of the end-effector tool of the robot arm will change. After the motion, the spatial position parameters (X, Y, Z) of the robot arm's end-effector will be recovered to that before the motion, while the parameters of RX, RY, and RZ will change due to the pose adjustment of the robot arm.

There are three cases of Z-axis alignment directions.

a. When the included angle between the Z-axis positive direction of the tool coordinate system of the robot arm's end-effector and that of the current user coordinate system is an acute angle, after the Z-axis alignment of the robot arm, the Z-axis positive direction of the tool coordinate system and that of the user coordinate system will be in the same directions;

b. When the included angle between the Z-axis positive direction of the tool coordinate system of the robot arm's end-effector and that of the current user coordinate system is an obtuse angle, after the robot arm aligning to the Z-axis,

the Z-axis positive direction of the tool coordinate system and that of the user coordinate system will be in the reverse directions;

c. When the Z-axis positive direction of the tool coordinate system of the robot arm's end-effector is perpendicular to that of the current user coordinate system, after the robot arm aligning to the Z-axis, the Z-axis direction is uncertain, and the motion is carried out according to the principle of shortest movement distance;

Chapter 5 Configuration

5.1. TCP Setting

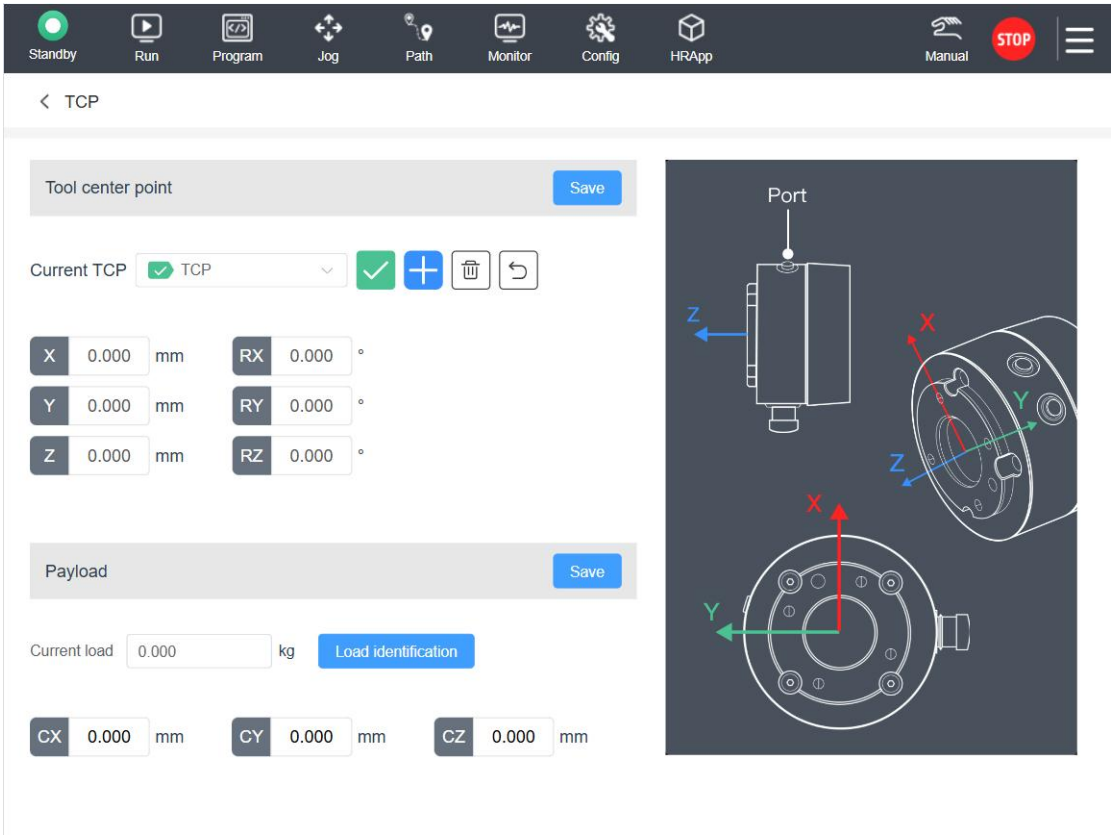


Fig. 5.1 TCP Setting

The tool coordinate point (TCP) is the center point of the tool at the robot arm's end-effector. Each TCP contains a set position translation and pose rotation relative to the center of the tool output flange.

The X, Y, and Z coordinates determine the position of the TCP, while the RX, RY, and RZ coordinates determine its orientation. When all values are zero, the TCP coincides with the center point of the tool output flange, as shown below:

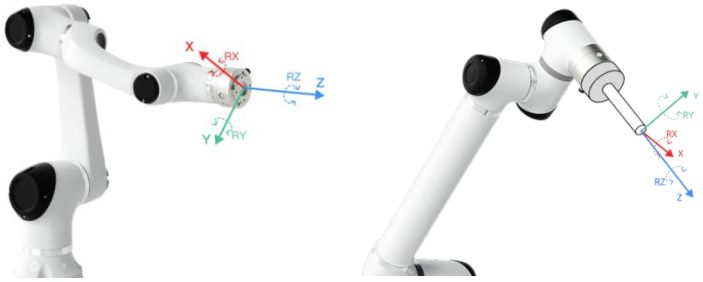


Fig. 5.1.2 TCP Demonstration

5.1.1. Create TCP

Click  , the Create TCP popup appears. There is a system preset name, while users can also customize the TCP name. It will take effect after clicking 【Confirm】. The newly created TCP will be displayed in the drop-down menu option of the current TCP. The TCP name can be entered in the input box. The name should be between 1 to 24 characters in length and can contain letters, numbers and underscores.

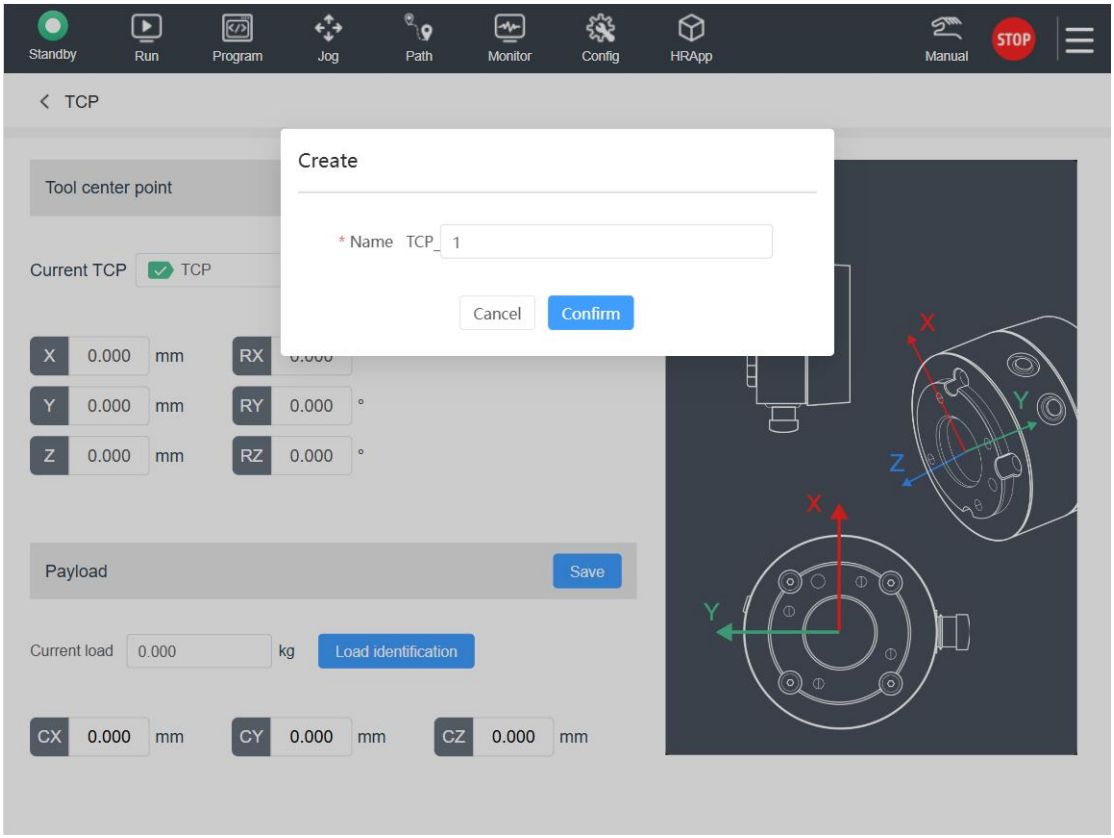


Fig. 5.1.3 Create TCP Interface

5.1.2. TCP Position Setting

The user can manually enter the value to set the TCP position, if the user already knows the offset value of the new TCP relative to the center of the flange (by measuring or viewing the size specifications). After entering the value, click【Apply】to take effect.

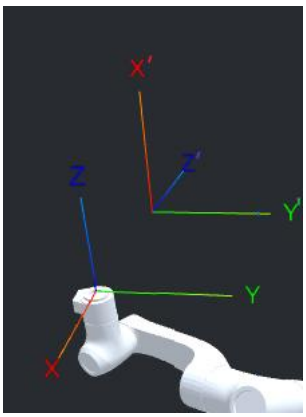


Fig. 5.1.4 TCP Offset

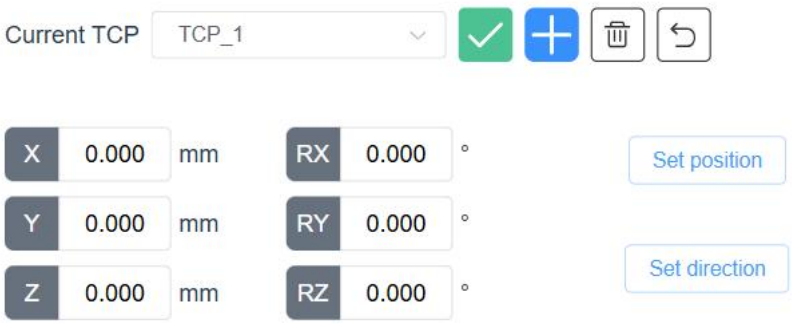


Fig. 5.1.5 TCP Position Setting

If the use does not know the offset value of the new TCP relative to the center of the flange, the origin position of the newly created tool coordinate system can be

determined by the four teaching points.

The teaching principle of the four-point method: Teaching by moving the end point of the tool from four different angle directions to the same position.

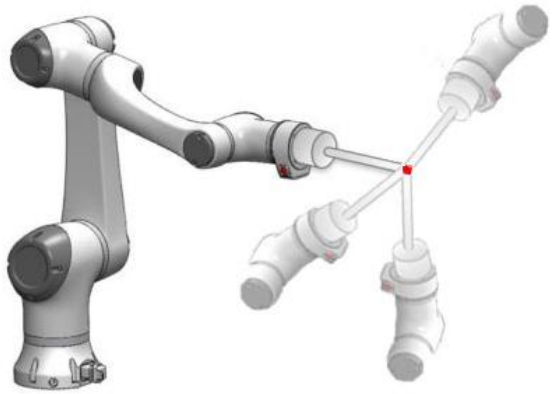
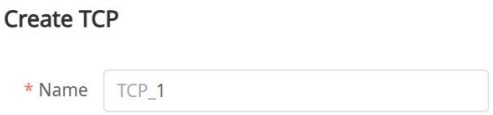
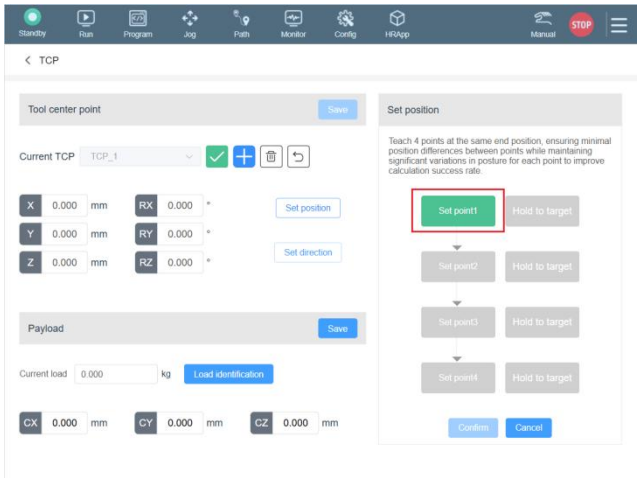


Fig. 5.1.6 4-point Teaching

Example:

SN	Operation	Example
1	Click the 【 】 button to generate TCP_1;	
2	Click the 【Set position】 button to enter the four-point teaching interface. According to the prompt, move the tool center point at the robot arm's end-effector to the same position from different angles. The upper right corner of the teaching box displays prompts during setting;	

3	When the first four points are taught, the green point button indicates that the points meet the standards; meanwhile, "High quality, available" is displayed directly above and the calculation result is displayed, indicating that it is usable;	

1) If after completing the teaching of the first three points, the Set point button becomes yellow, it indicates average point quality; meanwhile, the upper right corner prompts the calculation result, indicating that the point is available. Users can decide whether to re-teach the point of the yellow button based on the actual accuracy needs. After re-teaching, the green button color indicates improved point quality and higher accuracy of TCP;



Fig. 5.1.7 Low Point Quality

2) If after completing the teaching of four points, the Set point button becomes red, it indicates unqualified point quality; meanwhile, the upper right corner prompts "Unqualified and position cannot be calculated". The user needs to re-modify the point corresponding to the red button, so as to obtain green buttons for all or partially yellow buttons before successful setting. It is suggested that all points should be taught to become green in buttons, which means better point quality and higher TCP accuracy;



Fig. 5.1.8 Unqualified Point Quality

3) Users will continue to teach the 4th point to verify the TCP accuracy. When all points buttons are in green or partially yellow, click【Apply】to confirm the teaching positions;

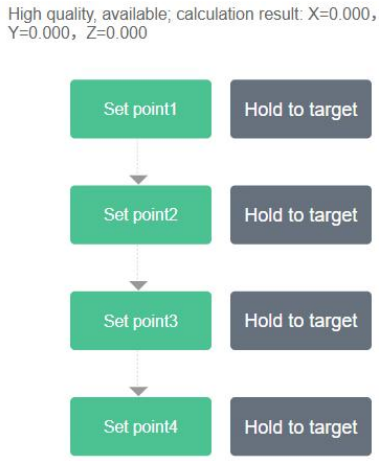


Fig. 5.1.9 Teaching Completed

4) To exit the teaching process, click 【Cancel】;

Note: The teaching points must be sufficiently diversified to ensure the accurate calculation results.

5.1.3. TCP Direction Setting

When the XYZ direction of the tool is inconsistent with the XYZ direction of the flange at the end-effector, the most obvious one is the Z direction, then the direction of the tool TCP needs to be re-taught; The orientation is set for the reference based on a coordinate system.

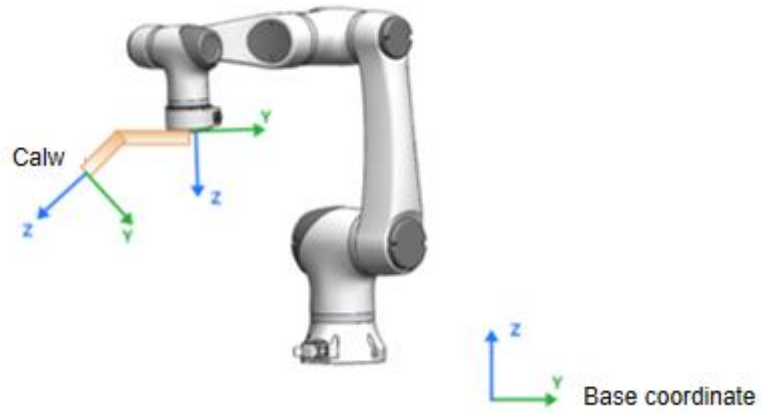


Fig. 5.1.10 TCP Direction Change

There are two ways to obtain the values of TCP position, one is by measurement, and the other is by the teaching calculation of teach pendant. The teaching calculation method is contained herein.

Teaching Calculation Method

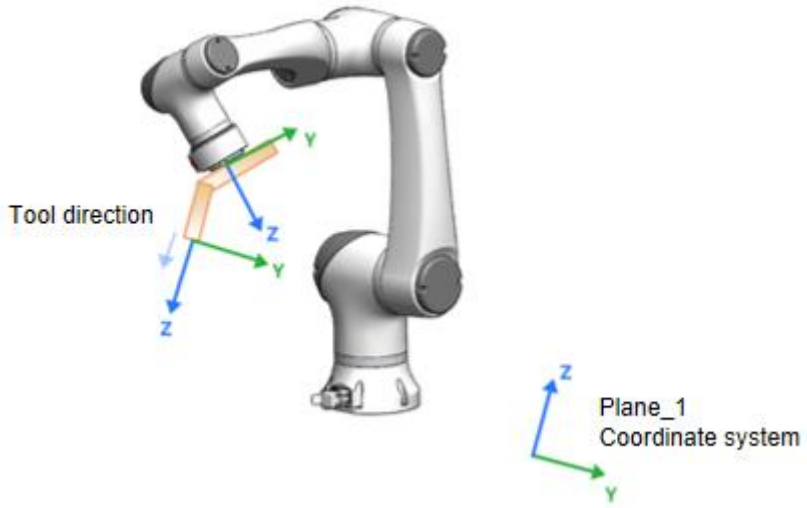
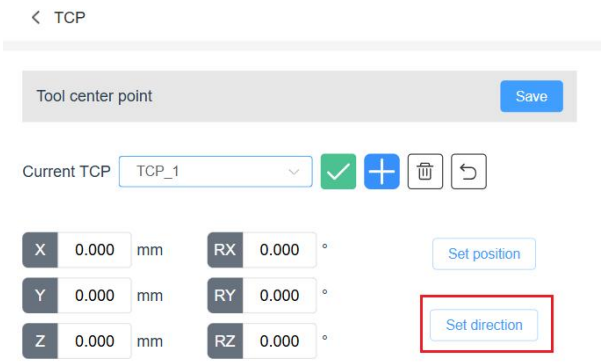
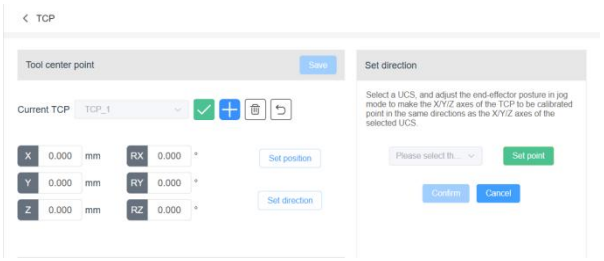
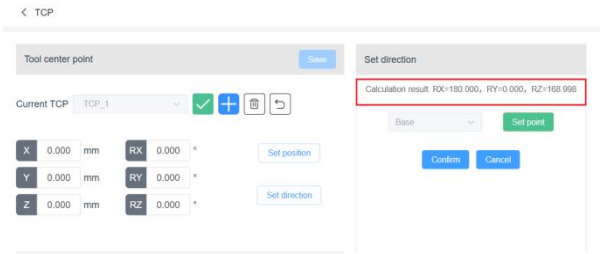
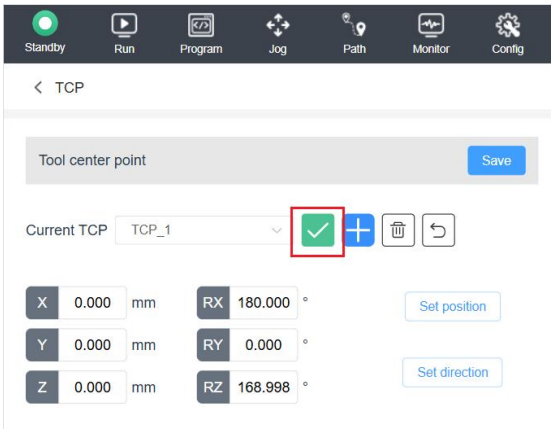


Fig. 5.1.11 Teaching Calculation Method

SN	Operation	Example
1	Click the 【Set direction】 button to enter the teaching page;	
2	Click the 【Set point】 button;	
4	In the point setting page, select a coordinate system from the drop-down list. For example, the "Plane_1" coordinate system means that the direction of the newly created TCP _2 is parallel to that of the "Plane_1" coordinate system, and the directions are completely consistent;	
5	Click【Confirm】to return to the teaching page to complete the teaching in TCP direction, and the calculation result will be prompted directly above;	

5.1.4. Default TCP

Only one of the configured TCPs can serve as the default TCP. In the TCP drop-down menu, there is a green icon to the left of TCP name set as default. To set the currently selected tool coordinates to the default TCP, click the【Set default】 button.



The TOOL mode in the jog interface uses the default TCP.

5.1.5. Effective Payload

Users can set the effective payload according to the actual load mass installed at the robot arm's end-effector and its offset from the center point of the flange in X, Y and Z directions. After clicking the corresponding input box, the virtual keyboard will pop up. Enter the correct value, then click OK. For example, the load setting range of a 5 kg robot is [0 ~ 5] kg.

Note: If the incorrect load is set, the operation of the robot arm will be seriously affected, and the functions such as free drive and force protection will no longer be accurate, and may cause danger!

5.1.6. Center of Gravity

The current center of gravity refers to that of the tool at the end-effector the robot arm. The user can specify the center of gravity position of the tool via CX, CY, and CZ. If not specified, the center of gravity of the tool is defaulted as 0. This setting is valid for all defined TCP.

5.1.7. Load Identification

The load refers to the tool at the end-effector of the robot. Without knowing the actual payload mass, the user can analyze the mass of the tool at the end-effector through the load identification function.

During the load identification, the robot arm continues to move, please reserve enough space. After clicking 【Load Identification】, the load identification popup appears. After clicking to confirm, the robot arm will start moving and load identification begins. In the process of load identification, the robot arm runs according to the established three trajectories, and the end-effector is analyzed and identified, which takes 6 minutes.

Note: Please pay consistent observation during load identification, and reach for emergency stop if there is risk of collision.



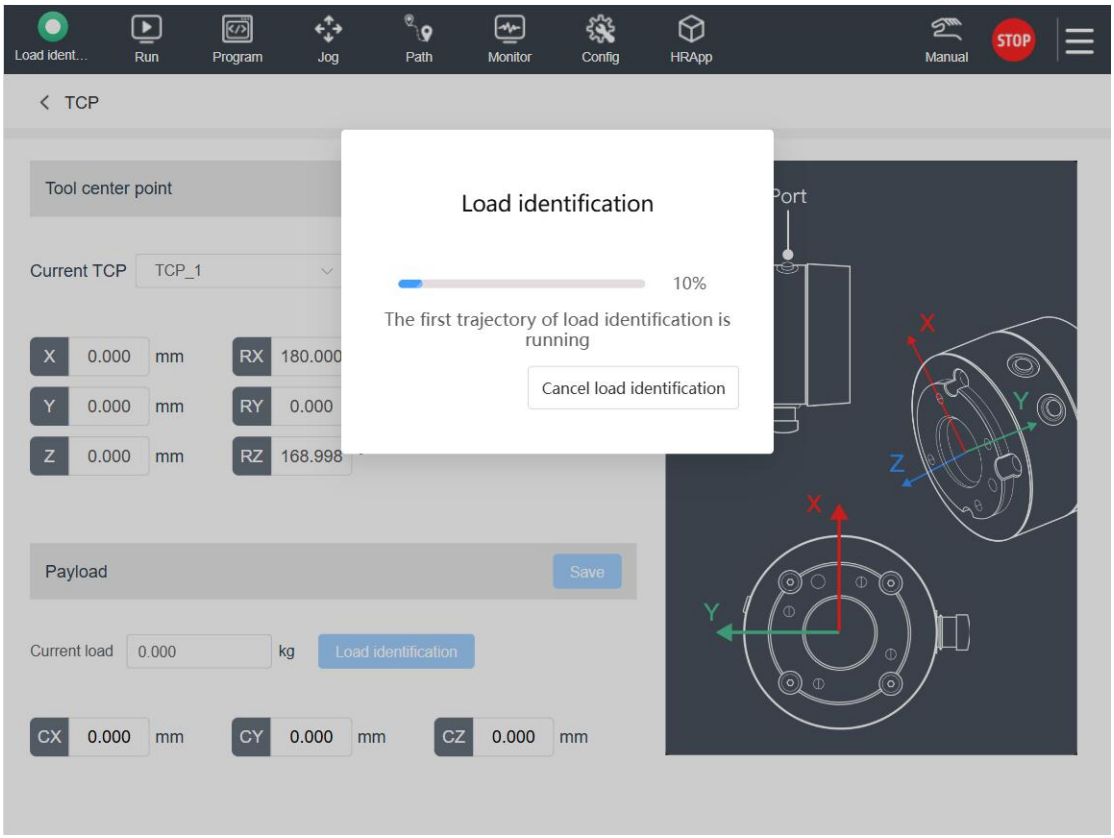


Fig. 5.1.20 Load Identification

During load identification, manually click 【Cancel Load Identification】 to exit the identification process. If an error occurs, the robot arm will stop motion and terminate the load identification, and an error will be notified for restarting the load identification.

After completing the load identification, a completion popup appears, the robot arm ends its movement, and automatically loads numerical values within the parameter settings of the effective payload.



Fig. 5.1.22 Load Identification Successful

Note: When the robot arm is idling, the parameters obtained by load

identification operation may be inaccurate, especially the centroid and mass data, but the moment (product of centroid and mass) results remain accurate.

Note: When the robot arm is set with an installation, the load identification will be unavailable and inaccurate even if identified.

5.1.8. Others

- **Modify:** After selecting the current TCP, users can modify the position and direction offset of the TCP;
- **Delete:** Delete the tool coordinates selected in the current TCP;
- **Cancel:** Cancel the currently unsaved teaching process or entered value;
- **Apply:** Click 【Apply】 to effect the edited data;
- **Set As Default:** To set a named TCP as the default TCP;

Note: If the TCP used by a waypoint is deleted, the data in the waypoint will be deleted synchronously. The waypoint becomes unusable, and the user needs to re-edit the waypoint and the corresponding TCP.

5.2. Base Mounting Angle

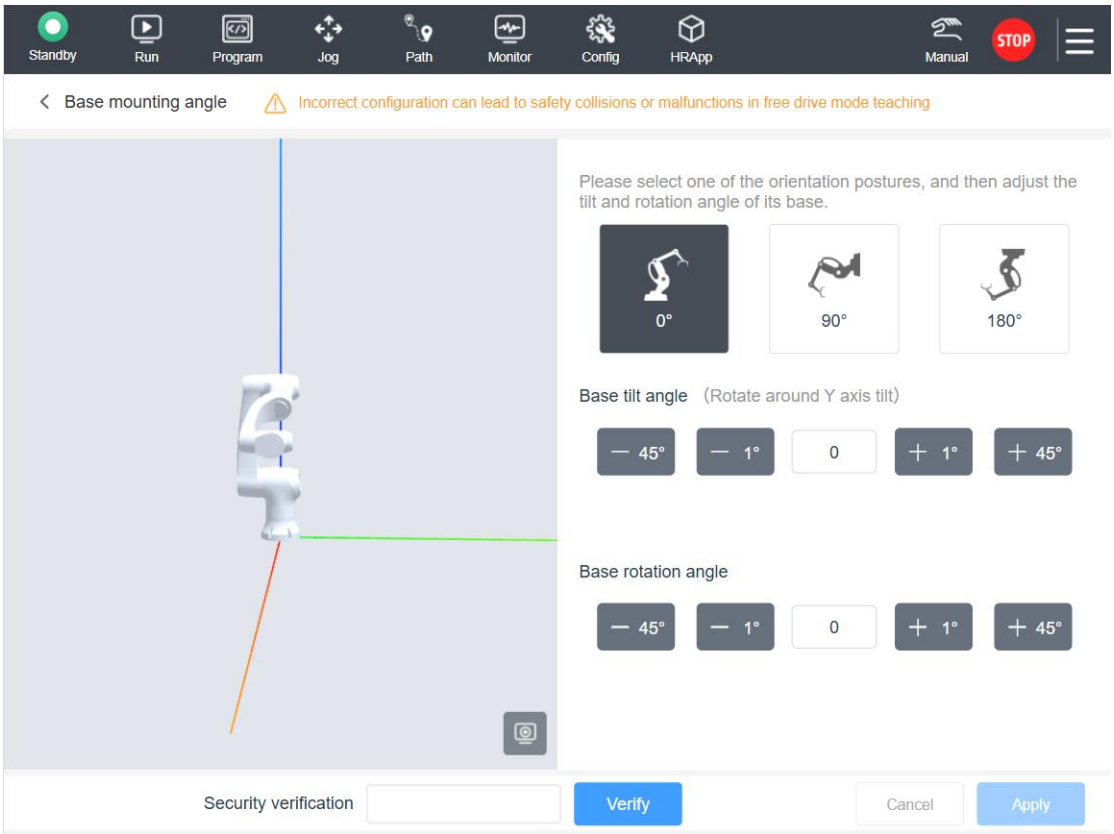


Fig. 5.2.1 Security Verification

This page configures the installation position and angle of the robot arm. The security password (consistent with the currently logged in administrator password) must be verified before operation.

The world coordinate system is a global three-dimensional rectangular coordinate system, usually referenced by the ground or installation platform, which is used to describe the absolute position and orientation of objects in space. Its origin is usually defined at the center of the robot arm base, with the X-axis point to the body cable directly, the Y-axis on the left side of the body cable, and the Z-axis vertically upward.

When the robot arm is mounted on the ground or on a table parallel to the ground, and the robot is in an initial pose, the base coordinate system of the robot is parallel to the Z axis of the world coordinate system.

If the robot arm is ceiling-mounted, wall-mounted or installed at a certain angle, the base coordinate system of the robot will no longer coincide with the world coordinate system. It is necessary to determine a new base coordinate system according to the actual installation mode. Through functions in the interface, users can redefine the user coordinate system of the robot to match the installation method, thus ensuring that the robot's motion direction and operating logic are as expected.

The controller uses advanced dynamics models to ensure the smooth and accurate movement of the robot arm, and enables the robot arm to support itself when it is in the basic free drive mode. Therefore, the installation angle and position of the robot arm must be set correctly.

Warning: Always apply the correct installation settings; otherwise, it may cause accidents, or the robot arm may lose control at the moment of pressing the free drive button!

5.2.1. Base Tilt Angle

The base tilt angle is relative to the horizontal plane.

Commonly used angles are that for ceiling (180°), wall (90°) and ground (0°). If other tilt angles are required, they can be adjusted by ±1° (hold and press/single click) and ±45° (single click). The angle for ground is by default (0°).

Tilt Angle: Rotate around the Y-axis of the world coordinate system, + corresponds to counterclockwise, the effective range is ±360°.

5.2.2. Base Rotation Angle

The base rotation angle refers to the angle at which the base rotates around its vertical line.

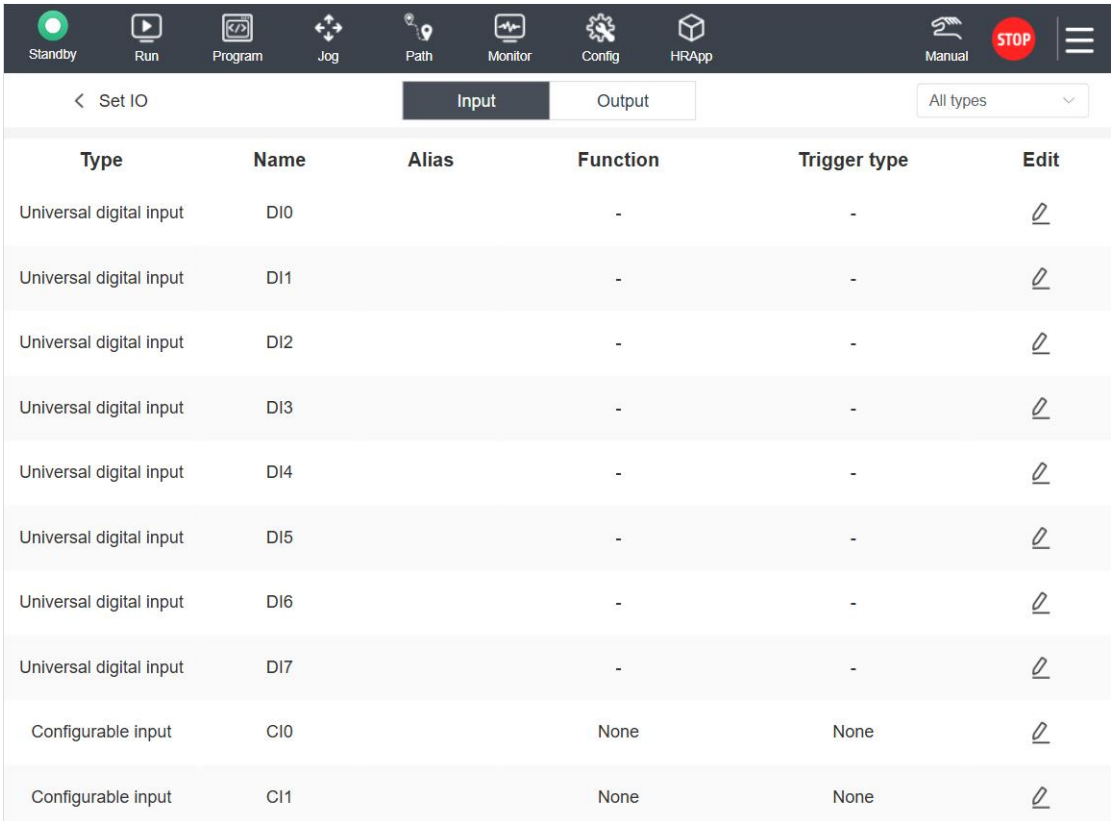
It can be adjusted by ±1° (hold and press/single click), ±45° (single click). The default value is 0°.

Rotation Angle: Rotate around the Z axis of the robot arm base coordinate system, + corresponds to counterclockwise, the effective range is ±360°.

5.3. I/O Setting

To configure input and output I/O interfaces. Input I/O includes general input, configurable input, analog input, the tool input at the end-effector, the analog input at the end-effector. While output I/O consists of general output, configurable output, the tool input at the end-effector and the analog input at the end-effector.

Among them, the "Configurable input or output" I/O can be set with aliases, functions and trigger types, while other I/O can only be set with aliases.



Type	Name	Alias	Function	Trigger type	Edit
Universal digital input	DI0		-	-	
Universal digital input	DI1		-	-	
Universal digital input	DI2		-	-	
Universal digital input	DI3		-	-	
Universal digital input	DI4		-	-	
Universal digital input	DI5		-	-	
Universal digital input	DI6		-	-	
Universal digital input	DI7		-	-	
Configurable input	CI0		None	None	
Configurable input	CI1		None	None	

Fig. 5.3.1 I/O Setting Interface

5.3.1. Configurable Input I/O

The I/O named CI0 ~ CI7 is a configurable input I/O. The input I/O signals can be set to trigger corresponding actions. Available actions include "Enable", "Clear error", "Switch to reduced mode", "Program pause ", "Program stop", "Continue program", "Run program" and "Stop motion".

Note: Each function can only be set with one I/O, and the same function is not available to multiple I/Os;

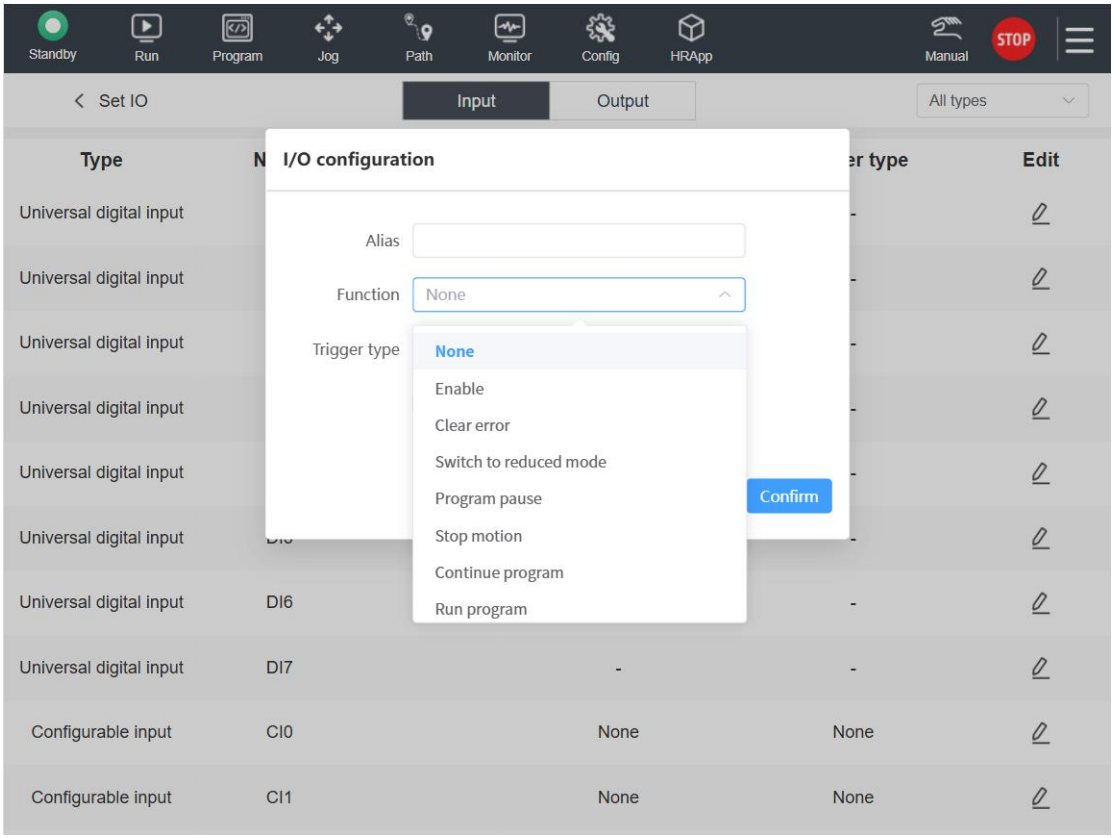


Fig. 5.3.2 Input I/O Configuration

The controller supports triggering a variety of operations through I/O signals, and the trigger type can be selected from"Rising edge" or "Falling edge". When the configuration function is enabled, the controller will continuously monitor the I/O status and execute the corresponding command when the selected trigger condition is met. The specific functions are as follows:

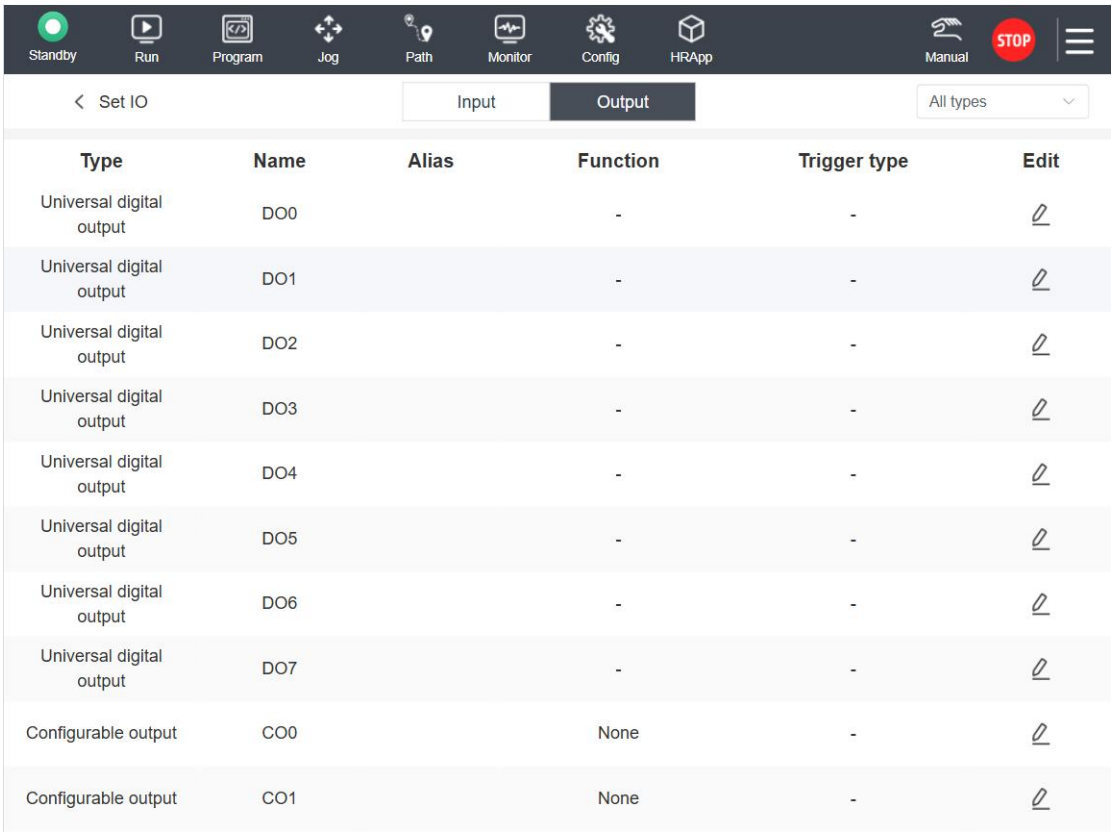
- **Enable:** Issue an enabling command after triggering;
- **Clear Error:** Issue an error clearing command after triggering;
- **Switch to Reduced Mode:** Switch to reduced mode after triggering;
- **Program Pause:** Issue a program pause command after triggering;
- **Program Stop:** Issue a stop program command after triggering;
- **Stop Motion:** Issue a stop motion command after triggering;
- **Continue Program:** Issue a continuation program command after triggering;
- **Run Program:** Issue commands to run the "Func _main" function and start the timer after triggering;
  - **Note: The following conditions must be met to use this function:**
    - **Write a function script named Func_main;**
    - **Compile the script and switch to program mode;**

For example, if the CI0 configuration activates the "Enable" function and selects the trigger type as "High level", when C/I0 inputs a high level signal, the robot is enabled; however, when CI0 inputs a low level signal, the robot is disabled.

Note: Each function can only be set with one I/O. If CI0 is configured with the "Enable" function, the "Enable" function will not be available for other CIs.

5.3.2. Configurable Output I/O

The I/O named CO0 ~ CO7 is a configurable output I/O, and the output I/O signal can be set to reflect the current state of the robot. When enabled, the robot default output is set to high.



Type	Name	Alias	Function	Trigger type	Edit
Universal digital output	DO0		-	-	
Universal digital output	DO1		-	-	
Universal digital output	DO2		-	-	
Universal digital output	DO3		-	-	
Universal digital output	DO4		-	-	
Universal digital output	DO5		-	-	
Universal digital output	DO6		-	-	
Universal digital output	DO7		-	-	
Configurable output	CO0		None	-	
Configurable output	CO1		None	-	

Fig. 5.3.3 Output I/O Configuration

The controller supports real-time monitoring of robot status and outputs high-level signals when conditions are met. The specific functions are as follows:

- Enable State: Monitor whether the robot is enabled, and output a high-level signal when the condition is met;
- Motion State: Monitor whether the robot is moving, and output a high-level signal when the condition is met;
- Error State: Monitor whether the robot is in an error state, and output a high-level signal when the condition is met;
- Reduced Mode: Monitor whether the robot is in the reduced mode, and output a high-level signal when the condition is met;
- Program Running State: Monitor whether the robot is in the program running state, and output a high-level signal when the condition is met;
- Program Holding State: Monitor whether the robot is in the program holding state, and output a high-level signal when the condition is met;
- Emergency Stop State: Monitor whether the robot is in an emergency stop state, and output a high-level signal when the condition is met;

For example, if CO4 activates the "Motion state" function, when the robot arm is in the motion state, the CO4 output signal is high; otherwise, the CO4 output signal is low.

5.3.3. I/O Configuration Editing

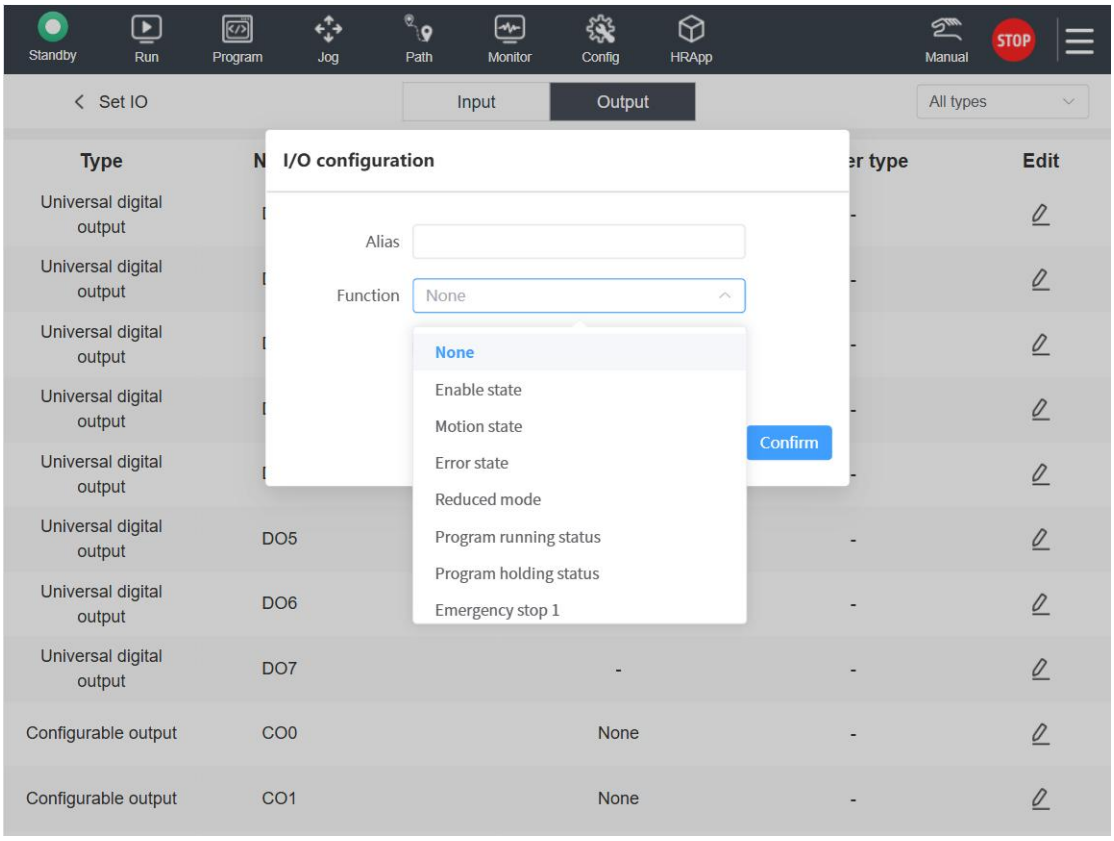


Fig. 5.3.4 I/O Configuration Editing

- **Alias:** Available for input and output signals, to facilitate memorizing the function of each signal. Click the edit button to modify;
- **Function:** Only for configurable I/O (input CI0 ~ CI7, output CO0 ~ CO7), used to configure trigger actions;
- **Trigger Type:** Determined only by the trigger action of configurable input I/O settings, matching【Rising edge】or【Falling edge】, or matching【High level】or 【Low level】 according to the corresponding action;
- **Enable:** function for configurable I/Os, unselected by default; when the "Enable" option is selected, the corresponding function is activated, and I/O information is input and output according to the selected function. If not selected, there is no difference between a configurable I/O and a general I/O. The name of the enabled I/O will be indicated by a small blue circle superscript;

Example:

Configurable Input I/O: If the CI0 configuration activates the "Enable" function, "High level", when CI0 inputs a high level signal, the robot is enabled; while when CI0 inputs a low level signal, the robot is disabled. If Enable is not selected, the function is not activated.

Configurable Output I/O: If CO4 activates the "Motion state" function, when the robot is in motion state, the corresponding CO4 is at a high level; otherwise, CO4 is at a low level. If Enable is not selected, the function is not activated.

5.4. Safety Limits

To set the general restrictions, joint restrictions, safety I/O, and safety collision of the robot arm, it requires verification of the security password (consistent with the administrator password) before operation.

The user can set the safety limit levels for normal and reduced modes in the general restrictions interface. Under normal conditions, the system will operate in one of the following safety modes, each with a safety limit level setting:

Normal Mode: The defaulted safety mode;

Reduced Mode: A mode with stricter safety restrictions within the range of normal mode. In this mode, the restrictions on speed and momentum are stricter for both the robot arm's end-effector and joints When activated, the robot arm's movement in joint space will be restricted;

Way to trigger the reduced mode: Through the configurable input I/O signals.

5.4.1. General Restrictions

Different safety levels correspond to different parameters of force, velocity, power, momentum, elbow velocity, and elbow force in the robot normal mode and reduced mode. Each parameter limits for each safety level is displayed in the text box below.

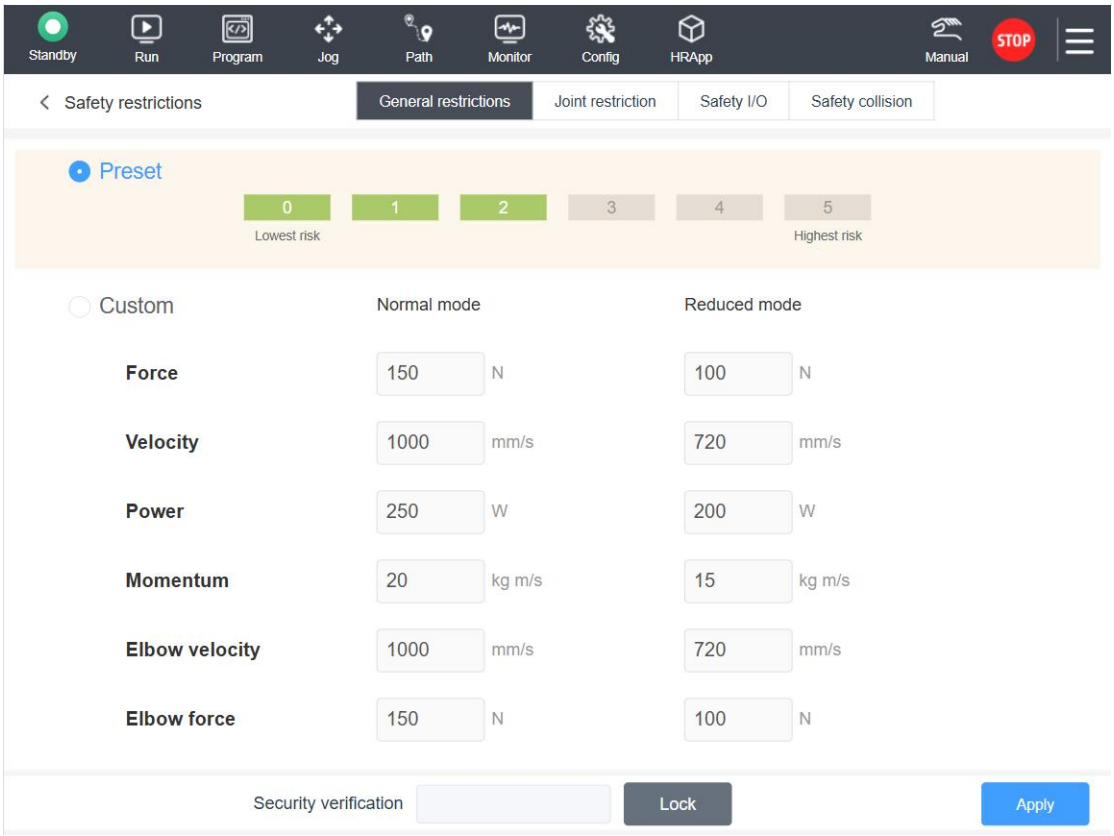


Fig. 5.4.1 Safety Level Interface

As shown in the figure above, after verifying the password, select the preset parameters to select the risk level. The parameters corresponding to each level are displayed below, including: force, velocity, power, momentum, elbow velocity and elbow force. Users need to perform a risk assessment and select the appropriate risk level according to the production environment.

Select Custom to enter the custom mode. In custom mode, the parameters can be manually modified to accommodate the actual scenarios.

Note: Parameters must not be altered arbitrarily, as this may result in significant safety risks.

- **Force:** To limit the maximum force exerted by the robot arm in the event of a collision;
- **Velocity:** To limit the maximum velocity of the robot arm;
- **Power:** To limit the maximum mechanical work done by the robot arm to the environment;
- **Momentum:** To limit the maximum momentum of the robot arm;
- **Elbow Velocity:** To limit the maximum velocity of the elbow (J3) of the robot arm;
- **Elbow Force:** To limit the maximum force exerted by the elbow (J3) of the robot

arm to the environment;

"High Risk" indicates that a smaller parameter limit is set,; for example, level 5. At this time, the robot arm has lower sensitivity to external collisions. It may require a greater external force to stop it, resulting in a lower safety factor;

"Low Risk" indicates that a large parameter limit is set; for example, level 0. At this time, the robot arm has higher sensitivity to external collisions. It may require a lower external force to stop it, resulting in a higher safety factor;

There are a total of 6 levels on the safety preset axle to choose from, and the required limit gradually increases from left to right.

After the setting is completed, click 【Apply】 to take effect. Please 【Lock】 before exiting.

Note: Do not select the stage with the least restriction unless necessary, as which guarantees low safety for the operator and robot.

5.4.2. Joint Restriction

Joint restriction is used to limit the motion of the robot arm joints in the joint space, including the joint range and joint velocity. There are also normal mode and reduced mode restrictions. The safety level limit does not affect the settings of the joint restriction.

Joint range defines the range of positions for each joint.

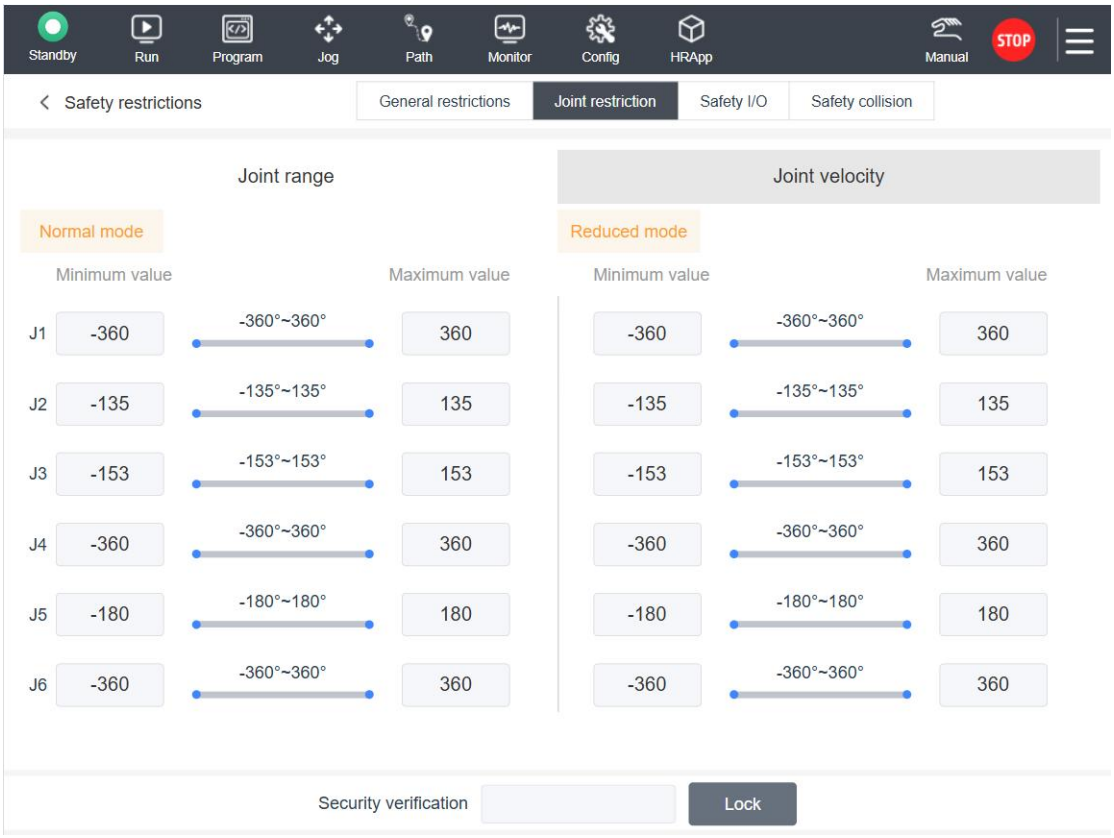


Fig. 5.4.2 Joint Range Restriction Interface

◆ The default joint angle range of the robot arm is shown in the interface.

Joint velocity defines the maximum angular velocity of each joint.

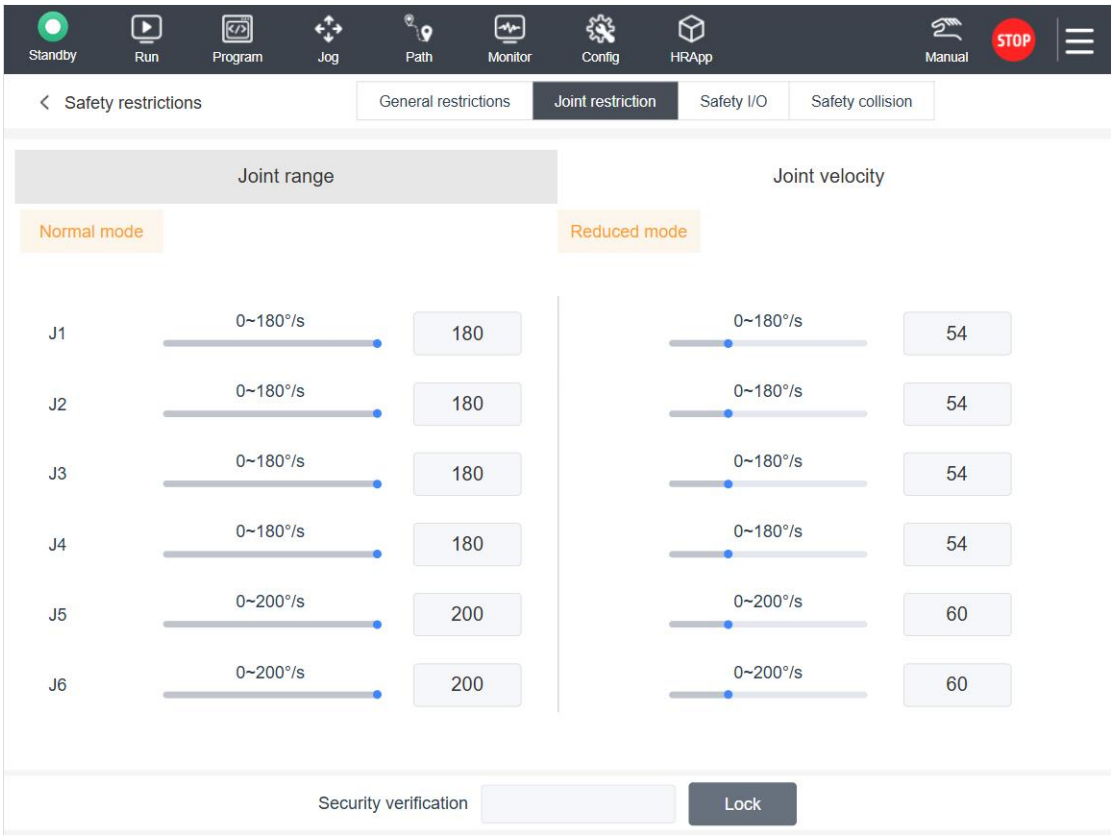


Fig. 5.4.3 Joint Velocity Limits Interface

- The default maximum joint angular velocity of the robot arm is shown in the interface.

After the setting is completed, click 【Apply】 to take effect. Please 【Lock】 before exiting.

5.4.3. Safety I/O

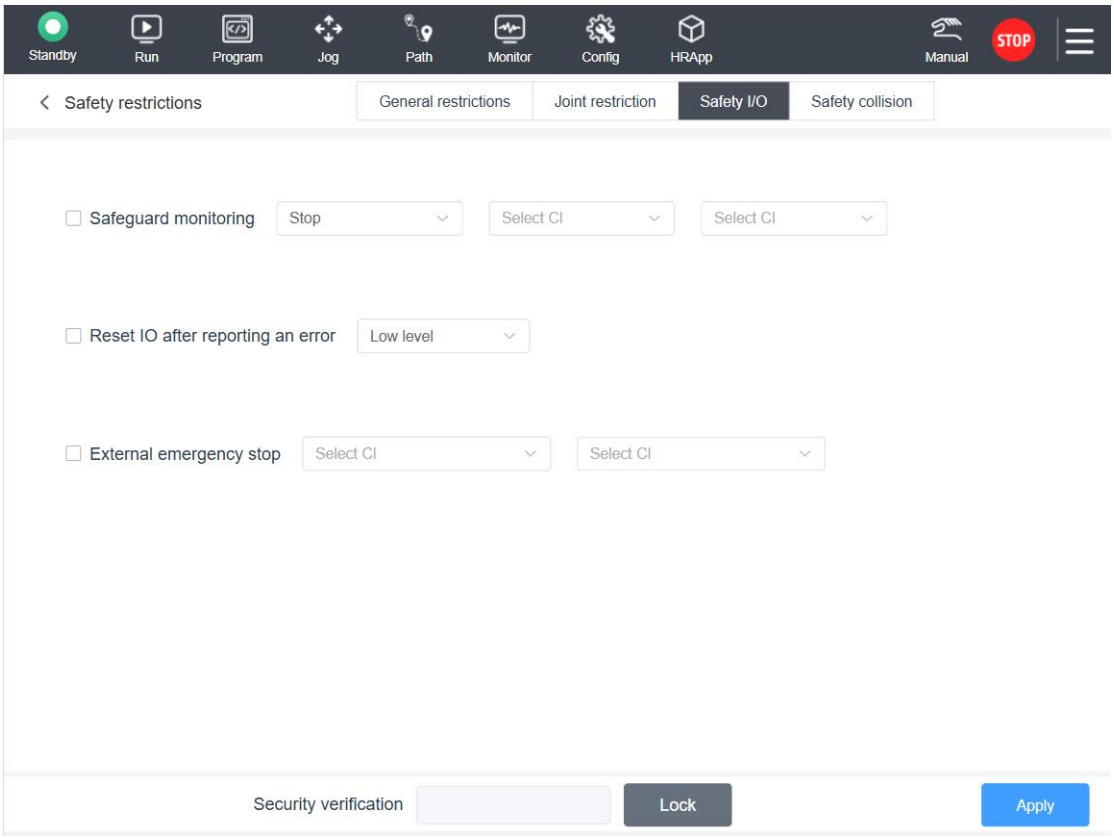


Fig. 5.4.4 Safety I/O Interface

◆ **Reset I/O After Reporting an Error:** After checking, if the robot reports an error, the output I/O of the electrical control box will be uniformly set to high or low;
◆ **Safeguard Monitoring:** Check to start safeguard monitoring, and configured to stop, pause program running or stop jogging when triggered;

【Enable Safeguard Monitoring Stop】: When the safety I/O signal is monitored, the motion stops and a popup appears to indicate the safety protection mode. The user needs to release the protective switch or turn off the signal, and then click【Reset】to clear the error before operating the robot arm again. After reset, the robot arm remains powered;

【Enable Safeguard Monitoring with Jogging Allowed】: When the safety I/O signal is monitored, the robot arm stops moving and the safeguard monitoring state is displayed, and jogging and configuration editing are allowed. The user needs to release the protective switch or turn off the signal, and then click 【Reset】 in the running interface;

【Enable Safeguard Monitoring Pause】: When the safety I/O signal is monitored, the

program will be paused. The user needs to release the protective switch or turn off the signal, and then click 【Reset】 to continue running the current program;

- **External Emergency Stop:** When selected, configured to stop or pause program running if triggered;

When the safety I/O signal is monitored, the motion is stopped and a popup appears to indicate the safety protection mode. Only when both preset detection ports are restored to the high levels can the emergency stop be reset, and then click the 【Reset】 button to operate the robot arm again.

After the setting is completed, click 【Apply】 to take effect.

Note: All types of electrical control boxes can be configured with this function; among which, the fourth generation electrical control boxes can flexibly designate a certain I/O as a dedicated I/O for safeguard monitoring.

5.4.4. Safety Collision

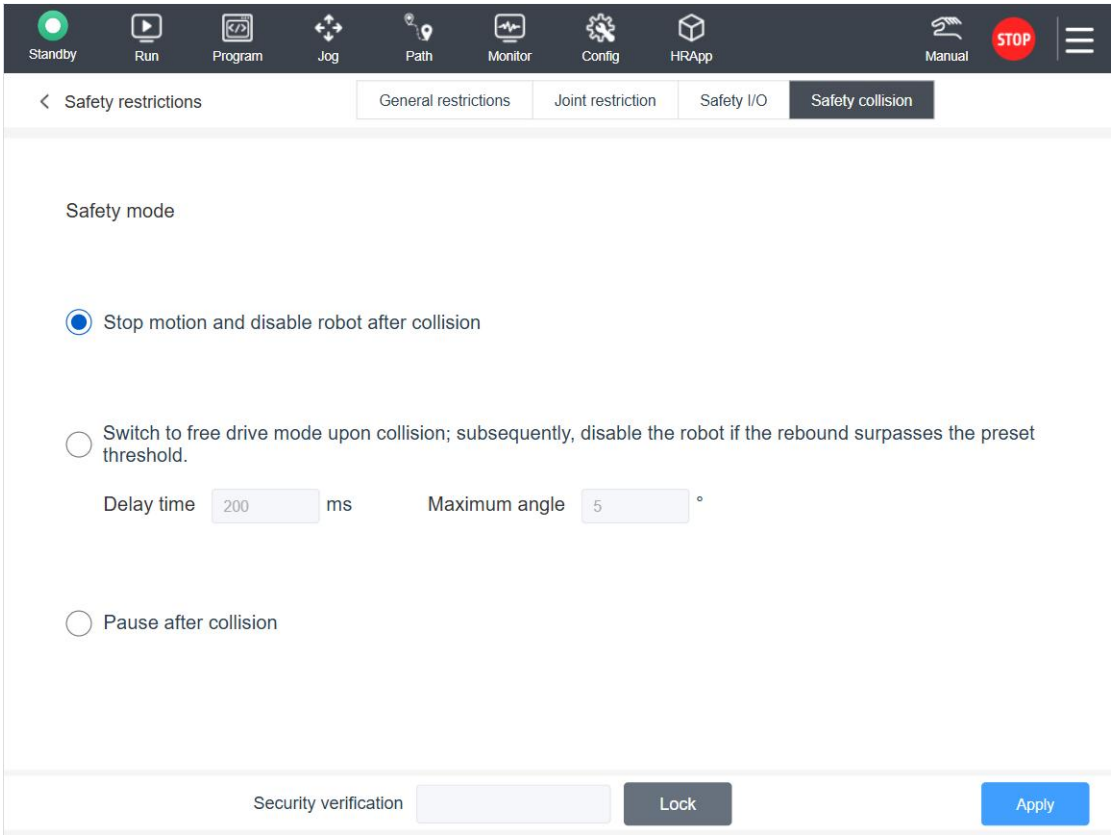


Fig. 5.4.5 Safety Collision Interface

When the robot arm encounters an obstacle, it will select different motion states

according to the force fed back by the obstacle. The threshold value of the force is determined by the level configured in the safety limits, and the user can configure the safety risk according to the actual needs and refer to "5.4.1. General Restrictions". Meanwhile, the user can also select the motion state of the robot arm after the collision. The optional modes are as follows:

1) Stop Motion and Disable Robot After Collision: After collision, the robot arm stops moving and pop up to remind the user click 【Reset】 to clear the error before re-operating the robot arm. The robot is disabled after reset.
2) Switch to Free Drive Mode Upon Collision; Subsequently Disable the Robot If the Rebound Surpasses the Preset Threshold: In this mode, the robot arm encounters and collides with an obstacle during operation and enters the basic free drive mode. However, when the rebound caused by the collision exceeds the preset value (the default value is a delay of 200ms, a maximum angle of 5°), it will stop moving and pop up to remind the user to click 【Reset】 to clear the error before re-operating the robot arm. After reset, the robot is disabled.
3) Pause After Collision: After collision, the robot arm stops moving and a popup appears to notify the user to click【Reset】 to clear the error and continue running the current program after clearing the obstacles.

The Mode 1 is the User Default Mode: "Stop motion and disable robot after collision".

5.5. User Coordinate System

The user coordinate system can be set by points, lines and planes. The user-defined coordinate system enables the robot to work in collaboration with different workbenches or fixtures.

Application Significance: In practical applications, the robot may need to work in different locations, and the ground in the new scenario may be tilted, or even the robot may be in a flipped state. Due to the tilt or rotary scenarios, the original motion path may no longer work.

In new scenarios, if it is necessary to continuously using the preset script program and ensure that the waypoints in the script remain unchanged, just change the corresponding user coordinate system to achieve this goal. The following illustration shows the user coordinate system impact on the path:

1) The original path of the robot teaching;

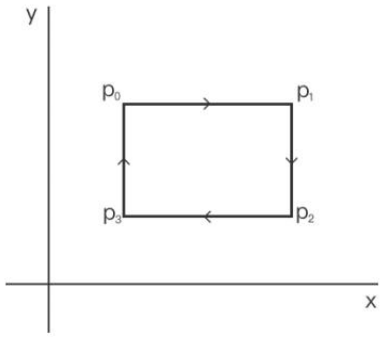


Fig. 5.5.1 Path in Default Scenario

2) After changing the robot scenario, if the current coordinate system is continuously used, the motion path of the script is as shown below, which is not desired by the user;

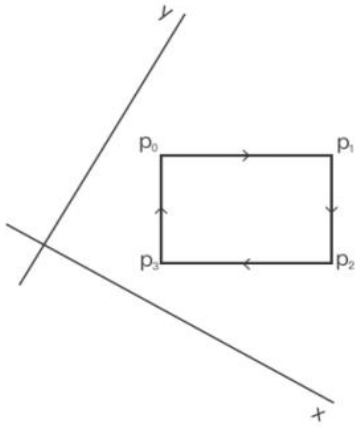


Fig. 5.5.2 Path after Changing Scenario

3) The significance of custom coordinate system: the robot shifts relatively as a whole with the change of coordinate system, and continues the original path operation;

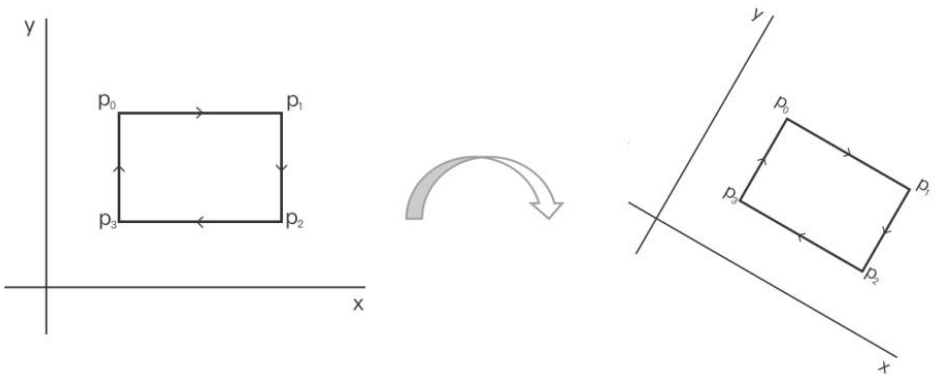


Fig. 5.5.3 Custom Coordinate System Effects

5.5.1. Point User Coordinate System

Click 【+ point】, and the "Point _N" point coordinate system for editing will be displayed in the right area, as "Point_1" as shown in the figure below: Click 【Set point】 to set the point coordinate system.

The point user coordinate system is a user coordinate system established based on a point, which is the origin of the user coordinate system; and the position and direction of the origin of the user coordinate system at this point are the values of the current TCP in the Base coordinate system.

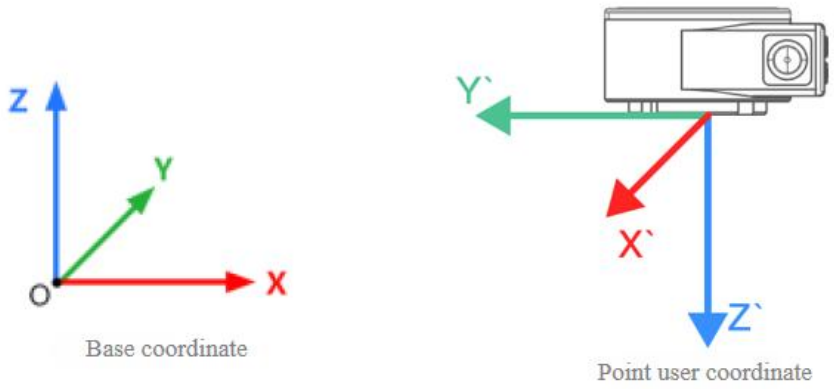


Fig. 5.5.4 Point User Coordinate System

The specific calculation steps are as follows:

1) Point1 is taught under the base coordinate;
2) Determining the position and direction of the taught point at the current end-effector;
3) The position and direction of the point user coordinate system are the values of the current TCP under the Base coordinates.

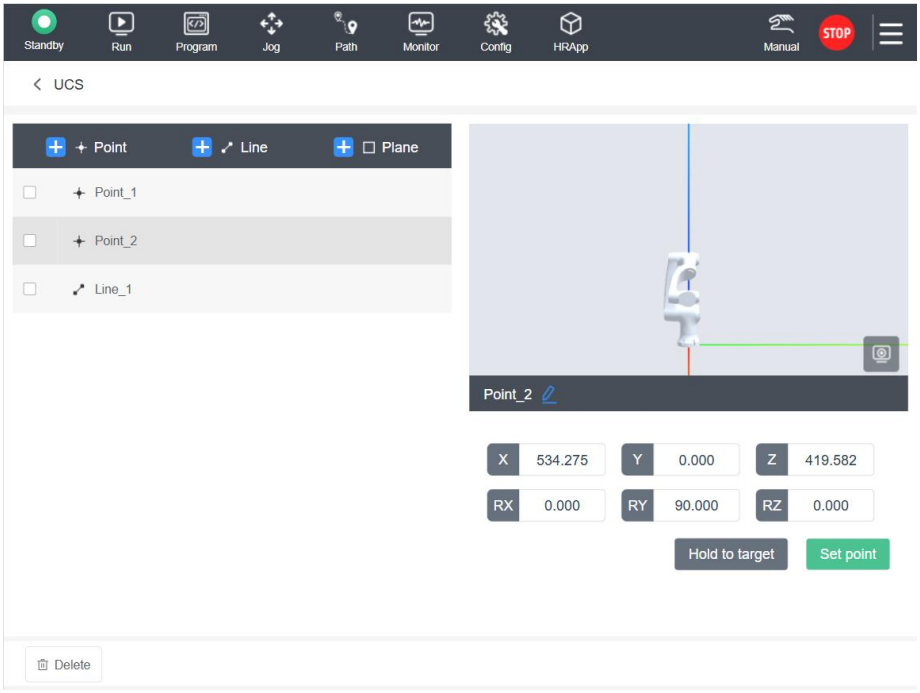


Fig. 5.5.5 Point User Coordinate System Design

- Rename: Click the edit icon next to Point_N, and enter a new name and confirm;
- Set Point: Click 【Set point】 to jump to the jog interface and set the point;
- Hold to Target: Press and hold this button to move the robot arm to the taught target point under the selected function. Once the movement is complete, the coordinate system of the function will align with the TCP (Tool Center Point) coordinate system;
- Delete: Check the coordinate system in the list and delete it; after deleting the coordinate system, if the coordinate system is used in the programs, the related parameters in the programs will be deleted synchronously;

Note: If the user coordinate system used by a waypoint is deleted, the data in the waypoint will be deleted synchronously. The waypoint cannot be called again, and the user needs to reedit the waypoint.

5.5.2. Line User Coordinate System

Click 【+ Line】, as shown in the figure below, to add the line function Line_1. The characteristics of the straight line can be calculated by two taught points, and the origin position of the line user coordinate system of the straight line is the same as that of the first taught point.

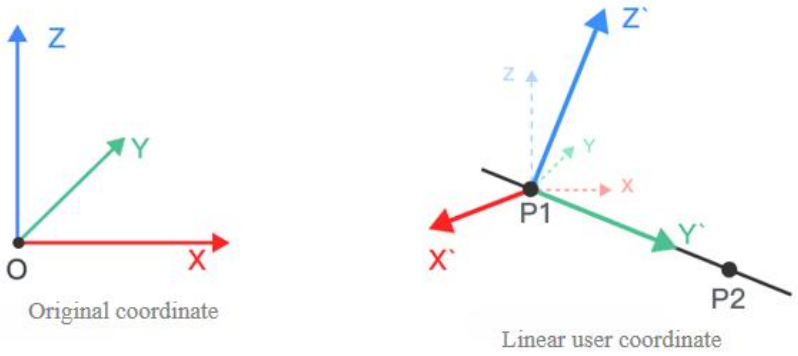


Fig. 5.5.6 Linear User Coordinate System

The specific calculation steps are as follows:

The first point P1 points to the direction of the second point P2, that is, the +Y' axis of the linear user coordinate system; the +Z' axis of the linear user coordinate system is related to the+Z axis of the original TCP coordinate.

According to the right-hand rule, the +X' axis of the line user coordinate system of the straight line can be calculated.

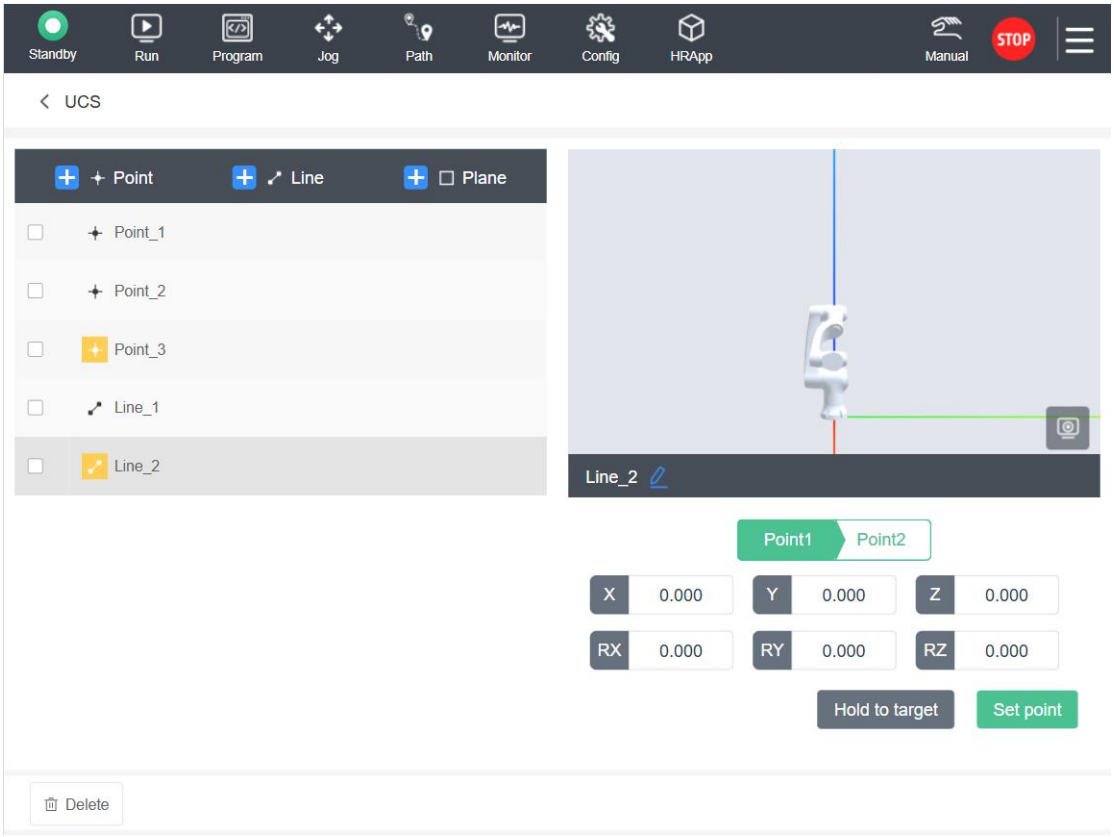


Fig. 5.5.7 Linear User Coordinate System Interface

5.5.3. Plane User Coordinate System

Click【 + Plane】, as shown in the figure below, to add the plane function Plane_1.

The plane is defined by the functions of three taught points. The position of the origin of the coordinate system is the same as that of the first taught point.

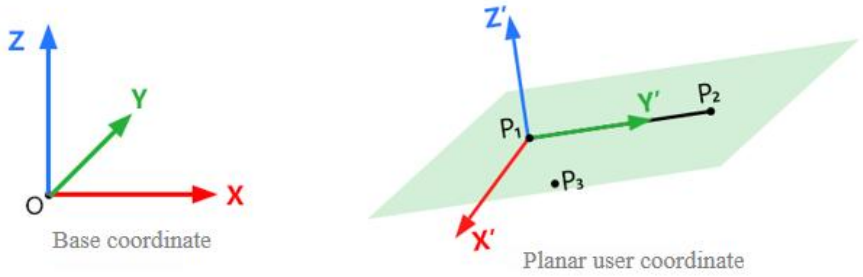


Fig. 5.5.8 Planar User Coordinate System

1) Moving the robot end-effector to point P_1;
2) Moving the robot end-effector to point P_2, the direction of P1 pointing to P_2 is the +Y axis of the user coordinate system;
3) Continue to teach point P3. At this time, the three points form a plane called Plane_1, and point P3 determines the +X axis (which is perpendicular to P_1 and P_2, and pointing to P_3);
4) The Z axis of the newly built user Plane_1 is perpendicular to the plane formed by the three point, and the +Z axis can be determined according to the right-hand rule.

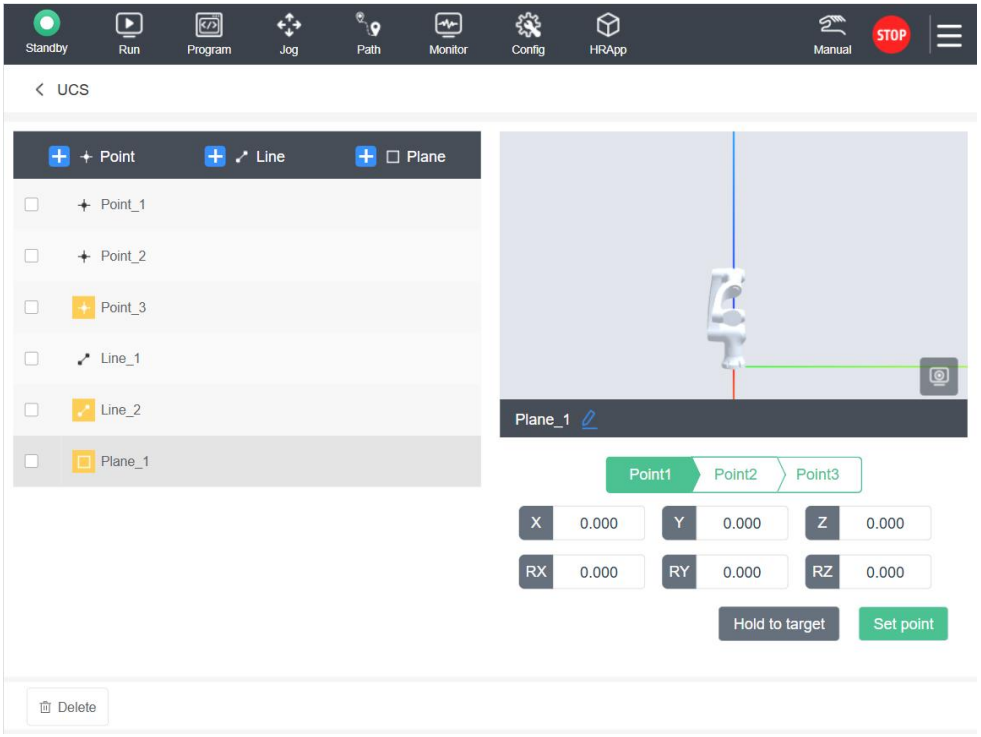


Fig. 5.5.9 Plane User Coordinate System Interface

5.6. Safety Range

The safety range can be set as a restricted area through plane mode or space mode, and the robot arm is prohibited from entering. This range further constrains the maximum space in which the robot arm's end-effector can operate.

When the robot arm's end-effector moves within the safe motion range, if the end-effector touches the restricted area, the robot arm will stop moving and report an error. Upon the occurrence of an error, click【Reset】 to retract the robot arm's end-effector and move it back to the safety zone before normal operation can resume.

When the robot arm's end-effector is within the restricted area, the system will monitor the motion trend in real time:

- Away From Restricted Area: If the end-effector motion trend is far away from the restricted area, the robot arm is allowed to continue moving until it enters the safety range;
- Deep into Restricted Area: If the end-effector motion trend is deep into the restricted area, it will be regarded as triggering the safety space restriction, and the robot arm will immediately stop moving and report an error.

Note: Security verification is required before setting.

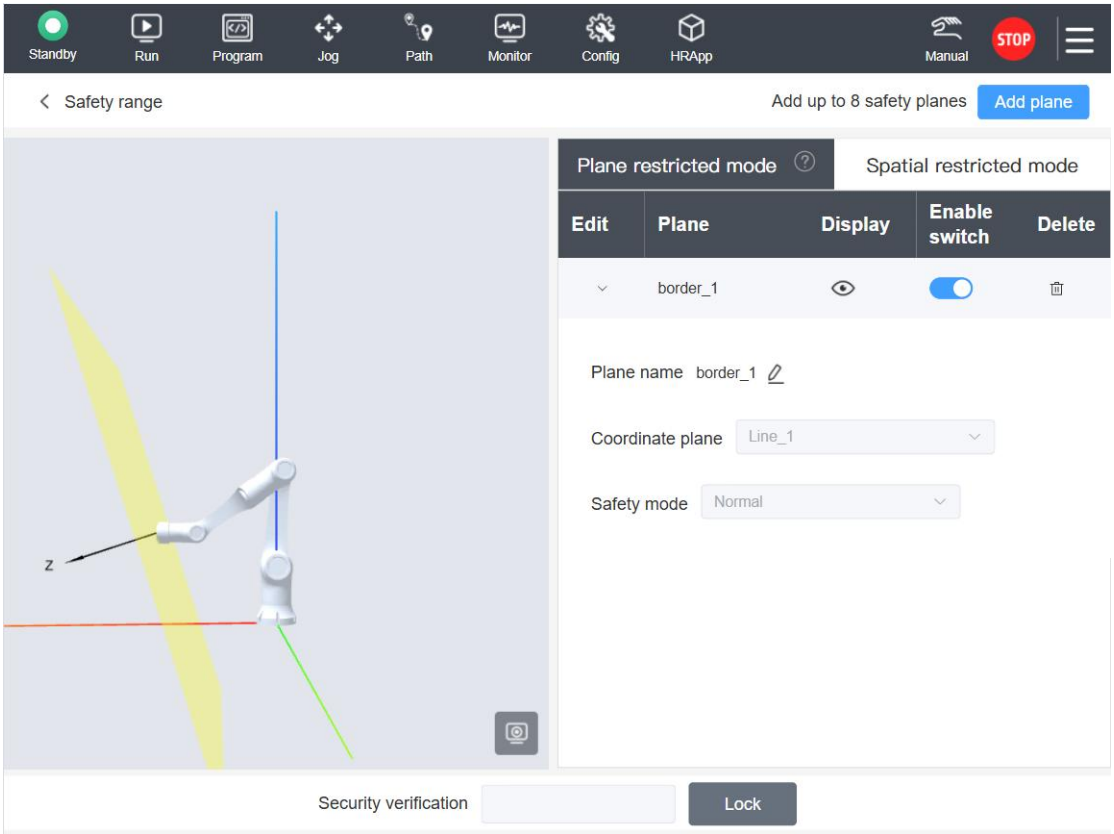


Fig. 5.6.1 Safety Range Interface

5.6.1. Plane Restricted Mode

Safety plane: A plane is defined by the user coordinate system, which is divided by zero point on the Z axis of the user coordinate system. The -Z direction is the safety area, and the +Z direction is the restricted area.

When enabled, if the robot arm's end-effector is close to or in the restricted area, the safety plane becomes red; while when the robot arm's end-effector is not close to or is not in the restricted area, or the safety plane is not enabled, the safety plane turns yellow.

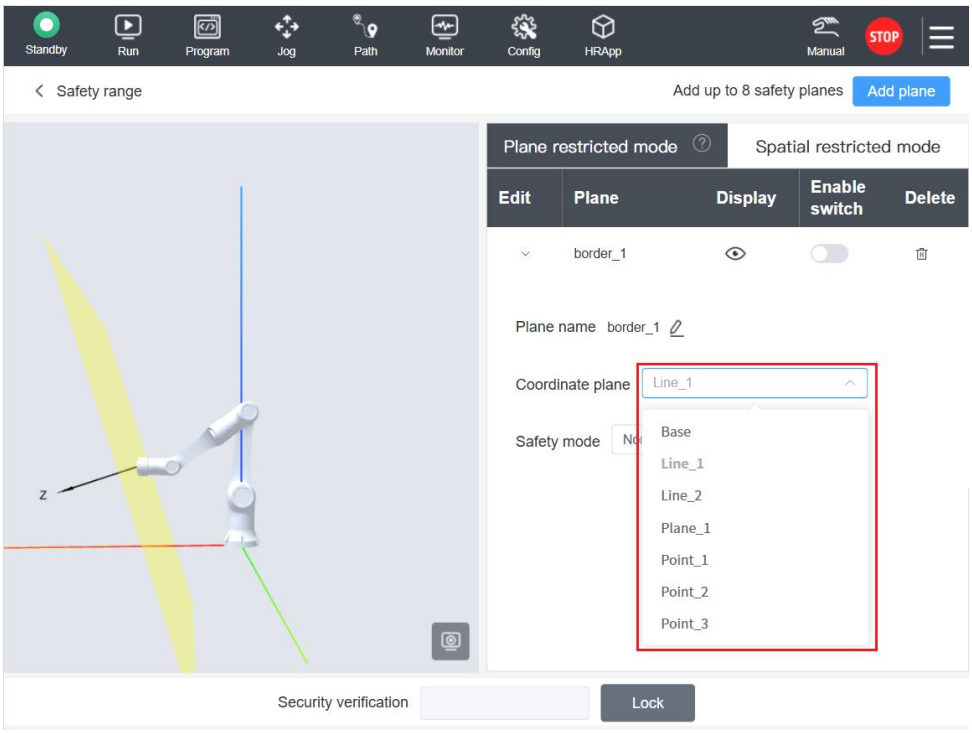
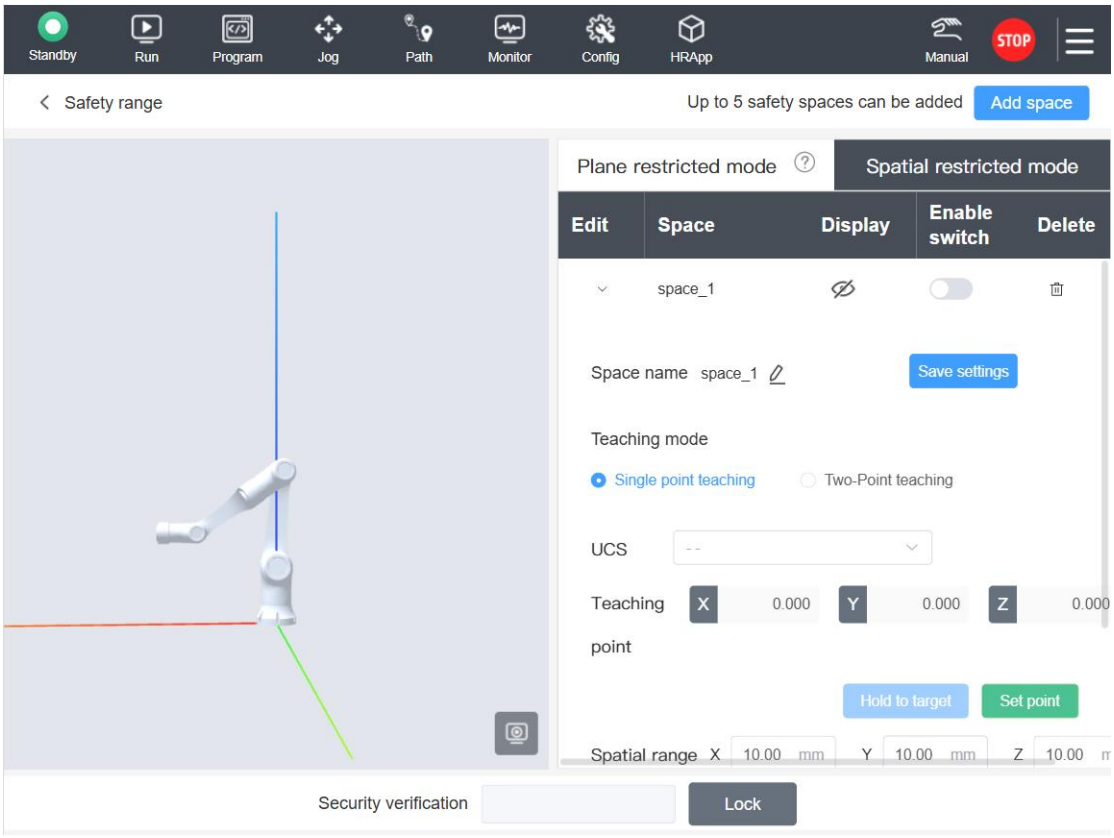


Fig. 5.6.2 Plane Mode

5.6.2. Space Restricted Mode

Within the range of motion of the robot arm, by creating a rectangular restricted space, the robot arm can only be allowed to move outside of it, which means the robot arm is prohibited from moving in it.



Restricted space can be created by teaching, which is classified as two modes, namely two-point teaching and single-point teaching.

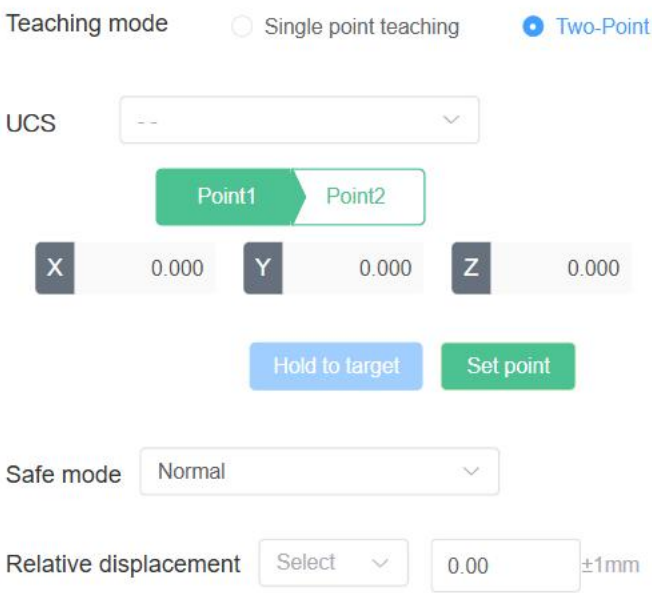


Fig. 5.6.3 Two-point Teaching in Space Mode

Two-Point Teaching: A rectangular restricted space is established by teaching two

points. When teaching points, they need to be in the same user coordinate system, and the user coordinate system is displayed in the user coordinate system list of the interface.

After teaching two points located in the same coordinate system, the system typically constructs a cuboid with these two points as diagonal points to define a restricted space range.

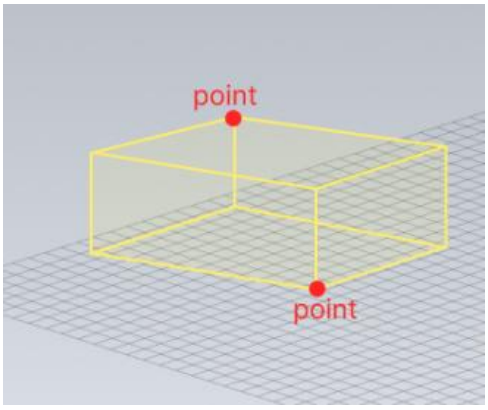


Fig. 5.6.4 Two-Point Teaching Space

In the process of teaching, the user must complete the teaching of point 1 before teaching at point 2.

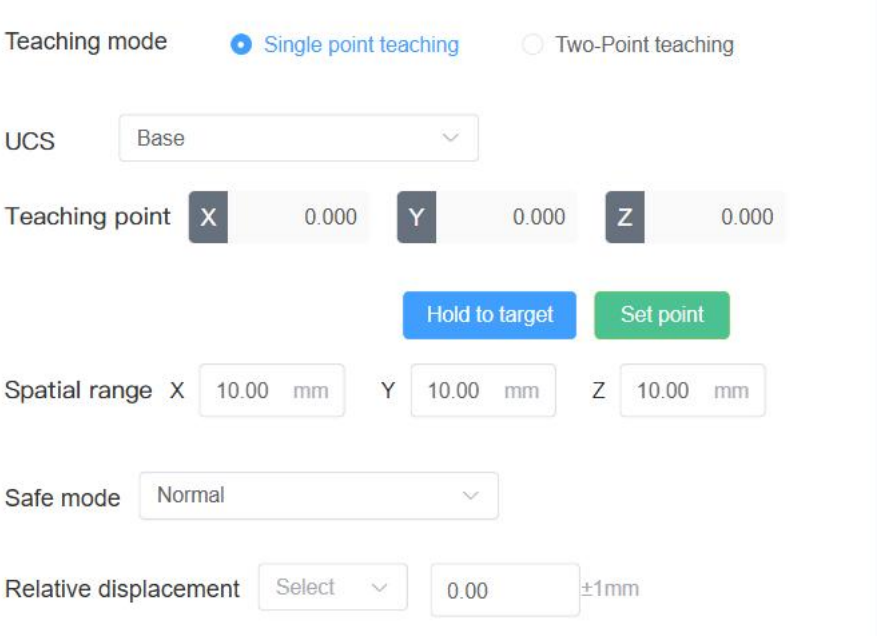


Fig. 5.6.5 Space Mode Single-Point Teaching

Single-Point Teaching: After teaching a point, the extension ranges of the XYZ axes of the current point is set, and the teaching of the restricted space is completed.

In the teaching process, the points are taught based on the user coordinate system. The system typically constructs a cuboid by using these two points, the taught point and the origin of the coordinate system, as diagonal points to define a restricted space range. This cuboid space corresponds to the current teaching - defined restricted space.

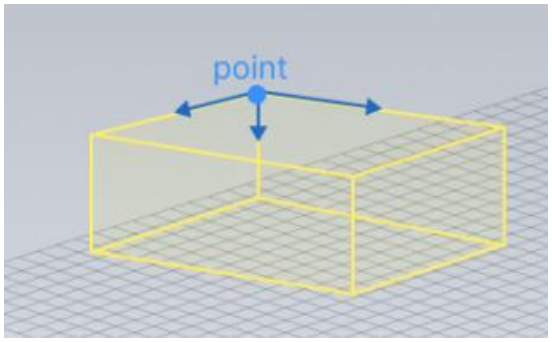
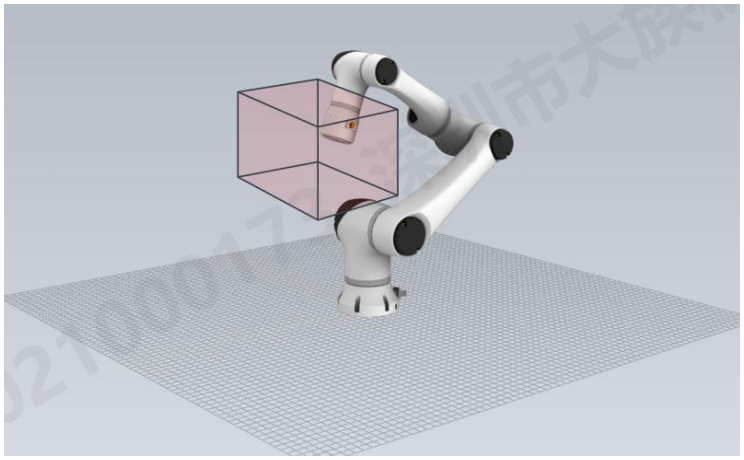


Fig. 5.6.6 Single-Point Teaching Space



End-Connector Contact Space

- When the robot arm's end-effector position is in a cuboid restricted space, the system will mark the center point of the cuboid space. If the motion trend of the robot arm's end-effector is further away from the center point of the cuboid space, it will be regarded as moving away from the cuboid and can move normally to a safe range.
- If the motion trend of the robot arm's end-effector moves toward to the center point of the cuboid space, it will be regarded as triggering the safety space restriction, the robot arm will be stopped and an error will be reported.

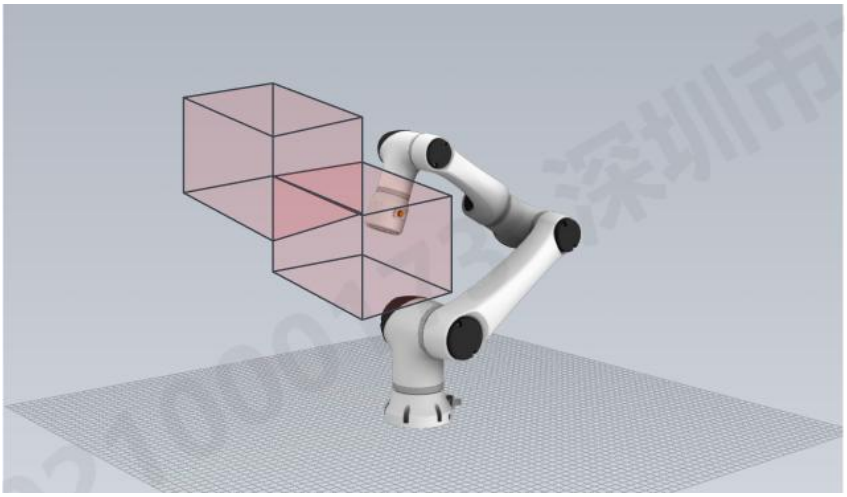


Illustration of Multiple Overlapping Spaces

- When the robot arm's end-effector is within a compound space formed by overlapping multiple cuboid spaces, it is regarded as a safety space error. The compound space's restrictive will be disabled, and. the robot arm can move freely. To re-enable the restrictive function of the compound space, move the robot arm back into the safety range.

5.7. Communication (Modbus)

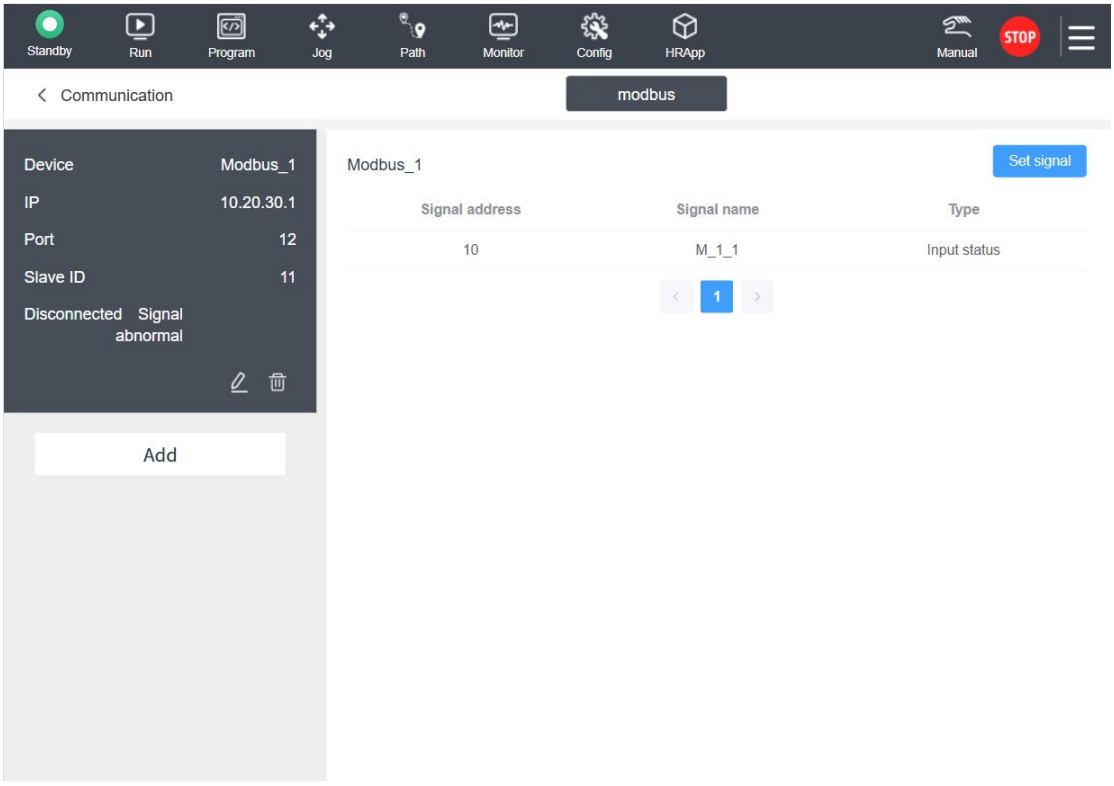


Fig. 5.7.1 Communication Interface

To set the Modbus external slave device signal. A connection to a Modbus server (or slave) of a specified IP address can be created by input/output signals (registers or numbers). Each signal address is unique.

5.7.1. Device List

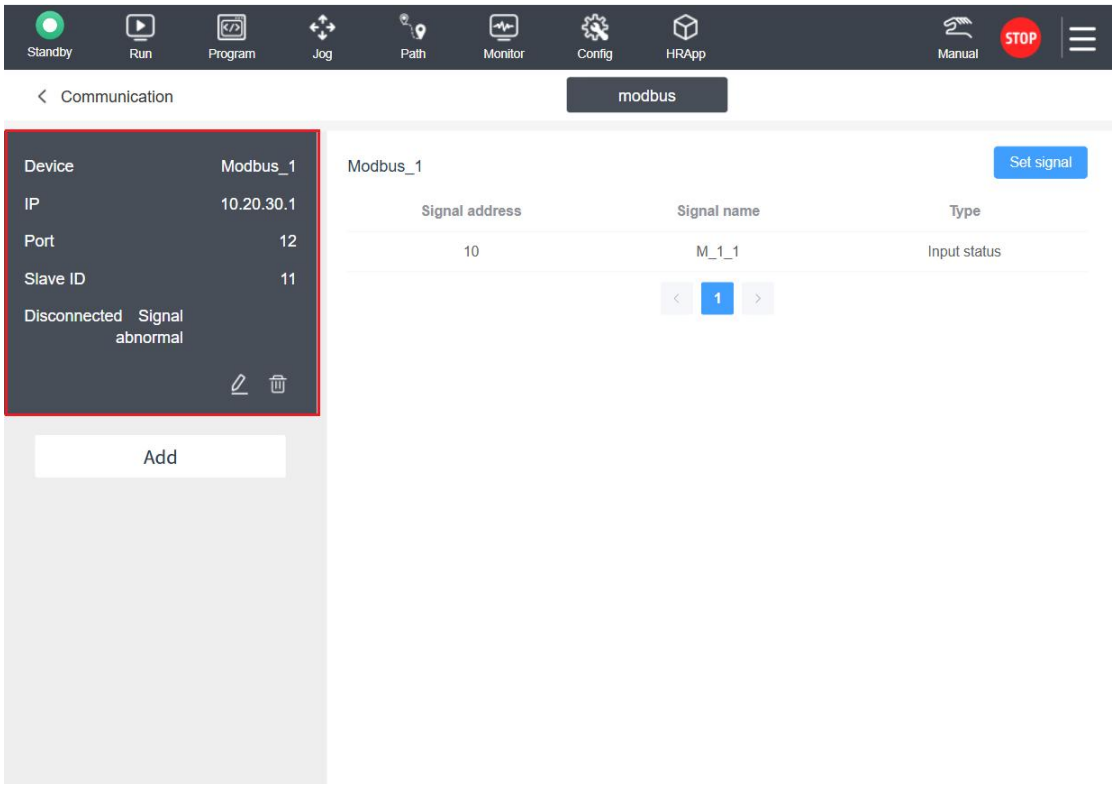


Fig. 5.7.2 Device List

- Add: Click Add device, and a popup appears for data entry, including the IP address, port, and slave ID; the Modbus name is not modifiable;
- Edit: To modify the Modbus device parameters;
- Delete: To delete the Modbus device;
- Select: Dark color means selected, and the signal list of the selected device is displayed on the right;

5.7.2. Signal List

The signals in the list are arranged by the signal group intervals added;

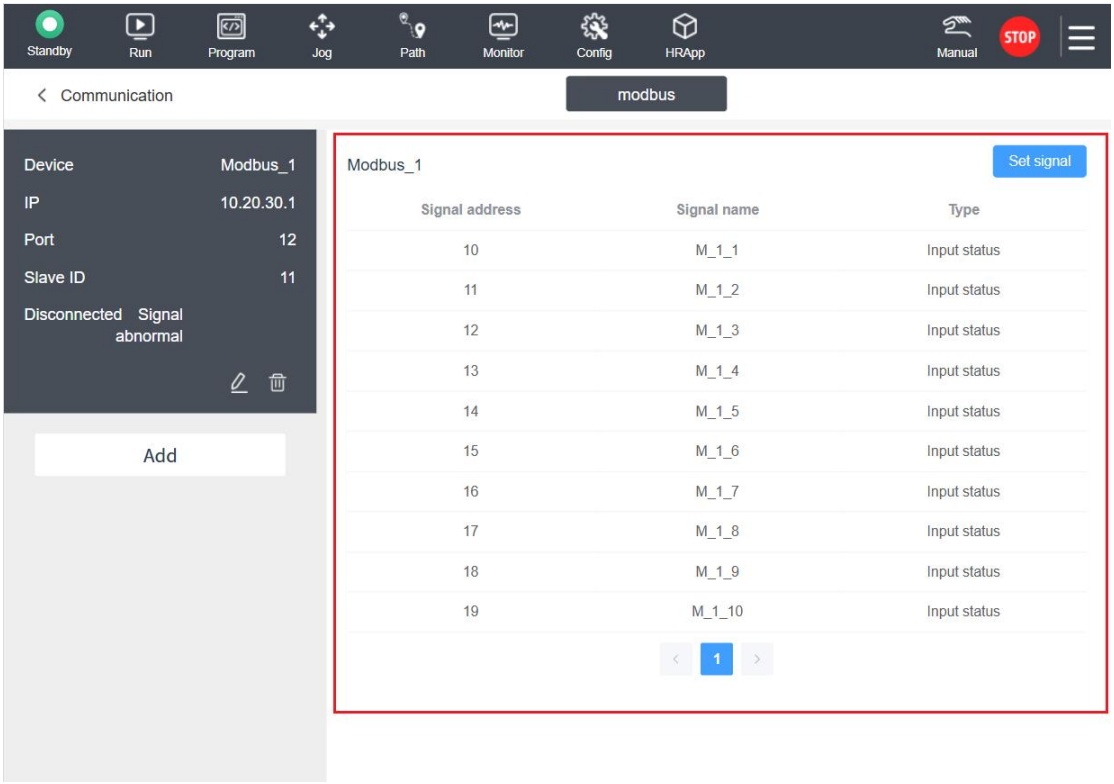


Fig. 5.7.3 Signal List

Set Signal: Click 【Set signal】 to edit signal group parameters, including signal group, signal type, address, read quantity and signal name;

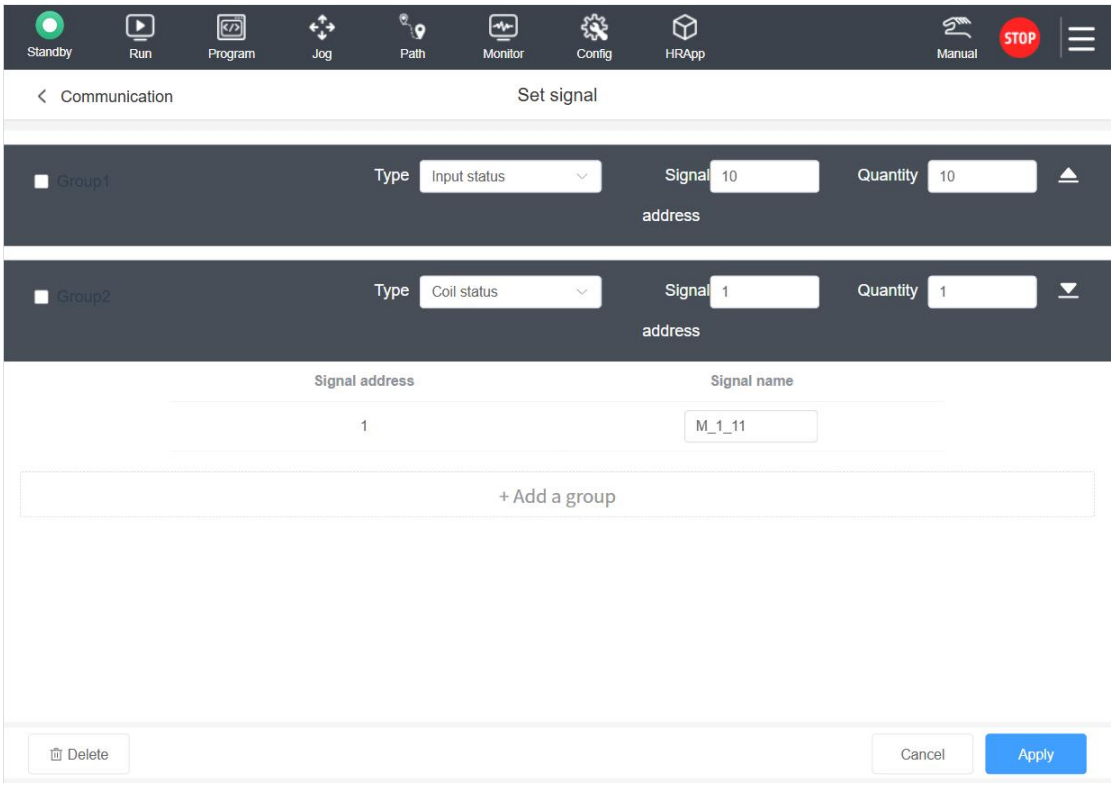


Fig. 5.7.4 Signal Setting

- **Add a Group:** Add a group of signals of the same type;
- **Type:** Use this drop-down menu to select the signal type.

Table 5.7.1 List of Signal Types

Signal Type	Target	Access Type	Application Scenario
Coil Status	Single bit	Read and write	For controlling/alarming switch status
Input Status	Single bit	Read only	For input diagnostic functions
Holding Register	16-bit	Read and write	For transmitting user-set parameters
Input Register	16-bit	Read only	For storing input signal values

- **Signal Address:** To enter the signal address to be read, which is the starting value

of the signal group;

- **Read Quantity:** To input the quantity to be read from the beginning to the end of the signal address; for example, if the signal address is 00100, and the number of reads is 3, then the signal within value range 00100 ~ 00102 is read;
- **Signal Name:** To facilitate user search, duplicate naming is not permitted;
- **Unfold:** Fold and collapse multiple signals in a group; click again to unfold;
- **Delete:** Check the box in front of the signal group name to delete the entire signal group;
- **Apply, Cancel:** Click apply, the configuration modifications are implemented immediately upon confirmation. Click cancel, it will revert pending adjustments and delete unsaved input.

Note: After the Modbus signal is added, the SetI/O command will be directly available; in addition, the monitoring interface will also be added a Modbus monitoring module accordingly.

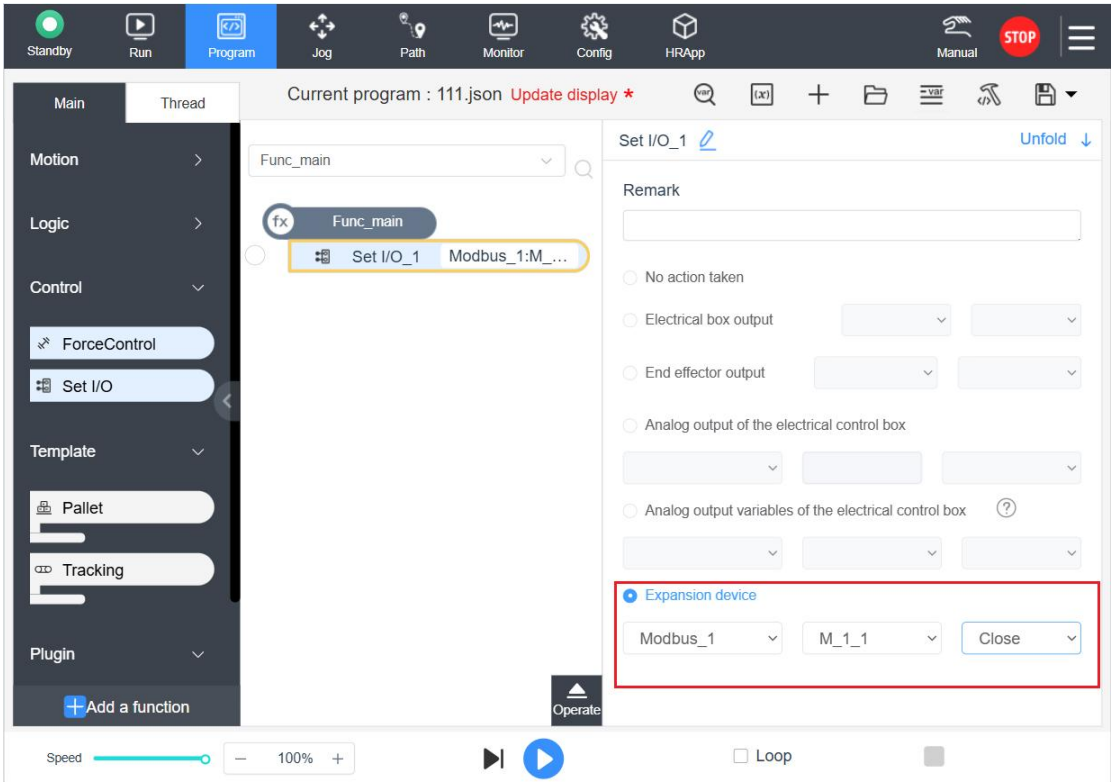


Fig. 5.7.5 Application Demonstration within the Program

5.8. Tracking Conveyor

The function of the tracking conveyor is to adjust the movement of the robot and realize the synchronous tracking of the robot and the conveyor.

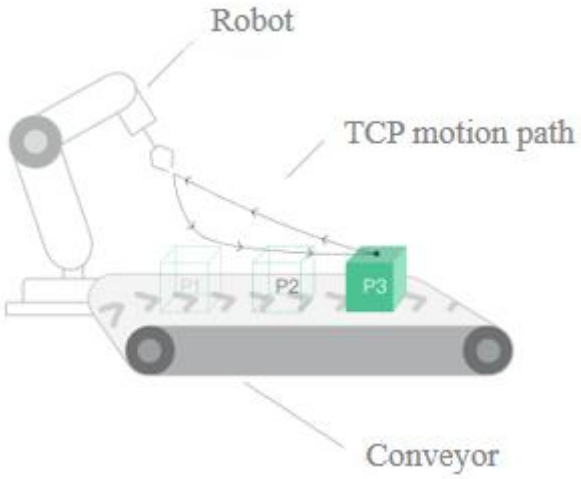


Fig. 5.8.1 Tracking Conveyor

5.8.1. User Coordinate System Setting

When the hardware conveyor is connected (please refer to the manual for how to connect the hardware conveyor), the conveyor needs to be calibrated in order to ensure that the direction and movement of the robot arm are synchronized with the conveyor. To calibrate the conveyor, it is necessary to ensure that the Y-axis motion direction of the robot arm is consistent with the motion direction of the conveyor, so it is necessary to teach a user coordinate first.

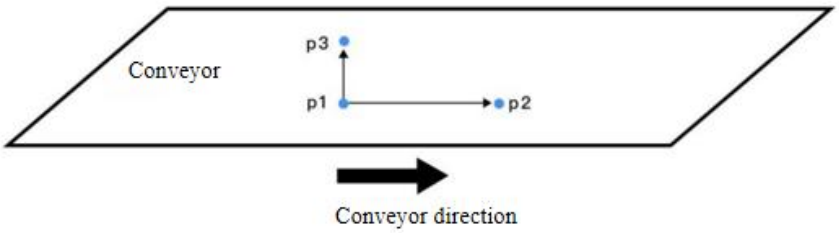


Fig. 5.8.2 Conveyor User Coordinate System

With the conveyor stopped, the user coordinate system can be defined by setting three points (P1, P2, P3):

1. Move the robot arm's end-effector to point P1 and set it as Point1;
2. To ensure that the +Y axis of the user coordinate system is consistent with the moving direction of the conveyor, the conveyor can be started;
3. Move the robot arm's end-effector to point P2 and set it as Point2;
4. Determine point P3 on the conveyor plane (P3 cannot be collinear with P1 and P2), and move the robot arm's end-effector to point P3 and set it as Point3.

After the above steps are completed, the user coordinate system teaching is completed.

Note: The positive direction of the Y axis of the user coordinate system is the same as the running direction of the conveyor, and the plane formed by the X and Y axes is parallel to the conveyor plane.

5.8.2. Conveyor Calibration

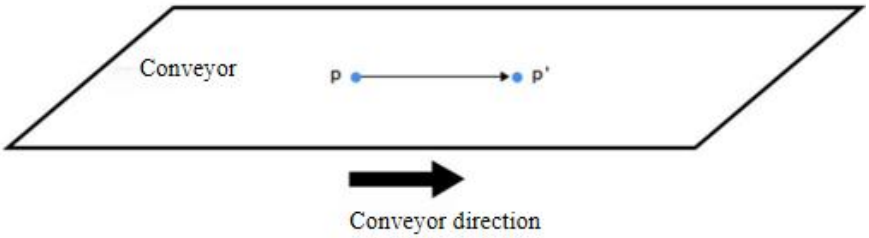


Fig. 5.8.3 Conveyor Calibration

Once the user coordinate system is set, it needs to be set to the current user coordinates. Before calibrating the conveyor, configure the conveyor type and follow direction.

When the conveyor stops, make a sign P on the conveyor, click Set point for the calibration point 1, move the robot arm's end-effector to the point P of the conveyor, and complete the setting of calibration point 1;

Continue to run the conveyor, allow its point P move to point P'; then stop the conveyor, click Set point for the calibration point 2, move the robot arm's end-effector to the point P' of the conveyor, and complete the setting of calibration point 2;

Finally, click【Start calibration】, and the conveyor calibration is completed. After the calibration is completed, a completion prompt will pop up. If the calibration error exceeds the allowable threshold, an error window will pop up.

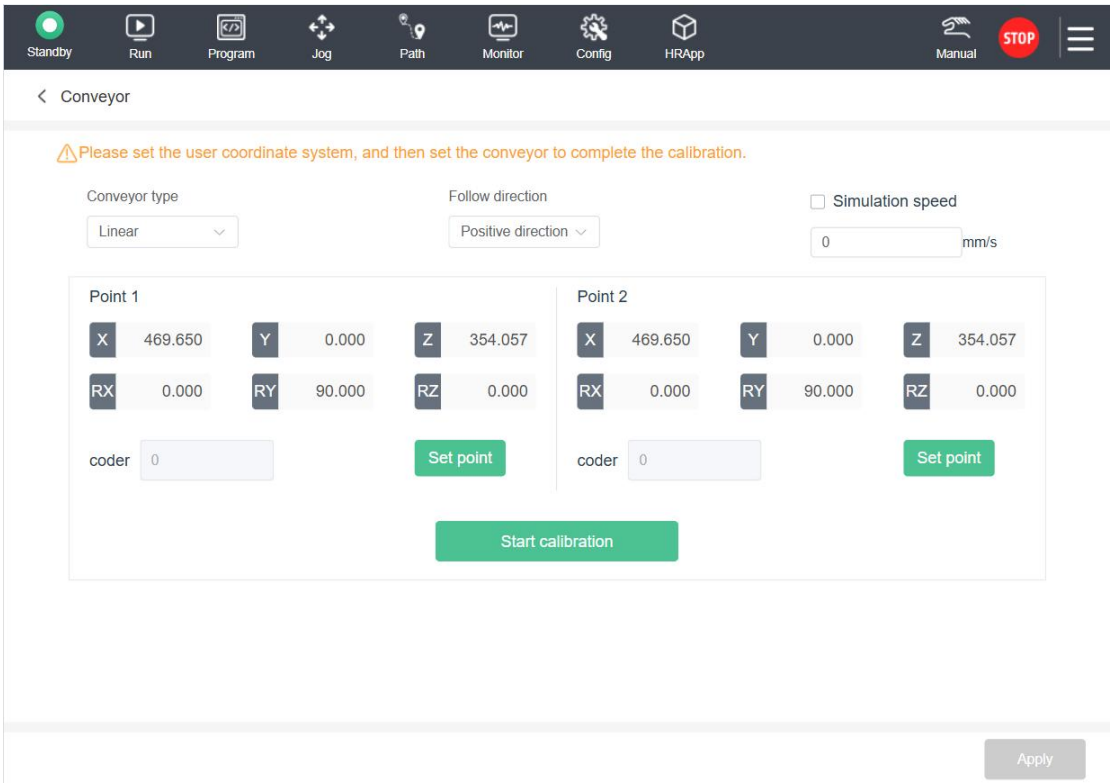


Fig. 5.8.4 Conveyor Calibration Interface

- **Conveyor Type:** Conveyor types are classified as linear and arc.
  - **Follow Direction:** The directions are positive and negative. When set to positive, the Y coordinate of the calibration point is positive; when the set to negative, the Y coordinate of the calibration point is negative.
- **Simulation Speed:** This is the operational speed used for debugging conveyor belts and is applied when selected.

Coder: Indicating the value read by the encoder.

Note: This function is only supported on 4th generation electrical control boxes.

5.9. Force Sensor

Traditional industrial robots are only capable of position control, limiting them to simple contact tasks like welding and painting. However, many tasks require

interaction with the environment and specific contact forces. This need gave rise to force control function, which adjusts robot trajectories using contact force feedback signals.

The Advanced Free Drive Mode is a control mode that allows an industrial six-axis robot arm to move freely under external forces. Typically used with torque sensors, it enables human operators or other external forces to directly push or guide the robot arm without traditional programming or controller input. This mode is mainly used in programming, teaching and debugging phases, making robot arm operation more intuitive and convenient.

In Advanced Free Drive Mode, each joint of the robot arm can move freely when external forces are applied. The control system does not actively apply torque. Instead, it uses force/torque sensors to detect external forces, calculates the needed drive current to counteract gravity and inherent resistance. This allows the arm to be easily moved by hand or other external forces when external forces are applied.

Note: This function requires external forces to be applied directly to the robot arm's end-effector. As the robot arm only has a force sensor at its end-effector and none on other parts, the robot arm will not respond and will remain stationary in Advanced Free Drive Mode if external forces are not applied there.

5.9.1. Advanced Free Drive

The Advanced Free Drive Mode includes position mode, attitude angle mode, free mode and custom mode, and users can choose according to their needs. The default displays are location mode and last set mode.

- Position Mode: Only movable along the X, Y, Z axes;
- Attitude Angle Mode: Only movable in RX, RY, RZ directions;
- Free Mode: Movable in X, Y, Z, RX, RY, RZ directions;
- Custom: The user can select one or more positions or rotation angles to move the robot arm;
- Compliance: By adjusting the compliance parameters, users can customize the damping of force-controlled dragging when enabling the Advanced Free Drive mode, thereby optimizing the feel of the operation hand;

Compliance can be classified as translational compliance and rotational compliance. By setting the total compliance, the user can simultaneously control both translational and rotational compliance, allowing them to be set to the same value. In contrast, custom compliance enables users to adjust the parameters of translational and rotational compliance individually.

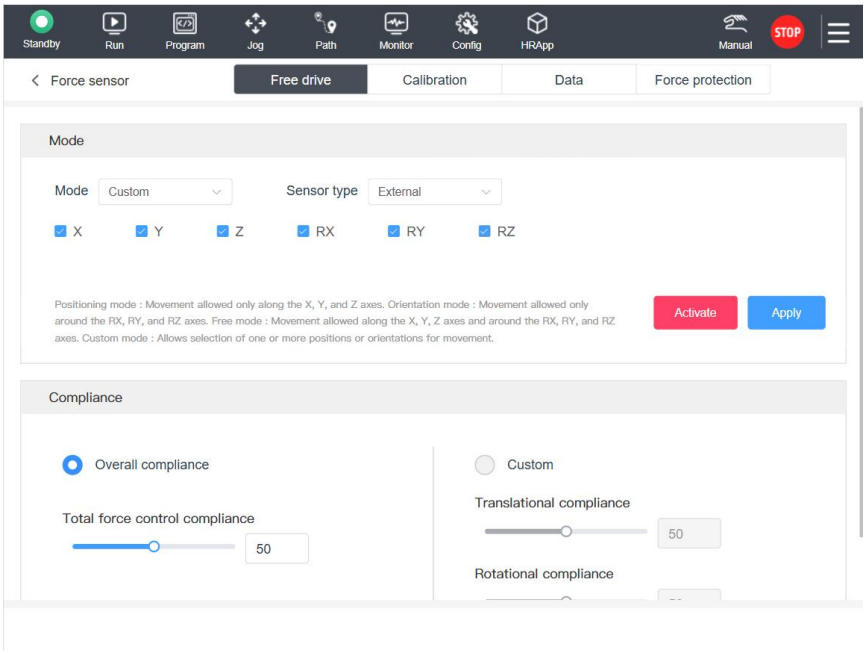


Fig. 5.9.1 Free Drive and Compliance

Click the 【Apply】 button to save the selected mode, implement it and obtain feedback prompts. After clicking the【Activate】button, the user can freely move the robot arm under the applied mode.

5.9.2. Calibration

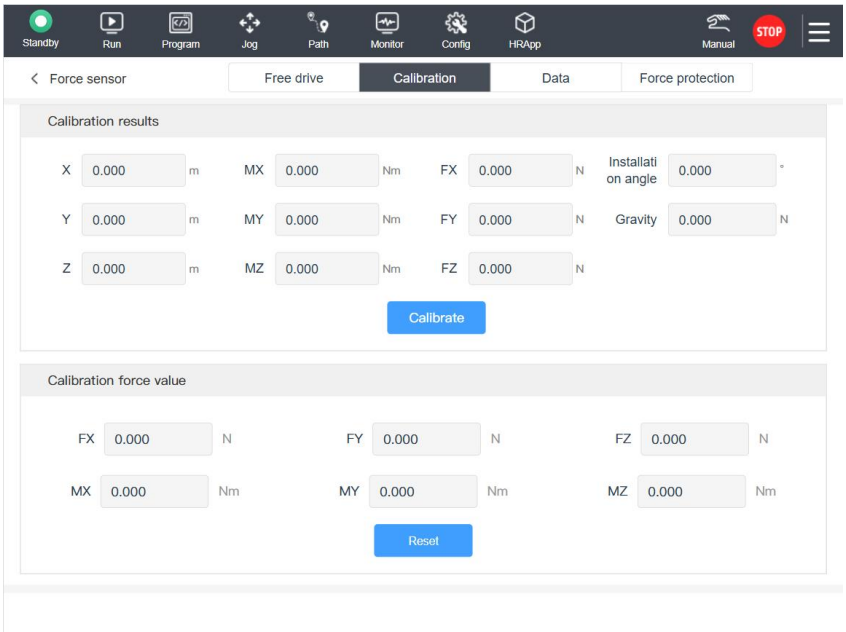


Fig. 5.9.2 Calibration Result Interface

Click the 【Calibrate】 button on the page to enter the calibration result interface.

Select the calibration accuracy. The user accuracy can be selected from eight to sixteen bits. After selection, the corresponding calibration quantity will be generated in the calibration list.

Next, click 【Calibration Point 1】 to navigate to the "Jog" interface, move the robot arm to the target position, confirm and complete the calibration for point 1. Repeat this process to complete the calibration for the remaining points.

Finally, click 【Apply】 to complete the calibration for the force sensor.

Click the 【Reset】 button to reset all currently calibrated points and clear all calibration parameters, returning them to zero, so that calibration can be performed again.

- The calibration accuracy can be selected from eight bits to sixteen bits. The higher the number of bits, the better the calibration accuracy;
- Click the【Calibration point n】button will navigate to the "Jog" interface, where users can use the jog buttons to move each joint of the robot arm;
- In the "Jog" interface, calibration points must cover multiple directions of RX, RY, and RZ;

Note: When a payload is attached at the robot arm's end-effector during calibration process, self-collision must be avoided;

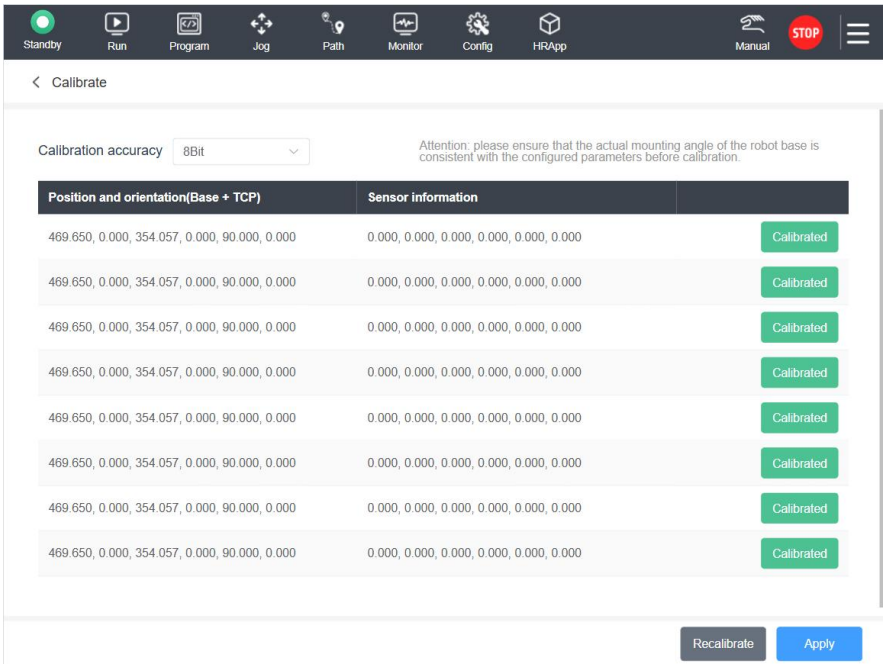


Fig. 5.9.3 Reset and Apply Results

5.9.3. Data

On the left is the data feedback from the force sensor, and on the right is the data calculated based on the calibrated accuracy.

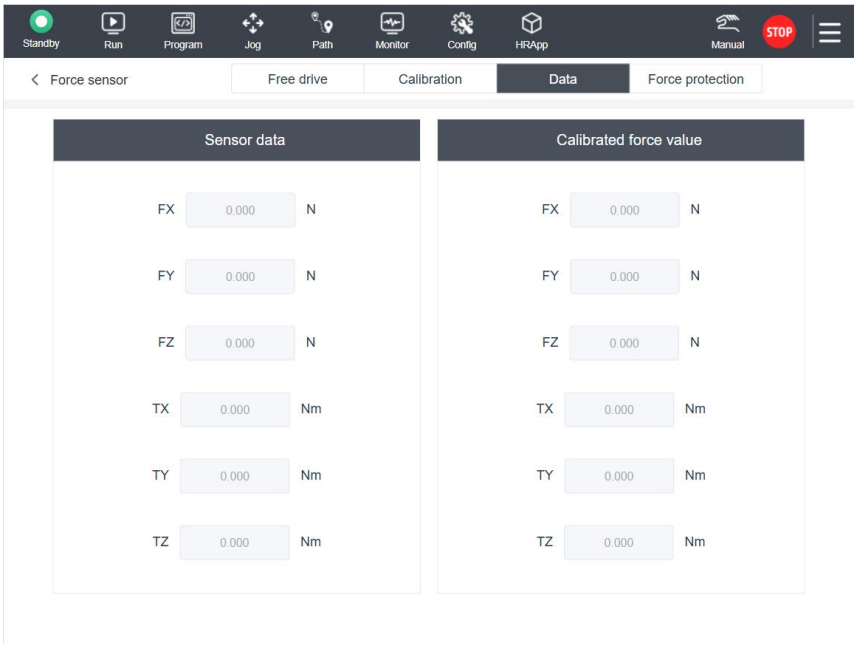


Fig. 5.9.4 Data Interface

5.9.4. Force protection

When the force sensor is subjected to an excessive impact, the force protection function will be activated.

Force protection types include two modes: power off and error reporting.

Users can adjust the force limit parameters in all directions, including the force magnitude, positive/negative values in the X, Y, and Z directions, and torque magnitude in the RX, RY, and RZ directions. After confirming the adjustments, the users must click 【Apply】 to implement them.

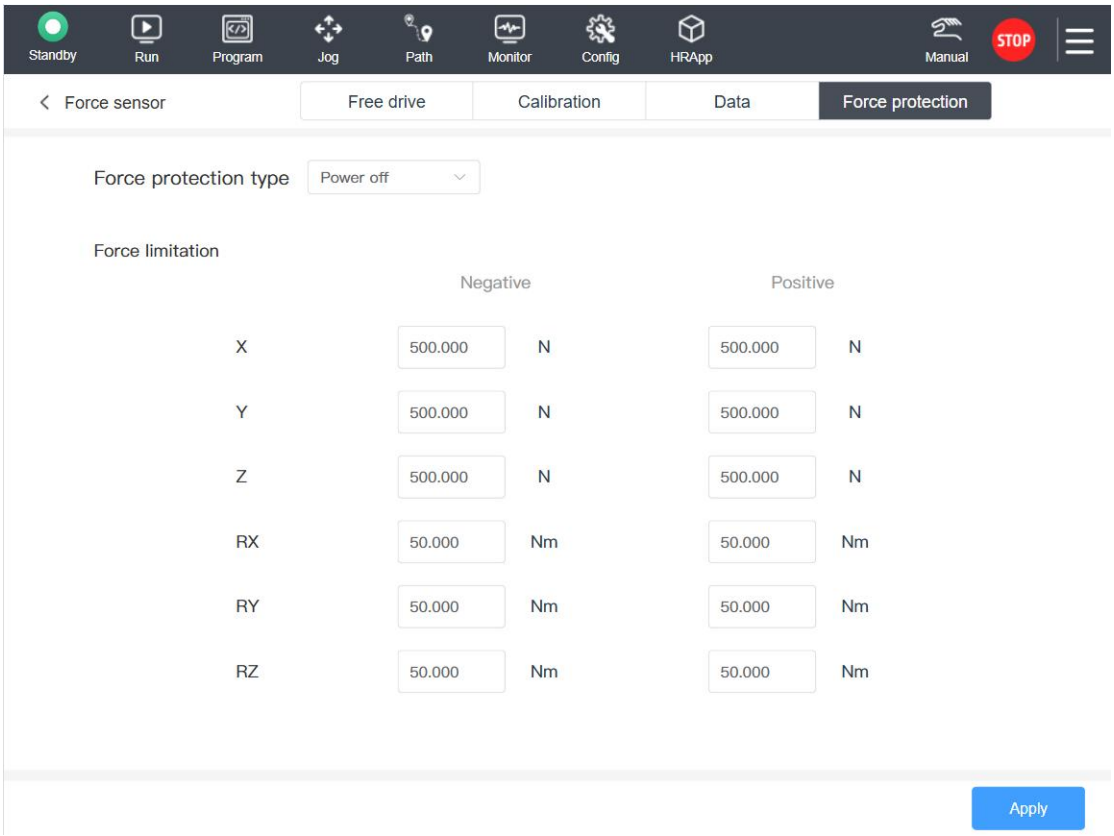


Fig. 5.9.5 Force Protection

5.10. Brake Operation

When the robot malfunctions or the joints of the robot arm need to be moved in a zero-gravity manner, for example, the robot arm is subjected to external forces, resulting in an error report and automatic disabling, and is impossible to activate the basic free drive function to change the robot arm's pose. Therefore, brake releasing operation of the joints of the robot arm is required. This function must be operated in the disabled state and is not available when the robot is enabled.

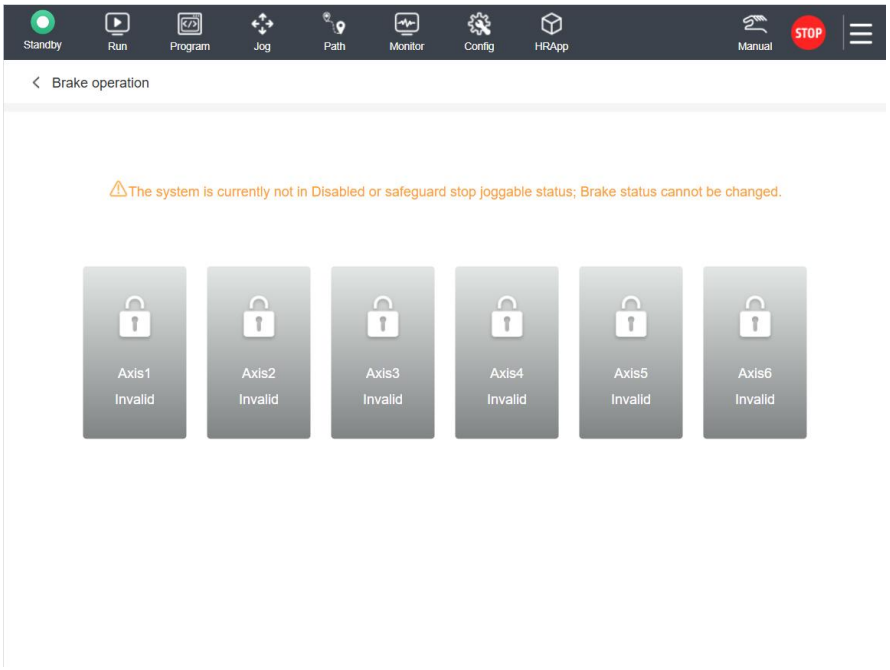


Fig. 5.10.1 Brake Release Prohibited When Enabled

Before using this function, the robot needs to be disabled. Once disabled, this function is available. Clicking on the corresponding joint will release the brake and unlock the relevant joint. The page displays the brake engagement or release status of the current individual joint of the robot arm.

Axis1 ~ Axis6 correspond to joints J1-J6 of the robot arm.

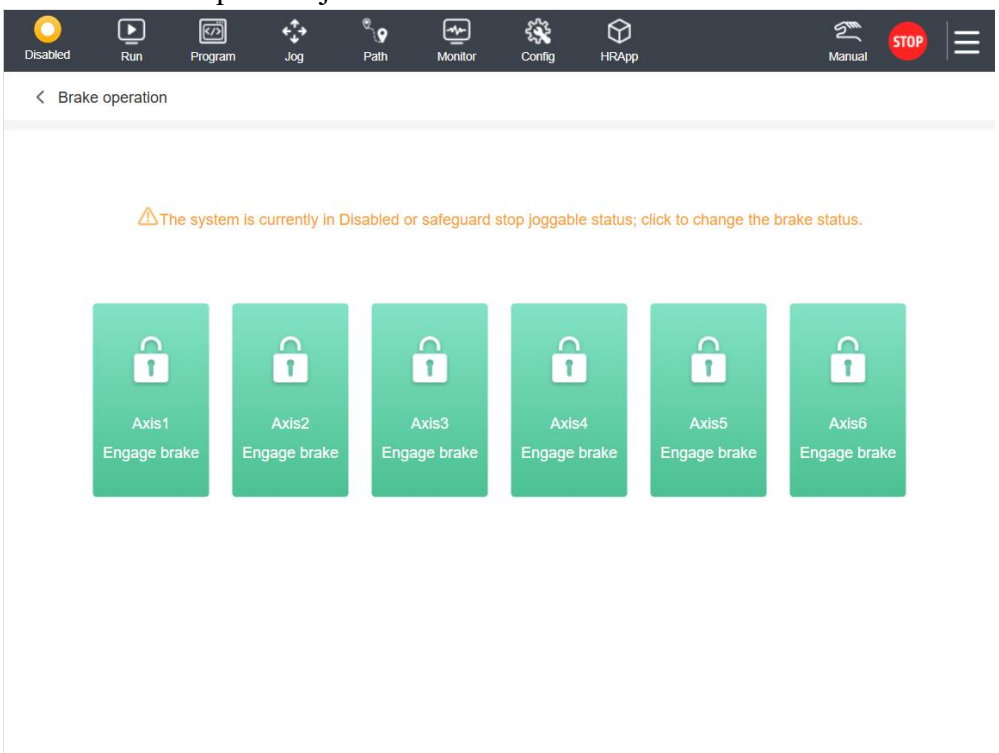


Fig. 5.10.2 Braking/Releasing Available in Disabled State

5.11. Boot Configuration

To configure the robot boot configuration, including temperature threshold, three-position button, Modbus slave station command, and remote shutdown. The user needs to verify the security password (consistent with the administrator password) before operating.

5.11.1. Temperature Threshold

This page is designed to set the low-temperature and high-temperature thresholds. If the joint temperature of the robot arm exceeds the low-temperature threshold or surpasses the high-temperature threshold, the robot will enter reduced mode.

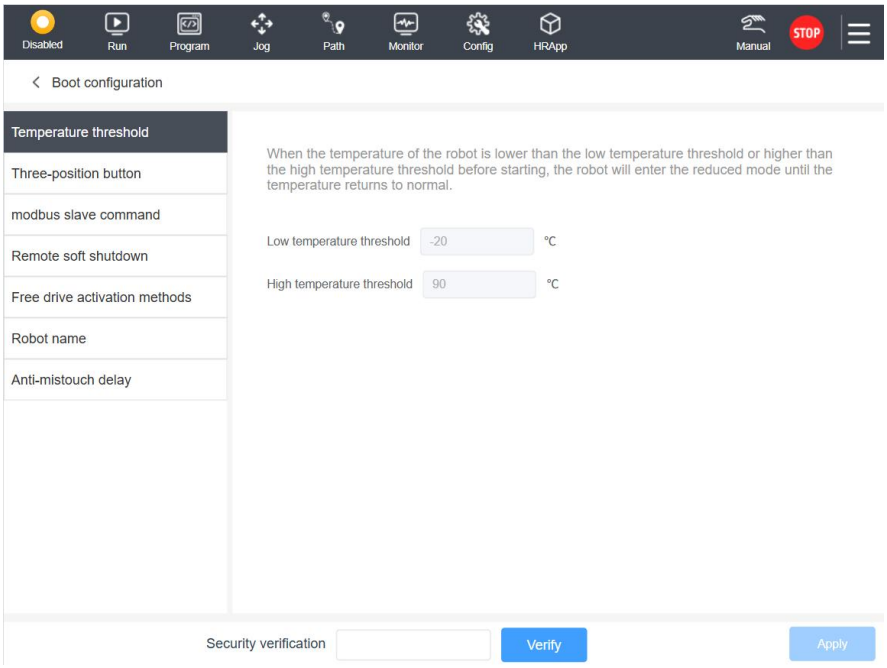


Fig. 5.11.2 Temperature Threshold

Note:

1. The low-temperature threshold is set within the range of -20 to 80 and must be an integer.

2. The high-temperature threshold is set within the range of -10 to 90 and must be an integer.

3. The high-temperature threshold must be higher than the low-temperature threshold, with a temperature difference of more than 10℃.

5.11.2. Three-position Button

The three-position button is located on the back of the teach pendant box and is a button with three positions. When the user releases the button, it is in the first position, and the button function is disabled; when the user lightly presses it to the middle (second position), the button function is activated; when the user presses the button firmly to the bottom (third position), the button function is disabled. During equipment debugging, the three-position button can effectively reduce personal injury caused by the equipment.

Whether the three-position button is enabled can be controlled by operating the corresponding switch on the "Boot configuration" page;

After the three-position button is enabled, some associated restricted functions can only be used when the three-position button is in the second position. If the three-position button is no longer in the second position during the normal use of these functions, the system will immediately stop the relevant movements or operations and return to the disabled state.

The three-position button only functions in manual mode and does not work in auto mode.

Note: This function is a standard safety function, designed to ensure safety during robot operation and prevent collision hazards caused by incorrect operations.

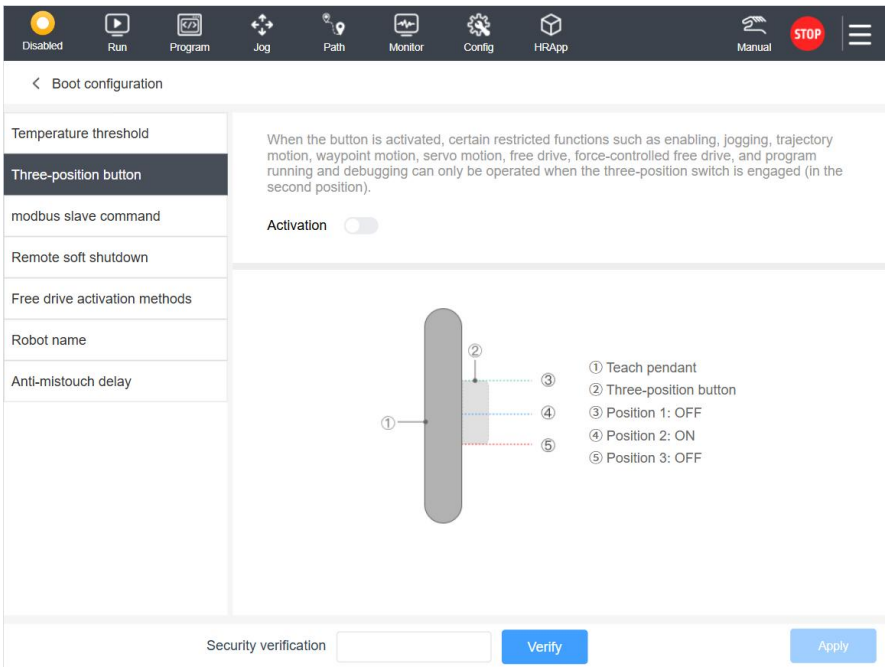


Fig. 5.11.3 Three-position Button of Teach Pendant

5.11.3. Modbus Slave Command

For detailed information, refer to section 5.7 Communication (Modbus). To set the commands sent by the Modbus slave station. This function is disabled by default.

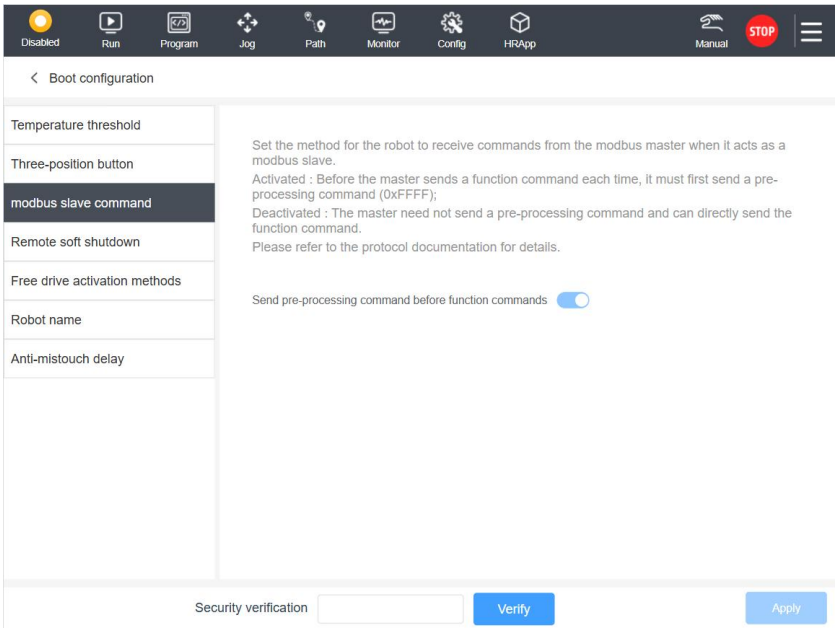


Fig. 5.11.4 Modbus Slave Command Interface

After applying the configuration, a system restart is required for the changes to take effect.

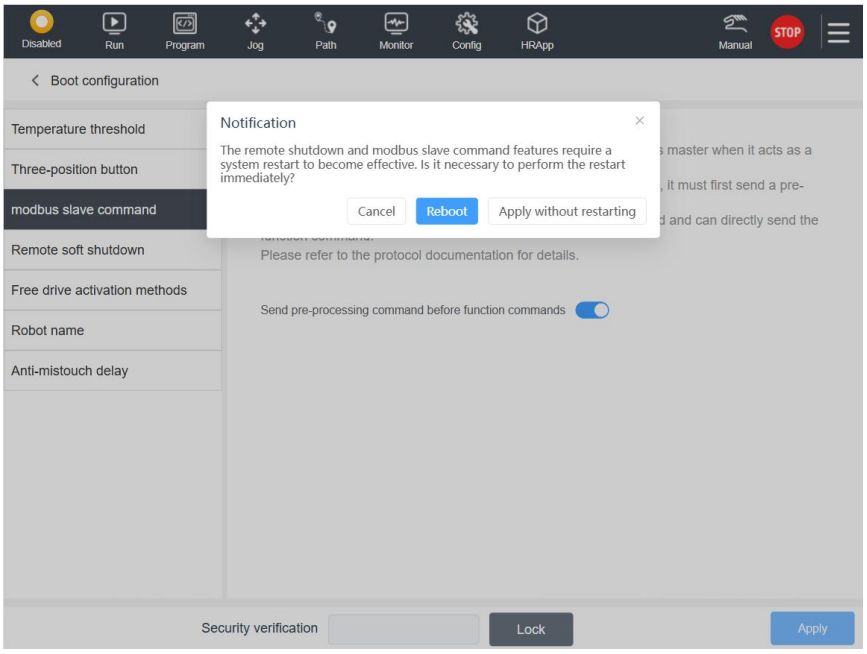


Fig. 5.11.5 Configure and Apply

5.11.4. Remote Soft Shutdown

The remote soft shutdown signal can be triggered in two ways: one is by briefly pressing the circular physical button on the teach pendant, and the other is by shorting the two wires extending from the REMOTE OFF/GND ports on the system control board.

When the system is running normally, it can respond to the software shutdown signal by first shutting down the operating system and then cutting off the power supply to the electrical control box. Activating this function requires a reboot to take effect.

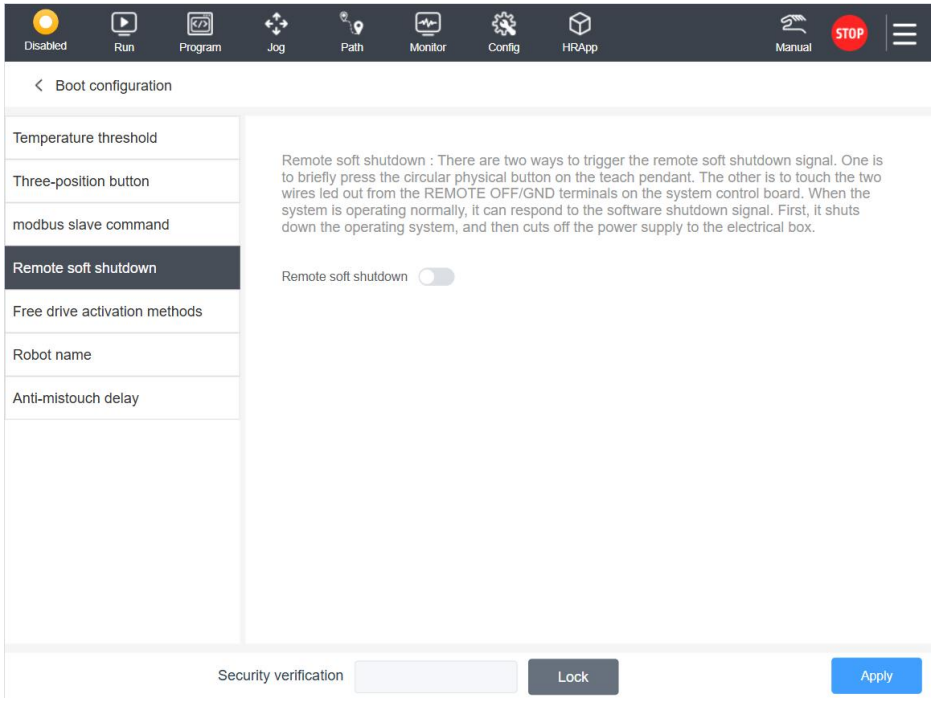


Fig.5.11.6 Remote Soft Shutdown

5.11.5. Advanced Free Drive and Basic Free Drive Activation Methods

There are several ways to activate the basic free drive mode:

Terminal Configuration: Users can access the "jog" interface on the terminal (teach pendant, PC, laptop, pad) and activate the basic free drive.

End-effector Button: After enabling the end-effector I/O function, press the end-effector Free button to activate the basic free drive.

IF/SDK: Using IF/SDK to activate the basic free drive.

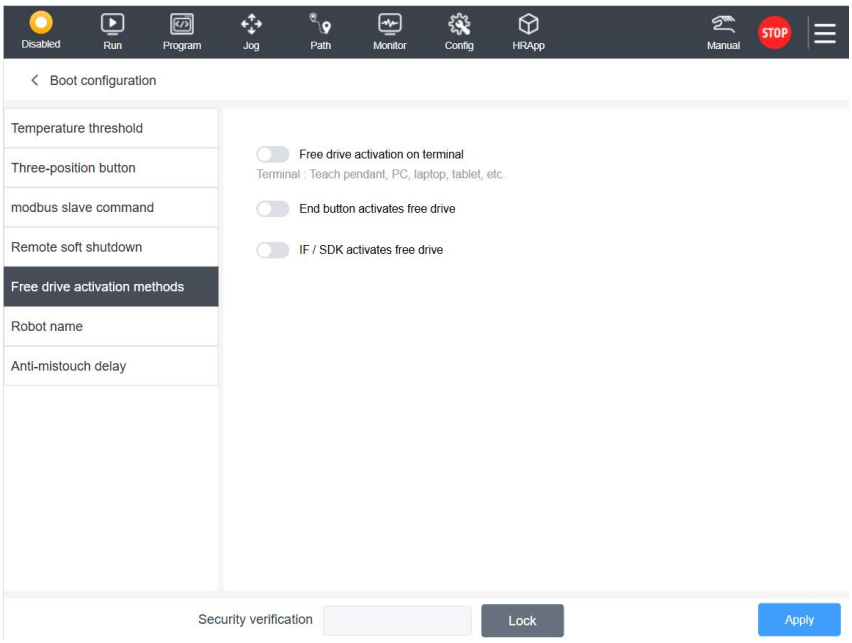


Fig.5.11.8 Advanced Free Drive and Basic Free Drive Activation Methods

5.11.6. Robot Name

The robot name can be customized to distinguish different robots.

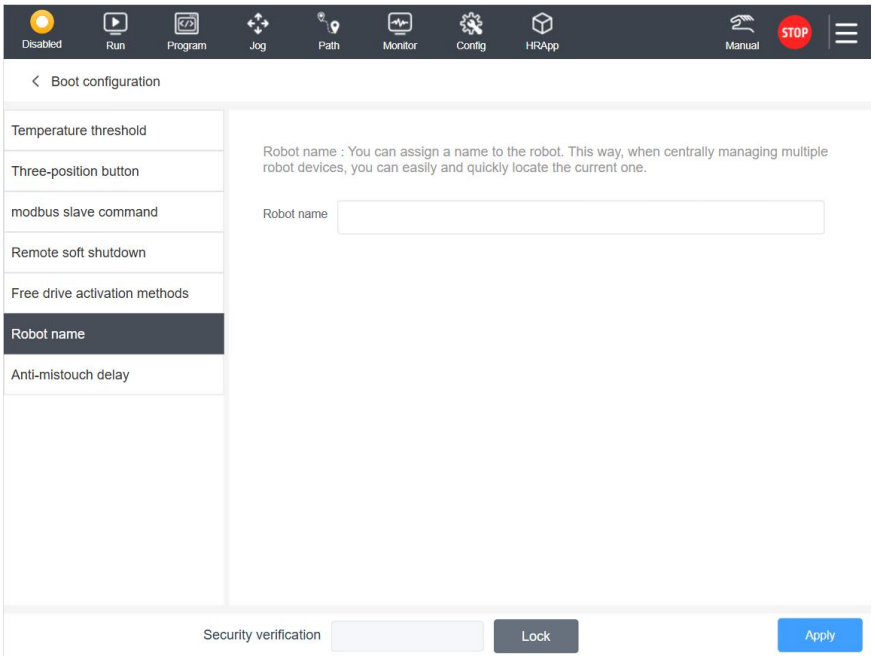


Fig.5.11.9 Robot Name Page

5.11.7. Anti-false Touch Delay

On the jog interface, the robot arm starts moving only after pressing and holding the button for the set time.

Security verification is required for setup.

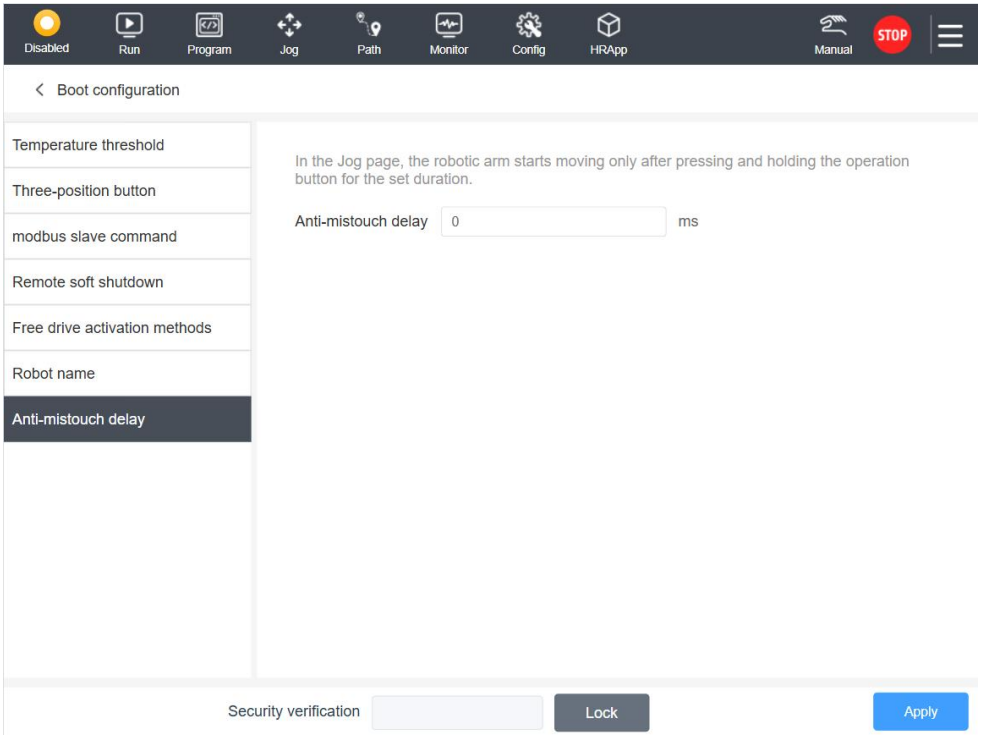


Fig.5.11.10 Debounce Time

5.12. Home

A customizable origin position can be set. Press and hold the 【Move to Home】 button in the "Jog" interface to reach it. In programming, a Home command can be added to move the robot arm to reach this safe position during back-and-forth motion.

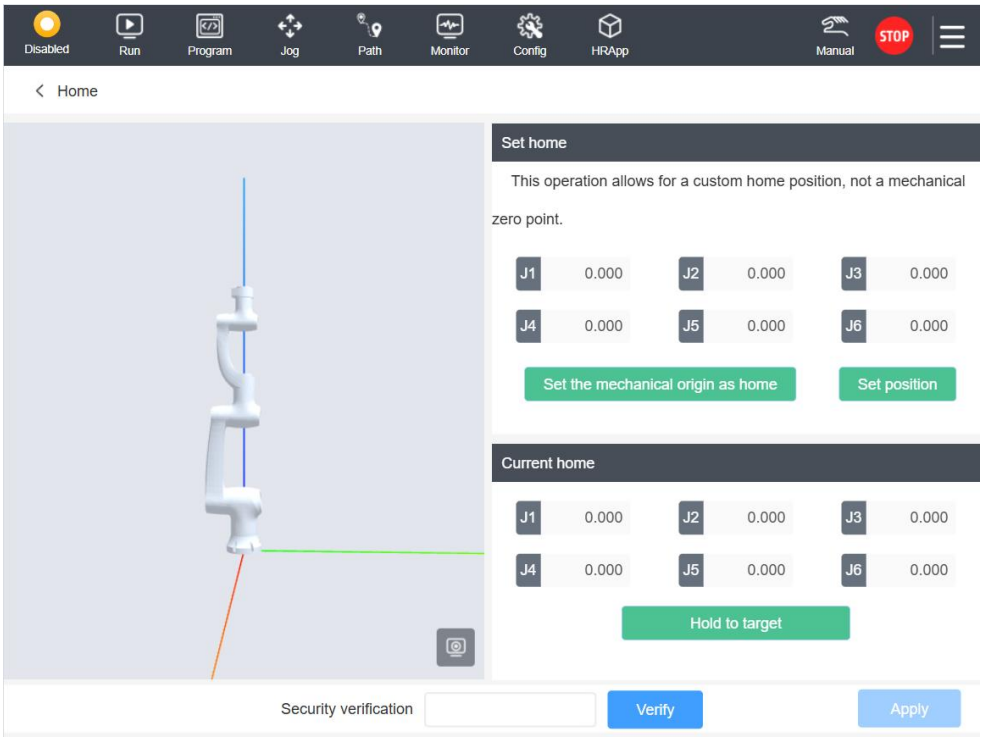


Fig. 5.12.1 Home Function Diagram

As shown in the figure above, after passing the security verification, click 【Set Position】 to set position. After the setting is successful:

- In the "Jog" interface, click 【Move to home】 to return home position;
- In the programming interface, executing the Home command will also return home position;

In addition, click 【Set the mechanical origin as home】 to set the mechanical origin (factory-set origin) as the current home. For detailed information on the default mechanical origin, please refer to 4.10 Move to Home.

In the "Current home" module, the joint angles of the current origin will be displayed. Press and hold the 【Hold to target】 button, and the robot arm will move to the currently set home position.

Press and hold 【Move to home】 on the "Jog" interface, and the robot arm will also return to the currently set home position.

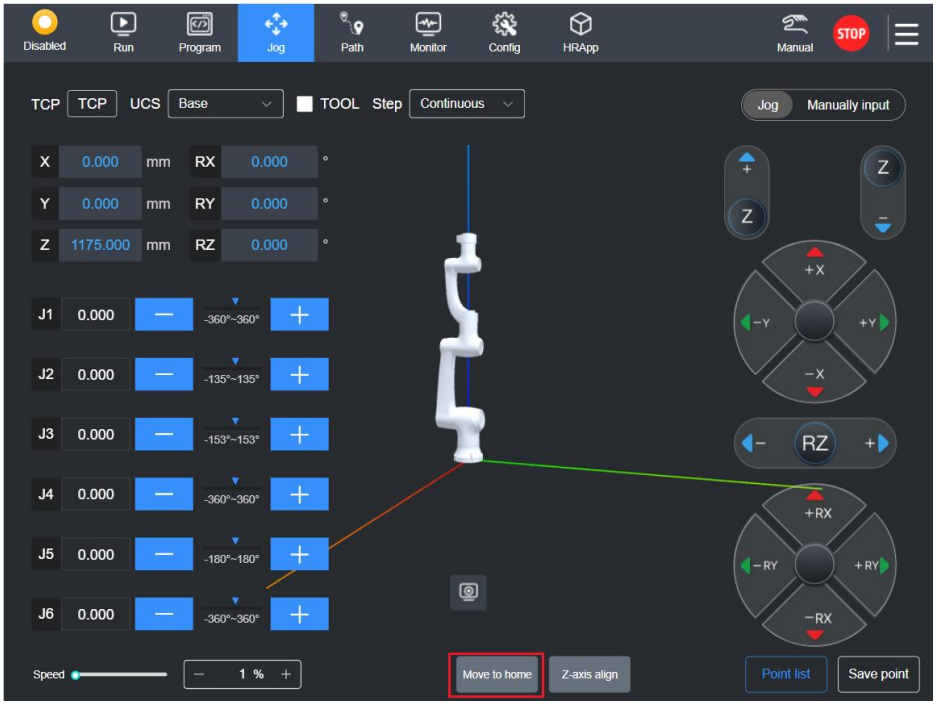


Fig. 5.12.2 Move to Home on the Jog Interface

5.13. Basic Free Drive Adjustment

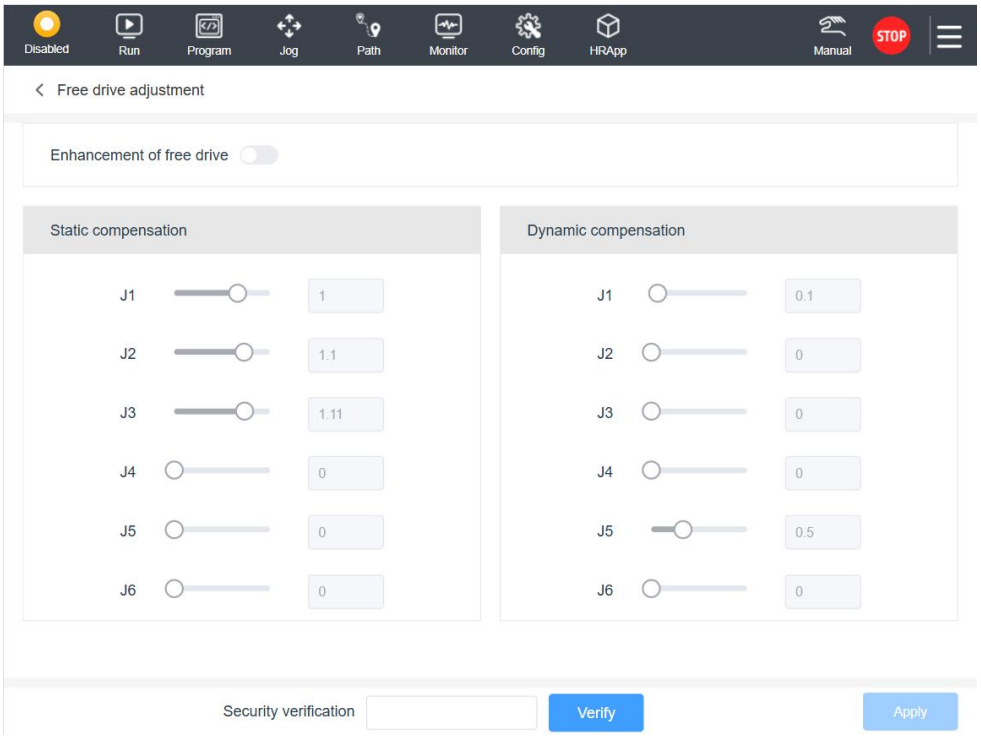


Fig.5.13.1 Basic Free Drive Adjustment

Adjust the compensation parameters to allow the operator to easily and smoothly manipulate the robot arm in the basic free drive mode while maintaining stability and accuracy.

Static Compensation: Used to adjust the static friction of the robot arm in a stationary state. Improves positioning accuracy and stability in a stationary or low-speed state. Directly affects the "lightness" of the robot arm in a stationary or low-speed state.

Dynamic Compensation: Used to adjust the dynamic friction of the robot arm in a moving state. Improves path tracking accuracy and stability during movement. Directly affects the "smoothness" of the robot arm during high-speed movement.

Chapter 6 Programming

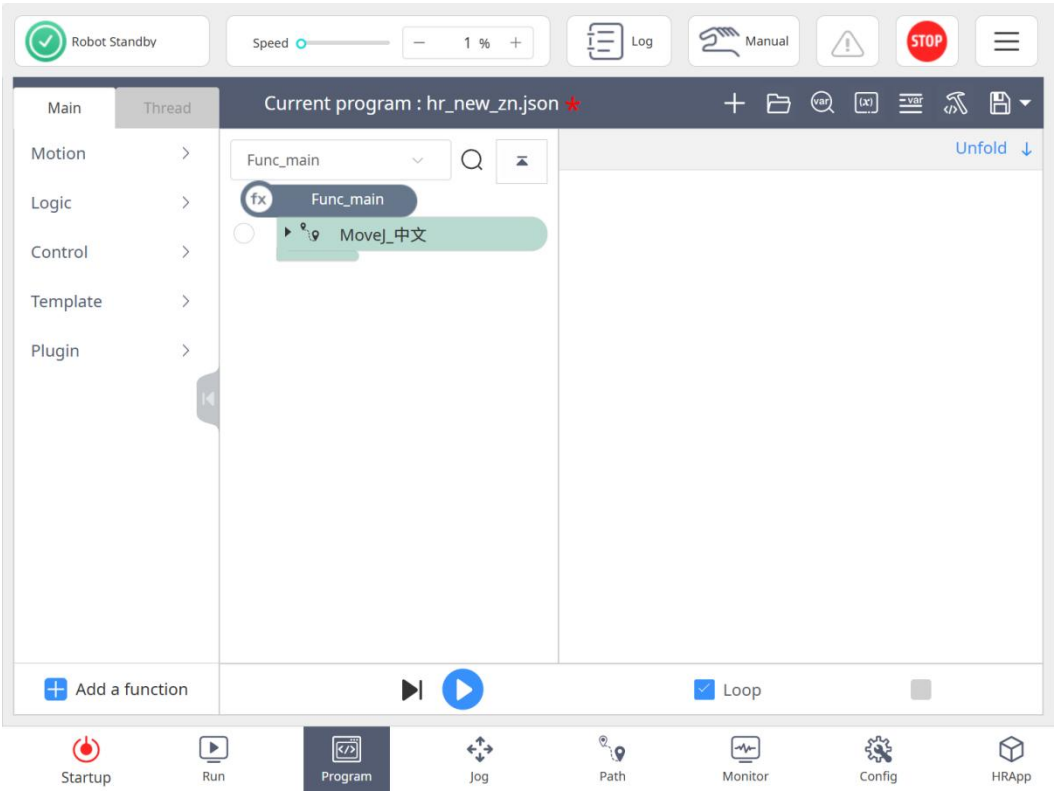


Fig. 6.1 Programming Interface

6.1. Getting Started

1. When enabled, add a program to the programming interface. 【Create → Name → Confirm】
2. In the Func_main function, drag-and-drop a Move command from the command menu to the function node. At this time, a WayPoint will be automatically added under the Move node and named WayPoint_1. The motion mode of Move is MoveJ by fault. Drag a WayPoint under WayPoint_1 again, it will be named WayPoint_2.
3. In WayPoint_1, click 【Set point】 to switch to the "Jog" interface to teach the position of this point, then, click 【Confirm】 to switch back; switch to WayPoint_2 and repeat the same steps to teach the position of WayPoint_2. WayPoint_1 serves as the starting point for the robot arm's movement, while WayPoint_2 is the endpoint.
4. Click 【 】 to complete the motion.

6.2. Interface Introduction

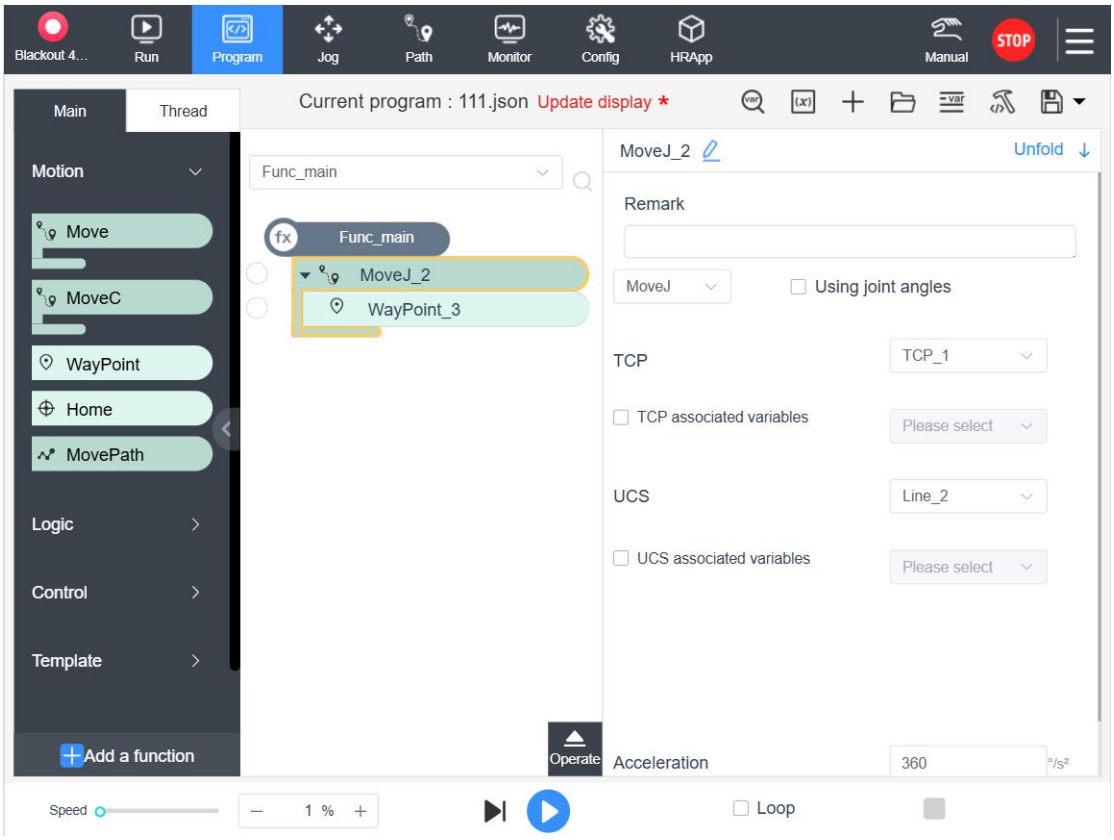


Fig. 6.2.1 Programming Interface Operation Buttons

Table 6.2.1 Overview of Button Functions in Programming Interface

	Lookup	Command Lookup;.
	Create	Create a new program;
	Debug Variables	Monitor variables in scripts, changes in debugging state is only displayed when the program is running;
	Open	Load programs, import and export files, delete programs;
	Variable	Manage variables, view variable values;
	Compile	Compile the current program;

	Save	After editing the program, click the 【Save】 button and select Save/Save As;
	Unfold	When editing command parameters, more areas can be expanded upward while the 3D model is hidden; click again to fold;
	Remark	Remarks for functions or commands to facilitate the interpretation of functions;
	Locate	Locate the currently executing command in the program;
	Expand	Expand all statements of the currently selected function;
	Shrink	Shrink all statements of the currently selected function;
	Comment	Select a command and click 【Comment】 to gray it out. The program will then skip it during execution.
	Copy	Click【Copy】, select one or more commands to be copied from the program list, and click 【OK】; used in conjunction with pasting;
	Cut	Click 【Cut】, select one or more commands to be cut from the program list, and click 【OK】; used in conjunction with pasting;
	Paste	Select a command and click【Paste】, then the copied command will be copied below the selected command;
	Undo	Click 【Undo】 to withdraw the last modification;
	Redo	Click 【Redo】 to restore the previous modification;
	Delete	Drag and drop the command back to the menu area, and the red trash can will appear. Release your finger to delete the command;
	Search	Click the 【Search】 button to turn the function drop-down menu into a search input box. Enter the command you want to find. To exit the search mode, click【Cancel】;

6.2.1. Create

Click 【Create】, enter the file name in the pop-up box, and click 【OK】 to create a new program file, as shown in the figure below. If there are other current files unsaved, a popup will appear to ask whether to save the edited files or not;

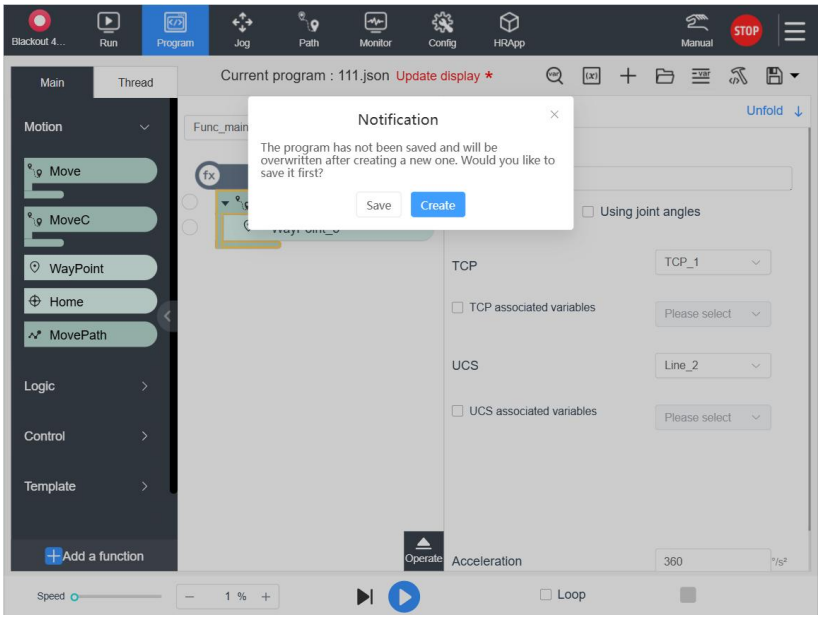


Fig. 6.2.2 Creating a Program File

6.2.2. Program List

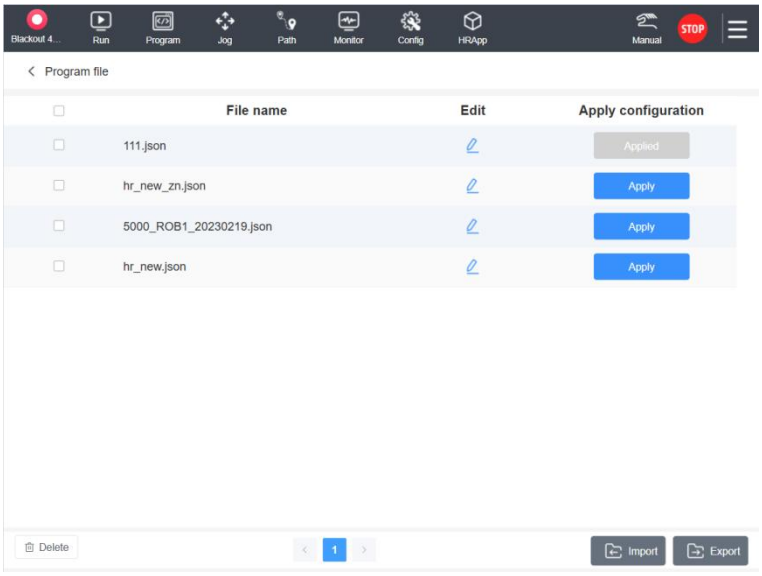


Fig. 6.2.3 Program List

In the file list, the saved files are recorded, and the files can be renamed, deleted, imported, exported and applied;

- **Rename:** Click 【Edit】, the file name setting popup appears, and click【OK】 to take effect;
- **Delete:** Check the box in front of the file and click 【Delete】; the system will prompt: Delete the program or not?
- **Import:** Import files from the default path in the system or from the USB flash drive to the list. After importing, the imported files will be displayed in the list;
- **Export:** Check the box in front of the file and click 【Export】 to export to the system default path or a USB flash drive;
- **Apply:** To open the program file; Click 【Apply】, and note that the currently running program remains in a temporary state until saved. If the currently edited program is unsaved, a prompt box will pop up to remind saving the current file. Users can choose to save the current file or apply the selected file directly, and the latter will not save the current file;

6.2.3. Function

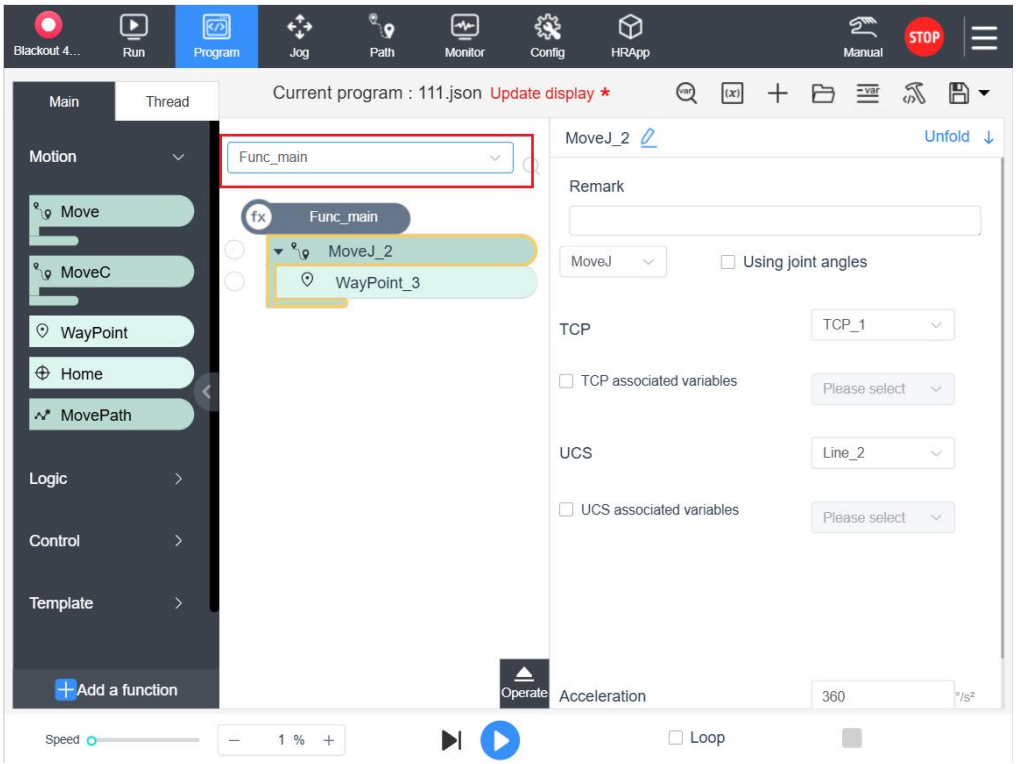
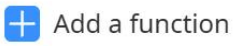


Fig. 6.2.4 Main Function Call

Func_main: The main function, which is automatically added after creating a new program, and cannot be deleted, indicating that when the【Run】interface is running, the program can only run from the Func_main. When debugging on the programming

interface, the user can directly select and execute a function via the function drop-down menu;

Func_N: 

Click 【Add a function】, and the function drop-down menu will add a function accordingly, named Func_1, 2, 3...; common functions can be deleted by dragging it to the left menu. The user can create a repeatable function for repetitive operations and repetitively call it using the Call logic command.

In order to facilitate debugging and reading scripts, different processes can also be divided into separate function modules, which can be called and executed through the Call command.

6.2.4. Command



Fig. 6.2.5 Program

- **Drag-and-Drop and Move:** Drag-and-drop the command from the command menu to the program list node;
- **Command Color Block:** Classified according to menu category. Green is the motion commands, yellow is the logic commands, blue is the control commands, gray is the template commands, and red is the third-party plugin commands;
- **Command Style:** The command being edited is in the selected style with yellow border, and the length of the command color block changes with that of the parameters and titles;
- **Command Title:** In the case of If command, it displays "if_remarks" after the remarking, with a maximum of 24 characters.
- **Parameter Box:** Some commands have parameter box prompts, indicating the current parameter settings of the command. Up to 24 characters are displayed, which facilitates browsing the command parameters set in the list.

6.2.5. Definition of Components

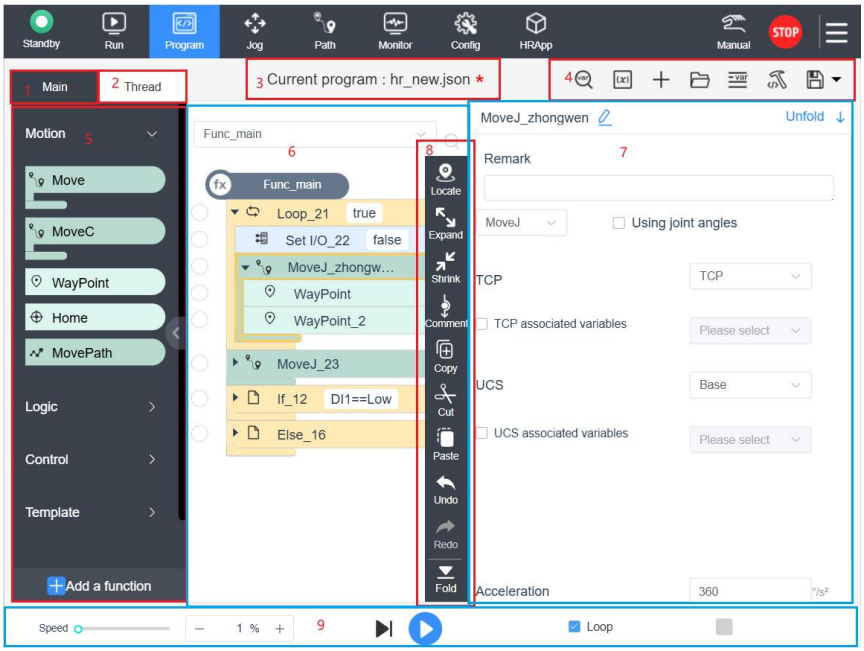


Fig. 6.2.6 Definition of Components

1) **Main Program:** The main thread running by the robot, including functions such as motion control, logic processing, template calling and third-party plugin reference.
2) **Thread:** Run in parallel with the main program, usually used to control external devices (such as signal status monitoring) and communicate with the main program through variables and output signals.
3) **File Name:** Display the name of the current program file.
4) **Functional Operations:** To look up, debug variables, create, load, variables, compile and save
5) **Command Menu:** Display available commands in categories, and support drag and drop to the program list for use; the menu is collapsible to hide.
6) **Command List: The core command sequence of the program. Users can drag and drop commands from the command menu to the list and freely adjust the order of commands; the list must contain a command to run the program.**
7) **Parameter Setting:** Display and set the attributes related to the selected command, and support adding remark information.
8) **Operation Area:** Support copying, pasting, skipping, etc. of commands in the command list.
9) **Bottom Running Area:** Provide debugging functions (such as single-step debugging and loop debugging), support adjustment of motion speed, and can set the current file as the default program file.

6.3. Motion Command

6.3.1. Move

The motion modes include joint motion command (MoveJ), spatial linear motion command (MoveL), and circular motion command (MoveC). The motion classification is expanded from the command menu on the left, drag the Move command to the Func_main node, and MoveJ is added by default. Users can switch among motion modes MoveJ, MoveL, and MoveC from the drop-down menu;

When the user selects Move or other related commands that can add waypoint commands, click the "Point" button at the robot arm's end-effector, and a new waypoint command containing the current robot arm position will be added under the selected command.

Note: Switching between MoveJ and MoveL does not affect the waypoints under their nodes. However, switching to other motion types, such as from MoveJ to MoveC, may result in the deletion of all waypoints under the node, leaving only the start, intermediate, and end point.

6.3.1.1. MoveJ Joint Motion

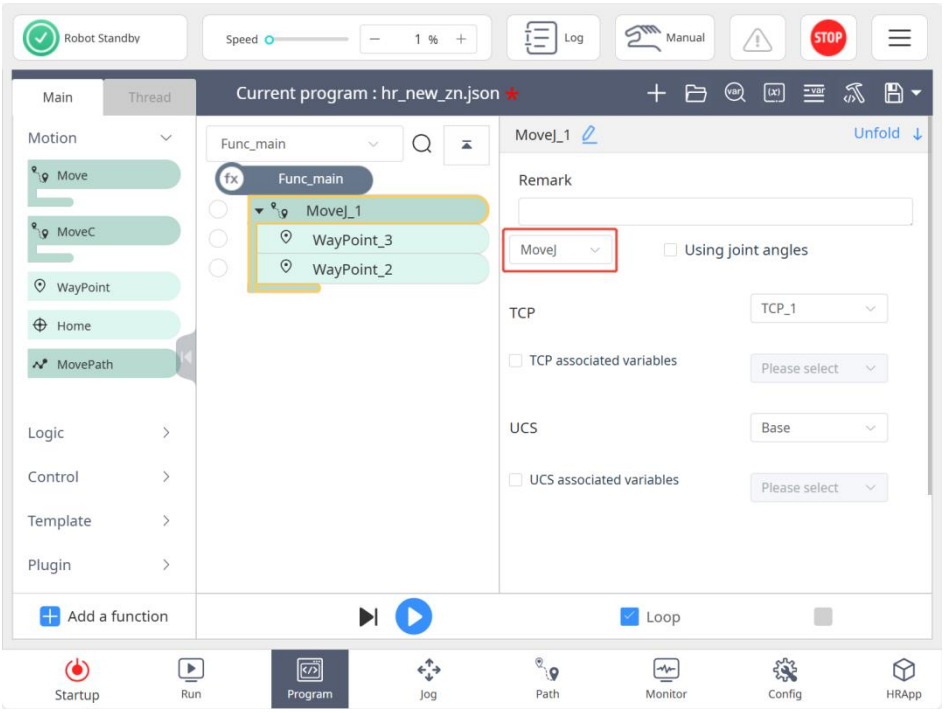


Fig. 6.3.1 MoveJ Motion

The joint axes of the robot arm move synchronously and quickly from one point to another. It is suitable for scenarios where there are no requirements for the motion path of the robot arm's end-effector. The robot arm's end-effector moves along an irregular curve. Please pay attention to whether there is a danger of potential collision.

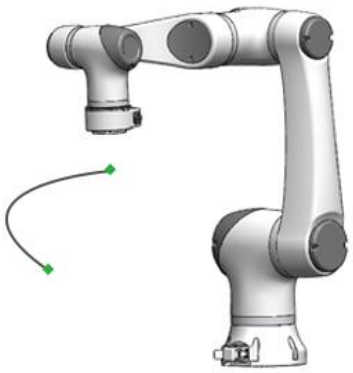


Fig. 6.3.2 MoveJ Motion

Parameters include enabling "Using joint angles", Set TCP, Acceleration, and Joint velocity;

◆ **Using Joint Angles**

If the "Using joint angles" function is turned on, the "Set TCP" option is not available. In this mode, the motion of the robot arm is determined only by the joint angle of the waypoint, independent of the selected TCP (tool center point).

◆ **Set TCP**

The user can switch to the appropriate TCP (tool center point) according to the tool at the end-effector. Select the TCP that matches the current tool at the end-effector from the drop-down menu. When setting a waypoint, the program records the spatial coordinate value of the tool function corresponding to that point. The motion path of the robot arm is determined only by the spatial position of the tool at the end-effector of the waypoint.

By default, the TCP option uses "Factory TCP".

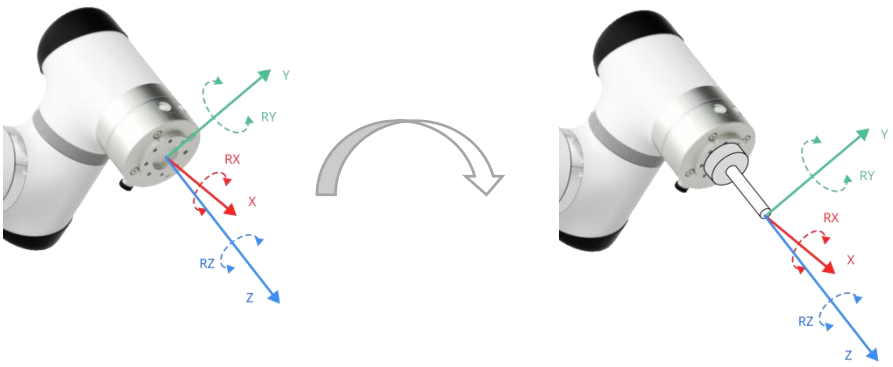


Fig. 6.3.3 Comparison of Tool Coordinates After Switching TCP

◆ **Acceleration**

The user can set the joint acceleration of the robot arm when moving from the current waypoint to the next waypoint.

◆ **Joint Speed**

The user can set the joint speed of the robot arm when moving from the current waypoint to the next waypoint.

Note: These parameters will be used as the parent properties of the waypoints under the MoveJ motion path, and the same applies to other motion modes.

6.3.1.2. MoveL Spatial Linear Motion

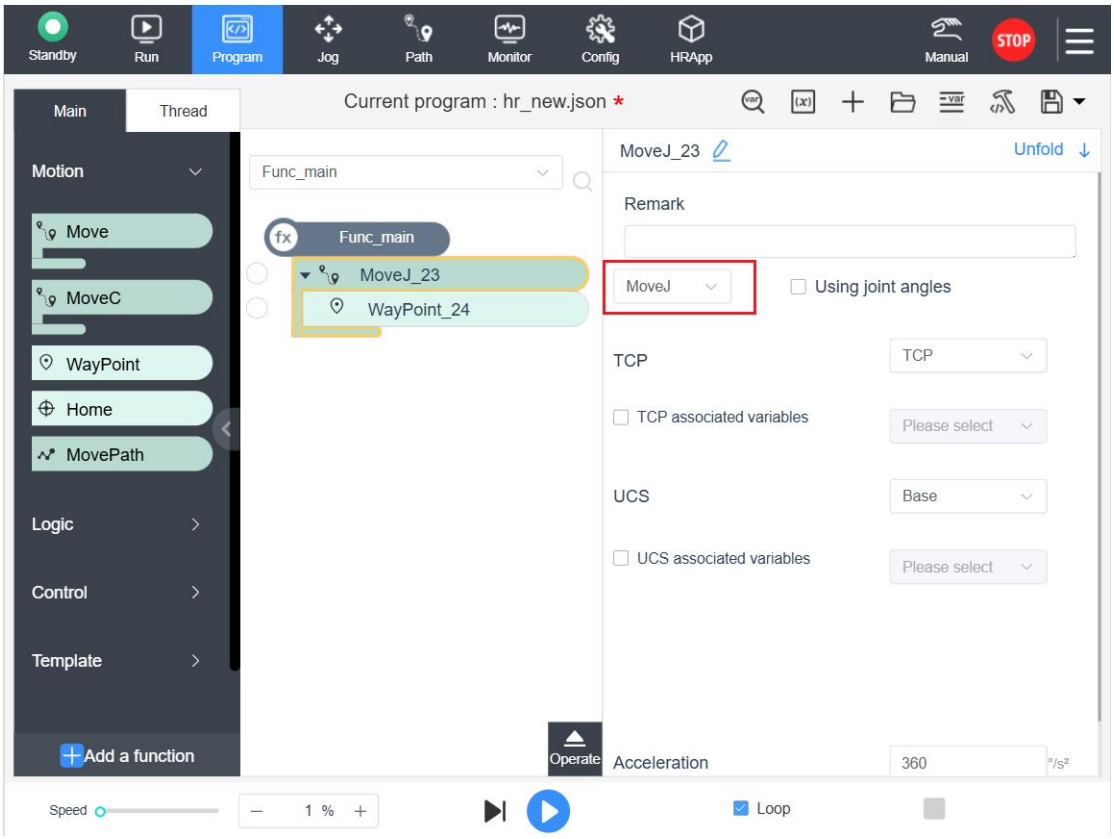


Fig. 6.3.4 MoveL Motion

If the robot arm is desired to move in a straight line, a spatial linear motion command can be utilized. This command enables the tool center point (TCP) to move along a straight-line path from the start point to the end point. When the orientations of the starting point and the end point are different, the TCP orientation will synchronously rotate with the position movement, eventually aligning with the orientation at the end point.

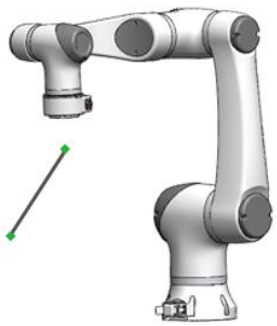


Fig. 6.3.5 MoveL Motion

Parameters include: Set TCP, Features of the Coordinate System, Acceleration, and Tool speed;

◆ **Set TCP**

The user can switch the appropriate TCP according to the tool at the end-effector. Select the TCP that matches the current tool at the end-effector from the drop-down menu. When a waypoint is set, the program records the spatial coordinate value of that point in the selected tool coordinate system. The TCP option is set to "TCP" by default. This parameter is a factory initial value, its spatial coordinates are (X = 0, Y = 0, Z = 0), and its orientation parameters are (RX = 0, RY = 0, RZ = 0).

When editing, the TCP associated variables can be selected, and **the called variable is of string type**.

◆ **Coordinate System**

Select the suitable coordinate system from the drop-down menu. After switching the coordinate system, the path formed by the waypoint will be relatively translated as a whole based on the coordinate system switched to.

When editing, the coordinate system associated variables can be selected, and **the called variable is of string type**.

◆ **Acceleration:** The joint acceleration of the robot arm as it moves from the current waypoint to the next waypoint.

◆ **Tool Speed:** The joint speed of the robot arm as it moves from the current waypoint to the next waypoint.

6.3.1.3. MoveC Arc Motion Command

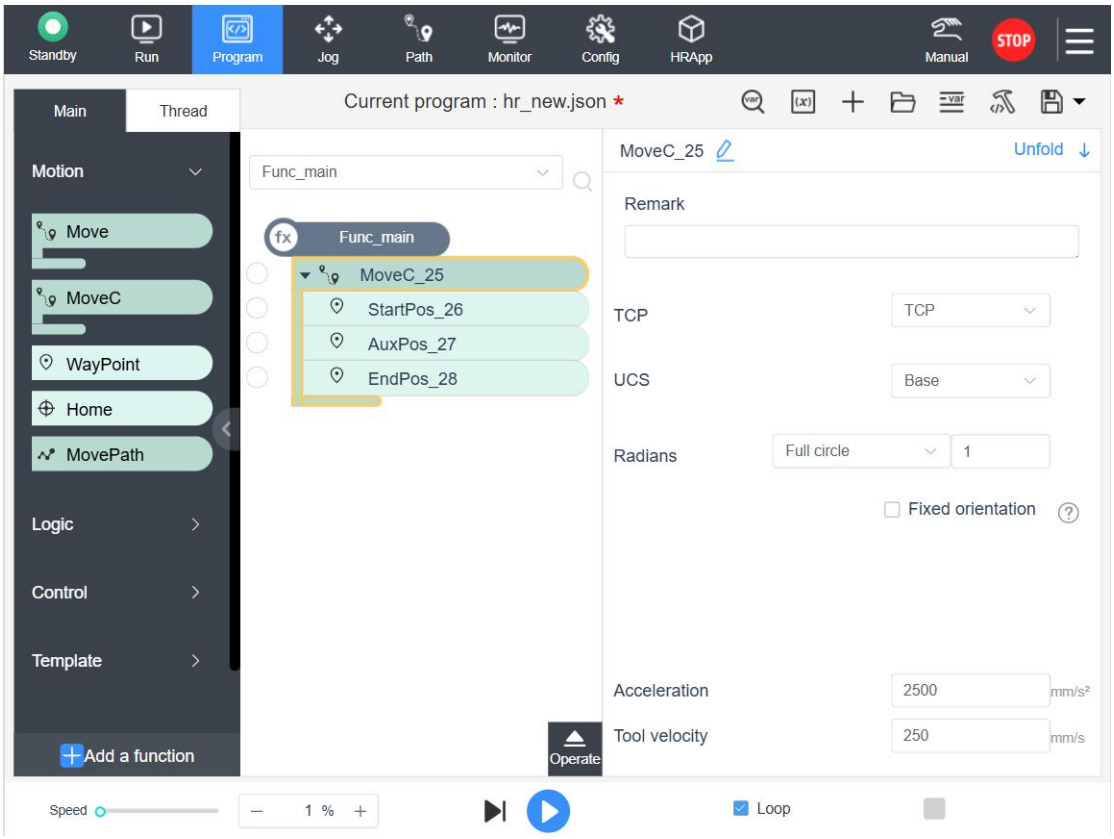


Fig. 6.3.6 MoveC Motion

The MoveC command is used to generate a circular or arcuate path. It is based on the geometric principle of defining a circle or arc using three distinct points. The three points must not be colinear and any two points must not coincide. If two points are at the same position or if all three points lie on a straight line, a circular path cannot be determined.

◆ **Set TCP**

Please refer to the TCP setting description of "6.3.1.1 MoveJ Joint Motion" for details;

◆ **Coordinate System Function**

Please refer to the coordinate system function description of "6.3.1.2 MoveL Spatial Linear Motion" for details;

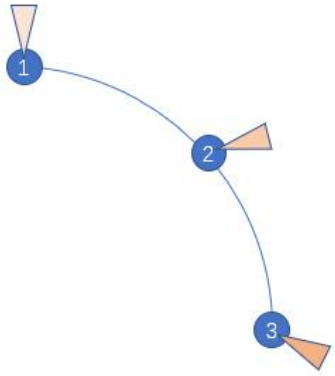
◆ **Blending Radius**

Please refer to the blending radius description of "6.3.2 Waypoint" for details;

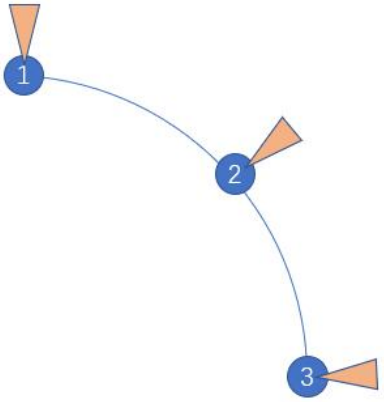
◆ **Function Description**

Table 6.3.1 MoveC Motion function Description

Feature	**Description**
Circular Arc	There are two types: arc, full circle
StartPos	The current position of the robot arm, that is, the starting point of the arc
ThroughPos	A point on the circular path can be set by either "manual teaching" of the PCS coordinate values or by "selecting a variable" to input the coordinate values.
EndPos	The endpoint of the circular path, which is also considered a waypoint in the case of a full circle, can be either manually taught as a PCS coordinate value or input via a variable.
Circular Number	When the arc type is set to full circle, this specifies how many times the robot arm repeats the circular path.
Fixed Direction	There are two types: 1) When the "Fixed direction" option is not selected, the orientation transitions from the start point to the end point, irrespective of the second waypoint's orientation. As shown in the figure below, points 1, 2, and 3 represent the start point, intermediate point, and end point of the arc, respectively.



2) When the "Fixed direction" option is selected, the orientation remains consistent with the initial TCP orientation throughout the motion. Points 1, 2, and 3 represent the start point, intermediate point, and end point of the circular arc, respectively. The tool at the end-effector maintains the initial orientation as it moves from the start to the end point along the path, as shown in the figure below.



6.3.2. WayPoint

WayPoint is a point on the motion path of the robot arm. A waypoint is a core element of a robot program and is used to indicate a specified position to which the robot arm is moving towards. The fixed position waypoints are taught by physically moving the robot arm to the desired position.

Waypoints are only used within Move nodes and cannot become first-level nodes alone (except within templates). Affected by the parameters in the Move function, waypoints should be taught according to the relationship between the functions applied in the Move function and TCP to achieve the desired TCP position.

Three waypoint position modes are available: fixed position, translation, and variable. Users can switch between these modes according to practical needs. The default mode is "fixed position".

6.3.2.1. Fixed Position

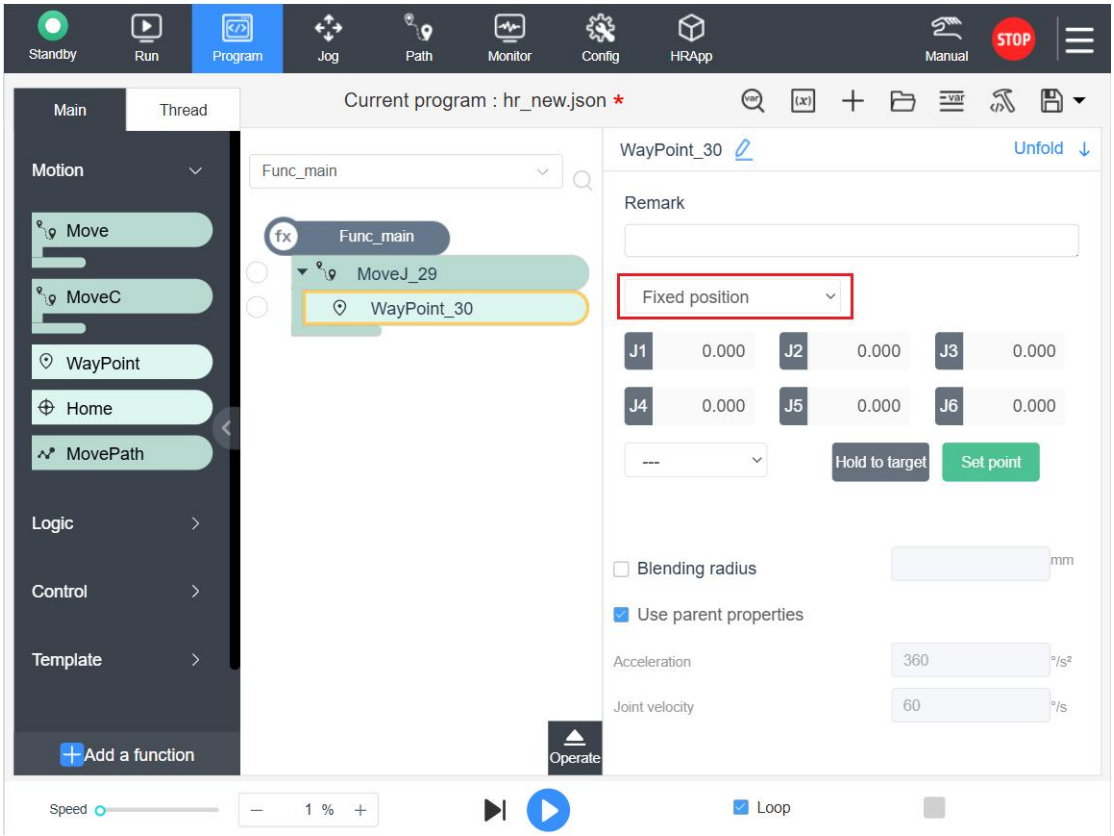


Fig. 6.3.7 Fixed Position

◆ **Point Information**

If the waypoint is within the MoveL function, the position information of the current

point is displayed as P [X, Y, Z, RX, RY, RZ]; if the waypoint is within the MoveL function, P [J1, J2, J3, J4, J5, J6] is displayed;

◆ **Set Point**

To switch to the "Jog" interface to teach the target position of the robot arm, and set the target motion position of the waypoint called WayPoint_1 waypoint.

◆ **Hold to Target**

Press and hold the【Hold to Target】 button to move the robot arm to the current point position. A "Target point reached" prompt will appear when it arrives.

◆ **Point List**

Click the drop-down menu to select the point, and the point information will display the position information of the current point. To view the specific location of this point, please press and hold 【Hold to Target】. You can edit the point list on the jog interface.

Note: If a point being used for a waypoint is currently deleted from the waypoint list, the corresponding waypoint information in the waypoint will also be synchronously deleted, and an error will be reported. The waypoint can be restored by modifying the waypoint information.

◆ **Use Parent Properties**

By default, the "Use parent properties" is selected, the acceleration and joint velocity (tool velocity) in Move function will be used; if it is not selected, the currently specified acceleration and joint velocity (tool velocity) will be used instead.

◆ **Blending Radius**

When the blending radius is configured between two adjacent linear segments, the robot arm will transition smoothly via a circular arc, enhancing operational efficiency. This ensures continuous motion without pausing at waypoints. A smaller blending radius creates a sharper corner, while a larger radius results in smoother cornering.

The blending radius enables smooth transitions via arcs. **The set radius must not exceed the straight line distance, otherwise, a calculation error will be reported. If two straight lines can be connected into one straight line, the set radius is invalid.**

Note: In MoveL and MoveJ functions, selecting the checkbox enables blending radius; deselecting it makes the robot stop at the point before moving to the next. In MoveC function, selecting the checkbox uses the current point's blending radius, while deselecting it applies the blending radius set in MoveC.

Example:

A. Blending Radius in MoveJ and MoveL Commands

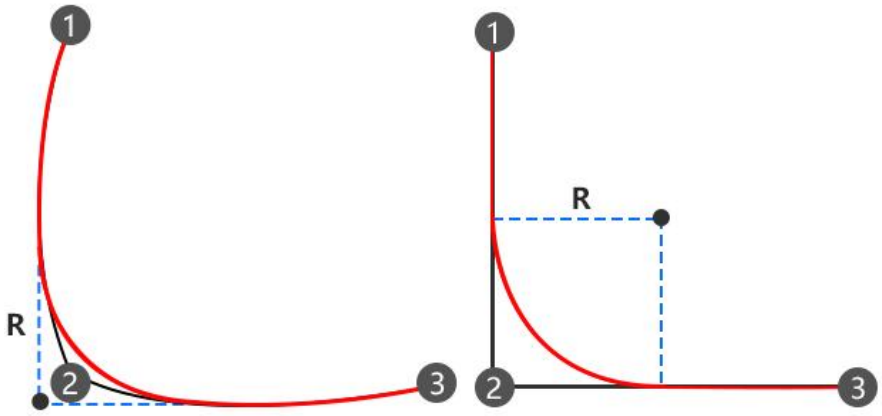


Fig. 6.3.8 Blending Radius in MoveJ and MoveL Commands

As shown in the figure, P1 is the current position of the robot arm, P2 is the target point of the first command, and P3 is the target point of the second command. The red path is the calculated path. The left and right figures are the MoveJ and MoveL commands respectively. **The setting of the blending radius should be done at the next target point after the skipped point.**

B. Blending Radius in MoveC Command

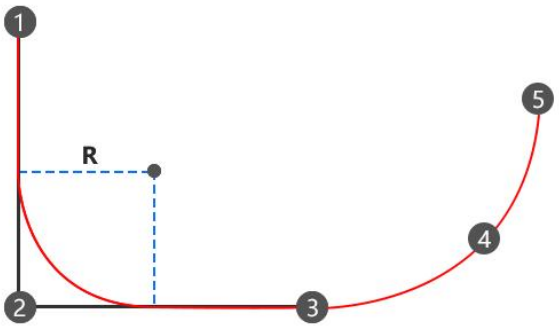


Fig. 6.3.9 Blending Radius in MoveC Command

P1 is the current position of the robot arm, P2 is the target point of the first MoveL command, and P3 is the target point of the second MoveL command, which is also the starting point of the arc. P4 is the intermediate point on the arc path, and P5 is the target position of the arc.

As shown in the figure, the red path represents the calculated motion trajectory. A blending radius is applied between the two linear segments to achieve a smooth arc transition. The combination of MoveL + MoveC ensures continuous motion between the linear path and the arc.

6.3.2.2. Translation

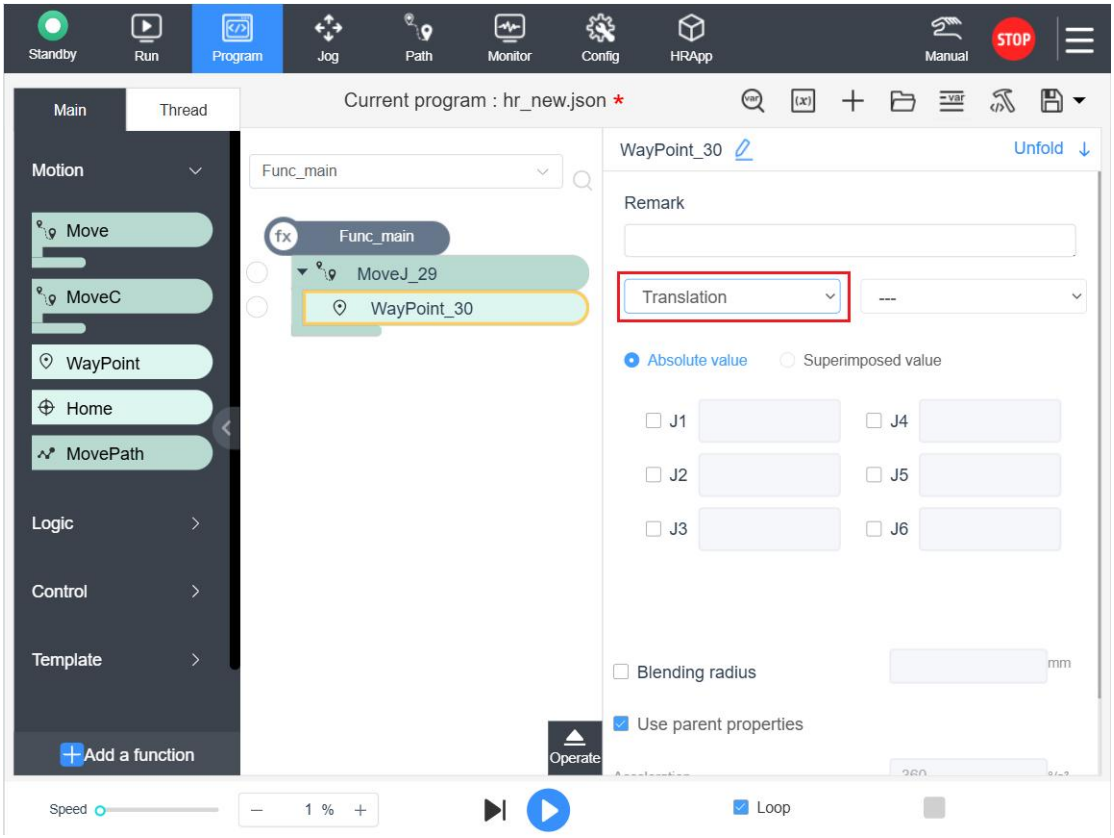


Fig. 6.3.10 Translation

- **Point List:** Select the specified point in the drop-down menu of point list in the upper right corner; for example, Point _1 is the waypoint 1;
- **Absolute Value:** Move to the absolute coordinate position in the currently selected user coordinate system (such as the Base coordinate system).
- **Superimposed Value:** The last point serves as the reference point, and the movement distance is based on the superimposed value;
- **Select Axes:** Specify certain robot axes for absolute or superimposed translation (J1, J2, J3, J4, J5, J6; X, Y, Z, RX, RY, RZ)
- **Blending Radius:** For details, please refer to the blending radius description of 6.3.2 Waypoint;
- **Use Parent Properties:** By default, the "Use parent properties" is selected, the acceleration and joint velocity (tool velocity) in Move function will be used; if it is not selected, the currently specified acceleration and joint velocity (tool velocity) will be used instead.

6.3.2.3. Variable

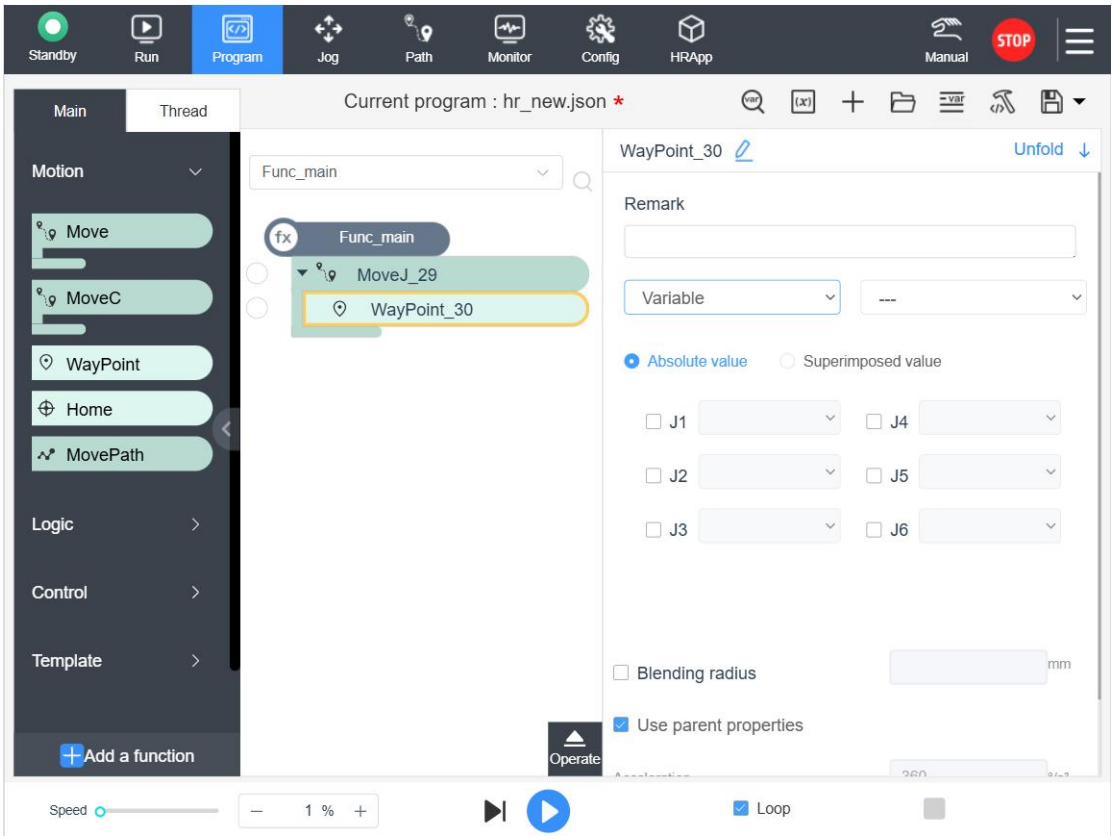


Fig. 6.3.11 Variable

Variables can be added to the variable list. Feedback point data can be received via the script command and assigned to the variable When the robot arm moves to the point, it moves based on the obtained variable position;

6.3.3. MovePath

MovePath is the robot arm's continuous path - control command, Users need to set up an executable path in advance to use this command.

Drag-and-drop MovePath to Func_Main, select the corresponding path name and motion mode in the path list (currently only MoveJ is supported). Click Edit path in the programming interface to modify the path speed ratio.

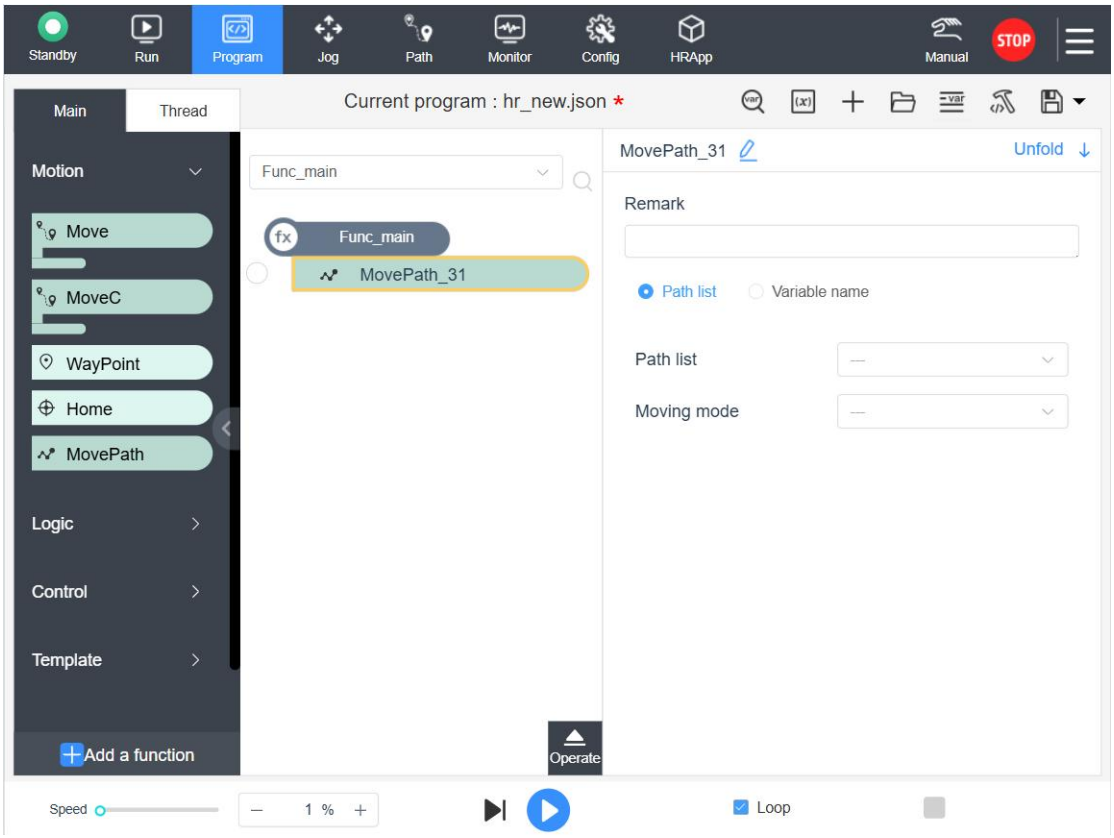


Fig. 6.3.12 MovePath Feature

6.3.4. Home

Home is the motion command for the robot arm to return to its home position. The home position can be modified by the user in Config > Home.

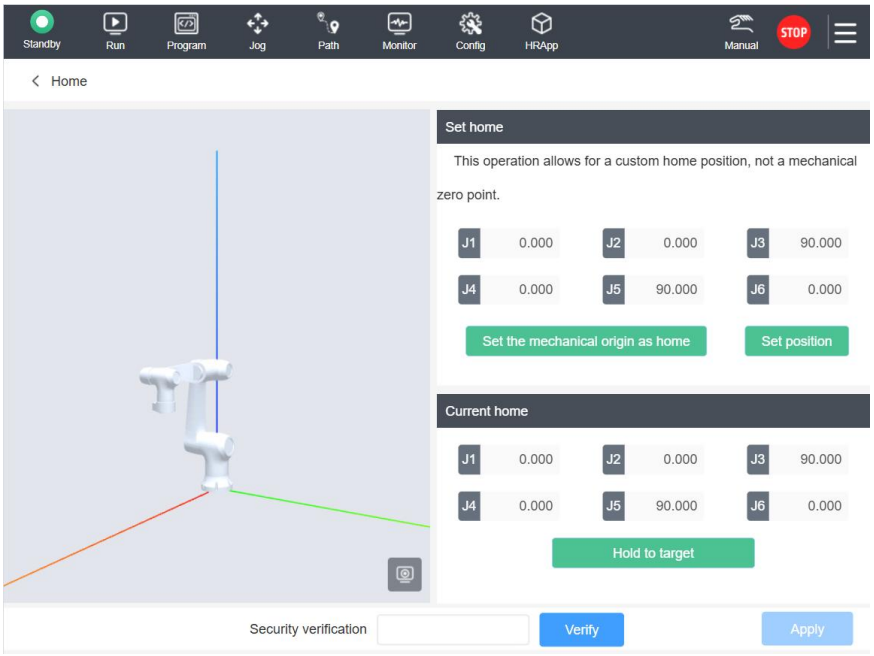


Fig. 6.3.13 Configuration Interface—Home

Drag and drop Home command into Func_Main, and the current home position will be displayed on the right, and the speed and acceleration can be set. When the program executes this command, the robot arm will return to its home position.

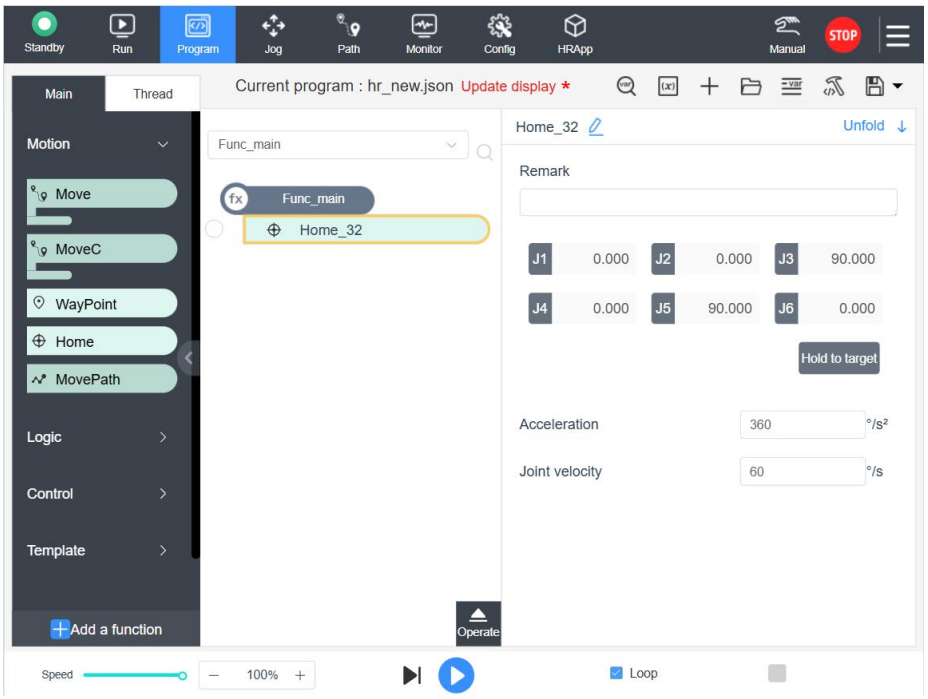


Fig. 6.3.14 Program Interface—Home

6.4. Logic Command

6.4.1. If

The conditional command guides the robot arm to perform corresponding motions based on different conditions. When the If condition is met, the robot arm executes the motion sequence under the If module. Once execution begins, the robot arm completes all commands within the If module, regardless of whether the condition remains true during execution.

Each If command can nest multiple Else If branches and one Else branch. When the If condition is true, the robot arm will execute the command in the If module; When the If condition is not met, it checks each Else If condition in order and executes the first matching branch. If no conditions match, it executes the Else branch commands.

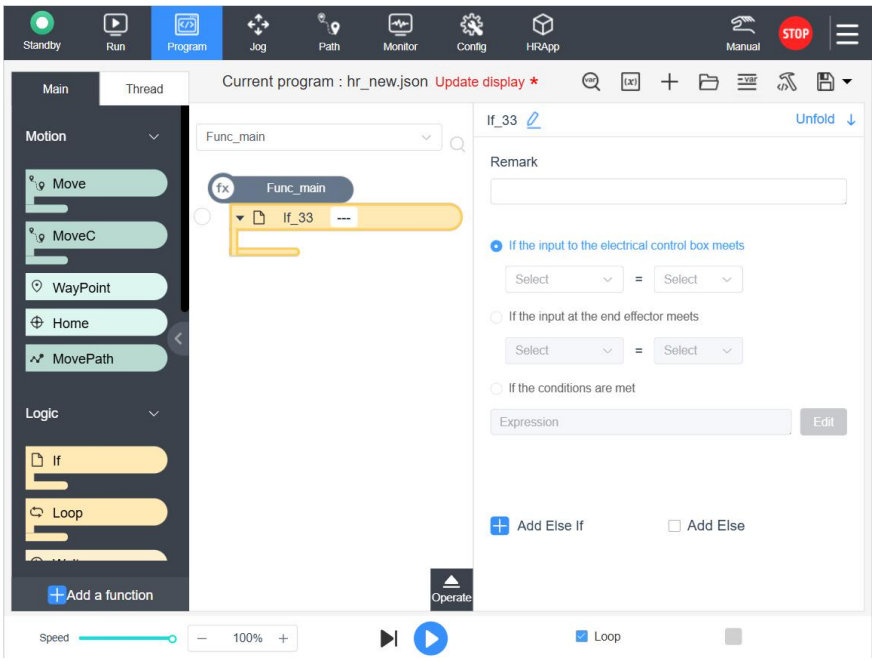


Fig. 6.4.1 If Condition Command

Three judgment conditions are provided:

- Judgement of the input I/O status of the electrical control box;
- Judgement of the input I/O status at the robot arm's end-effector;
- Judgment via custom condition expressions.

6.4.2. Wait

Wait Command: During program execution, the program will wait for a set time or until a condition is met before continuing.

Waiting for the specified time, I/O signals, or expressions.

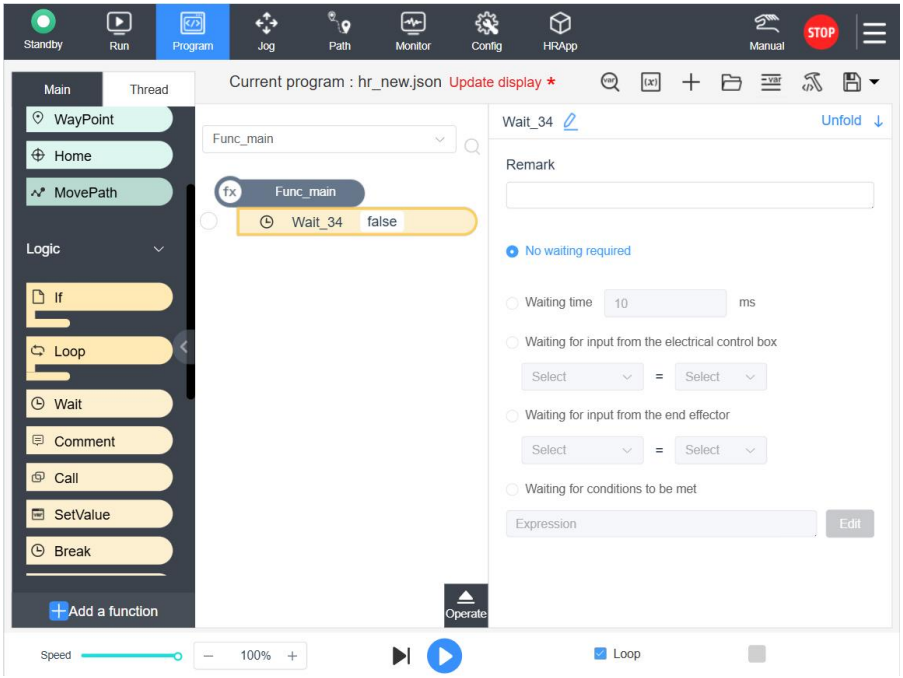


Fig. 6.4.2 Wait Command

6.4.3. Loop

Loop Command: This logic repeats a program sequence until the set number of iterations is exceeded or a preset condition is no longer met. The loop conditions can be set as infinite, a specific number of iterations, or a custom condition expression.

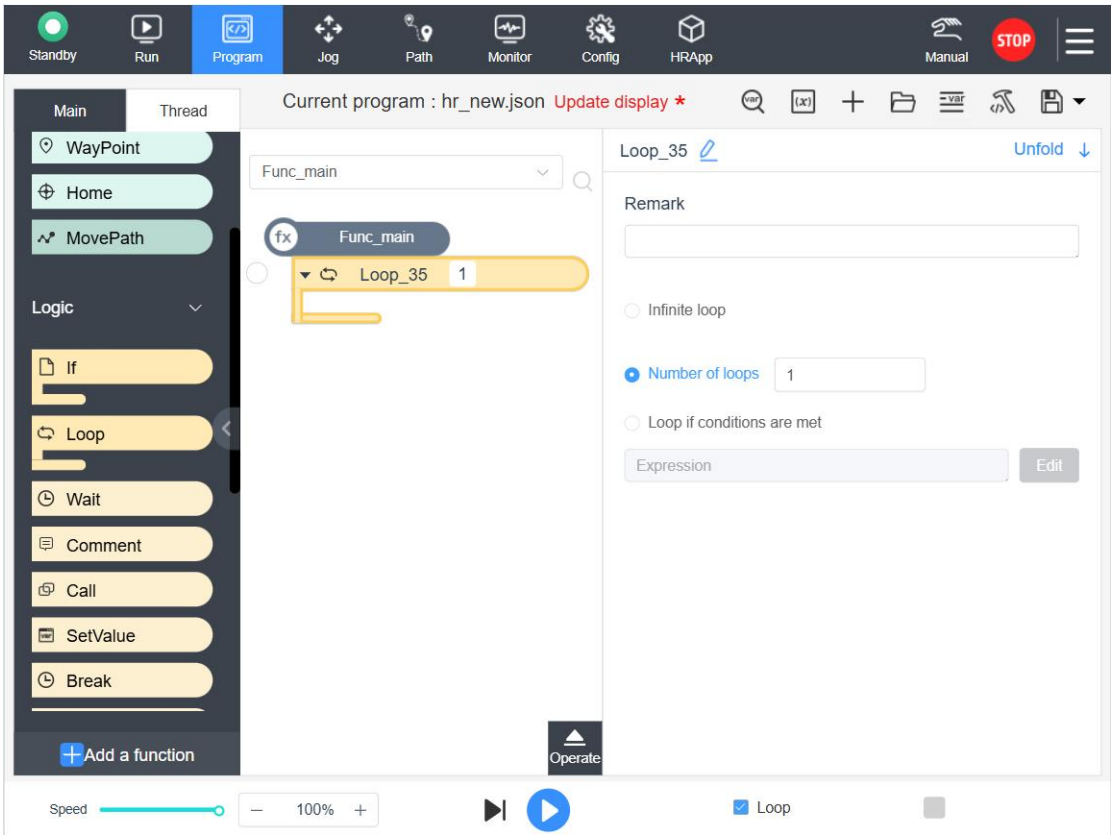


Fig. 6.4.3 Loop Command

6.4.4. Break

When a loop encounters a Break statement, it stops immediately and resumes execution with the next statement outside the loop. Any remaining statements in the loop body and the loop condition are ignored.

Note:

- **Break command exits only the innermost loop. For nested loops, use Break command in each loop layer to exit.**
- **Break command is only valid within a Loop node and can not be used elsewhere.**

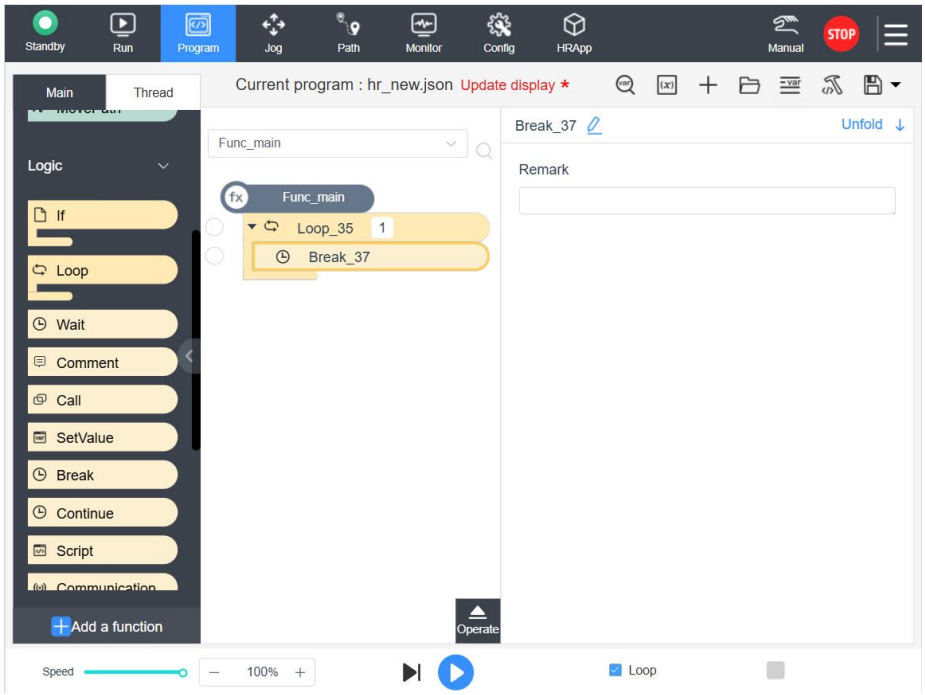


Fig. 6.4.4 Break Terminates the Loop Command

6.4.5. Continue

When a loop encounters a Continue statement, it skips the remaining codes in the current iteration and proceeds to the next loop iteration's condition check. If this occurs in the last loop iteration, Continue command behaves like the Break command, exiting the loop.

Note:

- **Continue terminates the current loop iteration but doesn't exit the entire loop. Instead, it proceeds to the next iteration's condition check.**
- **It only ends a single loop cycle and does not stop the loop's continuation.**
- **Continue is only used within a Loop node.**

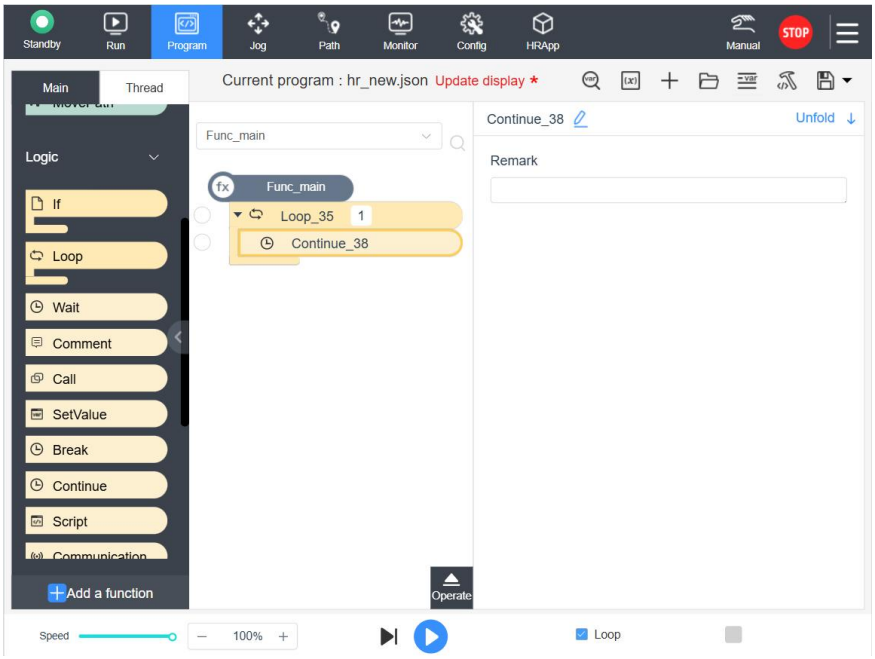


Fig. 6.4.5 Continue Skips the Current Iteration of the Loop Command

6.4.6. Comment

Comment is used to add explanatory notes and can be interspersed anywhere to explain and clarify function statements

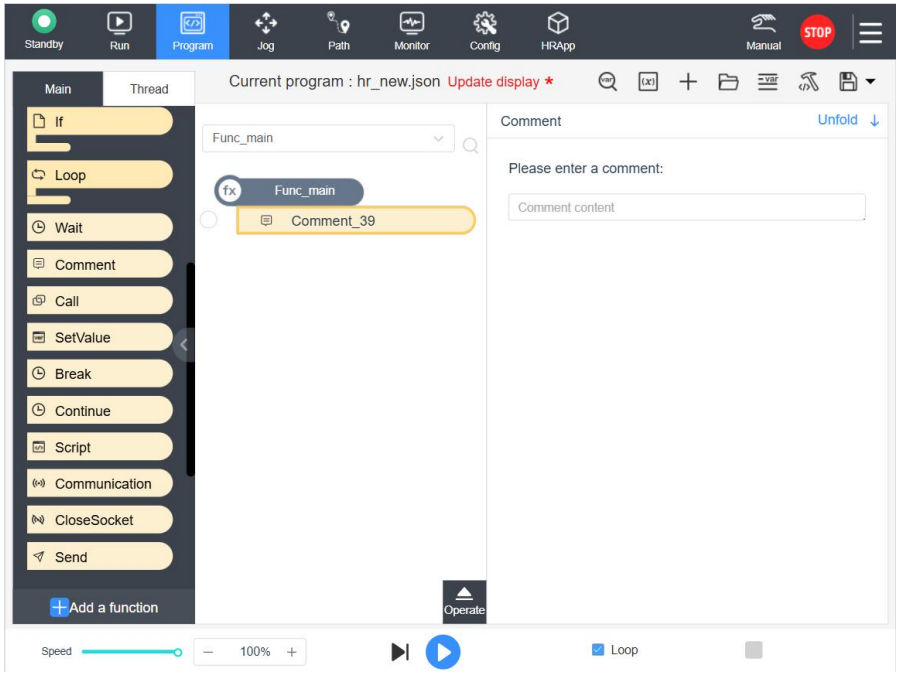


Fig. 6.4.6 Comment Statement

6.4.7. Call

Call is the function call command, used to invoke functions defined in the Function section. When the program encounters a Call command, it jumps to the specified function and executes its logic. After the function completes, the program returns to the statement immediately following the Call command and continues execution.

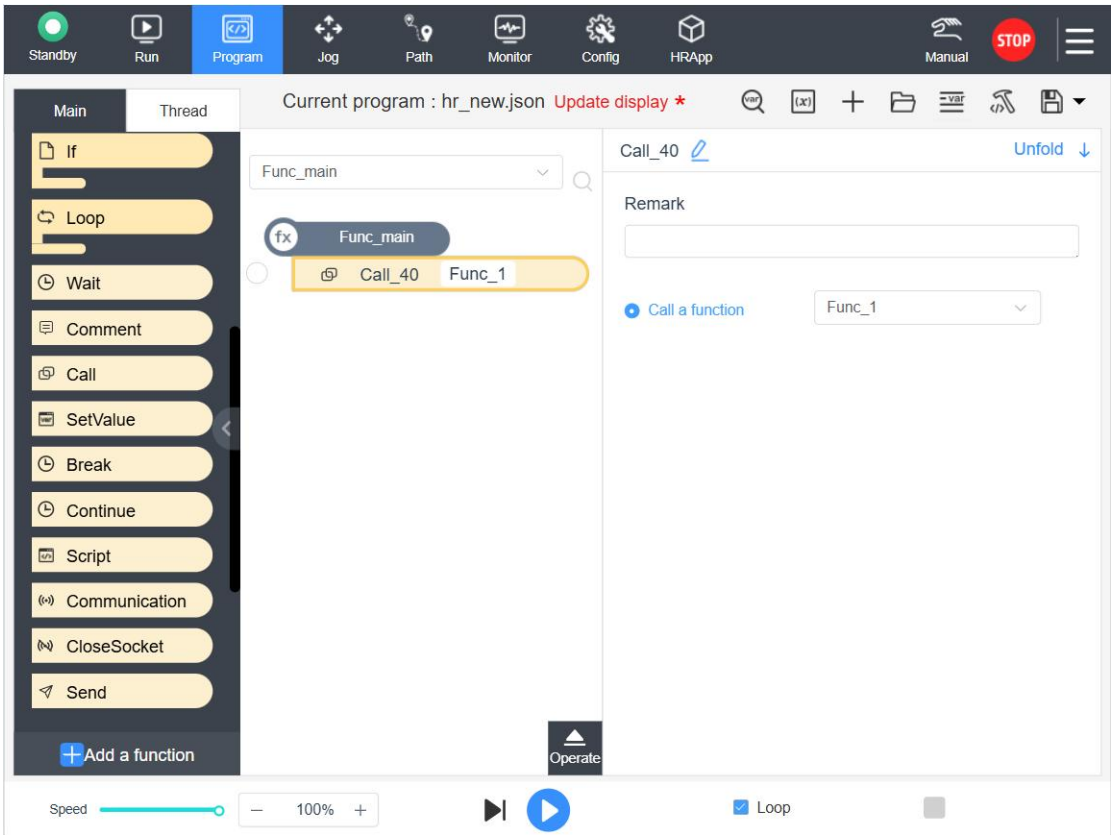


Fig. 6.4.7 Call Function Command

Call Types:

- Functions List: Select the desired function from a drop-down menu. When the program encounters the Call command, it jumps to and executes the selected function's script.
- Function Variable: Used to dynamically call functions based on job requirements. The program calls a function when the variable's value matches a function name. This simplifies complex decision-making logic. Function variables are of the string type.

6.4.8. Set Value

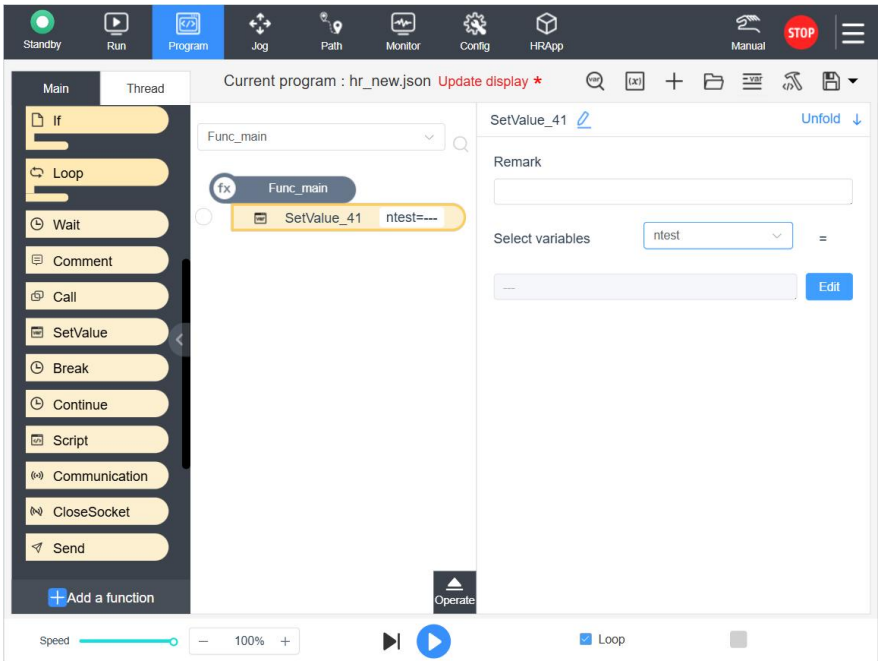


Fig. 6.4.8 Set Value for Variable Command

When assigning values to variables, users can edit expressions to calculate new values and assign the results to the selected variables. This method allows for flexible variable updates based on dynamic calculations or logical operations.

6.4.9. Script

Via script, robot position data acquisition and communication can be carried out;

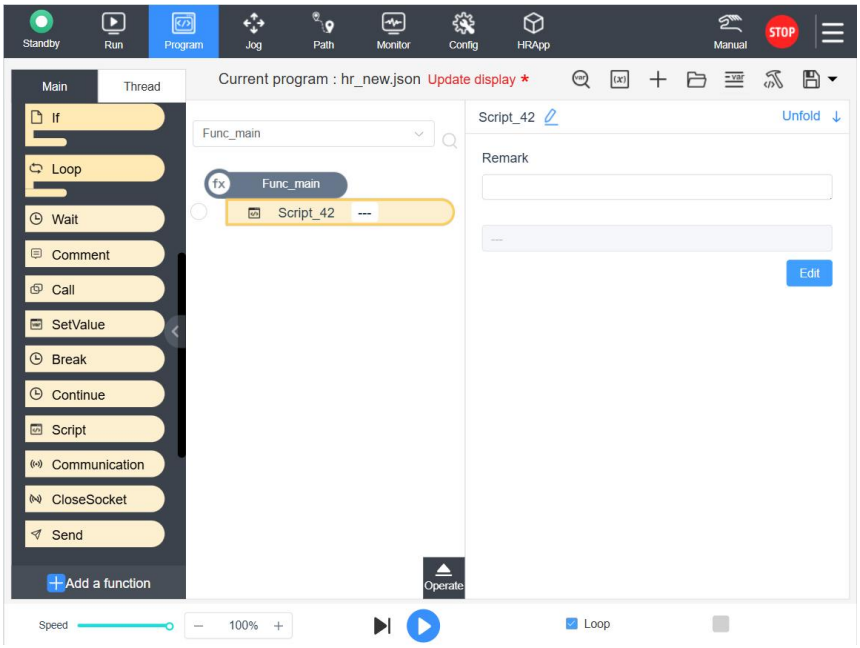


Fig. 6.4.9 Script

Users can import and export Python files from other sources into the script for editing, or create scripts directly. Editing must be done in the file editing interface. Edited files must be saved before they can be referenced in the program, and their names will appear in the Script property information. For more script-related instructions, please refer to the documentation titled "Huayan Script Library Manual".

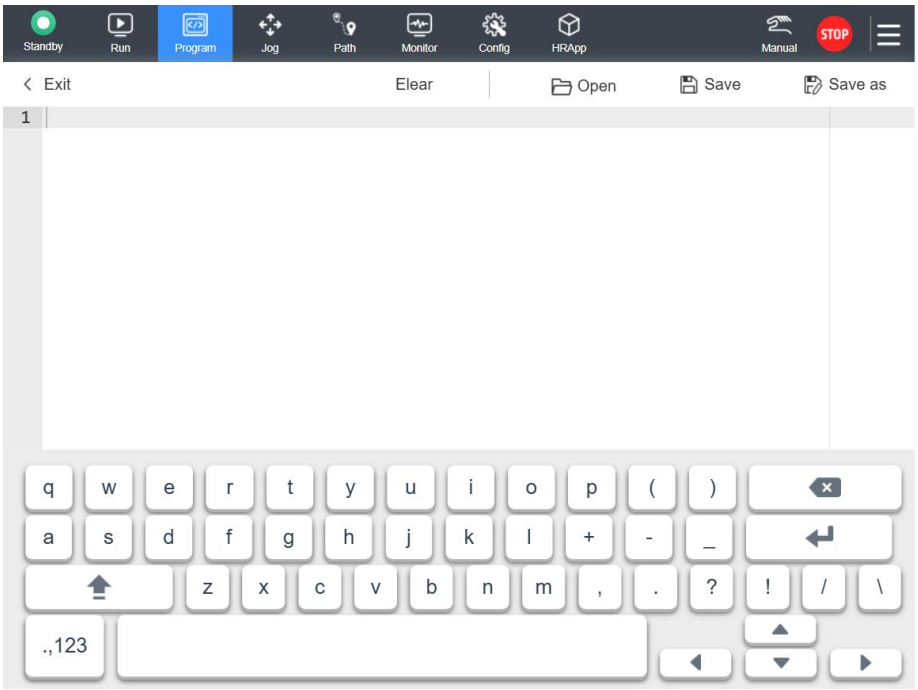


Fig. 6.4.10 Script File

- Exit: Quits the editing interface. When exiting, the system checks if the file is saved. If not, it prompts the user to save before exiting. After exiting, the Script properties display the file name, indicating successful program reference;
- Clear: Clears all codes after user confirmation; a pop-up window will appear to prompt whether to clear and then perform clearing;
- Open: Opens files from the default system path or other sources. If a script is being edited, it prompts the user to save before importing;
- Save: Saves files to the default path or exports via USB, with a prompt to name the file;
- Save As: Save as a file to the system default directory or exports via USB, with a prompt to name the file;

6.4.10. Communication

Used for TCP communication with external devices when the controller acts as a client.

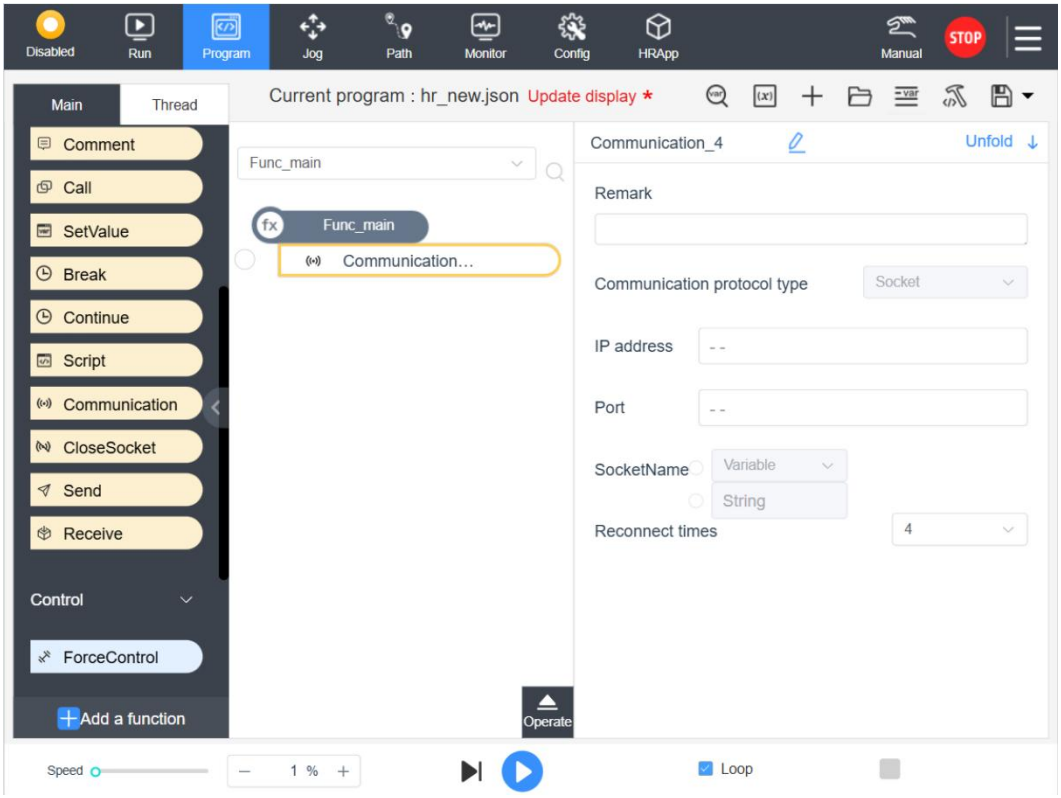


Fig. 6.4.11 Communication

Connect to the TCP server using its IP and port number, and assign a connection name (SocketName) via variables or an input box. If the Communication connection

succeeds, Send and Receive commands use this SocketName to interact. If reconnection attempts exceed the specified limit, an error pop-up window will appear and the error is logged. Multiple Communication commands can establish several connections to the same or different servers.

6.4.11. CloseSocket

Used to close the established Socket connection;

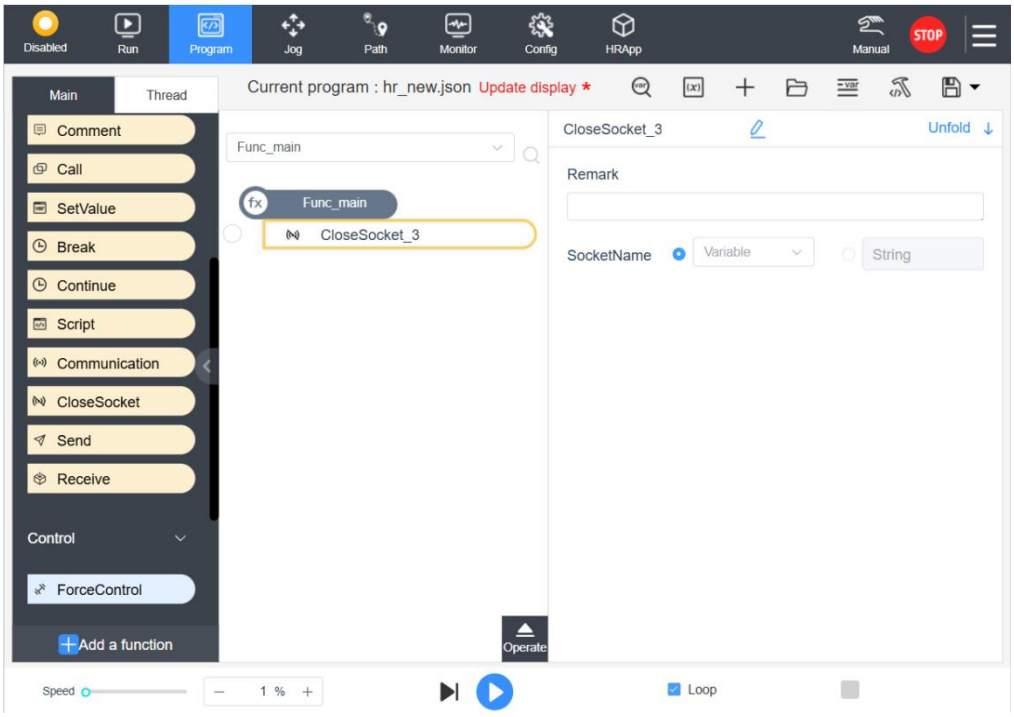


Fig. 6.4.12 CloseSocket

Specify the connection name (SocketName) of the connection to be closed. Set SocketName via a string variable or input box. Ensure the connection is established with the Communication command before calling this command.

6.4.12. Send

Sends string data to a server via an established Socket connection.

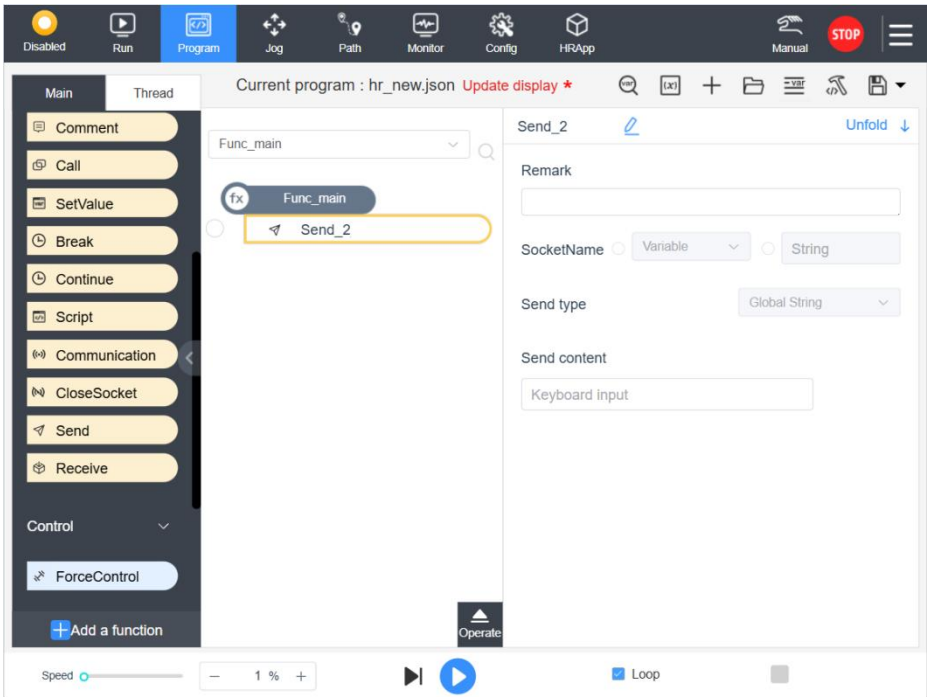


Fig. 6.4.13 Send Data

6.4.13. Receive

Receives string data from an established Socket connection and can split/process it with delimiters.

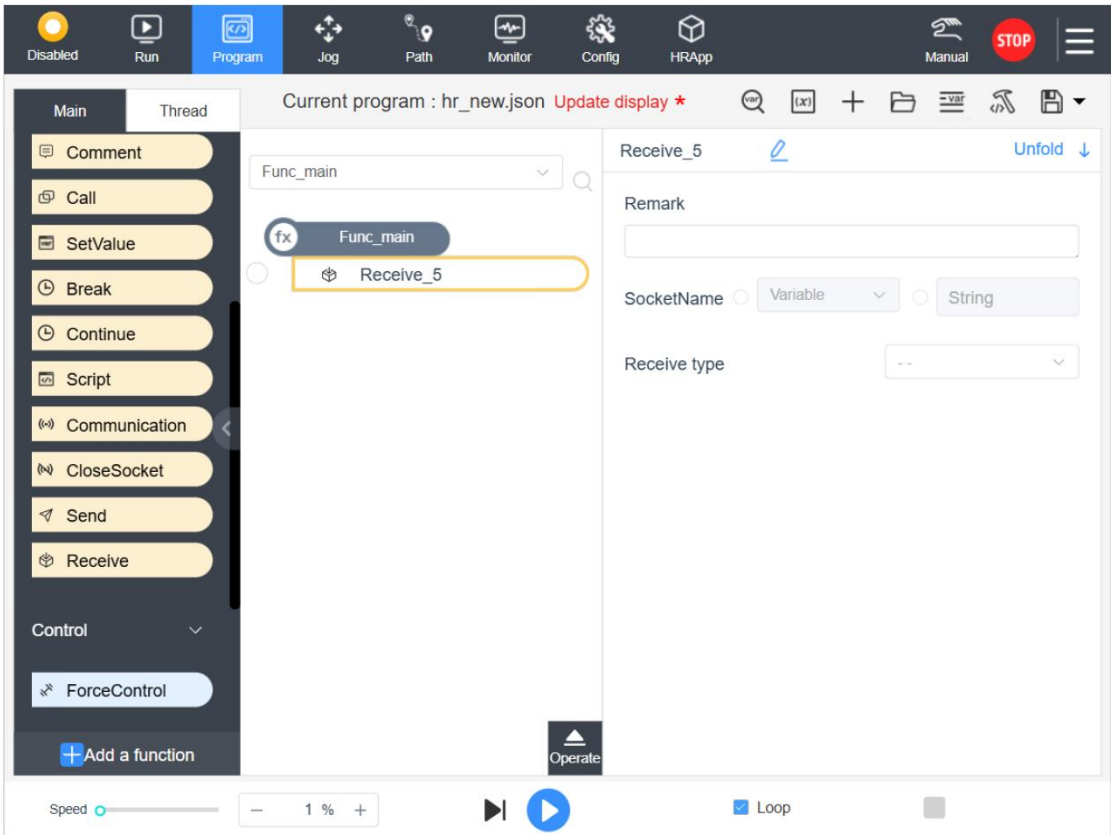


Fig. 6.4.14 Receive Data

- Global String: Receives and stores messages directly into the receiving variable without any processing. The stored string exactly matches the source string from the server.
- Split String: After receiving the message, it is first truncated according to the start and end fields, then split into segments based on delimiters. These segments are then sequentially stored in multiple receiving variables (e.g., Receiving Variable 1, 2, 3, etc.).

6.5. Control Command

6.5.1. Set I/O

Set the input/output (I/O) status for the electrical control box and the end-effector, as well as the current or voltage values of the electrical control box's analog output channels. Also, set the variable values of current or voltage for these channels. If Modbus is configured, set its output status to high or low.

Set the electrical control box's output I/O status, which involves setting the

corresponding I/O signals to either high or low. When the robot executes the Set I/O command, the I/O status changes, after which subsequent commands are carried out.

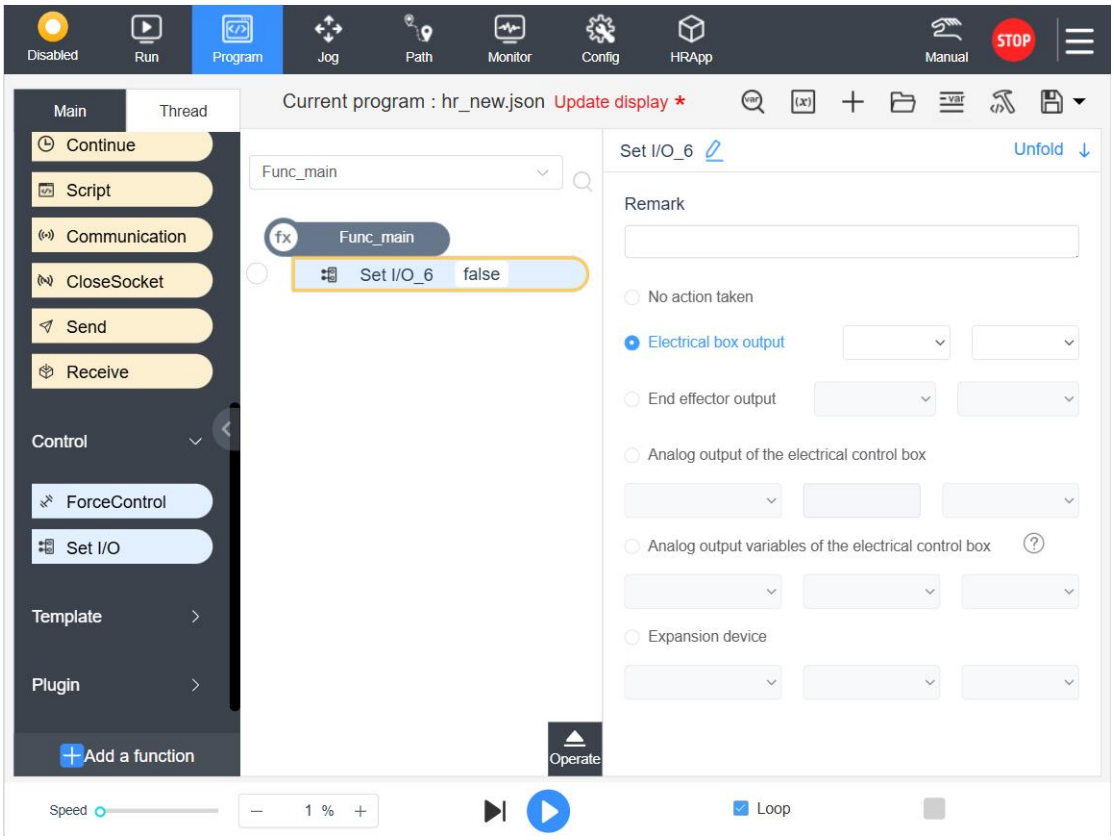


Fig. 6.5.1 Set I/O Command

6.6. Template

6.6.1. Pallet

The Pallet command enables continuous object (workpiece) picking, palletizing, and depalletizing on a fixed pallet. The user can set parameters including: coordinate systems feature of palletizing and depalletizing, pallet coordinate system, workpiece height, a workpiece counter, and options to remember the last workpiece position, run pre-start or post-end programs. The workpiece counter can be viewed in "Current Variables" during program execution. If "Remember Last Workpiece Position" is selected, the program resumes from where it stopped previously. Selecting "Pre-start Program" or "Post-end Program" allows adding extra logic or control commands before or after the Pallet command.

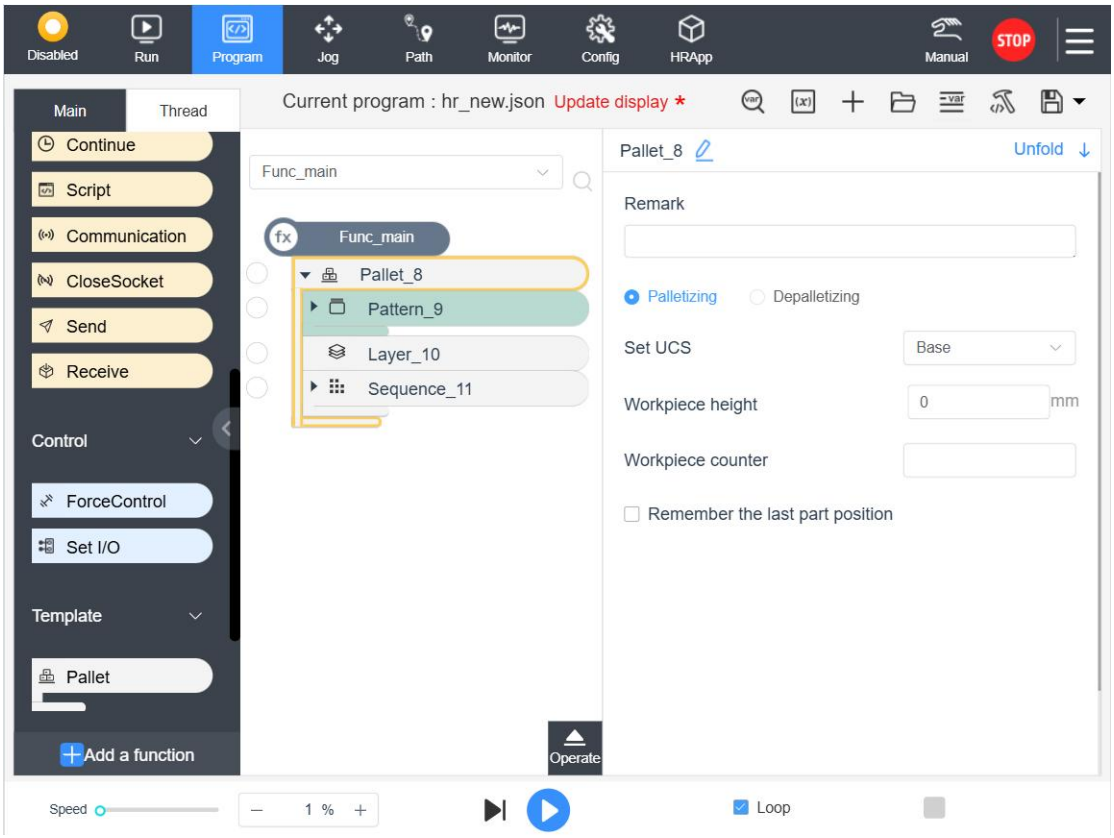


Fig. 6.6.1 Pallet Template

In addition, the pallet command includes three "sub-commands": "Pattern", "Layer", and "Sequence".

6.6.1.1. Pattern

Select a pallet pattern from three options: "Straight Line", "Block", and "Irregular". Take the straight - line pattern for example. After selecting it, define a start point (P1) and an end point (P2). Also, set the "Interval Count," which determines the total number of items for palletizing, as shown in the figure below:

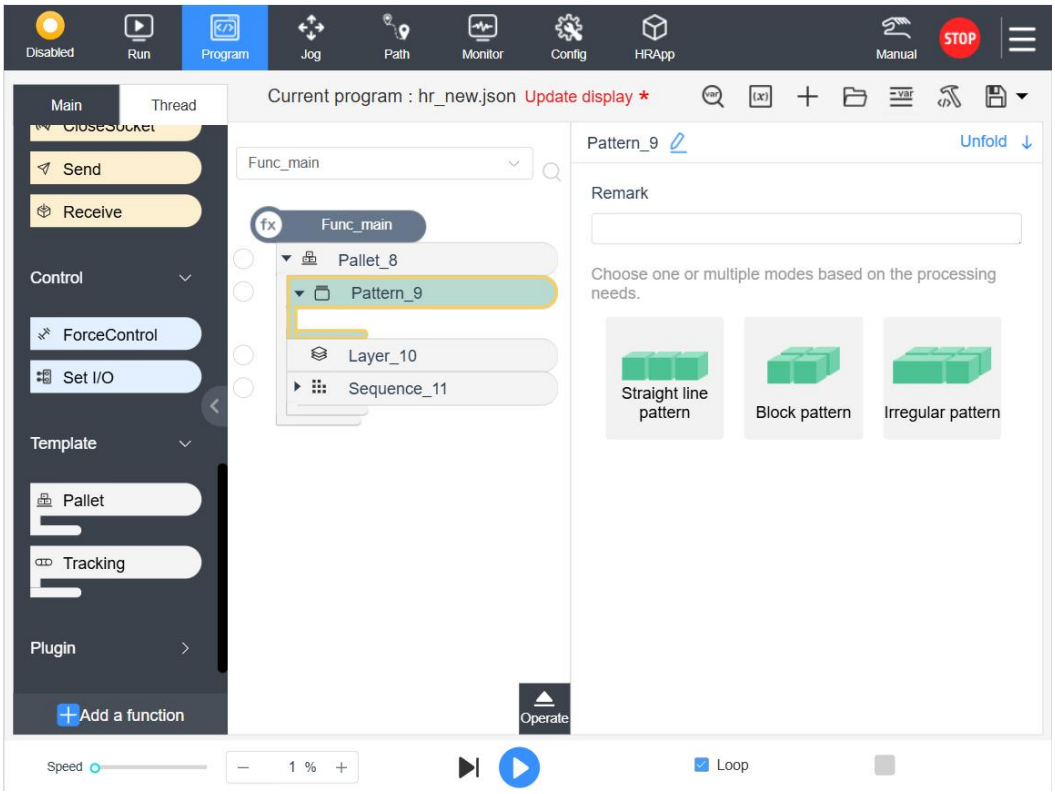


Fig. 6.6.2 Mode Selection

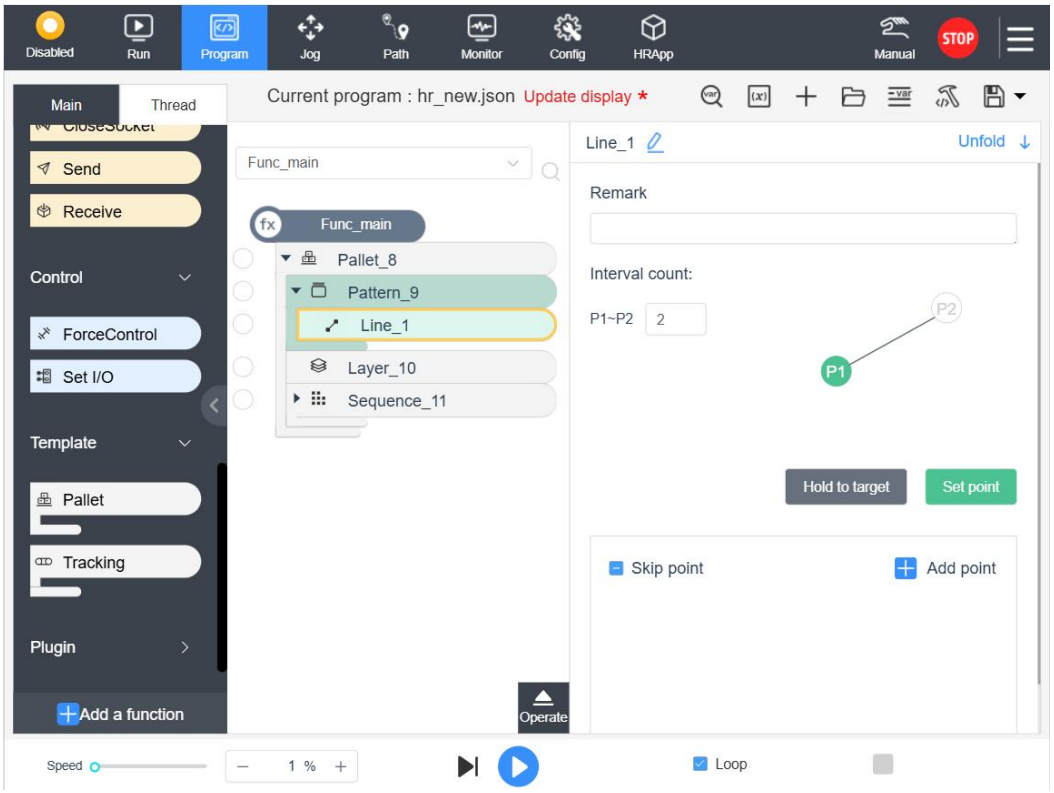


Fig. 6.6.3 Interval Count

6.6.1.2. Layer

In the "Layer" settings, you can specify the number of pallet layers and define the shape for each individual layer.

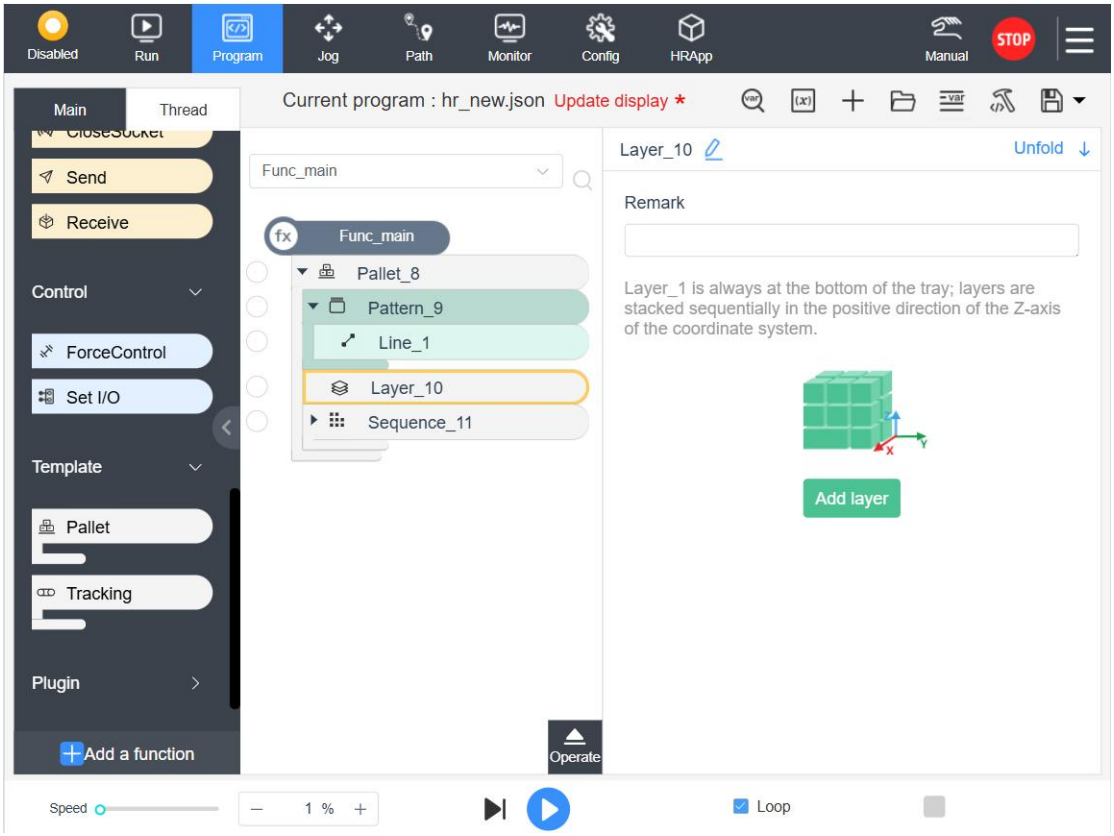


Fig. 6.6.4 Number of Pallet Layer

6.6.1.3. Sequence

To implement palletizing/de-palletizing, place ‘WayPoint' waypoints under ‘Sequence'. Set three waypoints as follows: WayPoint_1 right above the first object to palletize, WayPoint_2 as the palletizing anchor point, and WayPoint_3 as the lifting distance after palletizing completion. To set the pallet anchor point, simply select ‘WayPoint2' from the drop-down menu.

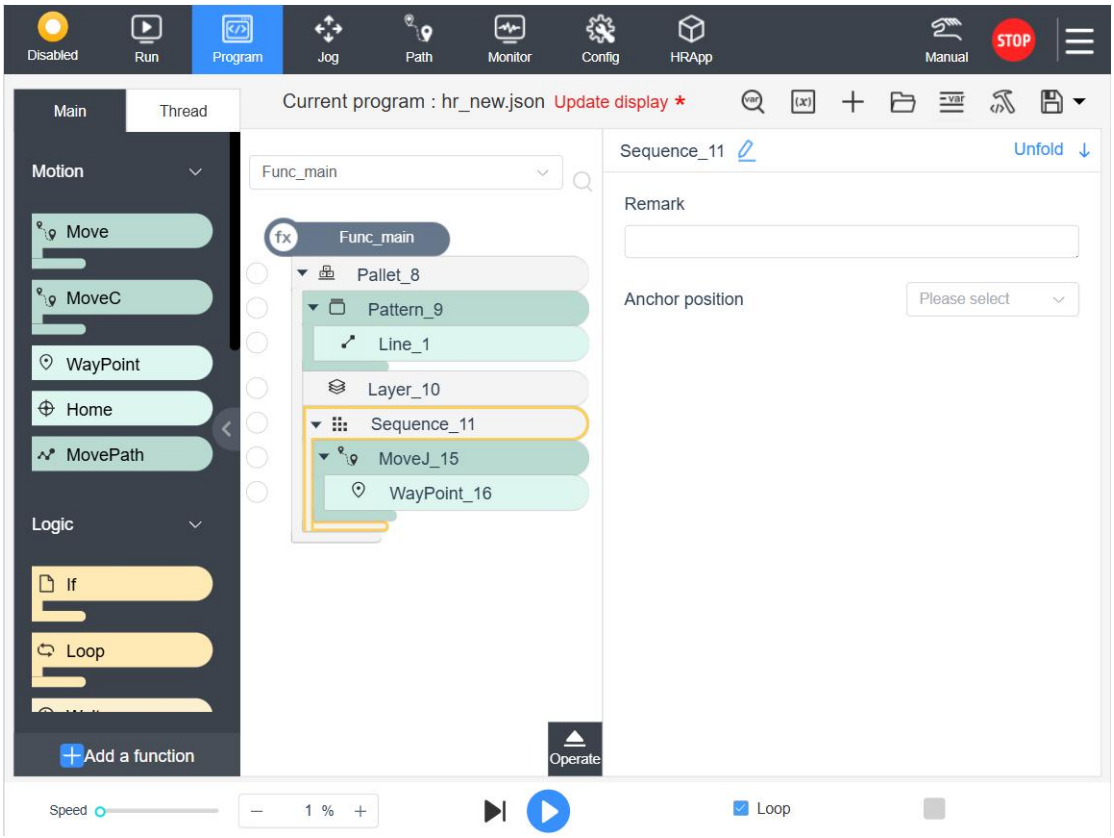


Fig. 6.6.5 Anchor Position Selection

To implement palletizing/de-palletizing, add a Loop command in the sequence based on the total number, as shown in the figure below.

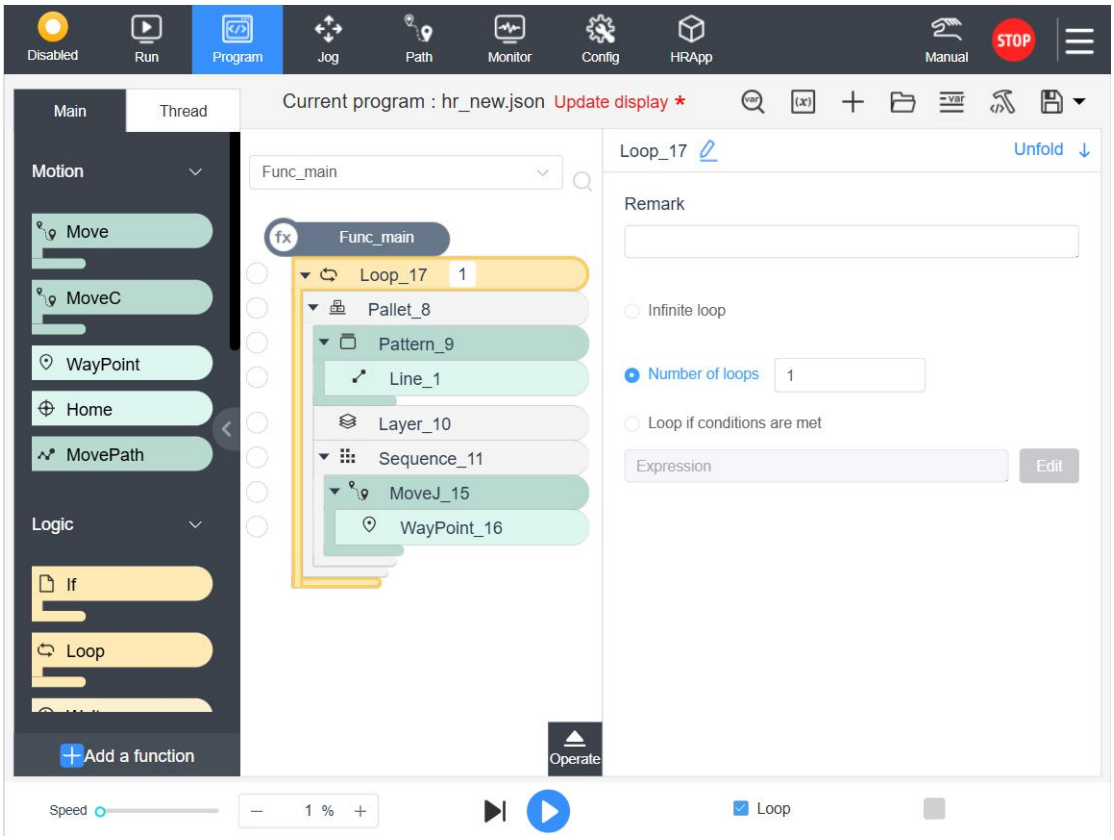


Fig. 6.6.6 Loop Selection

Note: The anchor point is a special "WayPoint". A "WayPoint" under the "Sequence" command is required to select the anchor point. During execution, the robot will move through these WayPoints at each pallet interval, forming a short path. The function of the anchor point is as follows: when a "WayPoint" is designated as an anchor point, its position will automatically align to each palletizing/de-palletizing point on the pallet. Since the path is fixed, the other WayPoints under the "Sequence" command will automatically adjust accordingly. They will undergo relative translation, and the entire path will also shift relatively.

6.7. Tracking

The Tracking feature adjusts the robot's movement to synchronize with the conveyor belt, enabling precise tracking.

6.7.1. Setting User Coordinate System and Tracking Conveyor

When the hardware conveyor belt is connected (for specific connection methods, please refer to the relevant manual), calibrate the conveyor belt to synchronize the robot's direction and movement with it. For setting up the user coordinate system and tracking the conveyor., please refer to Section 5.5 User Coordinate System User Coordinate System and Section 5.8 Tracking Conveyor.

6.7.2. Motion Mode Determination

Once the user coordinate system is established and the conveyor tracking is successfully set up, use the Move_L function (only this motion type is allowed under the tracking function). Select the corresponding user coordinate system, and the robot arm will synchronize with the conveyor belt's movement within the defined range.

6.8. Thread

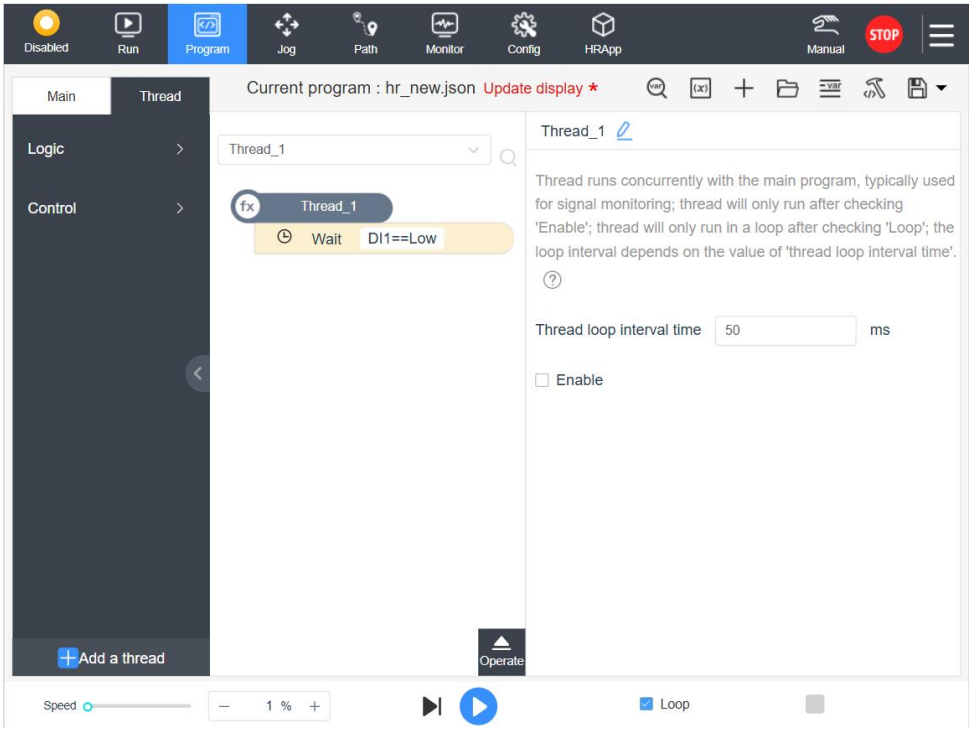


Fig. 6.8.1 Thread Interface

Threads are parallel execution units of the robot's main program, used for real-time signal logic monitoring. Click "Add a Thread" to create a sub-thread. It can run periodically, monitor logic signals such as I/O, Modbus status in real-time, and communicate with the main thread. Up to 5 sub-threads can run simultaneously for parallel multi-tasking.

◆ Thread Loop Interval: Enter 0 ms to run the thread continuously alongside the main thread in a loop; default is 10 ms per cycle.
◆ Enable: Selecting "Enable" activates the thread (marked with a "√" in the thread drop-down list); if not selected, the thread is disabled.

Note:
◆ **Motion commands cannot be added for sub-threads;**
◆ **All sub-threads terminate when the main thread stops.**

6.9. Variable

6.9.1. Variable List

Declare the variables as global variables in the variable list.

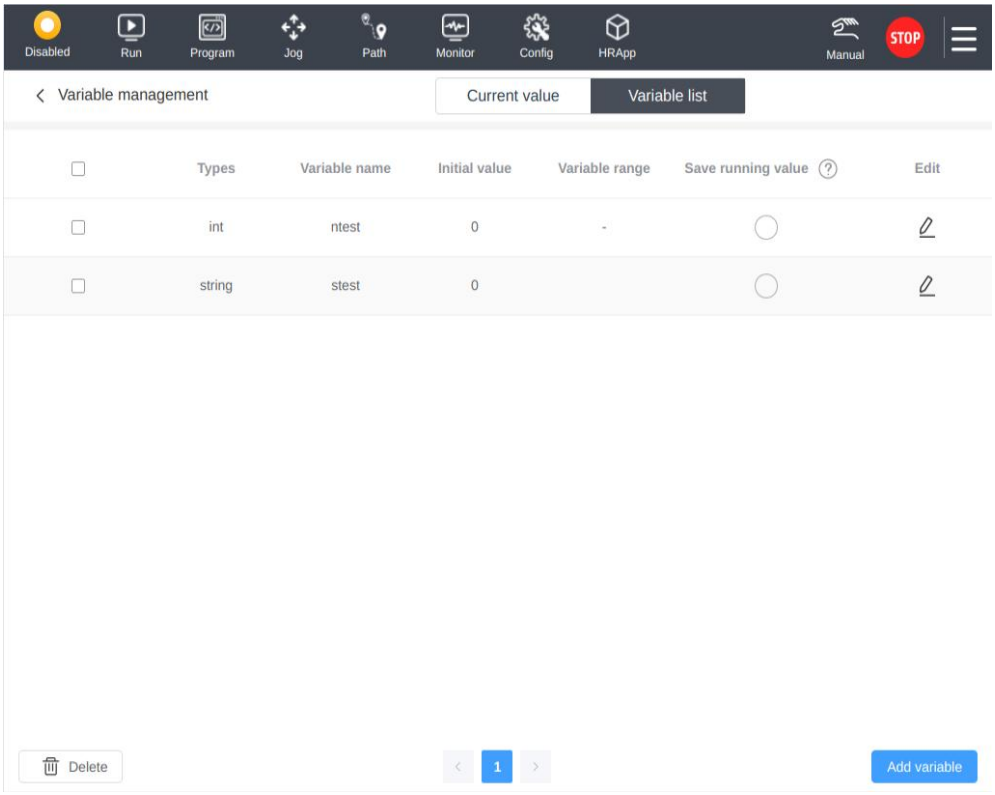


Fig. 6.9.1 Variable List

The variables can be edited, and there are no restrictions on the number of variables that can be configured.

- Add Variable: Used to create a new variable;
- Edit: 【 】 is used to modify the selected variables, including renaming, changing type, assigning an initial value, and configuring retention of the value from the previous execution cycle;
- Delete: Used to delete the selected variables. The user can update variable values during program execution by assigning values to variables through the SetValue command;
- Variable Range: Defines additional constraints on variable values by manually setting upper and lower bounds.

Variable type and description:

(1) int integer variable: The range is unlimited, and custom range is supported.
(2) double double-precision floating-point value: Floating-point number (decimal).
(3) string: Store a sequence of characters.
(4) joint_position: Consists of six independent floating-point values, separated by commas.
(5) tool_position: Consists of six independent floating-point values, separated by commas.

- **Variable Range:** Within the scope of the data type to which a variable belongs, its range can be further limited to meet specific needs.
- **Save Running Value:** In the pop-up window of editing a single variable, if the "Save" option is selected, the value of the variable will be recorded in real time and stored in the configuration file. When the program runs again, the saved value will be used instead of the initial value.

6.9.2. Current Variable Value

Displays variables used by the user during program editing. During script debugging, the current values of these variables are updated in real-time based on program execution, reflecting the latest assignments. Users can concurrently validate in real-time whether variables are correctly assigned while debugging.

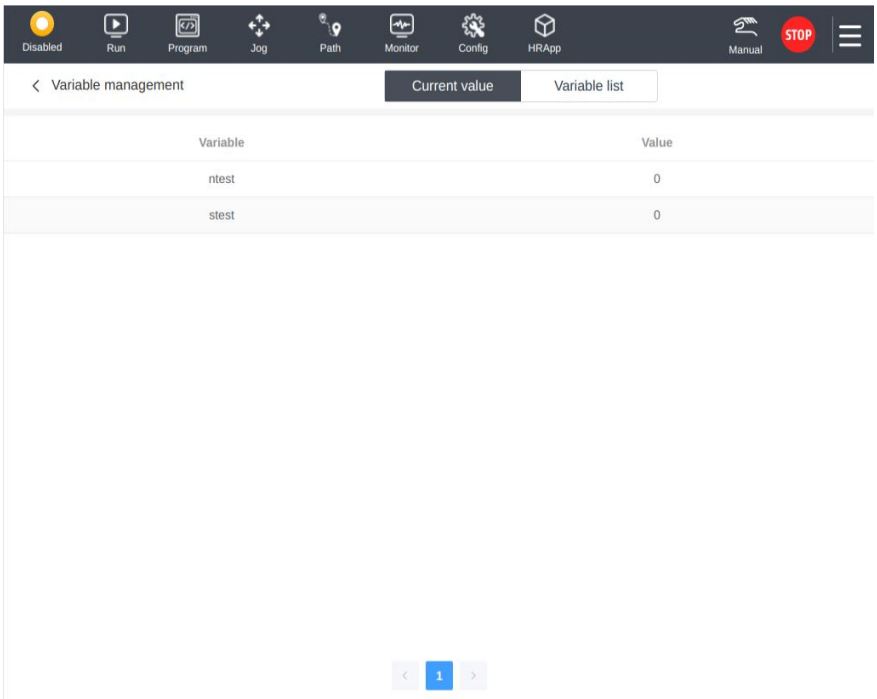


Fig. 6.9.2 Current Variable Value

6.10. Debugging and Running

After programming, debugging is required to verify if the program runs successfully. If it fails, a corresponding pop-up window will appear to prompt the user.

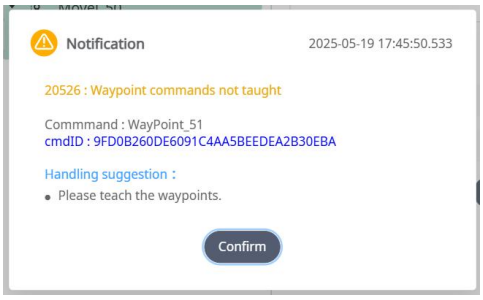


Fig. 6.10.1 Program Debugging and Running

6.10.1. Running

After completing program editing, select the function to debug from the drop-down menu. Debugging starts from Func_main.

If robot's end-effector is at the program's starting position when you click the【Run  】button, the program will start running directly. If you click the【Run  】button, step-by-step debugging of the program will start.

Note 1: The program cannot be executed successfully if there are commands with unset parameters.

When the user clicks【Run】, if there are unset commands in the program, a pop-up window will appear and warn, which prompts: "Program file compilation error"; meanwhile, the system popup will remind the specific contents and solutions of the command in issue, so that users can quickly identify and correct errors.

Note 2: The program may fail to execute due to parameter errors.

6.10.2. Executing

Upon successful program execution, the variable value changes can be viewed in the bottom-right window. A yellow arrow will appear before the current command to indicate its progress.

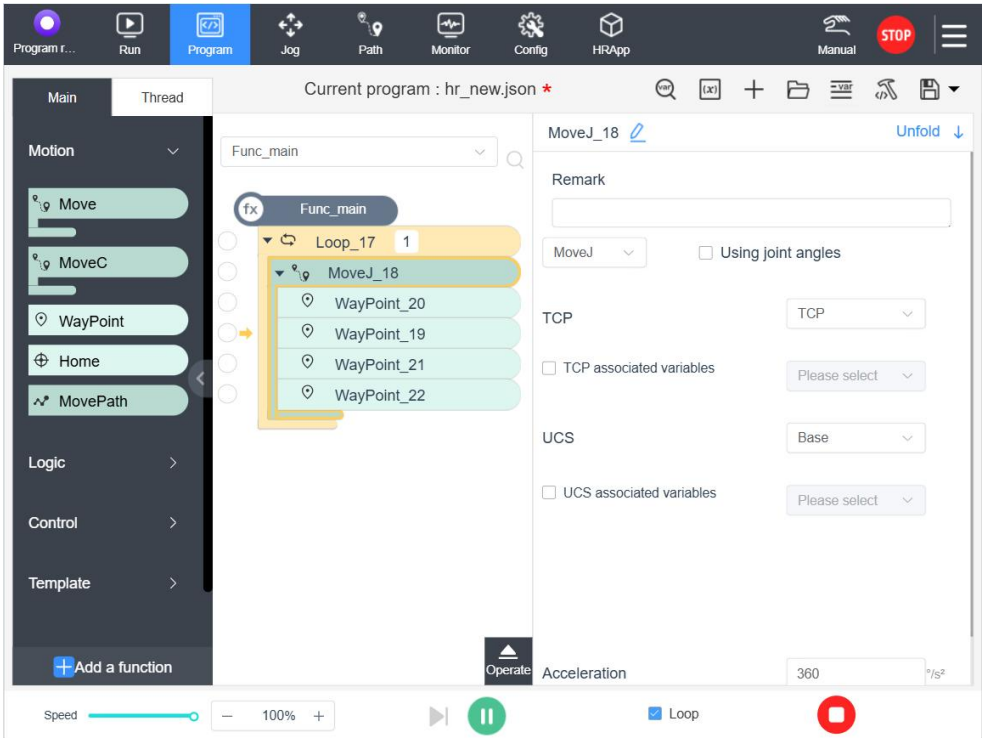


Fig. 6.10.2 Program Execution

- **Pause:** After the program starts, click the 【Pause  】button to interrupt execution. The button then switches to 【**Run**  】; clicking it again continues the program.
- **Stop:** Once the program is running, click the【Stop  】button to terminate program execution.

Note: When a program encounters an error during execution, a system pop-up window will show the error location and specific cause. Users can adjust parameter settings according to these prompts and re-execute the program for debugging.

6.11. Program Example

During robot operation, motion commands are the most basic and frequently-used commands. Below is a simple pick-and-place program. (The current end-effector is a gripper.)

1. When the robot is enabled, create a program in the programming interface: **File → New → Name the Program → Confirm Creation.**

2. Under the **Func_main** function, drag a **Move** command from the command menu to the function node. A **WayPoint_2** is automatically added under the **Move** node. Change the motion type of **Move** to **MoveL**. Select a suitable TCP for the gripper (end-effector) (pre-configured in settings). The coordinate system defaults to **Base**. Set appropriate acceleration and tool speed for the waypoint.

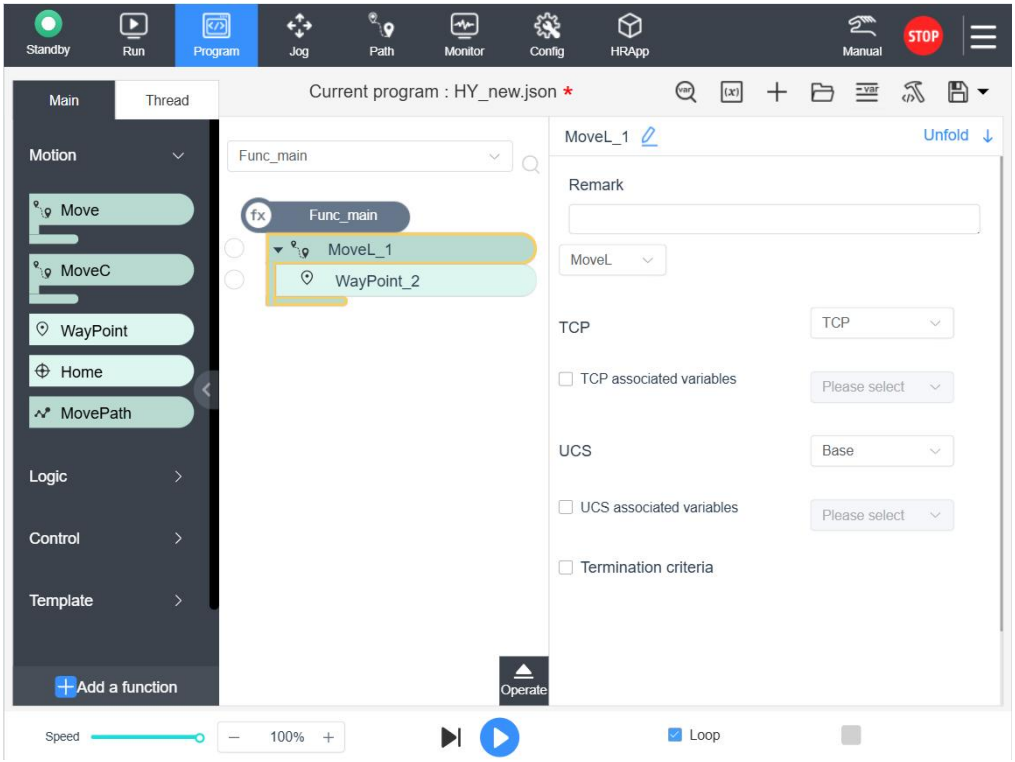


Fig. 6.11.1 Setting Status

3. Teach pendant **WayPoint_2**. If the desired point is already in the point list, select it directly from the drop-down menu in the current interface. **WayPoint_2** serves as the robot's starting point.

4. If you do not want the robot to stop between **WayPoint_2** and **WayPoint_3**, set and check the **blending radius**, ensuring it does not exceed the distance between the two points.

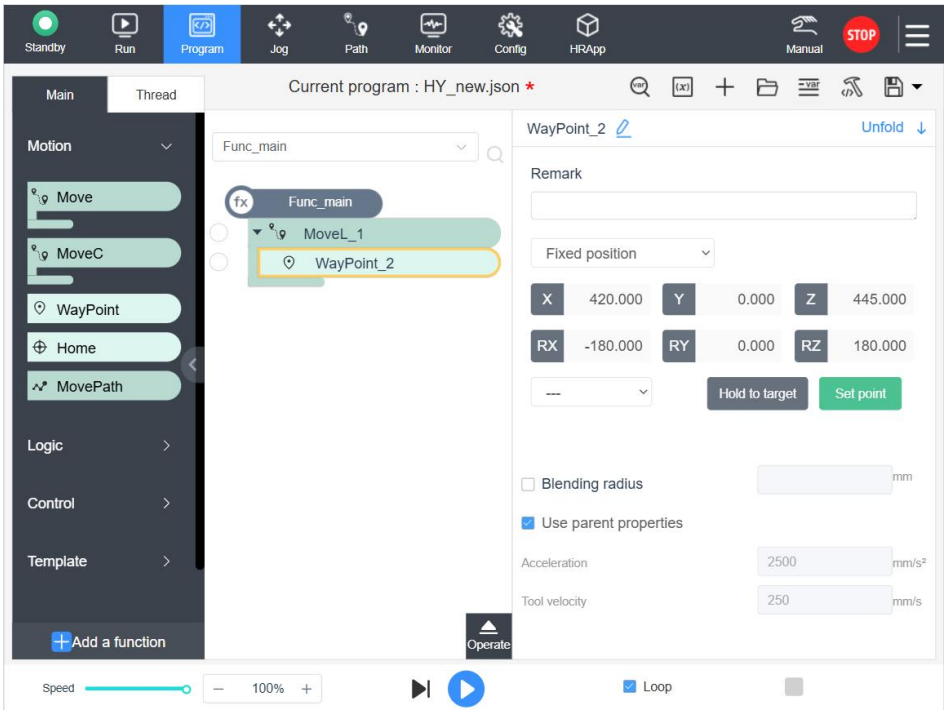


Fig. 6.11.2 Blending Radius Setting

5. Drag and drop a waypoint from the command menu to under **WayPoint_2** to generate **WayPoint_3**. Teach **WayPoint_3**, which is above the material's position. In the **MoveL** mode, the robot arm moves from **WayPoint_2** to **WayPoint_3** to prepare for material grasping.

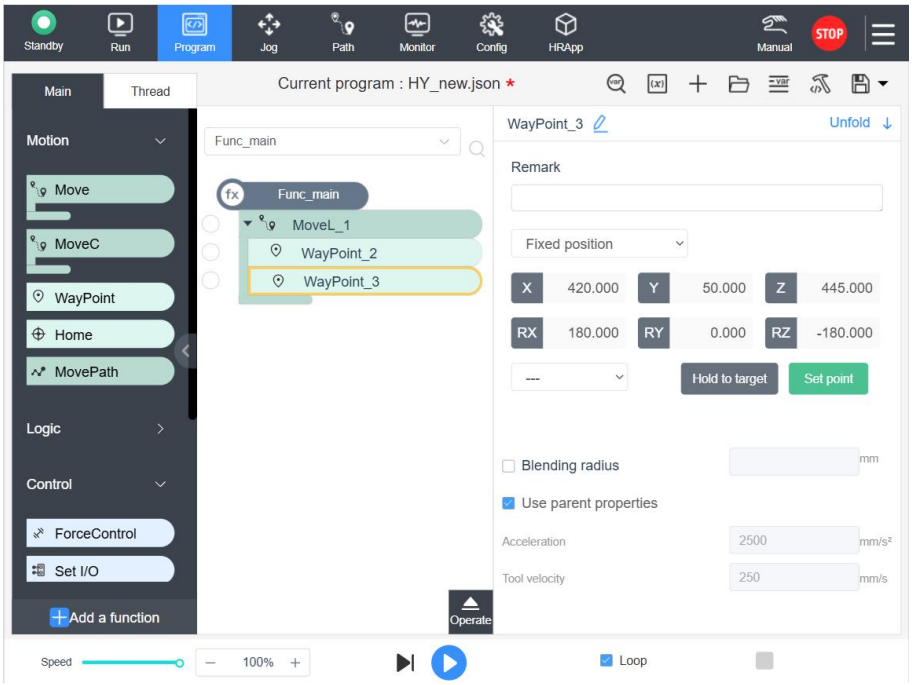


Fig. 6.11.3 WayPoint3 Setting

6. Drag and drop a waypoint from the command menu to under **WayPoint_3** to automatically create **WayPoint_4**. Teach **WayPoint_4**, which is the target point above the material, positioning the gripper above the material in preparation for grasping.

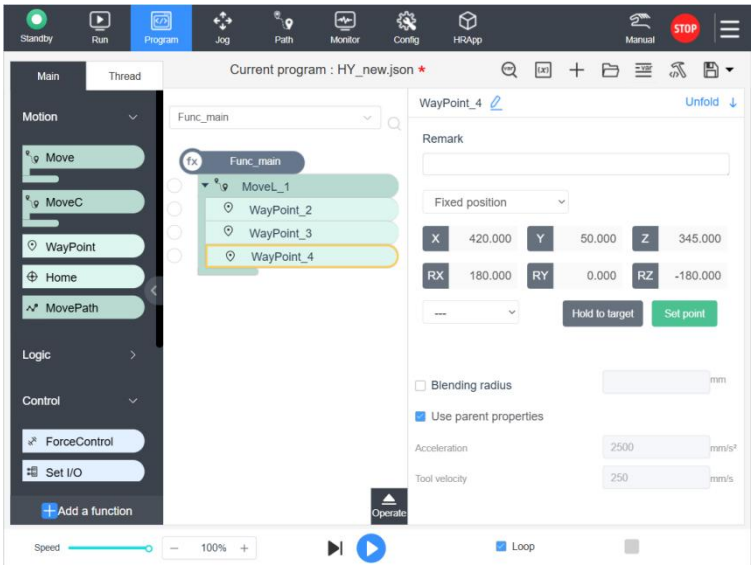


Fig. 6.11.4 WayPoint4 Setting

7. Drag and drop Set I/O from the command menu to under **WayPoint_4** and set the electrical control box output as DO0 = High, and close the gripper to pick up the material.

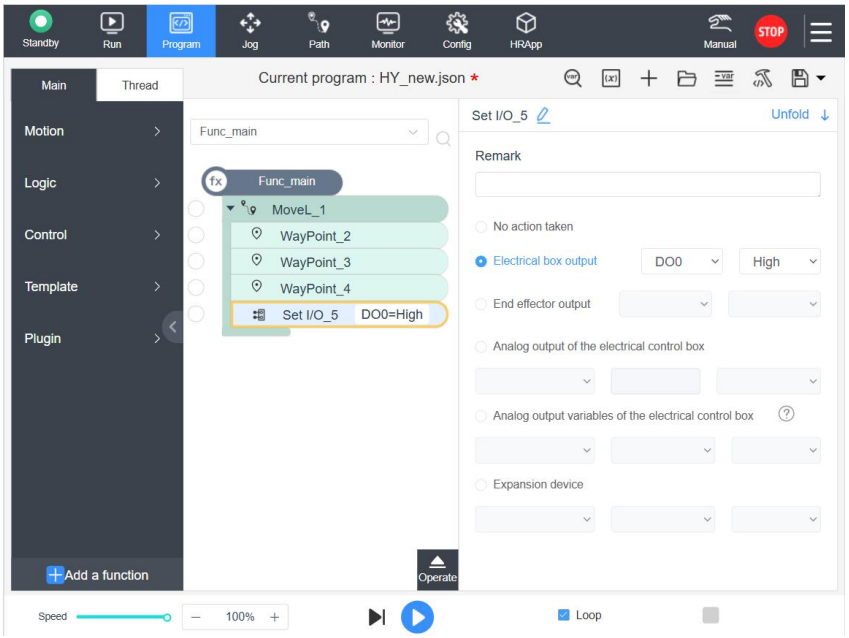


Fig. 6.11.5 Set I/O

8. Drag and drop a waypoint from the command menu to under SetI/O to automatically generate **WayPoint_6**. Teach **Waypoint_6**, which is a transitional safe position. The robot arm picks up the material and lifts it to the safe position.

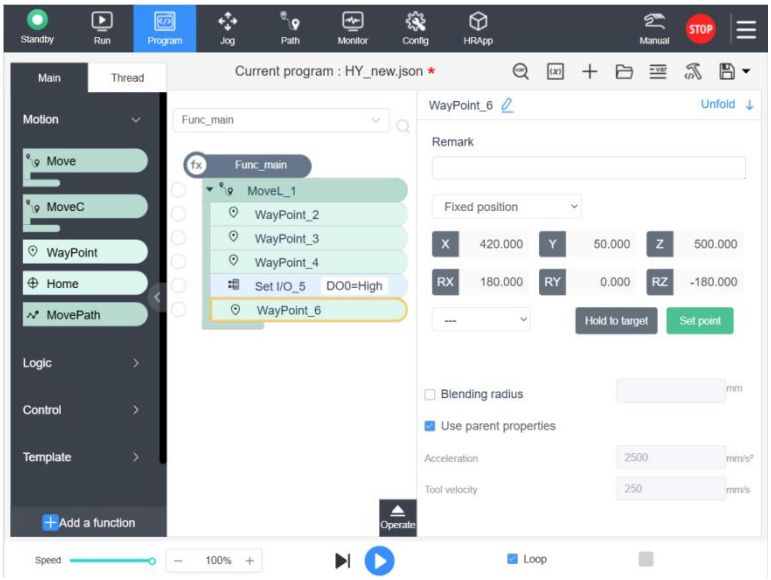


Fig. 6.11.6 Safe Position of Transition

9. Drag and drop a waypoint from the command menu to under **WayPoint_6** to automatically generate **WayPoint_7**. Teach **Waypoint_7**, which is the final workpiece placement position. In MoveL mode, the robot moves from **WayPoint_6** to **WayPoint_7** to reach the final placement position.

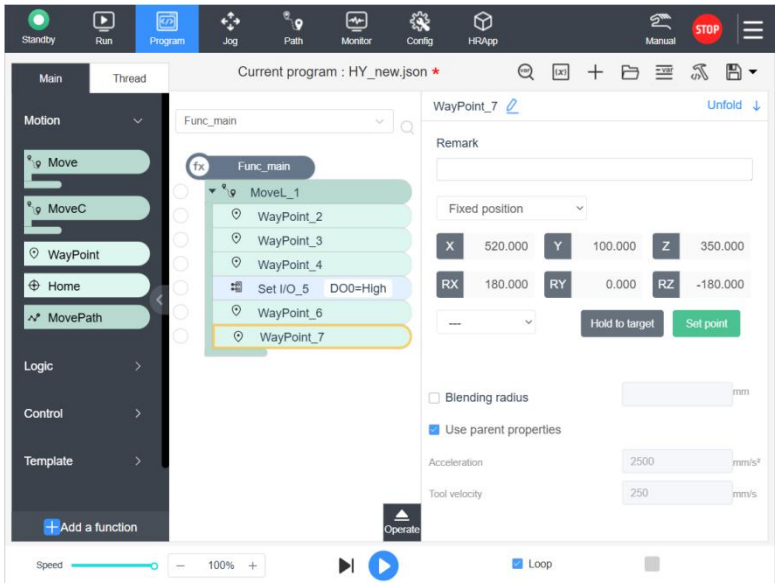


Fig. 6.11.7 Set Final Position

10. Drag and drop a **SetI/O** from the control command in the command menu to under **WayPoint_7**. Set the electrical control box output **DO0 = Low** (since the gripper is connected to I/O **DO0**). The gripper opens to place the material.

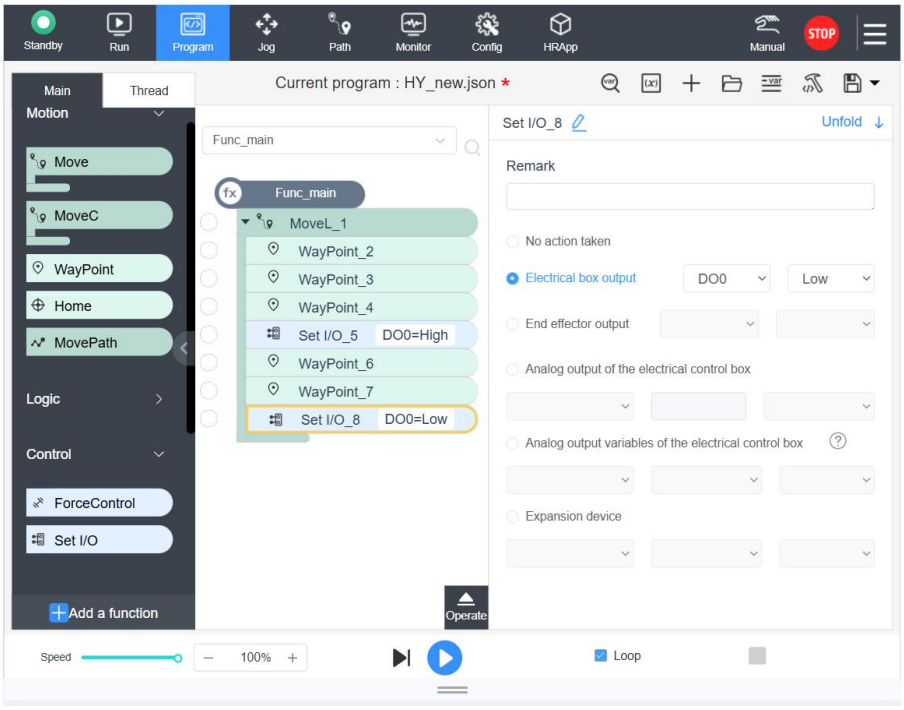


Fig. 6.11.8 Re-Set I/O

11. The robot arm returns to its safe home position.

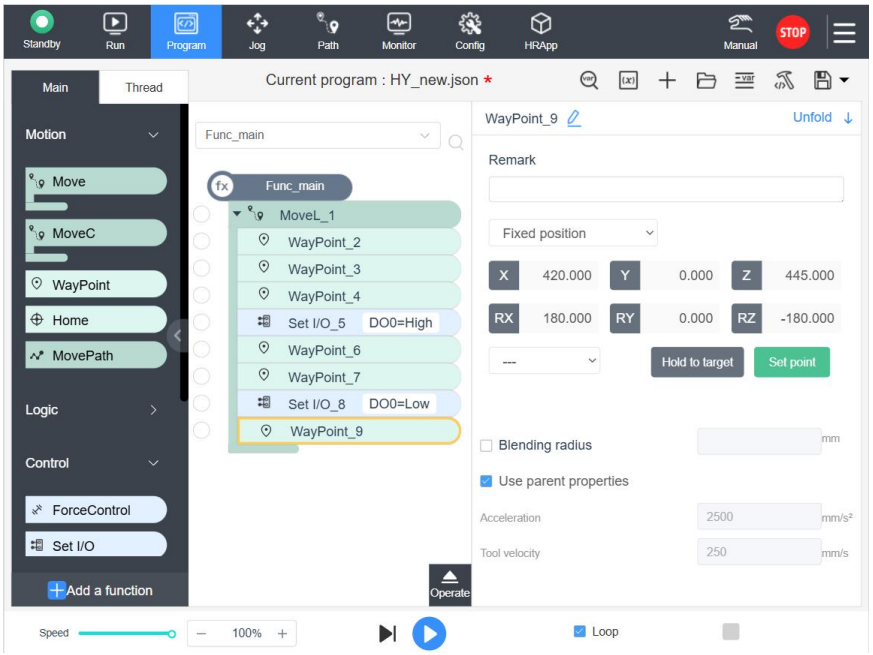


Fig. 6.11.9 Return to Safe Points

This program performs a pick-and-place operation with intermediate safe points to avoid obstacles during MoveL linear motion. After editing, click【Run】to execute it. Selecting the loop option enables continuous program executio

Chapter 7Running

The run interface allows users to directly execute pre-saved programs, enabling quick robot operation for general operators.

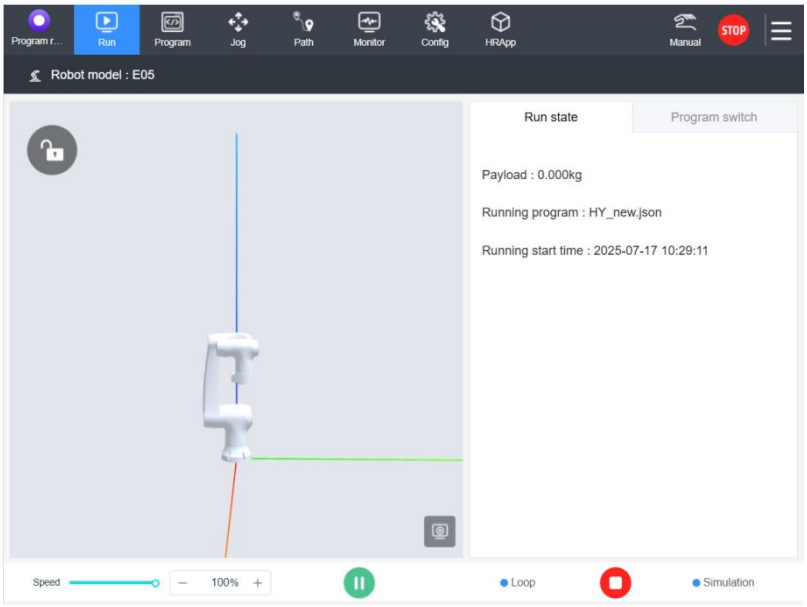


Fig. 7.1 Running Interface

◆ Robot Power Up and Enabling Operations

Please refer to "2.2.1 Initializing Robot" for details.

7.1. Initializing Robot

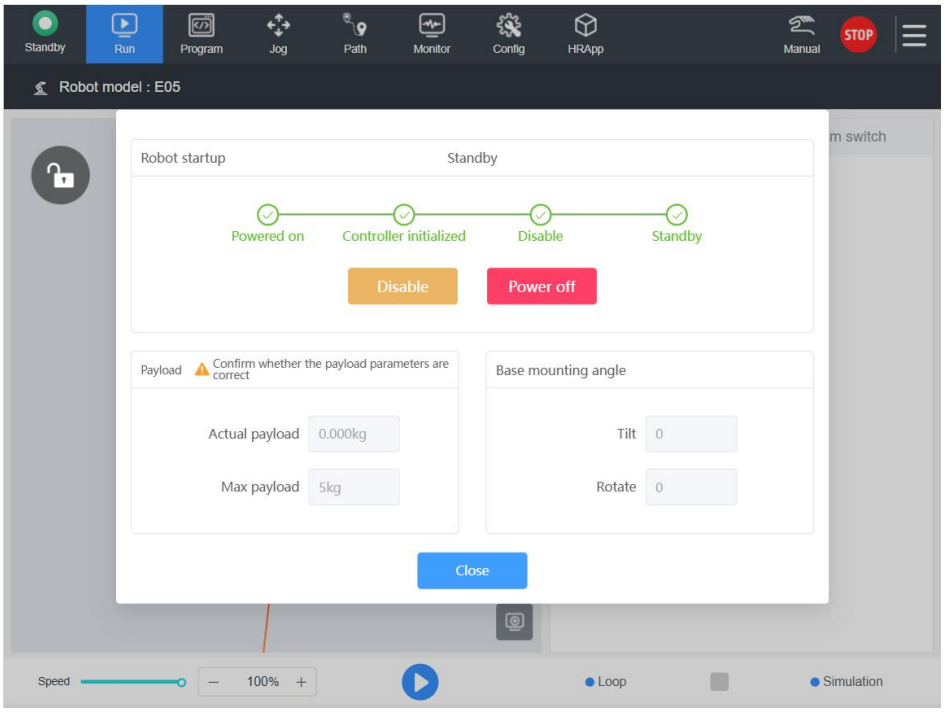


Fig. 7.1.1 Initializing Robot

7.2. Selecting the Program

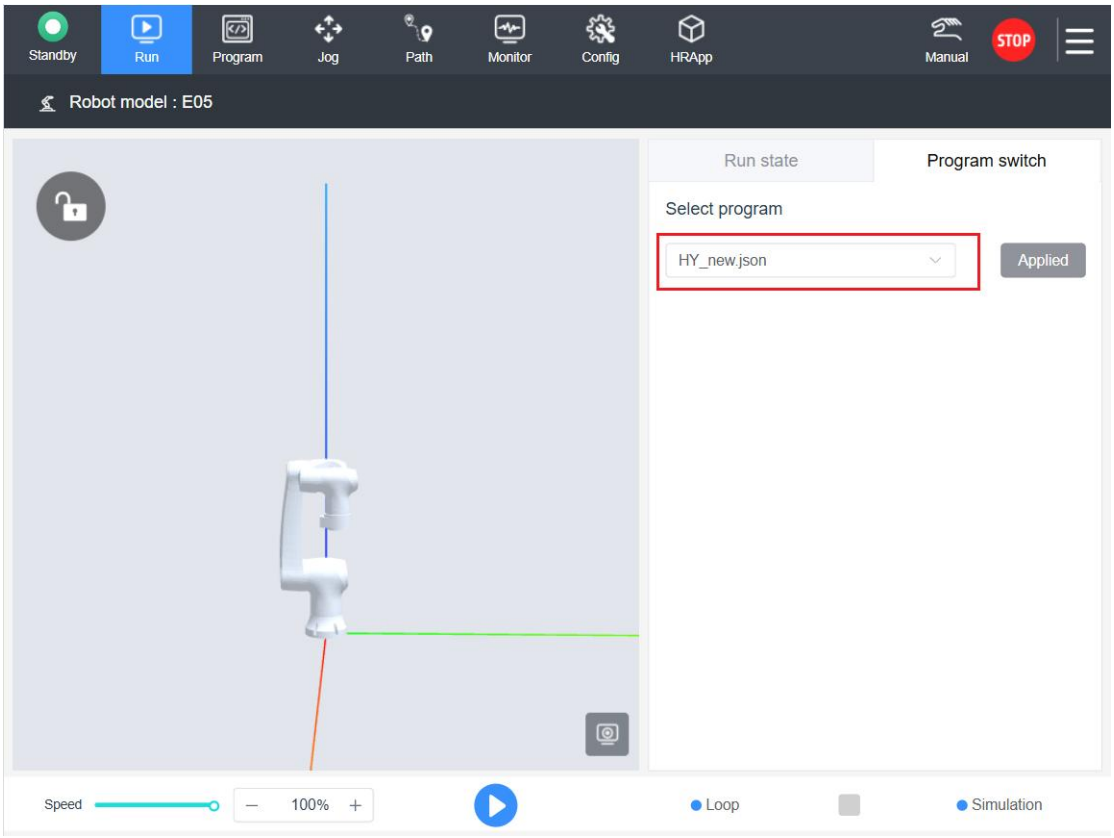


Fig. 7.2.1 Program Switch Interface

The drop-down menu lists all saved programs in the software. Users can select a program from it and apply it to the robot.

- **Run:** After the program is loaded, clicking the【Run】button will start the program directly if the robot's current position matches the program's starting point.
- **Pause:** After the program starts, clicking the 【Pause】 button to interrupt the program execution.
- **Stop:** Once the program is running, clicking the【Stop】button to terminate program execution.

Note: If the program includes enable commands, the users cannot run it when the robot is disabled. Manually enable the robot and wait for the robot arm to return to the start position before running the program.

7.3. Running Status

The runtime status displays key information about the robot arm's operation, including the script name, payload, runtime, execution time, and number of cycles, helping users monitor its working status.

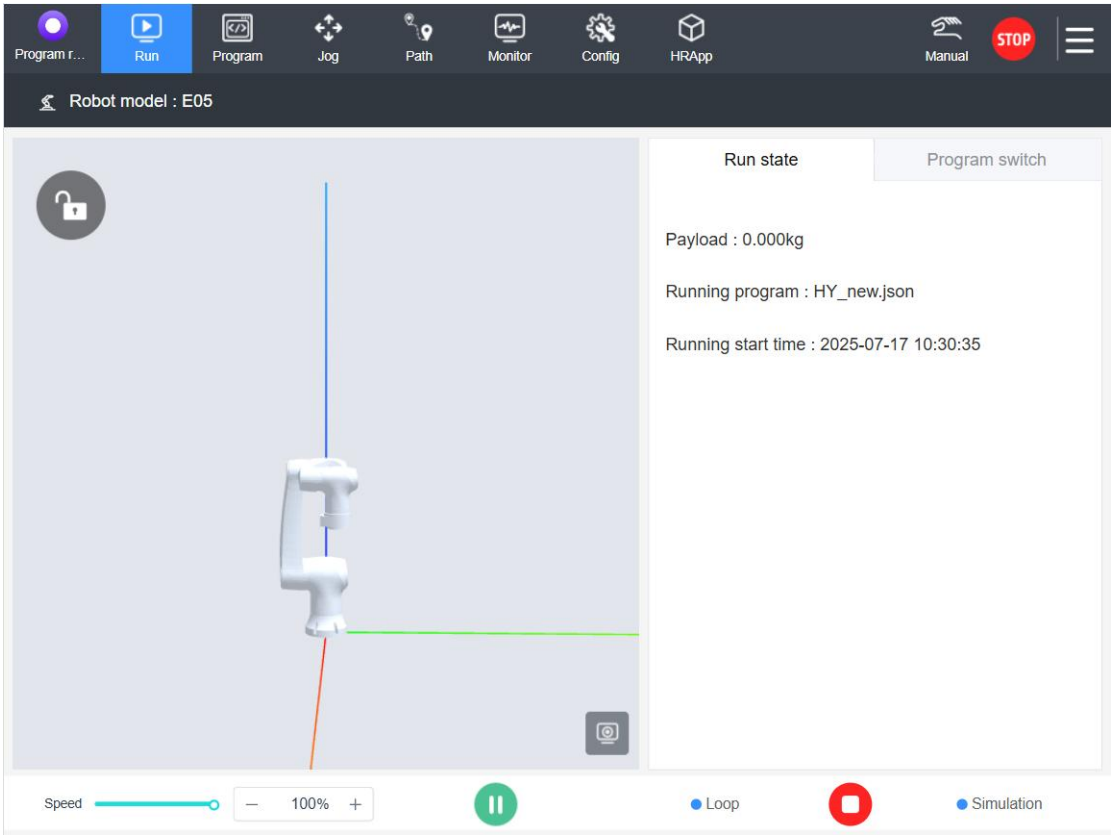


Fig. 7.3.1 Run the Program

7.4. Robot 3D Model

Synchronized with the robot arm.

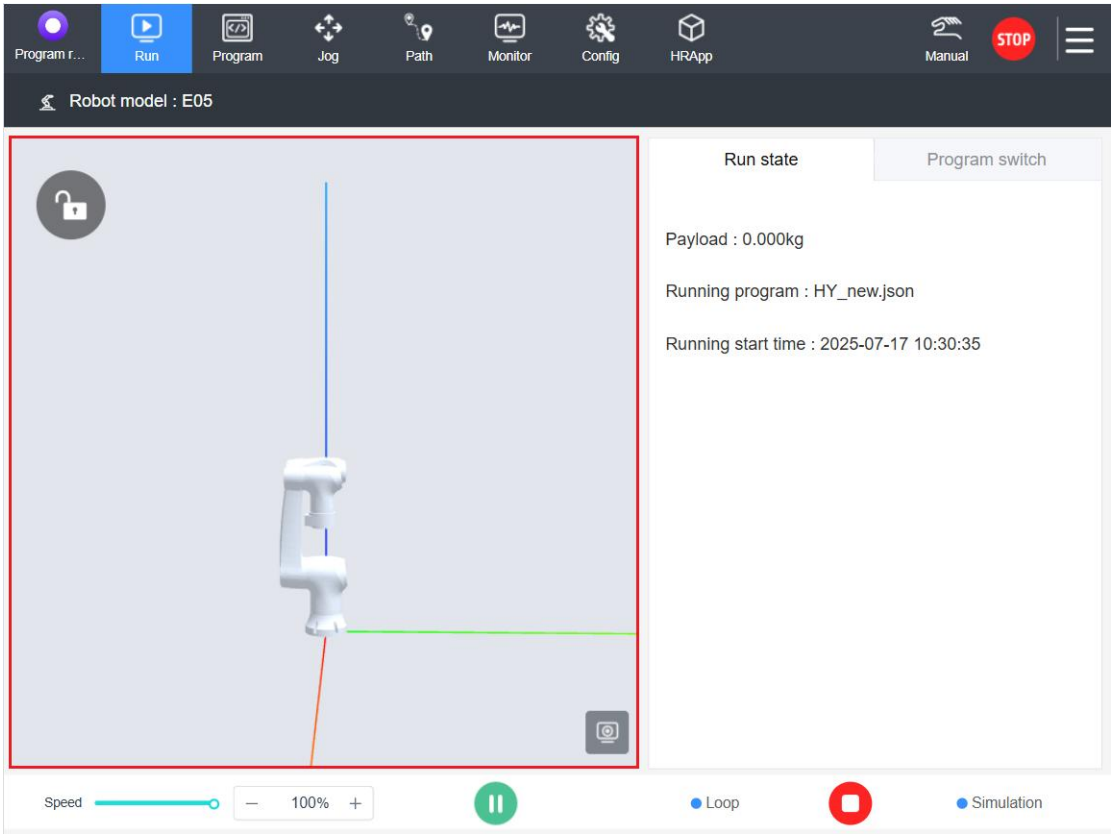


Fig. 7.4.1 Robot 3D Model

7.5. Recovery

After clicking  , the view will reset to the default perspective of the 3D graphic.

7.6. Locking

Once safety-locked, the interface is inoperable. Re-entry requires a password, which effectively prevents unintended operations during robot runtime.

7.7. Operation Area

The users can adjust the robot arm's speed, control script execution (run, pause, stop), and switch between simulated and real robot arm.

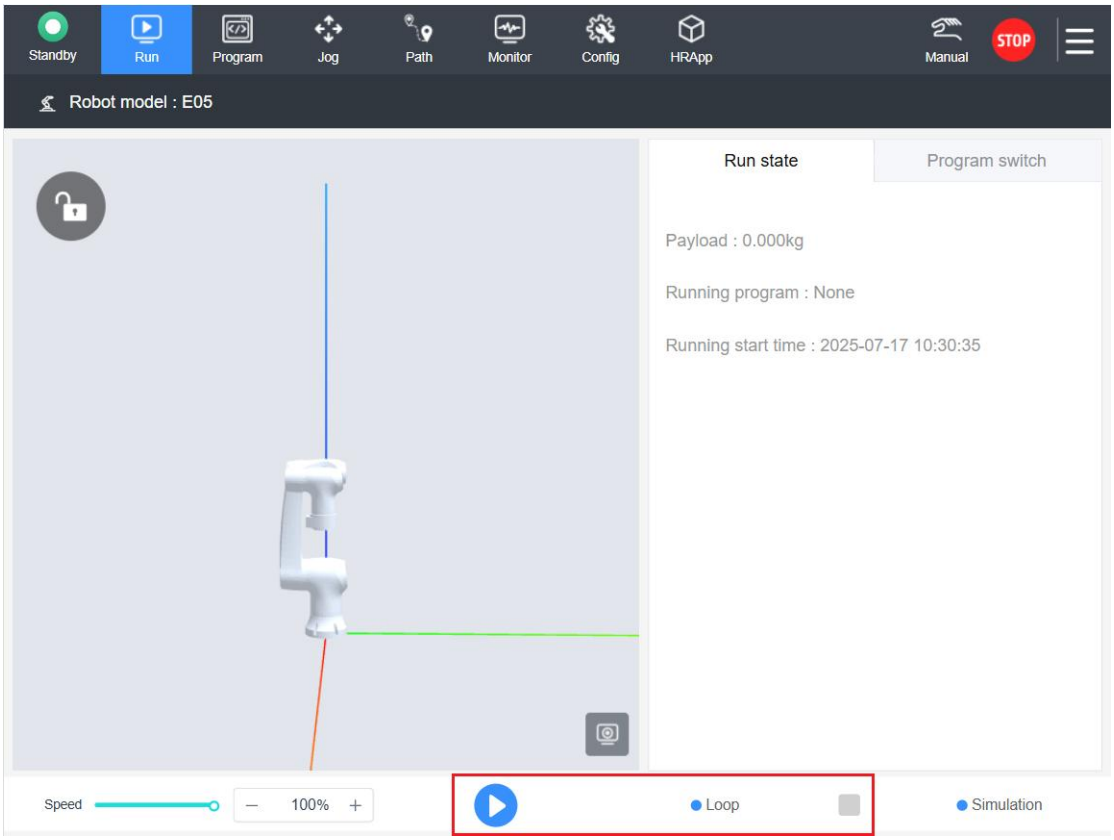


Fig. 7.7.1 Bottom Operation Area

- **Real Robot Arm:** Indicates connection to actual robot hardware. During teaching, ensure the robot and its surroundings are positioned safely;
- **Simulated Robot Arm:** Indicates a simulated robot connection. It mirrors the real robot in functionality but without hardware-related safety concerns, enabling faster and safer teaching;
- **Switching Real/Simulated Robot Arm:** Users can only switch between these modes by selecting the desired option when the robot is powered off. Switching is disabled in other states.

Chapter 8 Path

On the Path page, users can create, record, and save robot arm motion paths for repeated use in programming.

8.1. Page Introduction

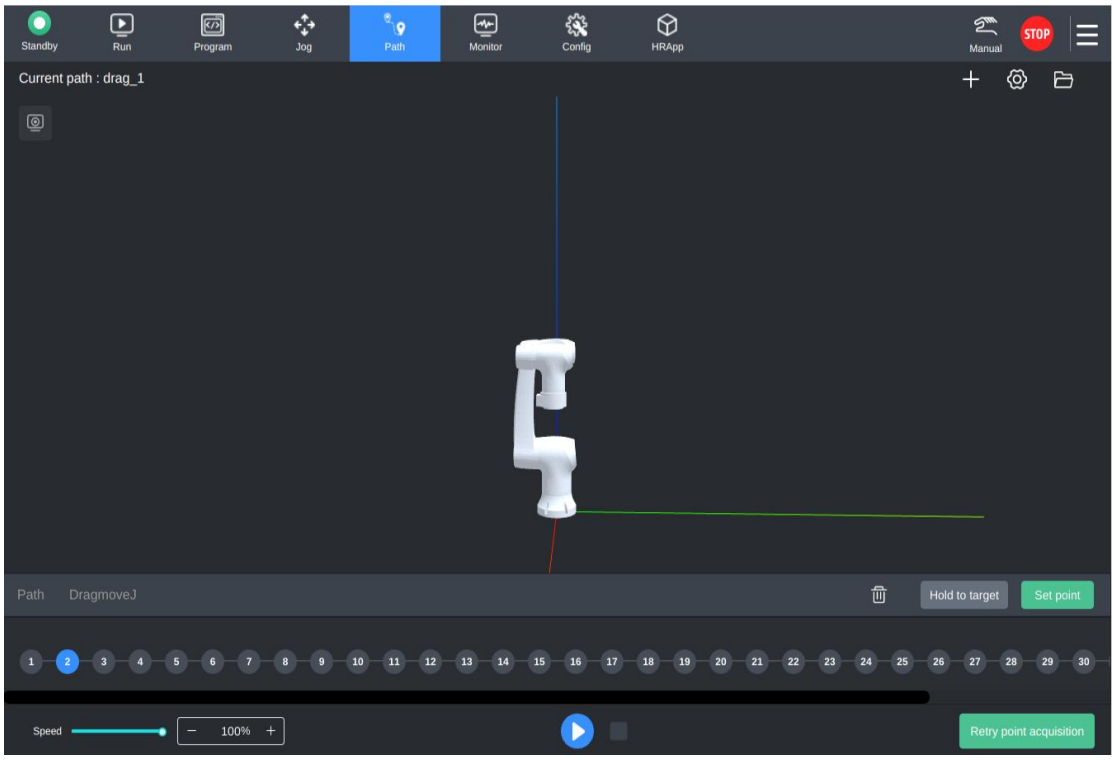
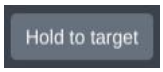
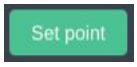
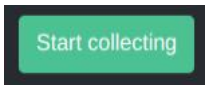
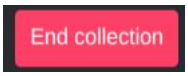
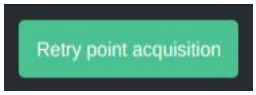


Fig. 8.1.1 Path Recording Interface

Table 8.1.1 Path Operation Interface Description

	Custom Name for the Current Path: When creating a new path, users can customize its name for easy identification and management;
	Restore: Restore the default perspective of the 3D view;
	Create Path: Click to initiate the creation of a new path.
	Path Collecting Settings: Configure the acquisition mode, blending radius, and speed ratio here;
	Path Files: Displays a list of created path files;
	Delete: Remove the currently selected path waypoint;
	Hold to Target: Press and hold to move the robot arm to the currently selected path waypoint;
	Set Point: Re-teach the currently selected path waypoint. Clicking this option navigates to the Jog interface.
	Path Waypoint: These are the waypoints collected for the current path, where the blue color indicates the currently selected waypoint. Operations can be performed on the selected waypoint;

  	Start Collecting: After creating a new path, click this button to begin collecting waypoints. End Collection: During the acquisition process, click this button to stop collecting waypoints. Retry Point Acquisition: After completing the acquisition, this button allows you to recollect all waypoints for the current path.
	Run: Run the path after acquisition;
	Stop: Stop running the current path;
	The Current Path Operation Mode: The trajectory will be executed using the MoveJ motion command. This mode cannot be altered;

8.2. New Path

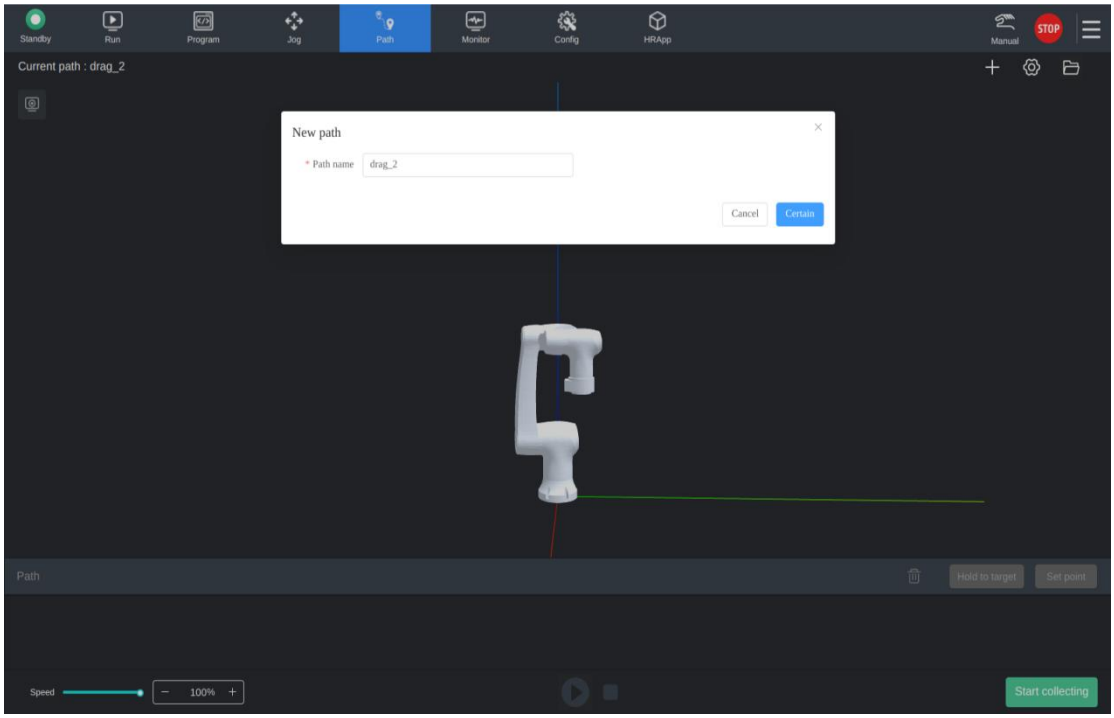
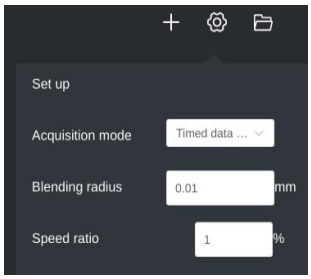


Fig.8.2.1 New Path

1) After clicking the button  , a pop-up window for creating a new path will appear on the page. The default path name is "drag_N", and users can modify "N". After creating a new path, the page will display the name of the current path.



2) After creation is complete, users can configure the path acquisition. Clicking on the button  will bring up the settings window. 

. In the window, users can configure settings such as the acquisition mode, blending radius and the speed ratio.

◆ Acquisition Mode: There are two modes, "Timed Data Acquisition" and "Specific Point Data Acquisition". In Timed Data Acquisition, the system collects robot waypoints every 200 ms by default. In Specific Point Data Acquisition, it collects waypoints based on the robot arm's real-time position and orientation.

- Blending Radius: Used to define the smooth transition area between points. For detailed information, please refer to the Blending Radius section of 6.3.2 Waypoint;
- Speed Ratio: The operating speed of the current path and is independent of the speed bar at the lower-left corner;

3) After completing the settings, click the【Start collection】 button to begin collecting path waypoints.
   Acquisition Mode 1: Timed Data Acquisition. The system collects the current position every 200 ms. If there is a certain distance from the last waypoint, the current position will be saved as the path waypoint.
   Acquisition Mode 2: Specific Point Data Acquisition. When the robot arm moves to the point to be collected by any method, click the【Add point】button or press the "Point" button on the end effector to collect the current position as a path waypoint.
4) After completing the waypoint collection, click the【End collection】button. A prompt "Collection completed" will appear. The system will then calculate and verify whether the path can operate normally. If it cannot, users can click the "Set points" button or the 【Retry point acquisition】 button to reset and collect the waypoints. Until the path is correct and it can operate properly.
5) After the collection is completed, the path is automatically saved to the path list.
6) Once the path calculation is confirmed to be normal, the users can click the【Run】button to start or stop the robot arm's motion along the current path.

8.3. Path List

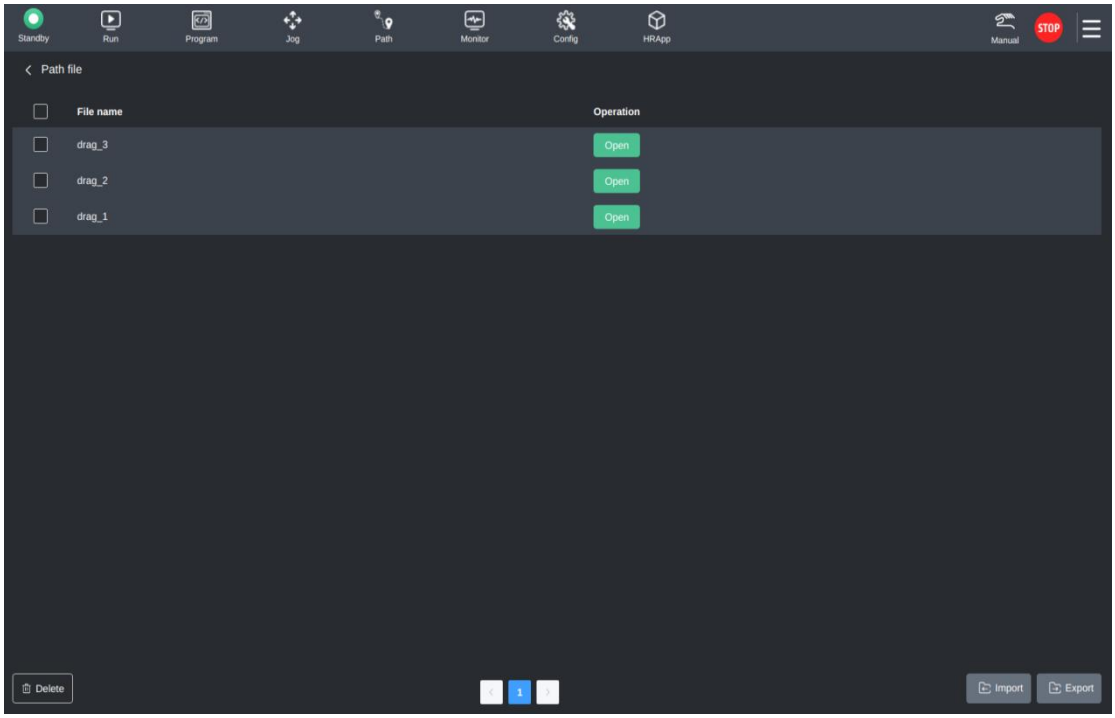


Fig.8.3.1 Path List

- Path Name: Name of the saved robot path file;
- Open: Click to open the current path file in the path page;
- Delete: Tick to select and delete the current path file;
- Import: Import path files from an external source. The imported files must match the system file format;
- Export: Export the currently selected path externally.

Chapter 9 Monitoring

On this interface, users can monitor and configure the real-time I/O signals of the electrical control box. It displays the current status of the I/O signals, including the I/O status during program execution. When the program stops, all output signals will remain as they are. Note that signals changing extremely fast might not be accurately displayed due to the interface's refresh rate limits.

9.1. Electrical Control Box I/O

9.1.1. General Input and Output

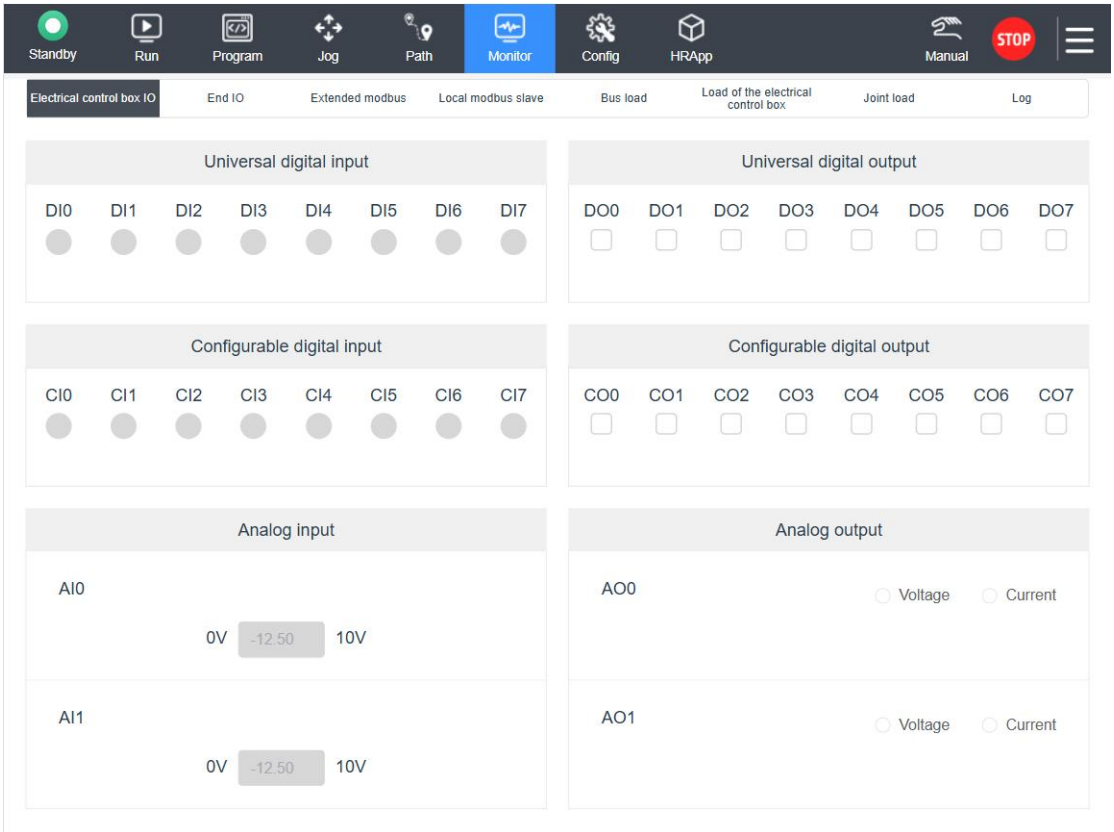


Fig. 9.1.1 Electrical Control Box I/O Interface

◆ **General Input:** DI0 ~ DI7, 24V power supply. If the input is at a high level, the corresponding circle turns blue; a gray circle indicates a low level;

◆ **General Output:** DO0 ~ DO7, 24V power supply. Buttons correspond to the output switches , the output level can be set as high and the switch turns blue .

9.1.2. Configurable Input and Output

- **Configurable Input:** DI0 ~ DI7, 24V power supply. If the input is at a high level, the corresponding circle turns blue, and the gray circle indicates low level;
- **Configurable Output:** DO0 ~ DO7, 24V power supply. Buttons correspond to the output switches , the output level can be set as high and the switch turns blue .
- **Configurable Input and Output Feature**
- Please refer to the "5.3 I/O Setting" configuration instructions for more details.

9.1.3. Analog Input and Output

- **Analog Input**: 2 channels, AI0 and AO0, with a voltage range of [0-10V];
- **Analog Output**: 2 channels, AI1 and AO1, with a current range of [4-20mA] and a voltage range of [0-10V].

9.2. End I/O

The user needs to perform security verification in advance, and then click the ”End effector I/O capabilities” button to enable the functions of the buttons at the end-effector.

Free: For activating the basic free drive mode;

Point: For saving positions.

Note: For safety consideration, once the "Free" button at the end-effector is activated, the basic free drive mode on the "Jog" interface will not be available.

The page defaults to displaying 4 buttons at the end-effector, but the actual number varies with the end-effector type.

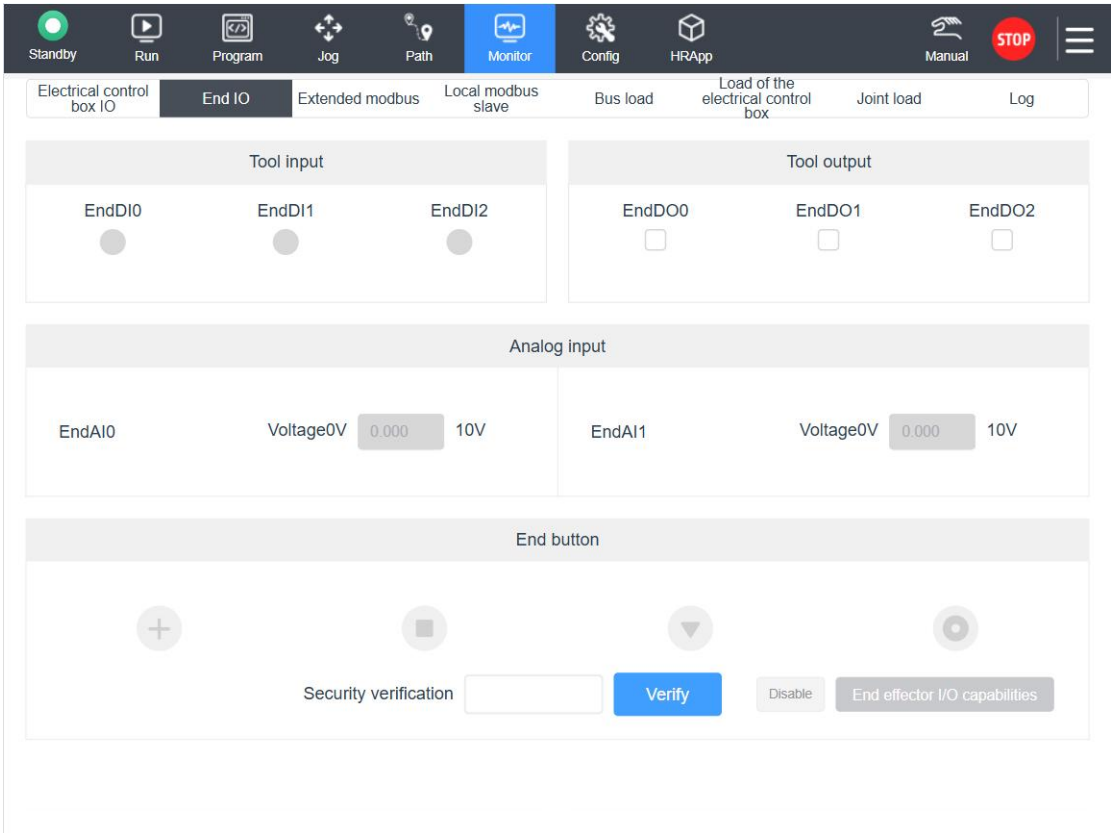


Fig. 9.2.1 End I/O Interface

9.2.1. Universal Input, Output and Analog Input

- **Tool Input:** EndDI0 ~ EndDI2, 24V power supply. If the input is at a high level, the corresponding circle turns blue; a gray circle indicates a low level;
- **Tool Output:** EndDO0 ~ EndDO2, 24V power supply. Buttons correspond to the output switches , the output level can be set as high and the switch turns blue ;
- **Analog Input:** 2 channels, EndAI0 and EndAI1. Voltage input only, within the range of [0-10V].

9.3. Extended Modbus

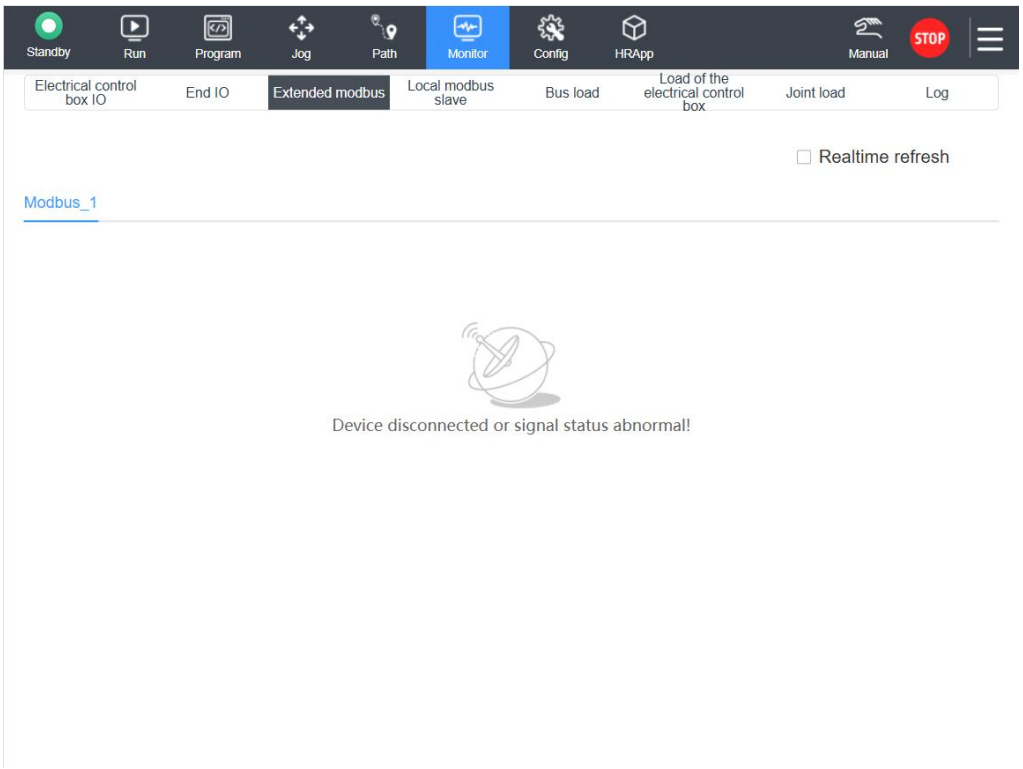


Fig.9.3.1 Extended Modbus

When a Modbus communication device is added to the configuration and successfully connected to the slave device, the signals of the added device will be displayed here for users to monitor the status. If data fails to be shown, please check whether the Modbus device parameters are properly configured in the settings.

Select【Real-Time Refresh】to view the latest data. To conserve system resources, the

refresh function will automatically terminate after a certain period of time. If the interface is exited and then re-entered, the Modbus data will stop updating. The user needs to re-select【Real-Time Refresh】to reactivate the data update function.

9.4. Local Modbus Slave

This page is designated for performing read and write operations on Coil Status, Input Status, Holding Register, and Input Register. Additionally, it offers two input boxes to set the register address and quantity, enabling flexible monitoring or operation of specific register ranges.

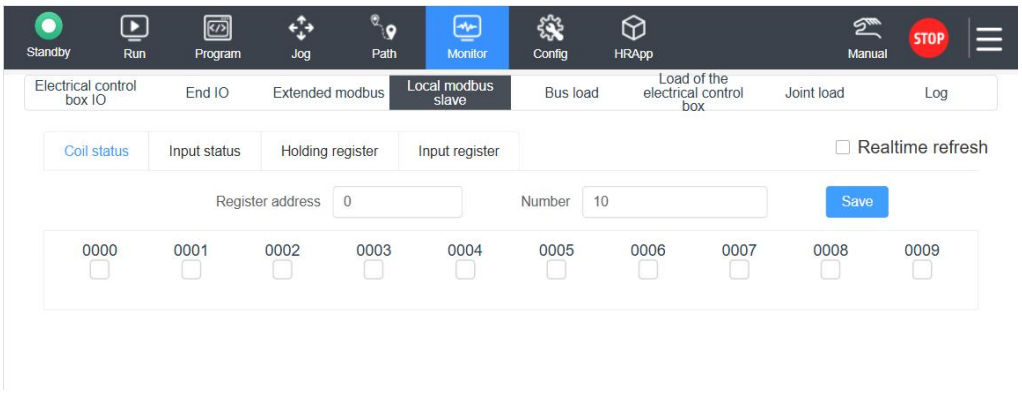


Fig. 9.4.1 Local Modbus Slave

9.4.1. Coil status

Indicates the output of digital signals, which are readable and writable with values of 0 or 1.

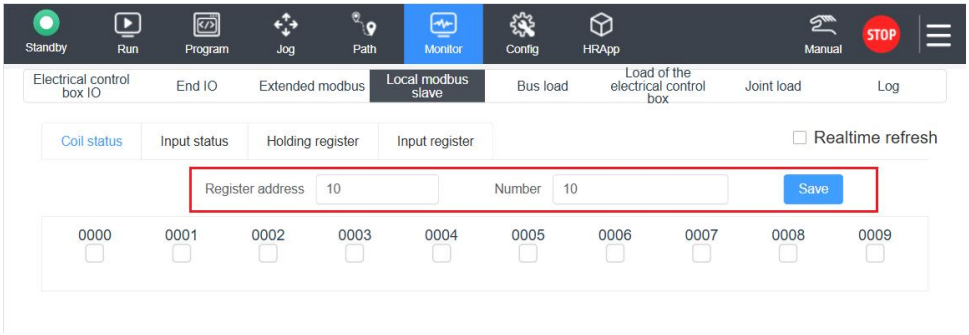


Fig. 9.4.2 Coil Status

9.4.2. Input Status

Indicates the input of digital signals, which are readable only with values of 0 or 1.

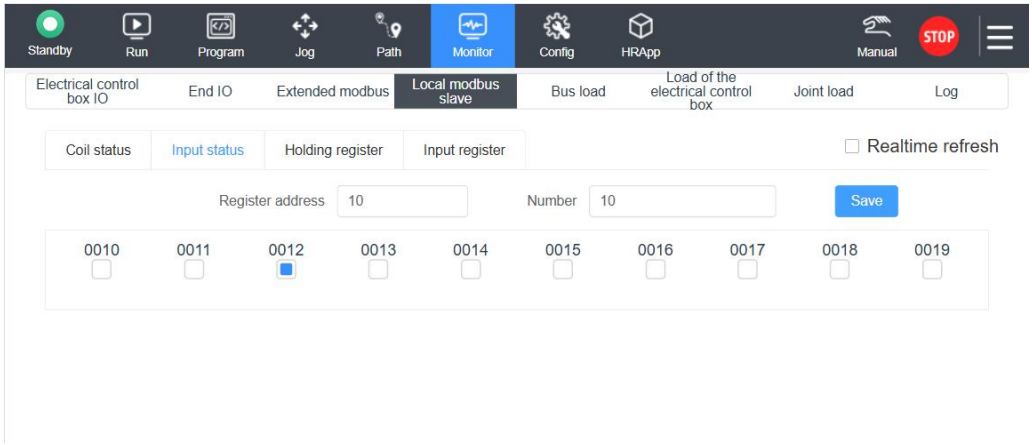


Fig. 9.4.3 Input Status

9.4.3. Holding Register

Stores numerical data, which is readable and writable in 16-bit integer format.

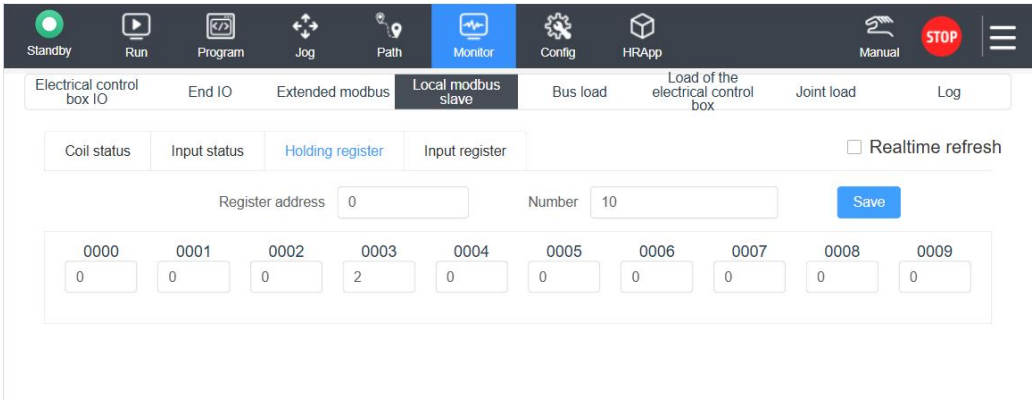


Fig. 9.4.4 Holding Register

9.4.4. Input Register

Stores numerical data, which is readable only in 16-bit integer format.

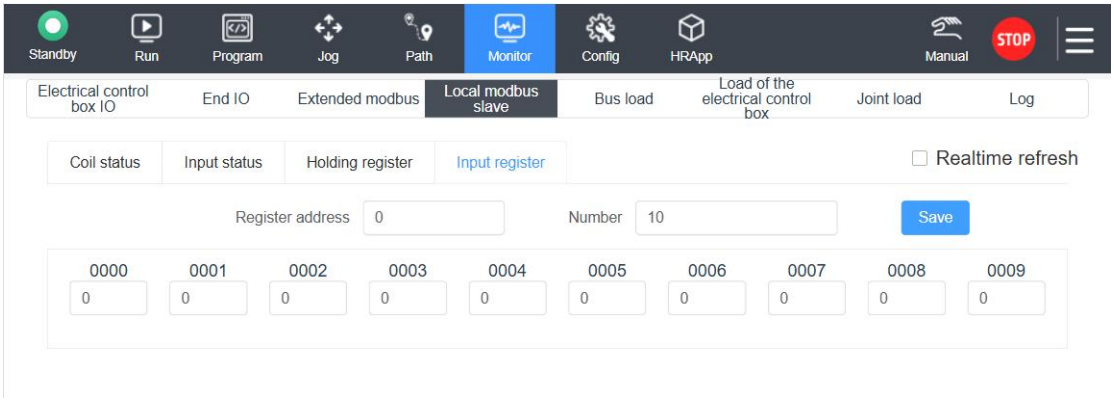


Fig. 9.4.5 Input Register

9.5. Robot Status

9.5.1. Bus Load

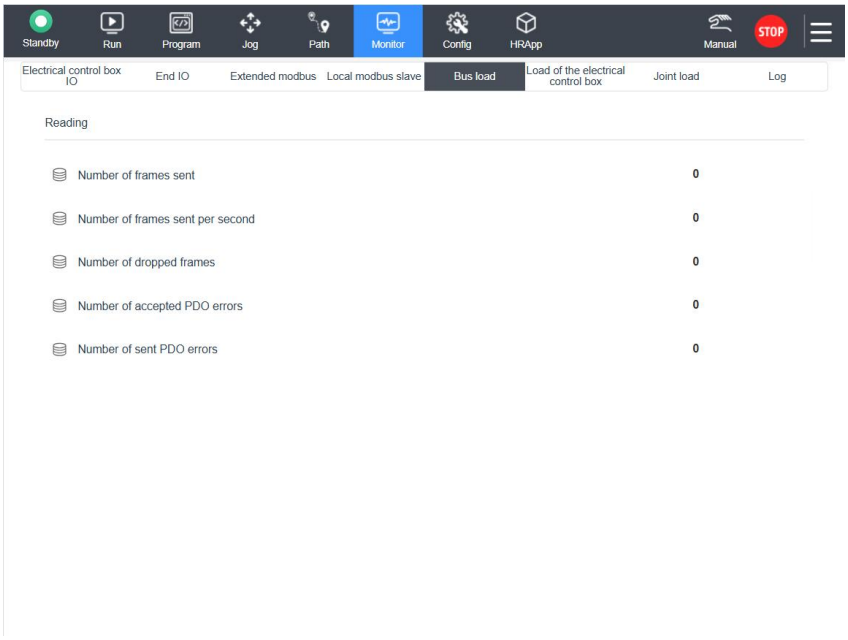


Fig. 9.5.1Bus Load

Displays real-time data frame transmission and reception status.

9.5.2. Load of the Electrical Control Box

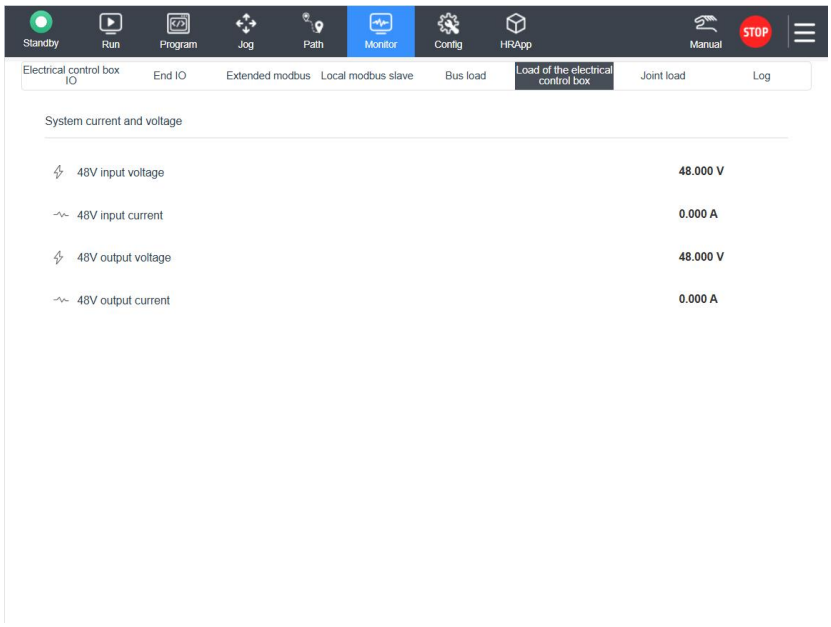


Fig. 9.6.1 Load of the Electrical Control Box

Displays real-time power supply status of the electrical control box.

9.5.3. Joint Load

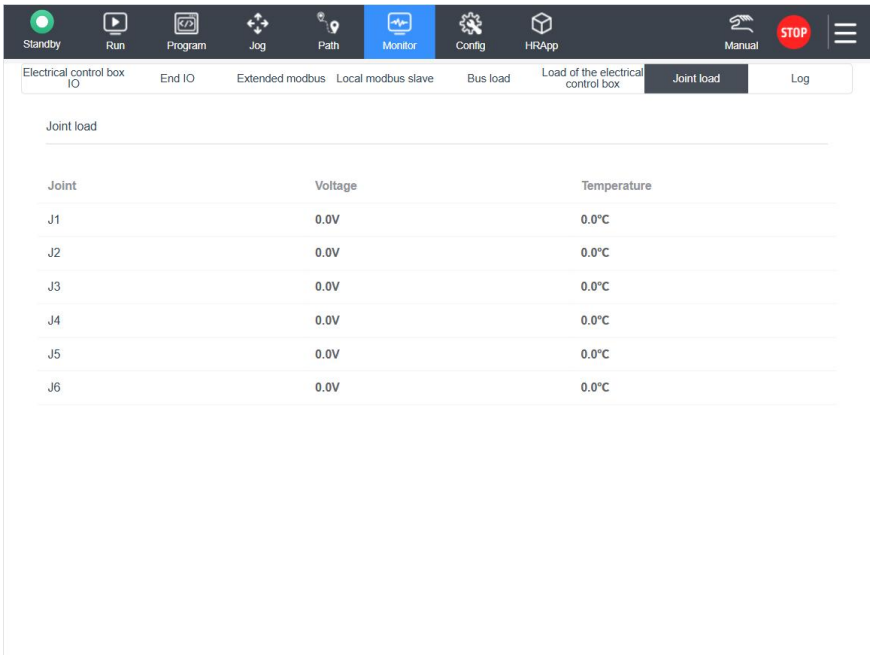


Fig. 9.7.1 Joint Load

Displays the joint-related information about the robot arm, including the voltage and temperature status of the joint load.

9.6. Log

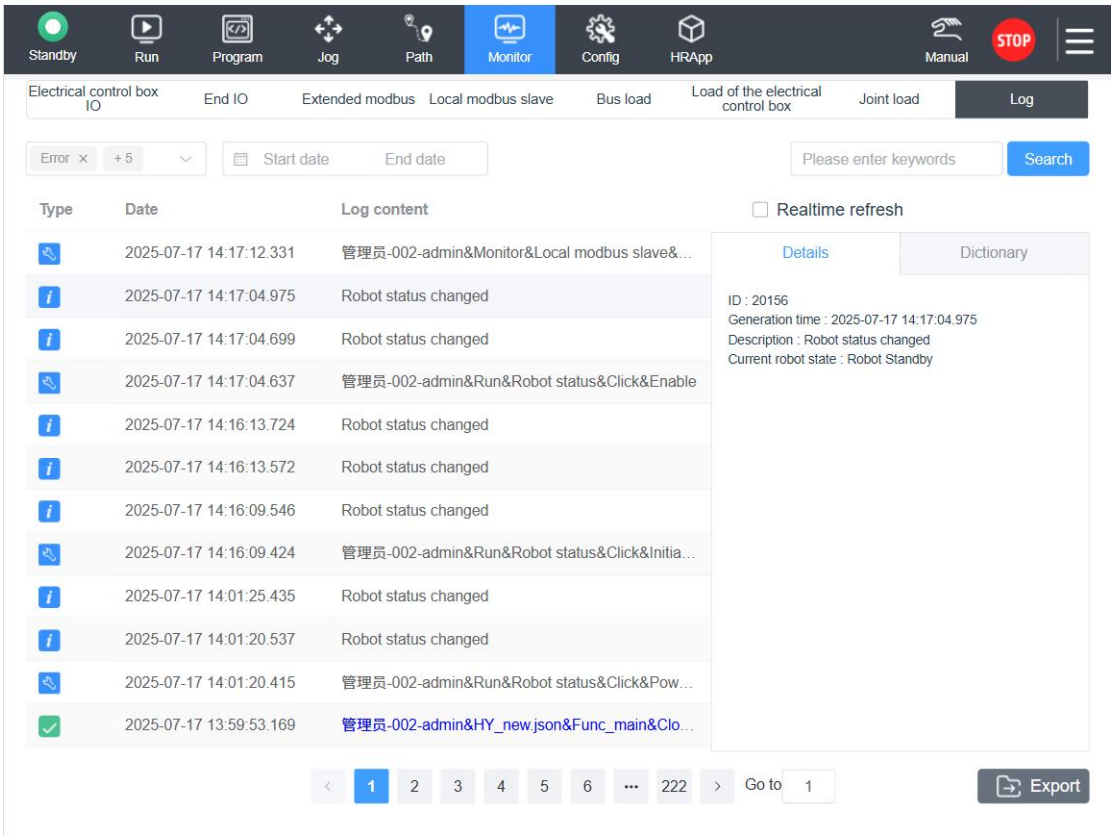


Fig. 9.8.1 Logs

Displays message severity classification, log message generation time and specific message content.

- **Type:** Log types include error, warning, event, operation record, parameter record, programming record, etc. Log lists can be filtered by querying types.
- **Log Display Content:** Log display includes user name, operation, operation parameters, command jump operations, etc. The operation logs cover user behaviors, such as login, script running, and robot jogging operation; the parameter logs record parameter modifications, including previous and new values of parameters such as TCP, command, and variable parameters, as well as command jump chain links for parameter changes.
- **Export:** Export all logs to external devices. When users have questions, logs can be exported and sent to the technical support team for navigating and problem-solving.
- **Details:** Clicking on a specific log entry will display its detailed information.

- **Dictionary: Enter the alarm and return code ID to query relevant errors.**

Chapter 10 Management

10.1. Authorization Management

10.1.1. System Authorization

If users need to operate the physical robot body using the programming functions, device authorization is required. To activate it, please contact the relevant supplier and the technical support team. Please provide the dynamic device code, MAC address, and serial number (SN code). After authorization, the page will display the authorization duration in real-time.

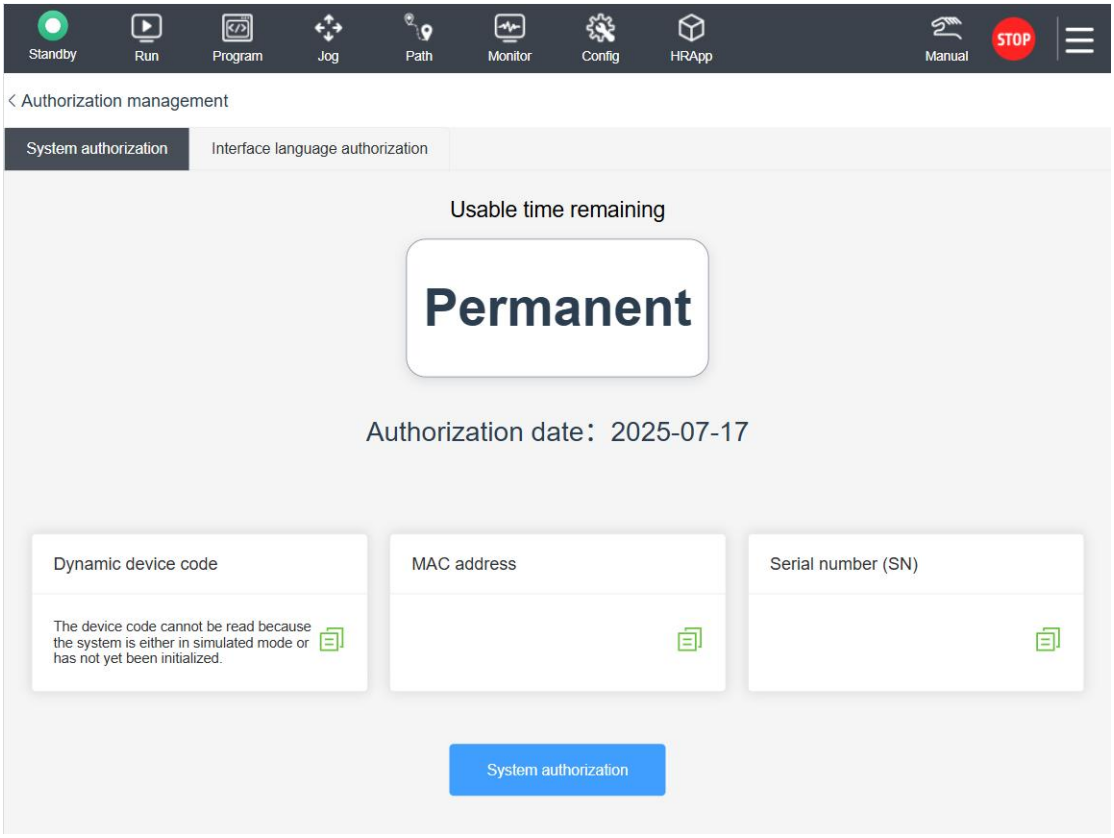


Fig. 10.1.1 Authorization Management

The custom permission level management allows users to comprehensively manage login permissions during usage.

SN Code: The unique device identifier distinguishing different units of the same type.

MAC Address: The hardware identifier for network devices, ensuring uniqueness in a

network.

Without international language authorization, the front-end only shows Chinese. However, in simulation mode, it is unrestricted by international languages and system authorization.

After three consecutive incorrect entries, system restart or waiting until the next day is required to re-enter. Authorization codes are valid for the day they are generated.

10.1.2. International Authorization

After two consecutive incorrect entries of the international language authorization code, the system must be restarted to input it again.

Once the authorization code is generated, it is permanently valid.

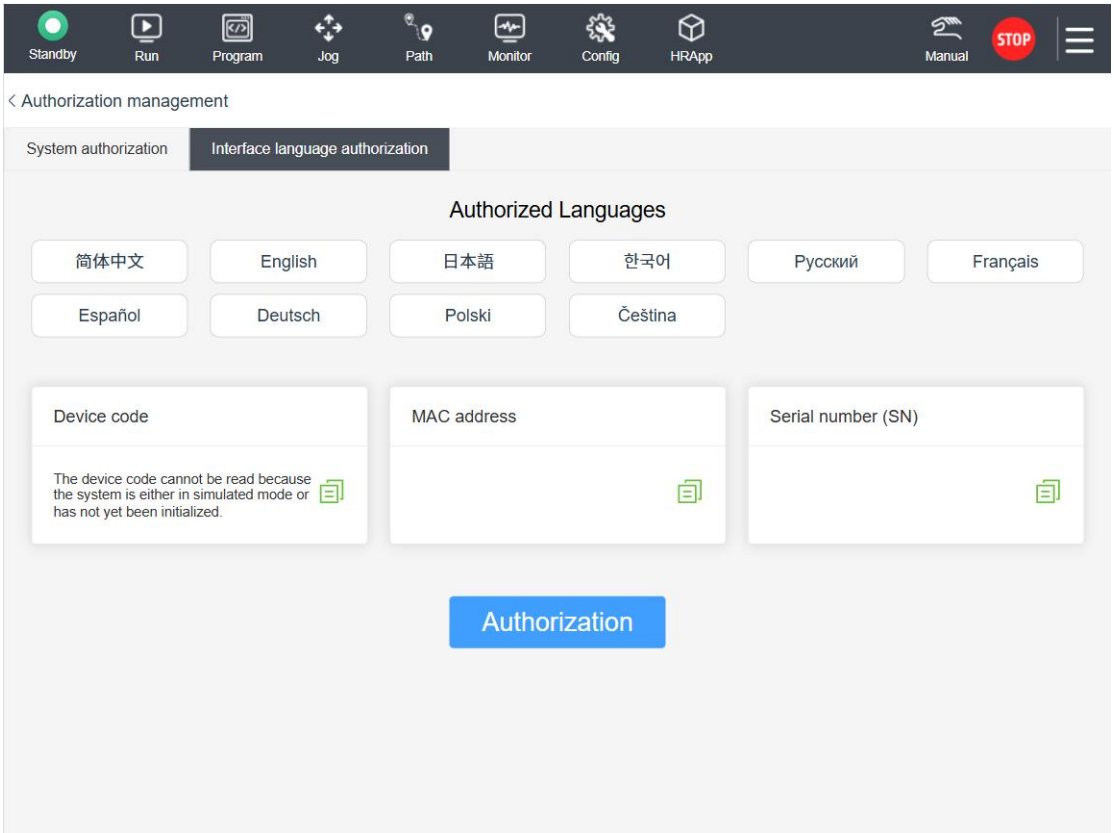


Fig.10.1.2 International Authorization Interface

10.2. Network Configuration

On the robot teach pendant, click Management-> Network to set a static IP address, as shown in the figure below:

Note: The static IP address must not be 192.168.0.0, 192.168.0.1 or 192.168.0.255.

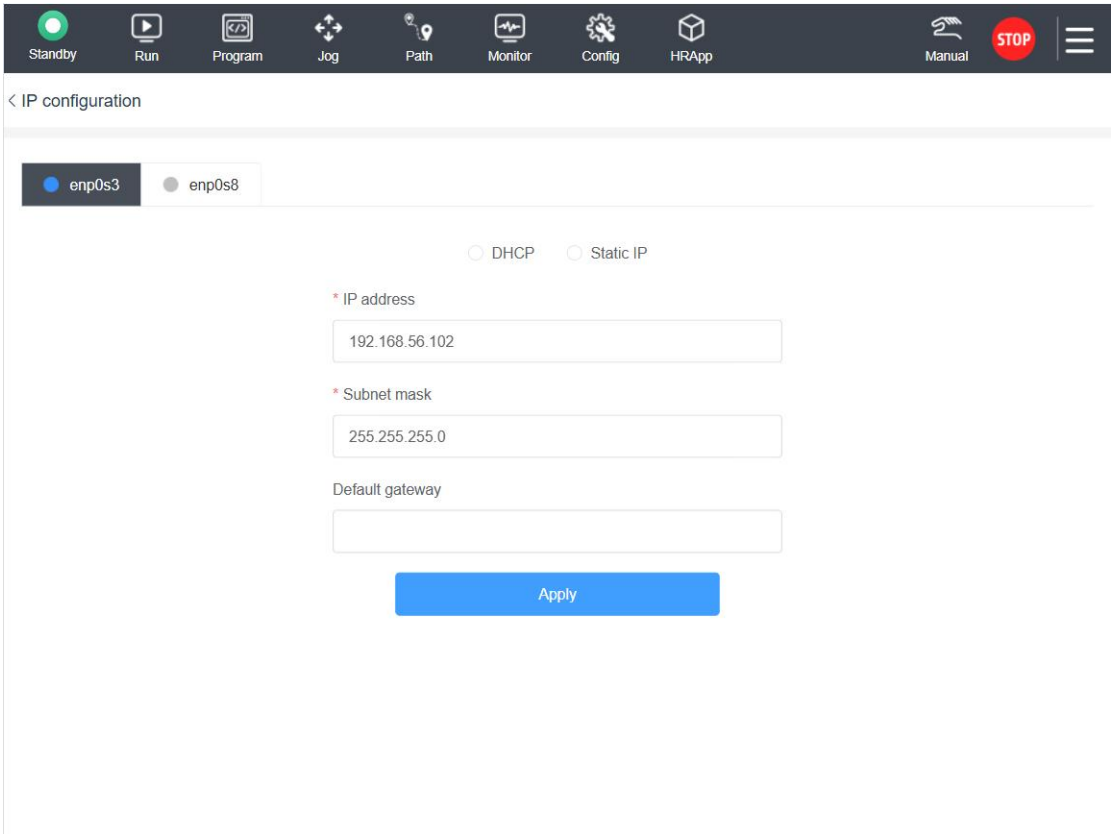


Fig. 10.5.1 Network Configuration Interface

The blue prompts show the current network IP in use. After clicking Apply, the system will automatically reboot. Click OK to confirm. Users can also opt for DHCP dynamic allocation. Once the robot network is successfully established, it will automatically assign IP addresses.

To use DHCP for IP address allocation, ensure the robot network and DHCP server are on the same physical network; otherwise, IP acquisition will fail. For DHCP-related queries, consult the relevant IT department of the client.

10.3. System Update

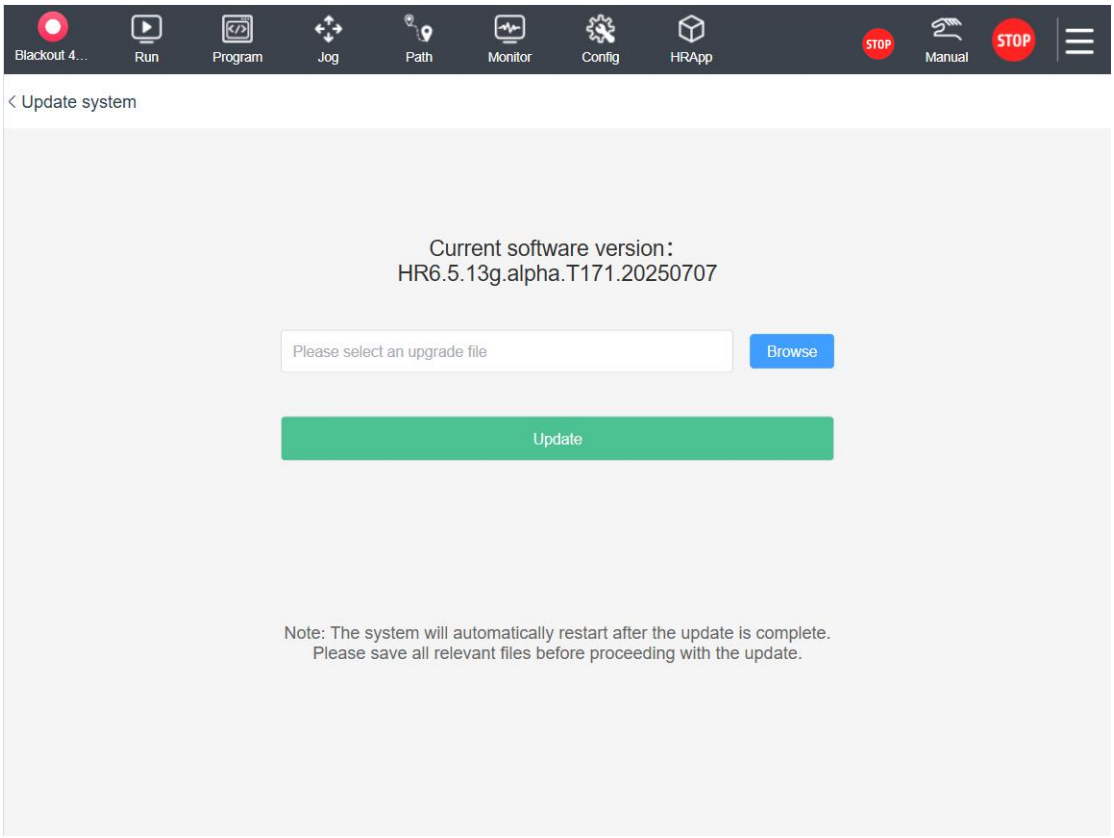


Fig. 10.4.1 System Update Interface

This function allows for the controller software system upgrades. The current software version is displayed, and an upgrade package can be imported to update the robot arm's controller software system.

- Browse: Open the folder to select the upgrade package;
- Update Package: Provided by technical support team or corresponding suppliers;
- Update: By clicking Update, the software system will be automatically upgraded, and it will take 2-3 minutes to finish upgrading;

Note: Before upgrading, please contact the technical support team for guidance. More importantly, please ensure the robot arm is powered-off and all the data is backed up. The system will reboot after the upgrade. If it fails, check the upgrade package or contact the technical support team for help. Upgrading from version 5.0 to 6.0 is irreversible.

Chapter 11 User Management

11.1. Accessing the User Management Backend

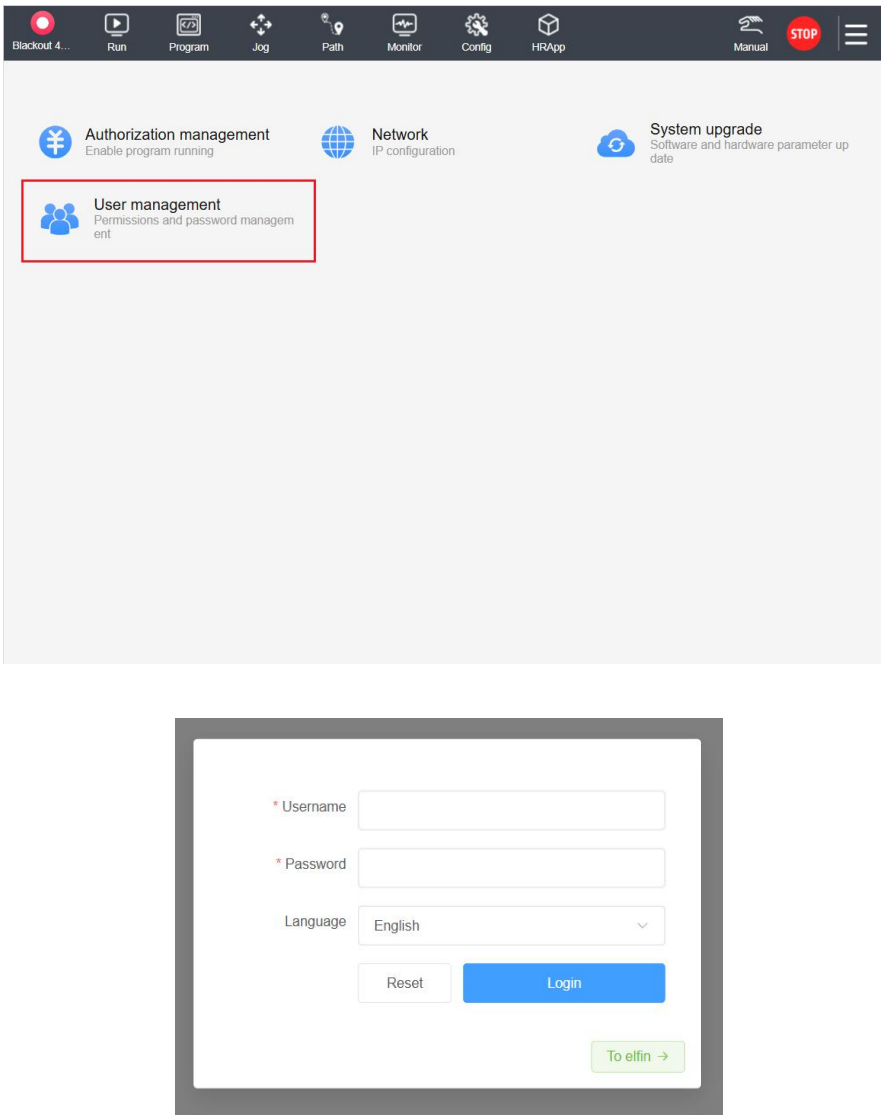


Fig. 10.2.1 Accessing User Management Backend Interface

Click 【User management】 from the stack menu to enter the user management backend. Log in using an account with administrator privileges.

Back to Main System: Click to return to the login page of the robot operating system, log in again and enter the system operation. Clicking "To Elfin" to return to the login page of the robot arm controller software system. Re-log in to enter the system and continue operations.

11.2. Role Management

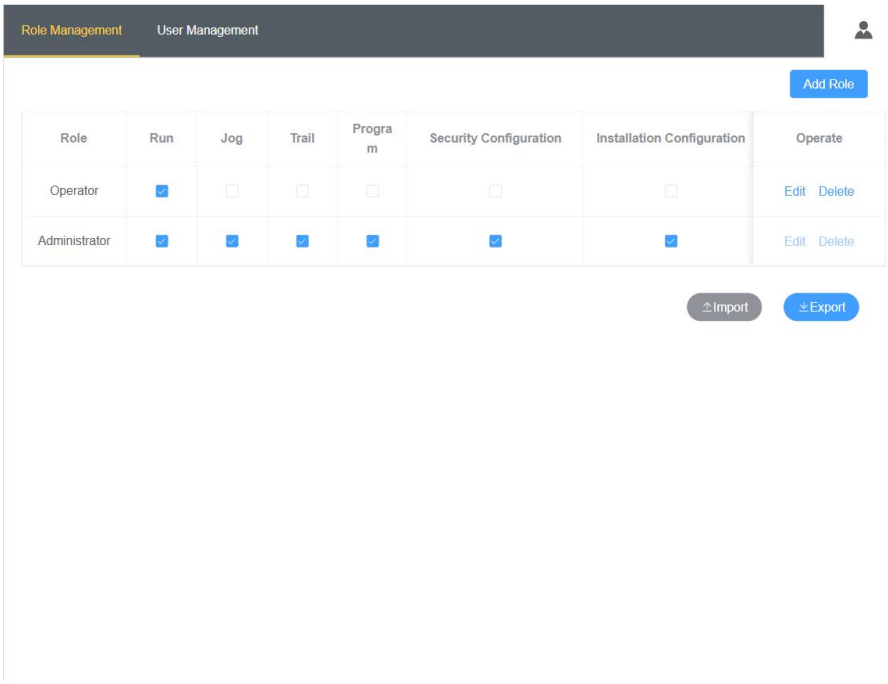


Fig. 10.2.2 Role Management Interface

Users can manage role permission groups on the page, including adding role, modifying role functions, deleting role and editing role permission names.

Add role permission: Click 【Add Role】, enter a role name in the pop-up window, confirm to activate it, and the new role will appear in the role permission list.

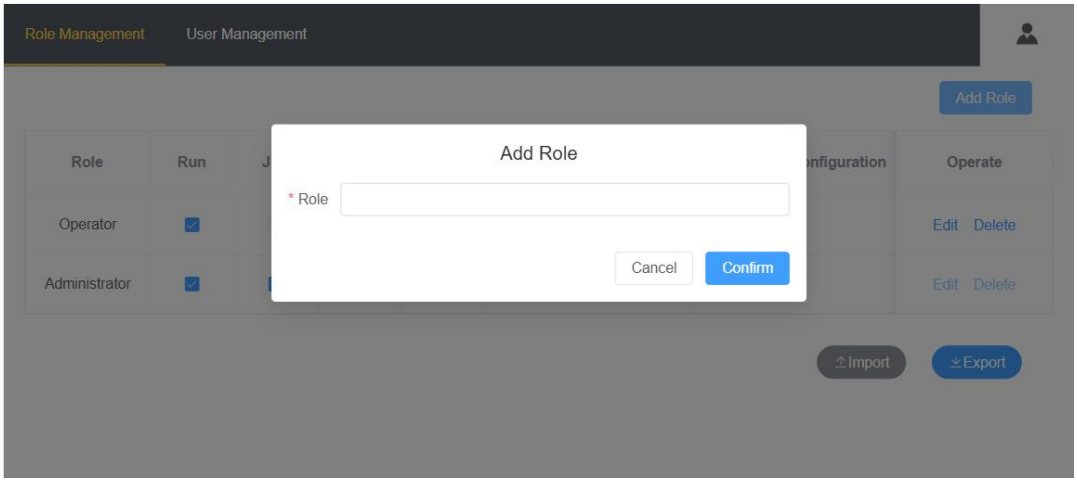


Fig. 10.2.3 Add Role

11.3. Role Management

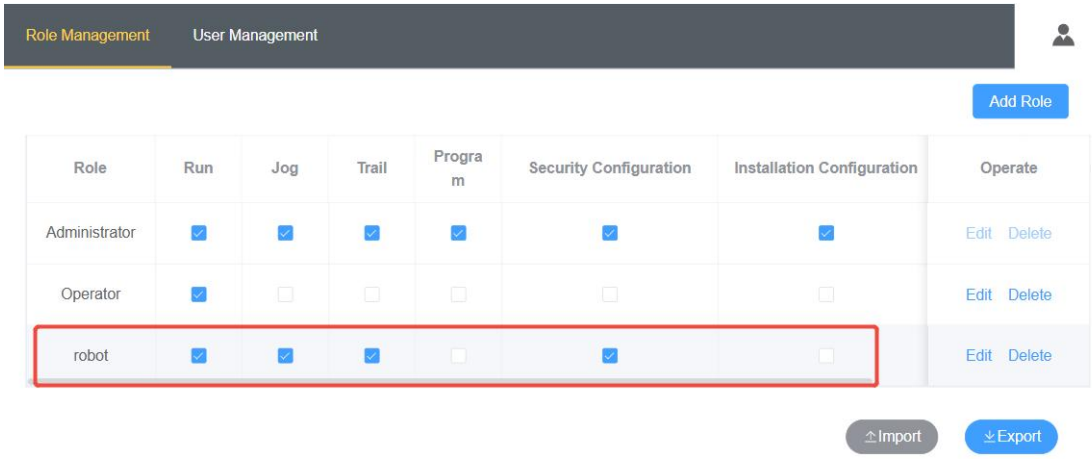
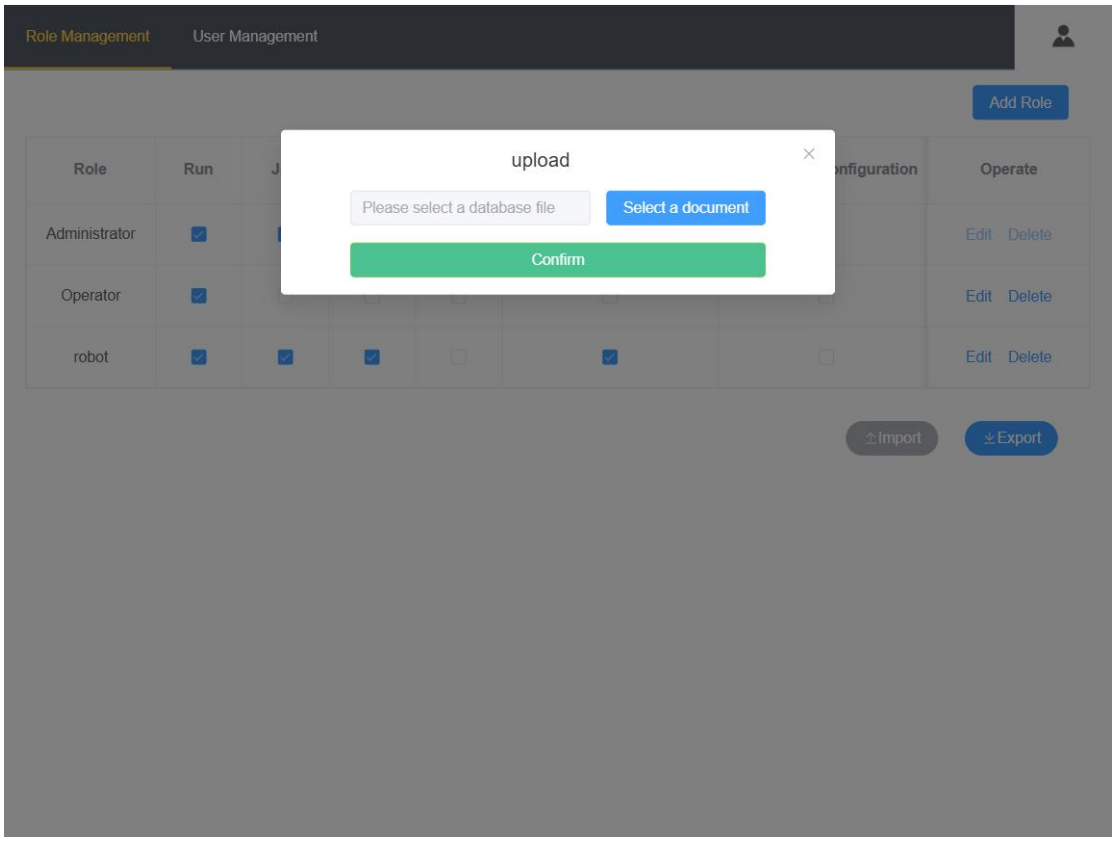


Fig. 10.2.4 Role Management

In the role permission list, users can customize the specific functions for each role permission group. Once the functions are selected, members of the role permission group will gain access to those functions.

- **Delete:** Click 【Delete】 to remove a role permission group. If users are assigned to the role permission group, the system will warn that they will lose their operational permissions. These users will not be able to operate the system until an administrator reassigns their permissions.
- **Edit:** Click【Edit】to change the name of a role permission group.
- **Export:** Click【Export】to save the current permission groups and their settings as a file of a fixed role.db type.
- **Import:** Click 【Import】 to upload files and restore previously exported role permission groups and settings.



The role management table outlines specific permission groups for different user roles, such as admins and operators.

Table 10.2.1 Role Management

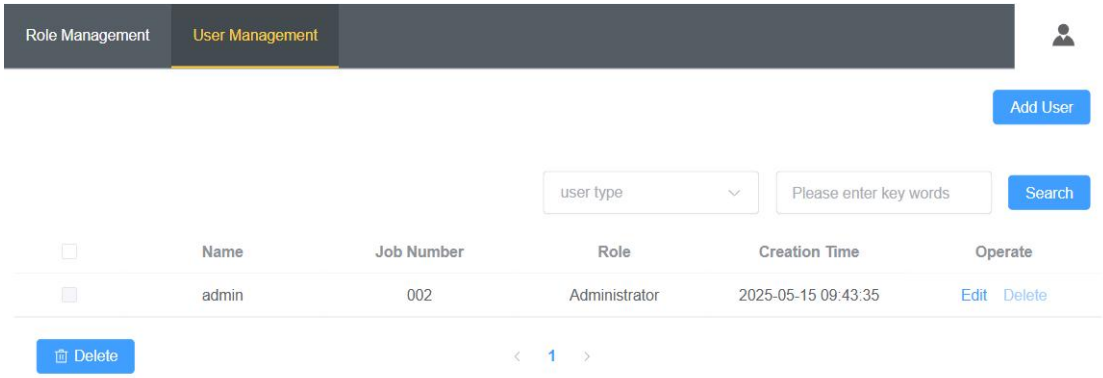
Role Type	**Role Description**	**Remark**
Admin	Access to full permissions, able to manage all accounts and secure passwords.	Manage all account passwords, including security passwords
Operator	Can only access to t he "Run" interface. Other operations require additional administrator authorization, and the administrator will provide the login account and password.	The administrator provides the login credentials (account and password).

◆ The default admin account and password are set before delivery. If forgotten or lost, please consult the corresponding supplier and the technical support team;

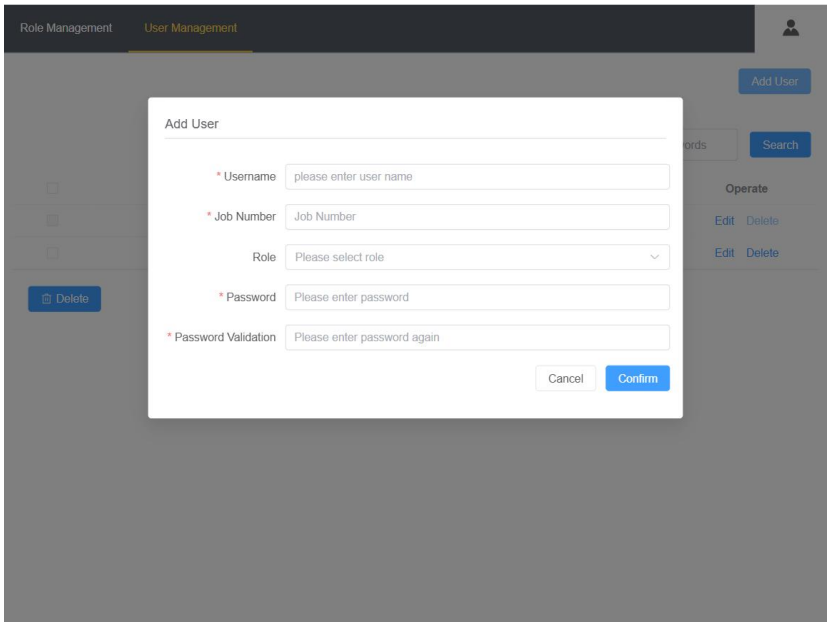
- Users can access robot functions based on their account permissions.;
- If the operator forgets the credentials (account and password), please request assistance from the administrator;

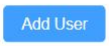
The administrator can assign specific permissions to particular users as needed.

11.4. User Management



On this page, users can manage system users, including adding new accounts, deleting accounts, and editing user details.

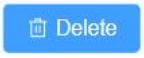


Add user: After administrator account login, click 

【Add user】. A popup will appear for entering user name, job number, role permission group, password, and password verification. After confirmation, the addition will be successful.

Edit user: Click 

and the user editing popup will appear for editing the user name, job number, permissions, password and other contents of the currently selected user. Click Confirm to activate it.

Delete user: Click 

to delete the currently selected user. The system will prompt for confirmation before deletion; the deletion will be finalized upon confirmation.

11.5. Login Setting

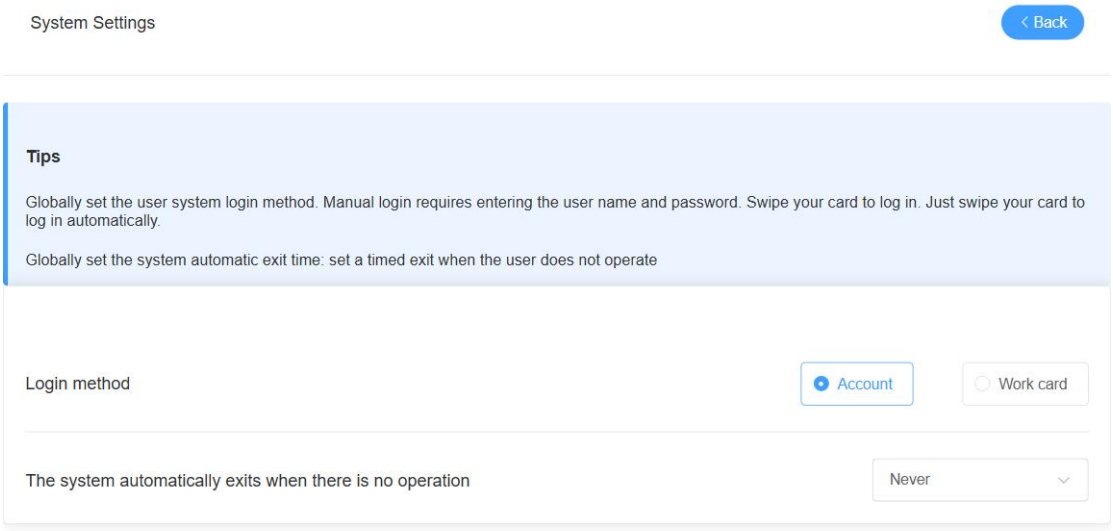


Fig. 10.3.1 System Login Settings

The system offers multiple login methods: credentials-based and card-based. Credentials-based login is the default. For card-based login, configure the card reader for one-tap access.



Fig. 10.3.2 Login and Auto Logout

In the settings, the users can configure the system's idle auto-logout time. If the system is unused for the set period, it will automatically log out the current user to prevent accidental robot operation.

The default is "Never" (no auto-logout). Users can adjust it based on their needs.

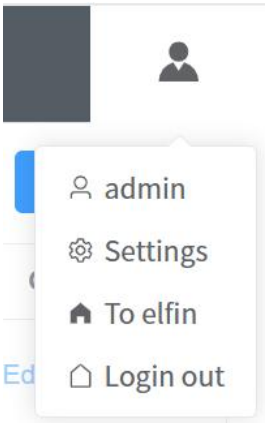


Fig. 10.3.3 Return to Main System

◆ Click 【Personal Center】on the upper-right-corner navigation bar, then select 【Main System】to exit the user management platform and return to the controller software operation interface. Re-login is required to operate the robot.

Chapter 12 HRApp

Based on the Elfin software architecture, this is a plugin framework designed to enhance the multifunctional capabilities of robotic systems. The platform enables users to develop and integrate custom functionalities, facilitating seamless interoperability between robotic units and external intelligent systems (e.g., IoT devices, automated workflows, or legacy automation infrastructure). By standardizing API connectivity and protocol adaptation, the platform streamlines system integration, improves operational throughput, and ensures backward/forward compatibility across heterogeneous robotic networks.

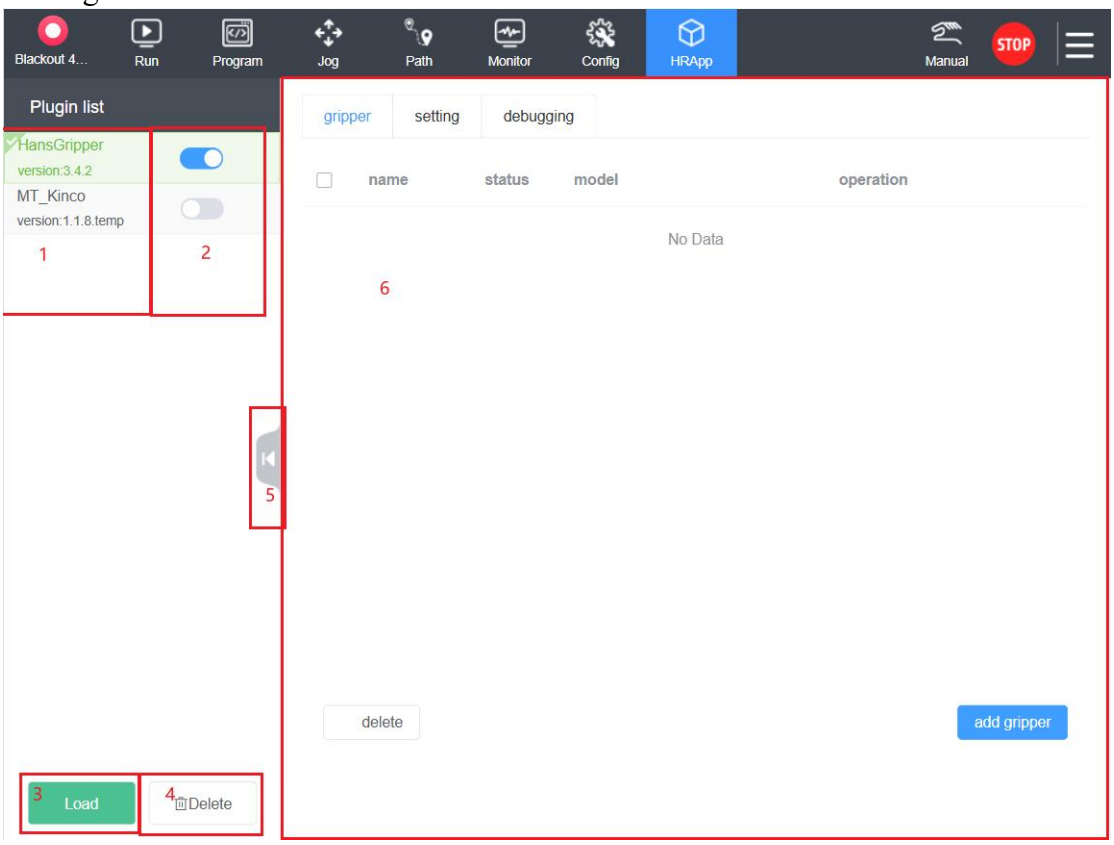


Fig. 11.1 Plugin Platform

Plugin Management Interface:

1) **Plugin List:** Displays all currently imported plugins, showing their names and status of the programming switches;
2) **Switch:** When enabled, the plugin's related instructions are visible in the Program interface for use. When disabled, the plugins are hidden;
3) **Load:** Import relevant plugin installation packages from external files. For obtaining specific installation packages, please consult the technical support team;
4) **Delete:** Removes the selected plugin. Upon confirmation, the plugin is permanently deleted;
5) **Expand/Collapse:** Controls the expansion or collapse of the plugin list for easier viewing;

6) **Plugin Details:** Displays configuration settings and details of the selected plugin;

Note: The system currently supports only plugins developed by the company. For any additional robotic functionality requirements, please contact the technical support team.

Chapter 13 Exception Handling

13.1. Protective Stop

During robot operation, errors may occur, leading to abnormal performance or other exceptions. To avoid such abnormal conditions, a protective stop is required. A protective stop is a cessation ("safe stop") triggered by the robot controller when it detects an error. This warning message will be displayed on the teach pendant or external monitor to notify the error state. In robotics, protective stops are classified as follows:

◆ Class 0 stop, also known as Stop 0: Immediately cut off the power supply of the motor.

Because the motor loses power during the deceleration process and cannot be controlled, the robot arm will deviate from the predefined running path, which is an uncontrolled stop mode. The emergency stop button is implemented by performing Stop 1 first; if Stop 1 does not take effect within the specified time, Stop 0 will be automatically started.

◆ Class 1 stop, also known as Stop 1: The controller controls the motor to slow down to stop, and then cuts off the power supply.

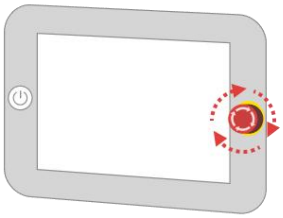
This is a controlled stop method, and the robot arm basically completes the deceleration according to a predefined path. In the latest robotic systems, due to the use of safety controller technology, after the emergency stop button is triggered, the system prioritizes the execution of Stop 1 stops (except in manual mode) to protect the braking system. If Stop 1 does not take effect within the specified time limit, the system will automatically upgrade to Stop 0 stop.

◆ Class 2 stop, also known as Stop 2: It is completely decelerated through the servo system. After stopping, the power supply is not cut off, and the motor is still electrified. It is a completely controlled stop mode. Nowadays, the program stops of robots all fall into this category.

The current controller software realizes protective stop operations through various functions, including emergency stop, safety range and collision.

13.2. Emergency Stop

For the "Class 0 stop" in protective stop, if the user predicts that a sudden hidden danger will occur during the robot arm motion, please press the emergency stop button immediately. After which, the mechanical arm and electrical control box are powered off and stop working, and reset can be done according to the prompt. After resetting, the robot is in a power-off state, and the user needs to power it on again to get the robot back to normal working state.



Reset Steps:

1) Unscrew the red emergency stop button on the teach pendant clockwise first.
2) Click the 【Reset】 button in the popup to clear the error.
3) Return to the running homepage and click the【Power on】button to return to the normal state.

Note: If the emergency stop button is not unscrewed, click【Reset】directly, and "Reset failed" will be prompted. The user needs to follow the reset process to return to normal. If the repeated reset fails, check for hardware issues.

13.3. Safe Collision Stop

When the robot arm hits an obstacle, it will judge whether to stop moving according to the force fed back by the obstacle. The magnitude of the force is judged according to the level configured in the safety limits. If the safety risk level is higher, the greater feedback force is needed to stop the robot; if the safety risk level is lower, the smaller feedback force can also stop the robot.

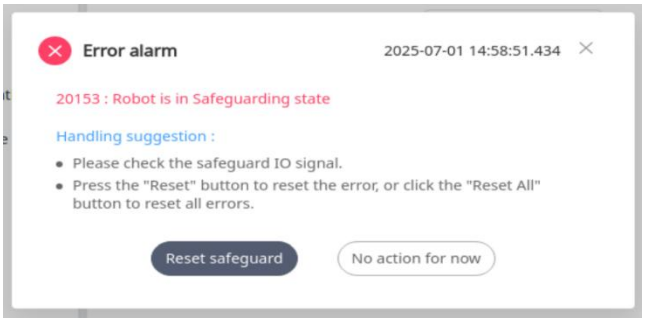
After stopping, it can be restored to normal by resetting. To be on the safe side, make sure the robot doesn't hit an obstacle when resetting, so as not to trigger a safe collision stop again.

13.4. Trigger Safeguard Monitoring

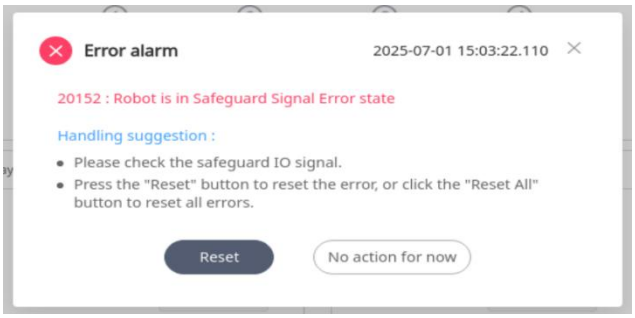
Enable the safeguard monitoring in the safety limits, configure the safety I/O signal to stop or pause, and when the I/O signal is monitored, the system will trigger a protective stop and pop up a prompt.

The two input channels of the safeguard monitoring cannot be the same. If the user select the configurable inputs CI0 and CI1, there are the following three possibilities:

1) If both CIs are high, the robot can move normally;
2) If both CIs are low, the interface prompts "The robot is in the safeguard monitoring state, please Reset". At this time, the robot is still enabled. Restore both CIs to high levels, and click reset, the and robot will be able to move normally;



3) If the two CIs are one high and one low, the interface prompts "The robot is in a state of inconsistent safeguard monitoring signals, please reset". At this time, the robot is in a power-off state. Restore both CIs to high levels, click Reset, re-power on and enable, the robot will be able to move normally;



4) If there is an error in one of the two CIs, such as one of the CIs timing out during use, resulting in the two CIs being unable to be set high or low at the same time, the interface prompts "The robot is in the safeguard monitoring error state, please reset". At this time, the robot is in a power-off state. Restore both CIs to normal, click Reset, re-power-on and enable, the robot will be able to move normally.

◆ Safeguard Monitoring Stops: When the safety I/O signal is monitored, it stops moving and a popup appears to indicate the safety protection state. The robot can only be operated again after resetting and clearing the error. After reset, the robot

is not de-electrified.

- Safeguard Monitoring Pause: When the safety I/O signal is monitored, the script will be paused. After resetting and clearing errors, the current script can continue to run.
- Safeguard Monitoring Stop and Jog: When the safety I/O signal is monitored, the script running will be terminated. If the safety light curtain is currently triggered, jog operation is available to adjust the robot position (while the script running is disabled). After resetting and error clearing, the current script can be run again.

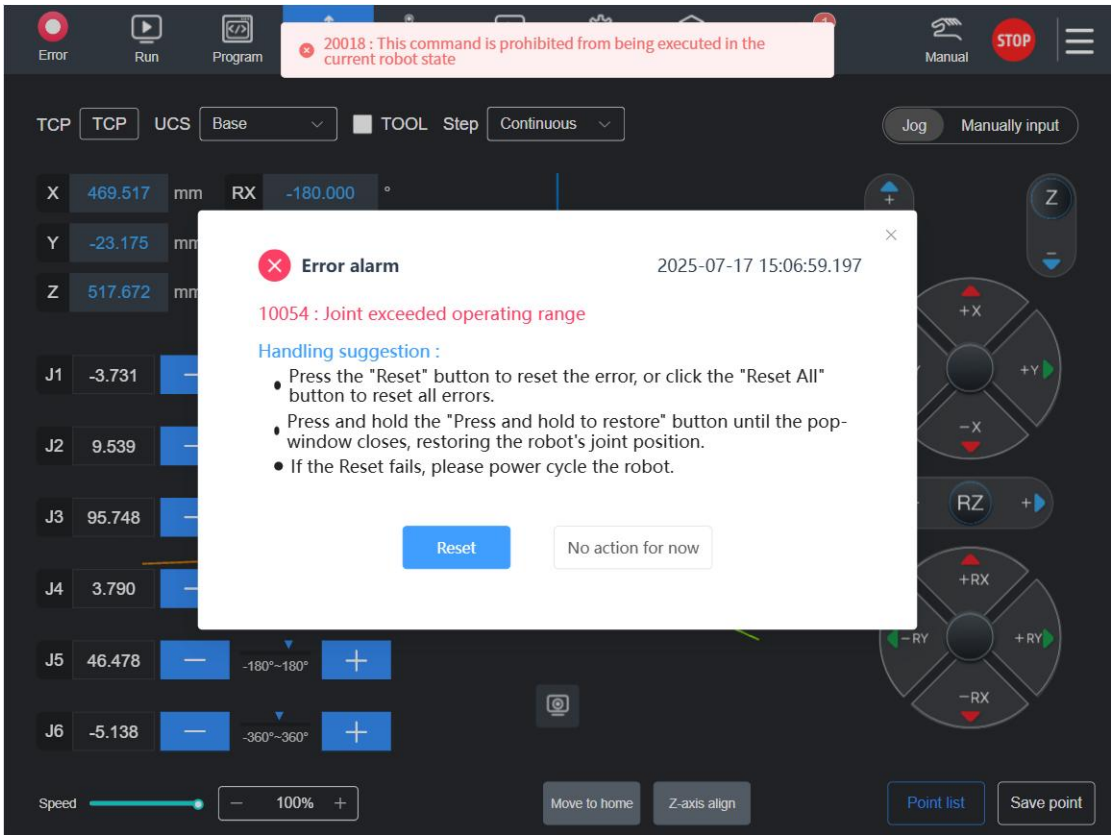
Reset Steps:

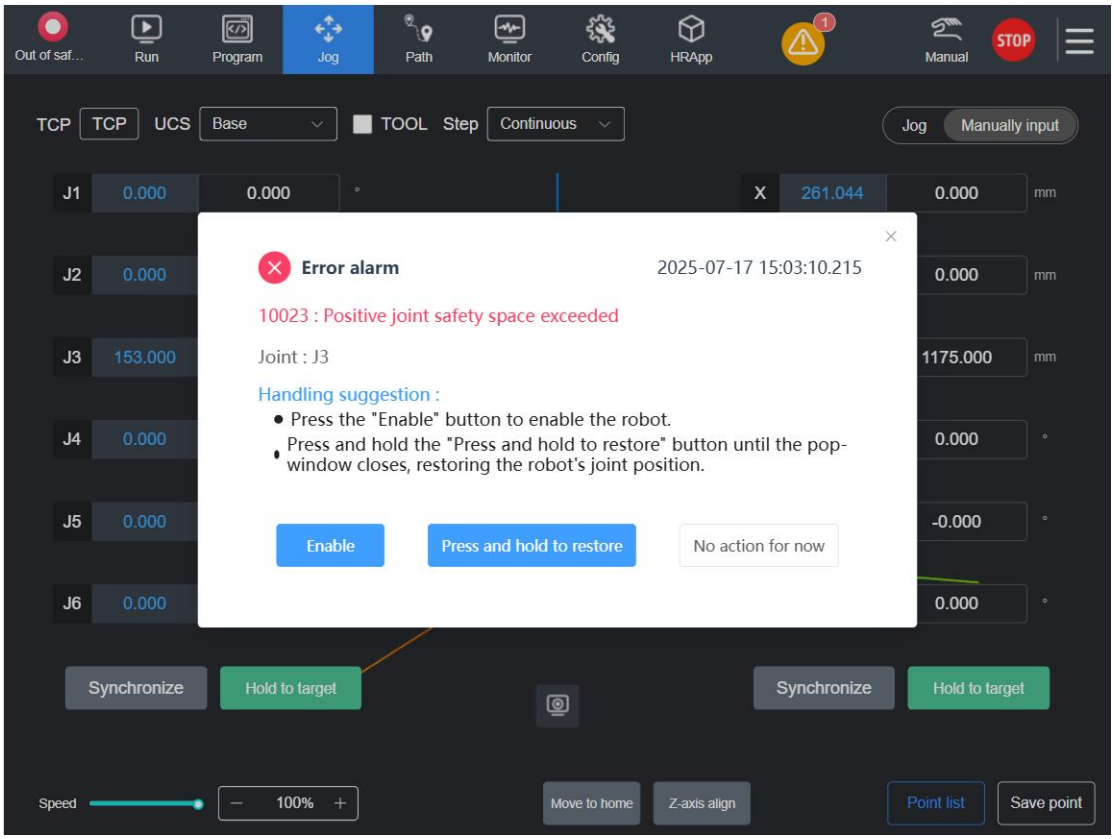
1) Release the protective switch first.
2) Click the 【Reset】 button in the popup to clear the error.

Note: If the switch is not released, click 【Reset】 directly, and it will prompt "Reset failed". Users still need to follow the reset process to return to normal.

13.5. Beyond Safety Range

During the movement of the robot arm, if the movement exceeds the restricted safe range (joint range), a prompt will pop up. The user needs to press and hold to restore. Operate the robot to move away from the restricted area, and the state return to normal when the robot exits the restricted area. Refer to "Safety Range" for details.





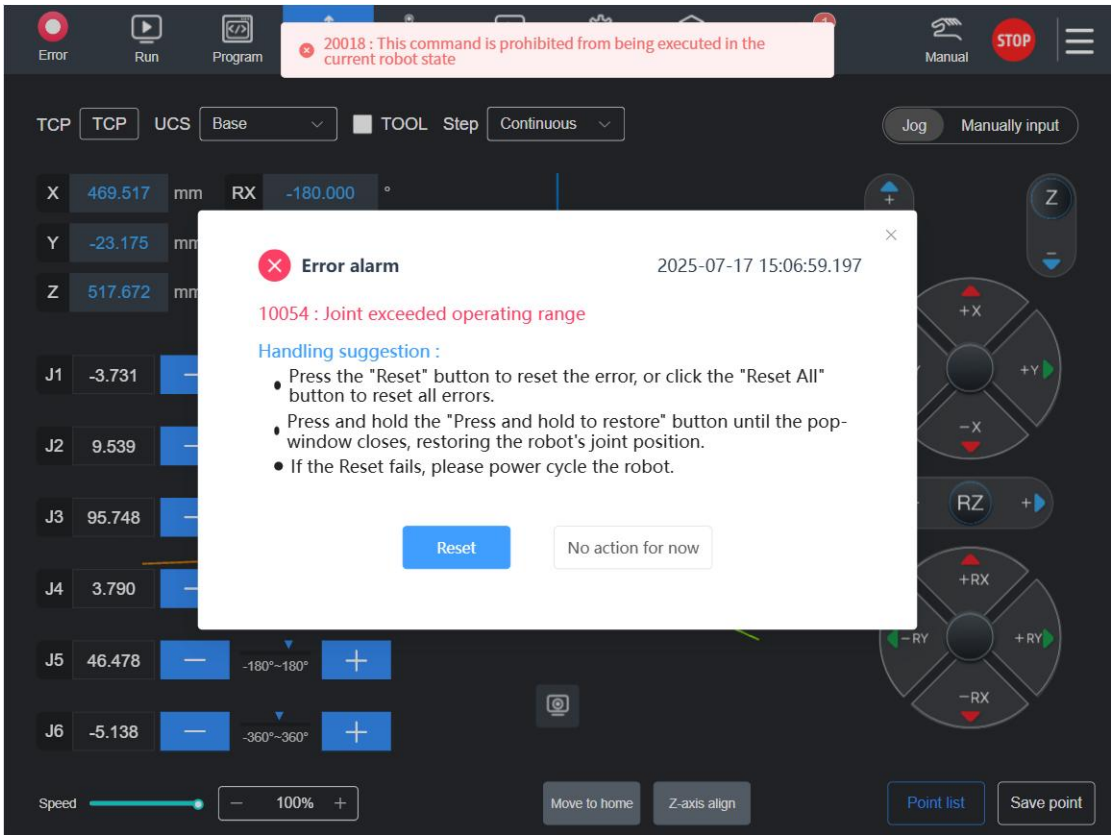


Fig. 11.5.1 Beyond Safety Range

13.6. Singularity Point States

"Singularity" is a common robot phenomenon, which is not caused by robot design, processing, or system defects. It is an inherent characteristic of six-degree-of-freedom non-redundant robots. Customarily, the configuration where strange phenomena occur is usually called "singular configuration", also called "singularity point".

Singularity point usually refers to some positions of the robot arm in its workspace, which will cause some performance parameters of the robot arm (such as velocity, acceleration, force, etc.) to be discontinuous or infinite. These positions may be due to the structural design of the robot arm, joint configuration, or characteristics of the kinematics equation. At the singular position, the robot arm may encounter the following situations:

A. Infinite Joint Velocity: At some singularity points, in order to keep the velocity of the end effector constant, it may be necessary to have an infinite angular velocity of a certain joint, which is impossible in practical applications.
B. Infinite Joint Acceleration: Similar to velocity, some singularity points may require infinite acceleration of the joint to achieve the desired end acceleration.
C. Infinite Moment: In some positions, in order to maintain the balance of the robot arm or generate a certain force, infinite driving torque may be required.
D. Discontinuity of Kinematic Solutions: At some singularity points, multiple kinematic solutions may exist for the robot arm, i.e., multiple different combinations of joint angles can lead to the same end position and pose. This will make it difficult for the control system to choose a stable solution.
E. Range of Workspace: At the edge of the workspace of the robot arm, singularity points may also appear, because some joints of the robot arm may have been extended to the limit and cannot be moved further. A proper understanding of the concept of singularity points is helpful to avoid these unfavorable positions when designing and controlling the robot arm, and ensure the stable operation and efficient work of the robot arm. In practical applications, engineers will reduce the impact of singularity points by optimizing the structural design of the robot arm and improving the control algorithm.

13.6.1. Wrist Joint Singularity

When the Elfin robot 【joint 4】 and 【joint 6】 are parallel, the robot is in wrist singularity. In this singular pose, 【joint 5】 of the robot is usually near 0°, as shown in the following figure:

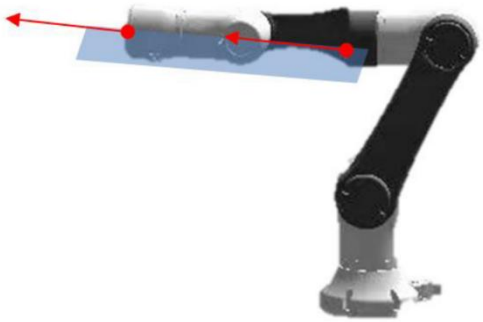


Fig. 12.6.1 Wrist Joint Singularity

When the user uses Cartesian space commands to control the robot to translate and rotate, and the robot reaches the position of the wrist joint singularity and its adjacent area, the controller will trigger the software emergency stop, and the user will be informed that "The robot is approaching【Singularity point】". When wrist singularity occurs, the motion velocity of 【joint 4】 and 【joint 6】 may approach the joint-limiting velocity.

13.6.2. Shoulder Joint Singularity

When the end of the robot arm 【ball joint center point】 (the ball joint center point refers to the intersection of the three axes of 【joint 4】, 【joint 5】 and 【joint 6】 of the robot arm) is above the rotation center of 【joint 1】, the robot arm is in the shoulder joint singularity configuration. Under conventional installation, the shoulder joint is usually located at the top of the robot arm.

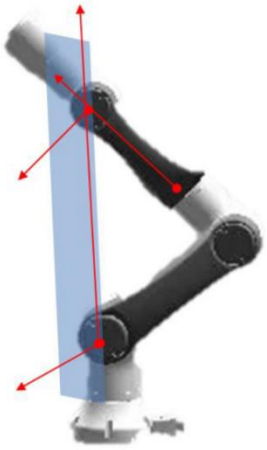


Fig. 12.6.2 Shoulder Joint Singularity

When the user uses Cartesian space command to control the robot arm to translate and rotate, and the robot arm reaches the singularity position of the shoulder joint and its adjacent area, the controller will trigger the software emergency stop. When shoulder singularity occurs, the motion of【joint 1】and【joint 4】may rotate rapidly by 180°.

13.6.3. Elbow Joint Singularity

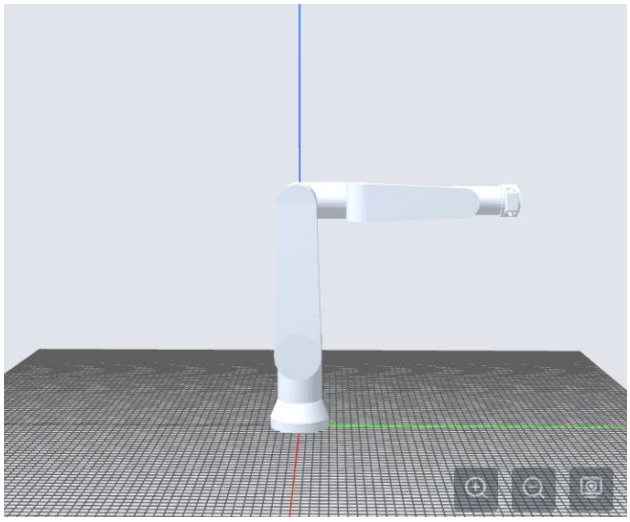
When the robot arm 【joint 2】, 【joint 3】, 【joint 4】 are coplanar (as shown in Fig. 3), the robot arm is in the shoulder singularity configuration. The elbow joint configuration only appears when the robot arm is fully deployed, which is relatively intuitive. In this singularity pose, the robot arm 【joint 3】 is usually near the 0° position.



Fig. 12.6.3 Elbow Joint Singularity

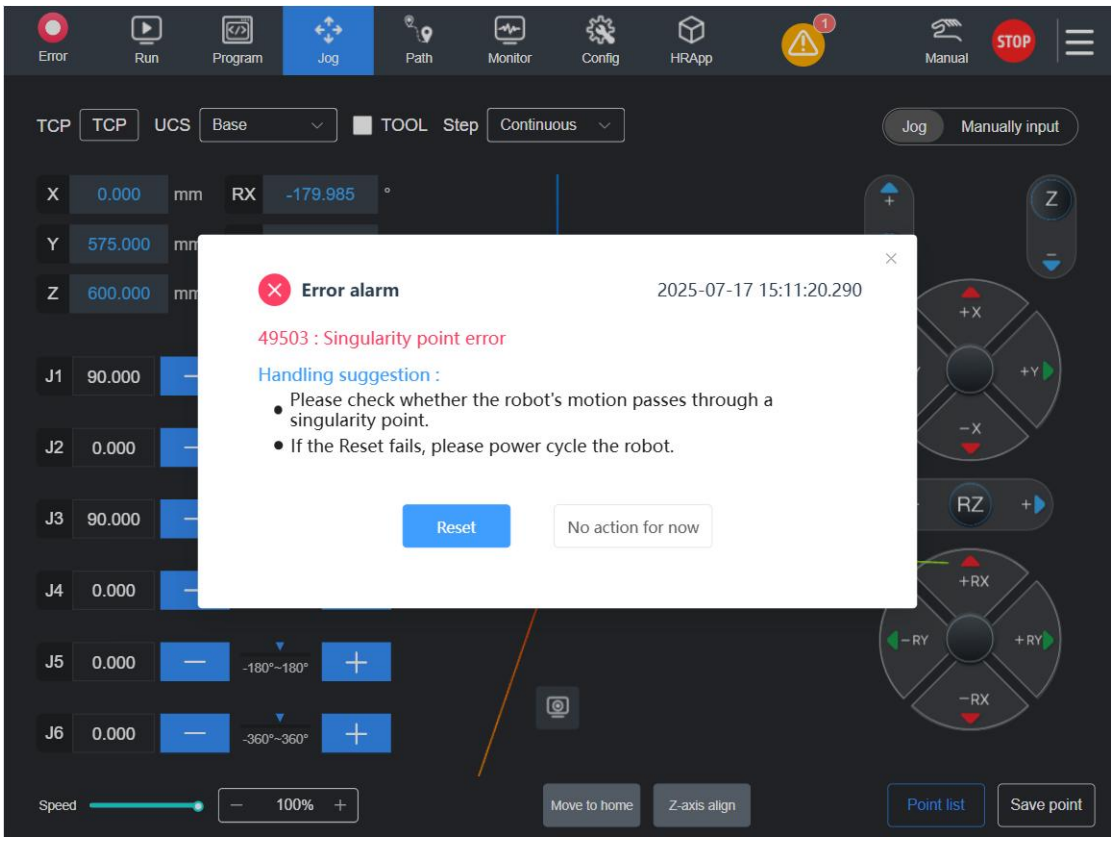
When the user uses Cartesian space commands to control the robot arm to translate and rotate, and the robot reaches the shoulder joint singularity position and its adjacent area, the controller will trigger the software emergency stop, and the user will be informed that "The robot is approaching 【Singularity point】". When shoulder singularity occurs, the motion velocity of 【joint 2】 and 【joint 4】 may approach the joint-limiting velocity.

13.6.4. Singularity Warning and Recovery



When the robot arm approaches the wrist joint singularity point, such as (90, 0, 90, 0, 0, 0), when the user uses Cartesian space commands to control the robot arm to perform position translation and pose rotation, the robot software emergency stop will be triggered, prompting "Singularity error".

When a singularity error occurs, the user can dismiss the singularity alarm as follows

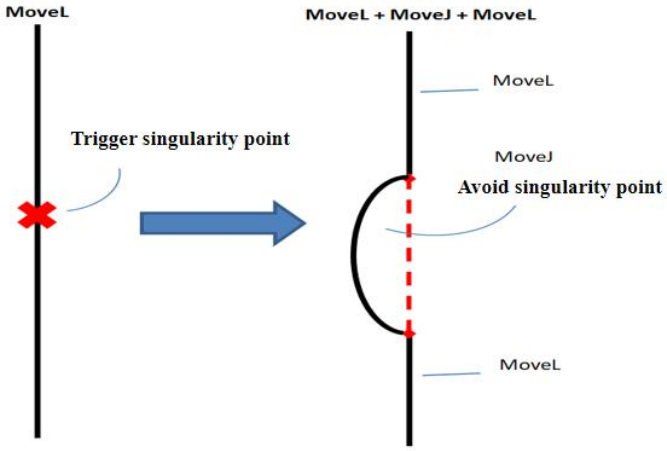


1) Click 【Reset】 to clear the error and the popup disappears;
2) Adjust any axis from J1 to J6 so that the robot pose is not at the singularity position.

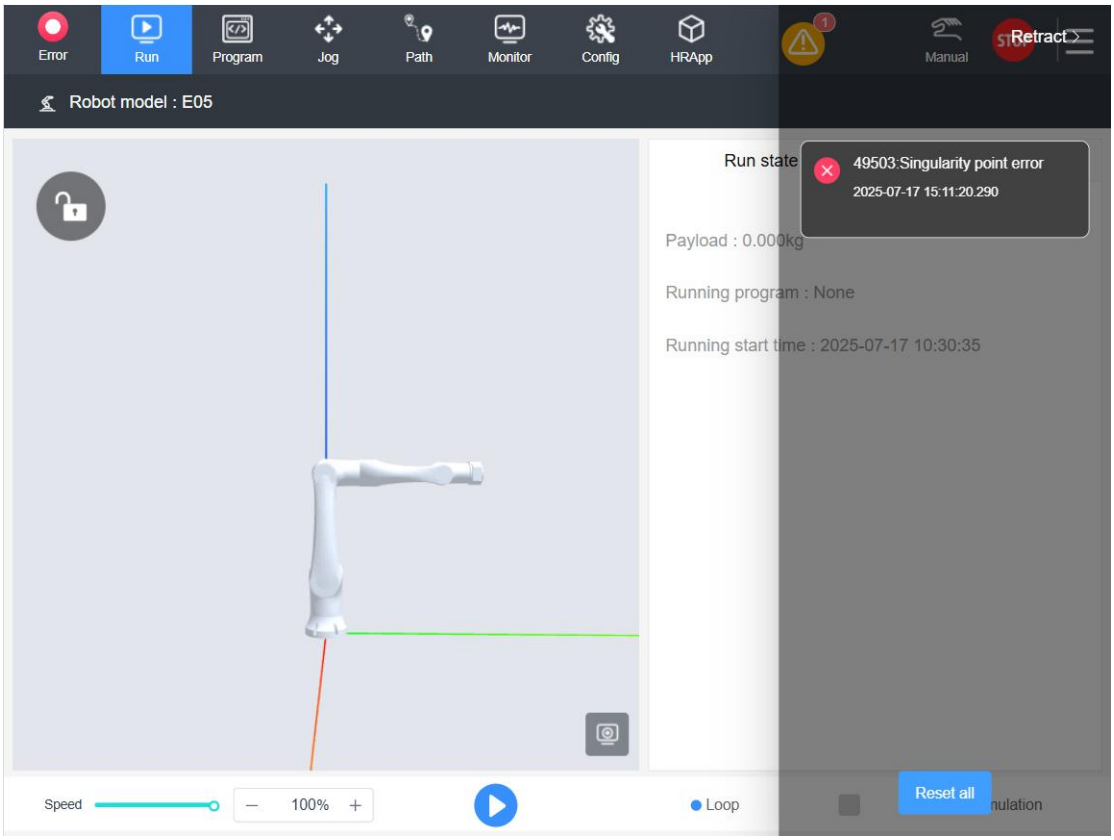
Note: When encountering a singularity point, the pose of the robot arm can only be moved away from the singularity point by adjusting the joint angle after reset. At this time, adjusting the tool position direction (X, Y, Z, RX, RY, RZ) is ineffective.

13.6.5. Singularity Avoidance

1) Try to keep the Cartesian space path away from the singularity point. Taking the wrist singularity as an example, keep 【joint 5】 as far away from the 0° position as possible. It is recommended that all points in a path meet the requirements of angle of 【joint 5】 less than 3° or greater than 3°.
2) Use joint motion 【MoveJ】 to transit the Cartesian space, and pass through the singularity points.



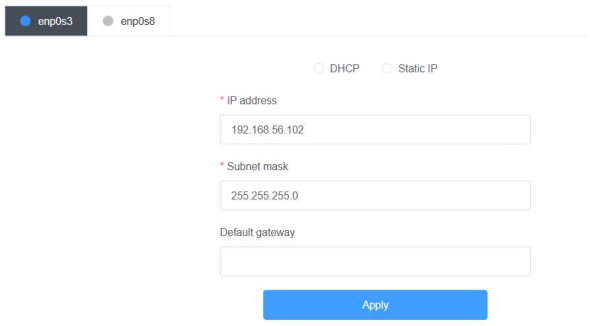
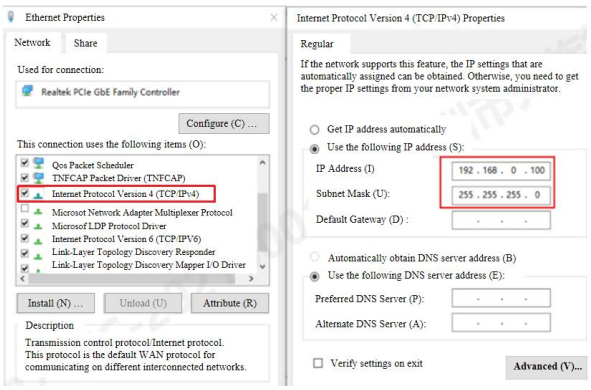
13.7. Warning Popup

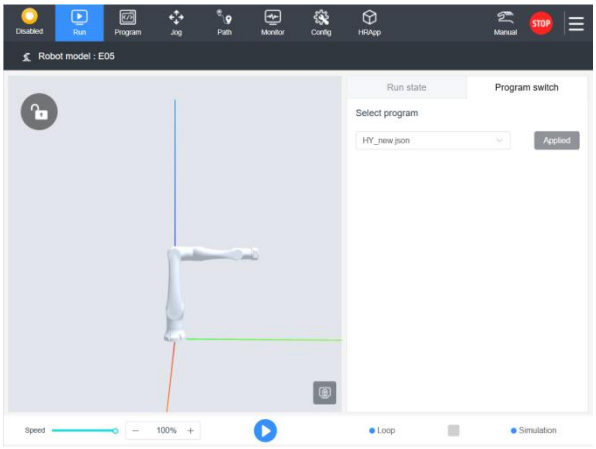


When an error prompt appears in the system, if the user selects "Do not process for the time being", the error alarm popup can be closed. The user can suspend processing of this error alarm, and a sign  will be displayed at the top right. After clicking on the sign, the user can view all the error prompts. Click the 【Reset all】 button, all error alarms can be handled until the popup disappears.

Chapter 14 End-effector Connection

The controller software operation interface can be remotely displayed on the personal computer through the network. But the computer and the robot need to be in the same subnet. In order to realize communication between the computer and the robot, the network can be configured in the following ways:

SN	Operation	Example
1	On the robot teach pendant, click Menu-> Management-> Network to set a fixed IP address. **Note: IP addresses cannot be 192.168. 0.0, 192.168. 0.1 or 192.168. 0.255;**	
2	Configure the laptop TCP/IPv4 network as a static IP address;	
3	After configuration, check whether the laptop and the robot are in the same network segment. If so, the robot can be controlled through the web page;	

4	Open the browser, type the address in the following format in the URL bar: Robot IP +/dist (such as 192.168. 0.10/dist), and enter the web page. After successful login, users can control Huayan S.p.A through Huayan S.p.A software;	

Chapter 15 Virtual Robot

15.1. Overview

Huayan provides a virtual robot function that allows users to operate the controller software without connecting to a physical robot. This function allows customers to become familiar with and master the Huayan controller software in advance.

Users need to pre-install virtual machine software. It is recommended to use a virtual machine:

1. VMWare
2. VirtualBox

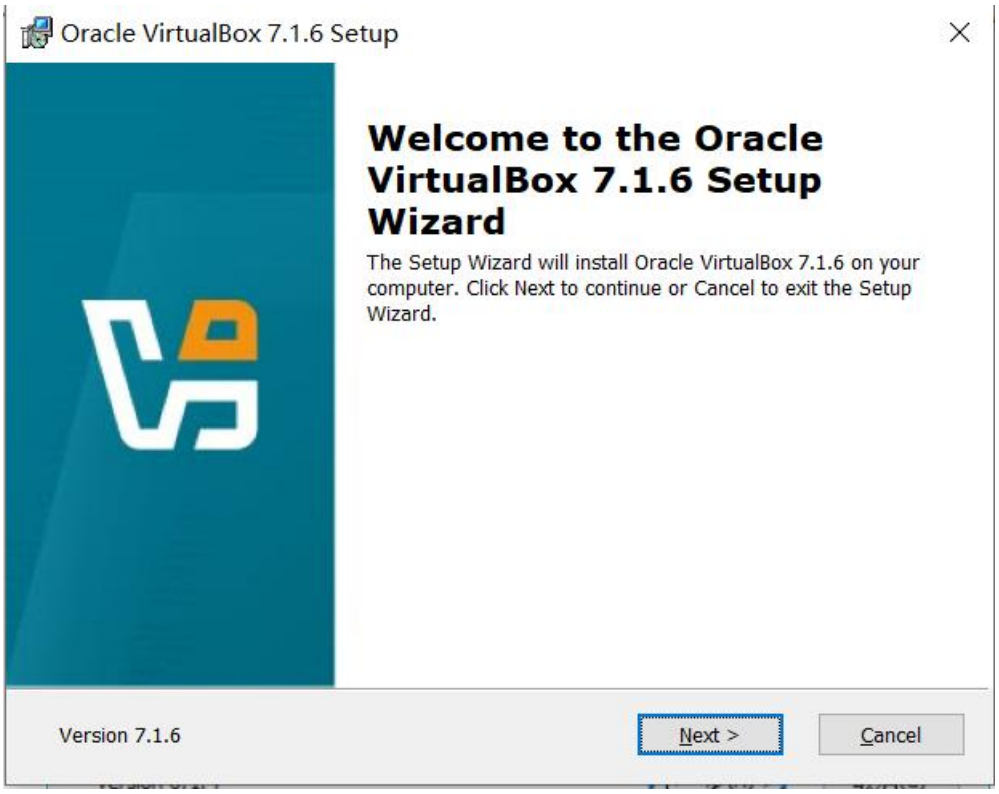
This document is based on the use of VirtualBox. VMWare can also load this system, but customers need to solve the VMWare authorization problem and create an Ubuntu virtual machine by themselves before loading the virtual disk.

15.2. VirtualBox Installation

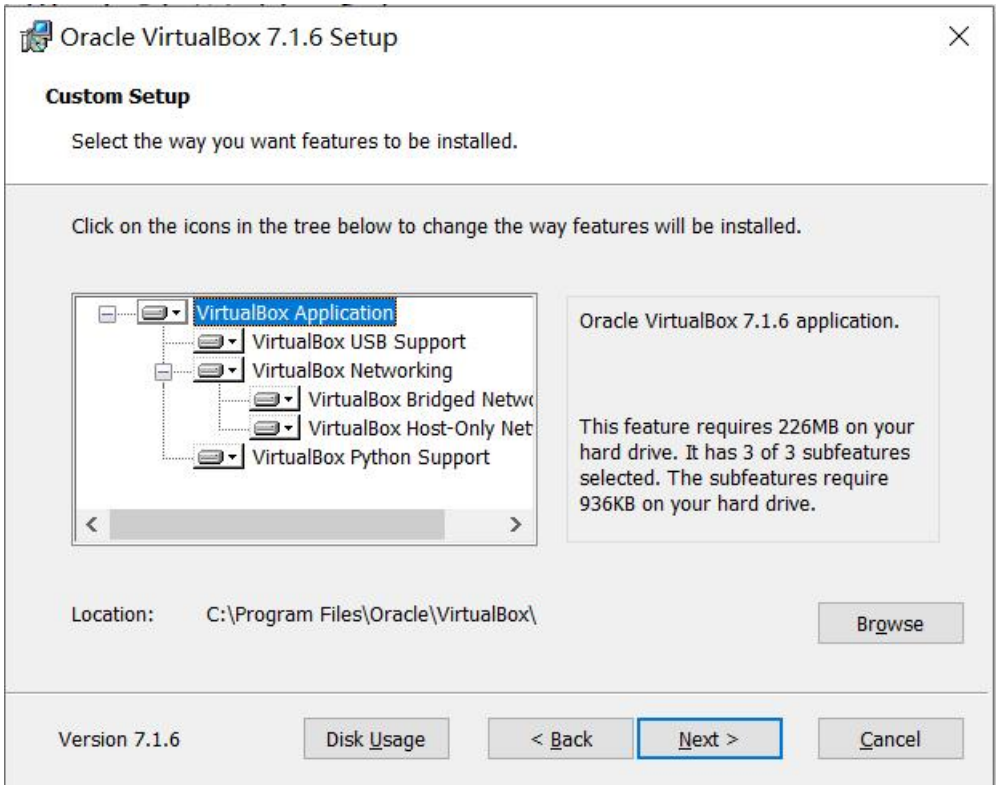
VirtualBox is an open source virtual machine software. Users can install and run Solaris, Windows, DOS, Linux, OS/2 Warp, BSD and other systems as client operating systems on VirtualBox.

VirtualBox software from the official website. The installation process is as follows:

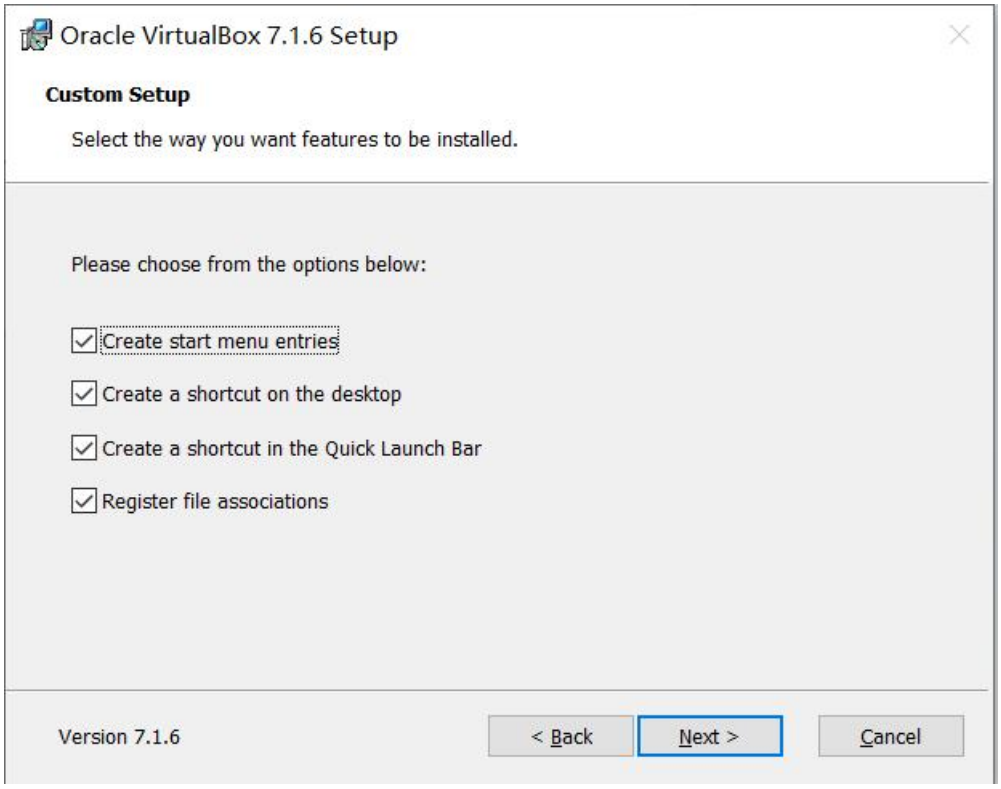
1. Double-click the software installation package to start the installation and click Next;



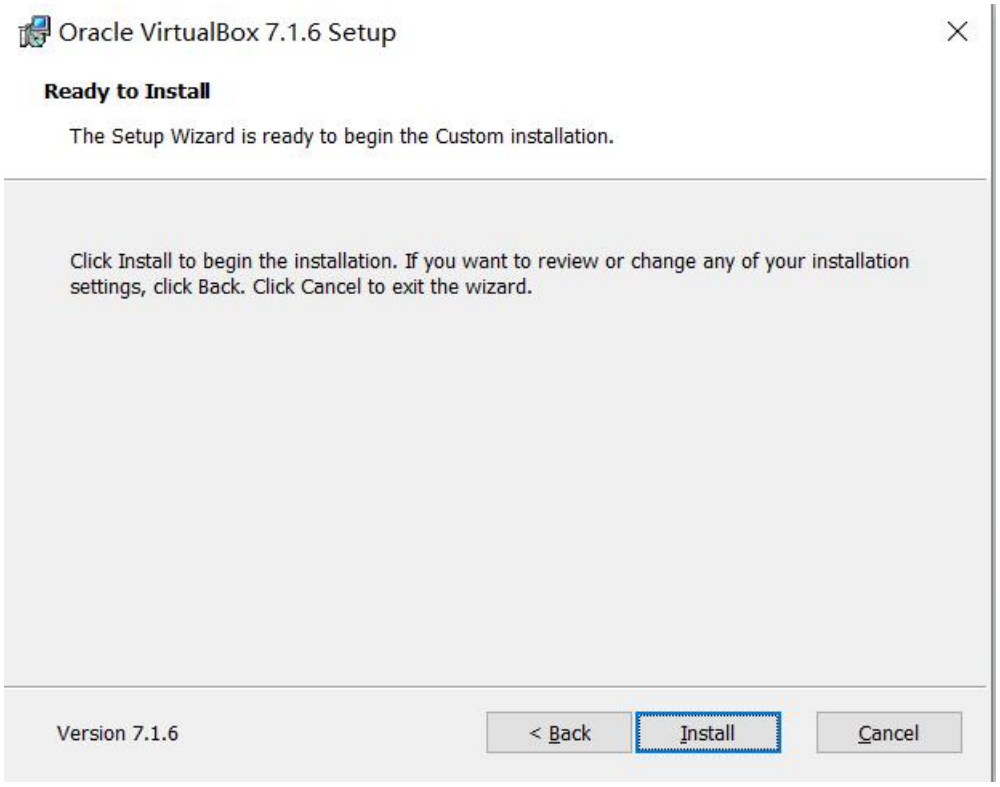
2. Select the default installation path and click Next;



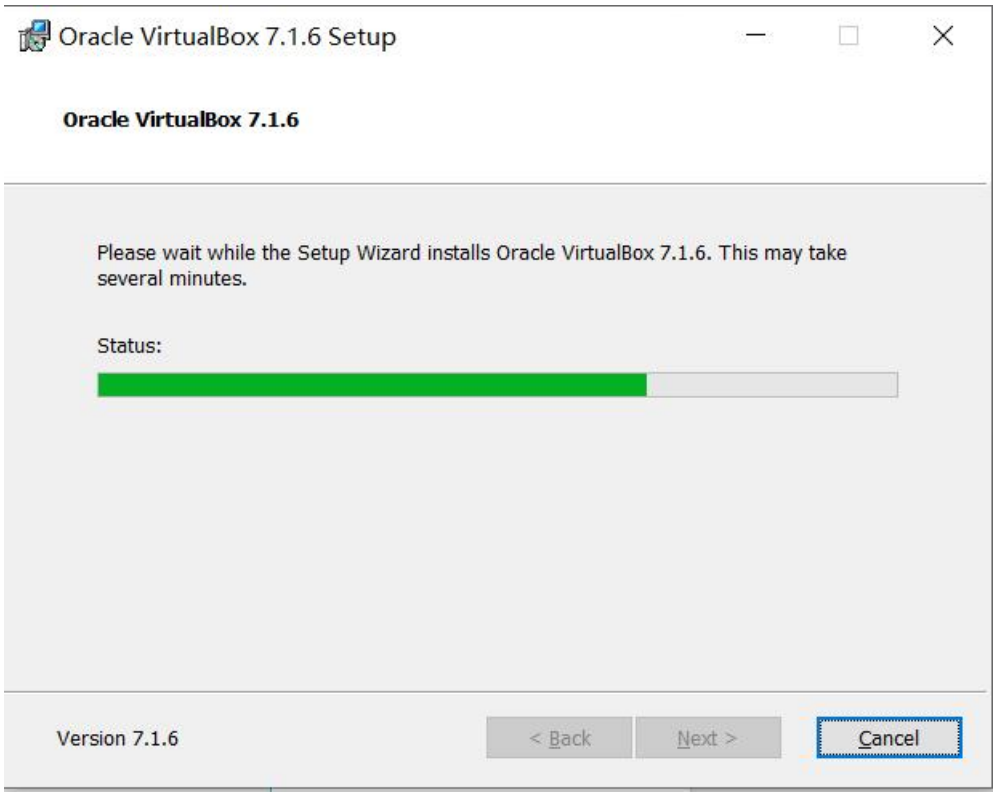
3. Check the following options and click Next;



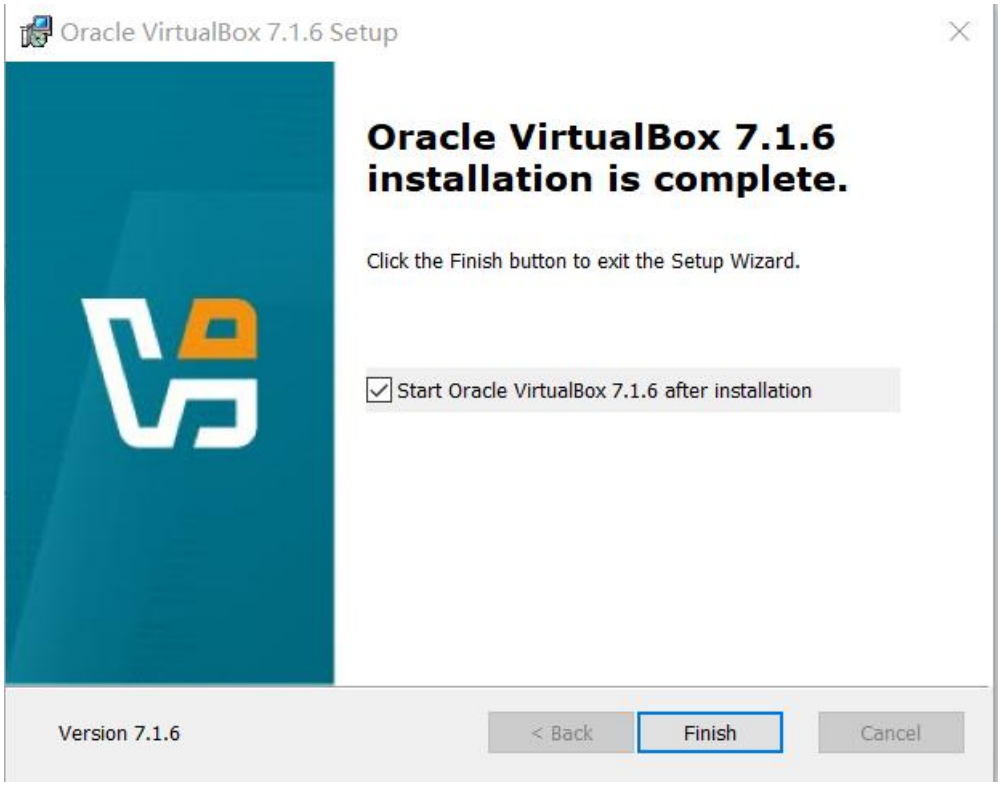
4. Click "Install" to start the installation;



5. Wait for the installation to complete;



6. Finally, click Finish to complete the installation successfully.

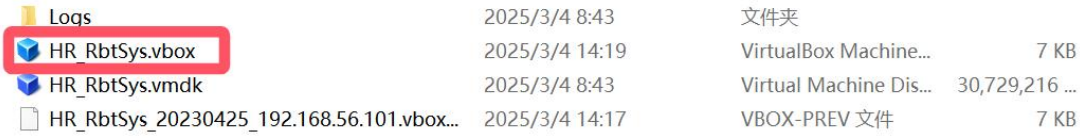


15.3. Virtual Robot

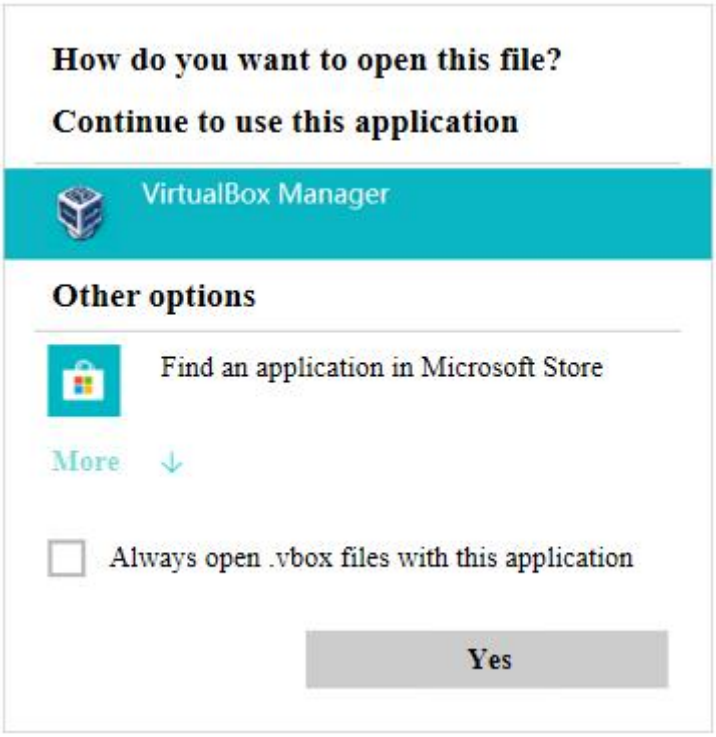
Users can use the virtual machine on their own PC to familiarize themselves with and master the Huayan controller software in advance, or they can write programs in the simulation software and import them into the actual robot for use.

15.4. Loading the Virtual Machine

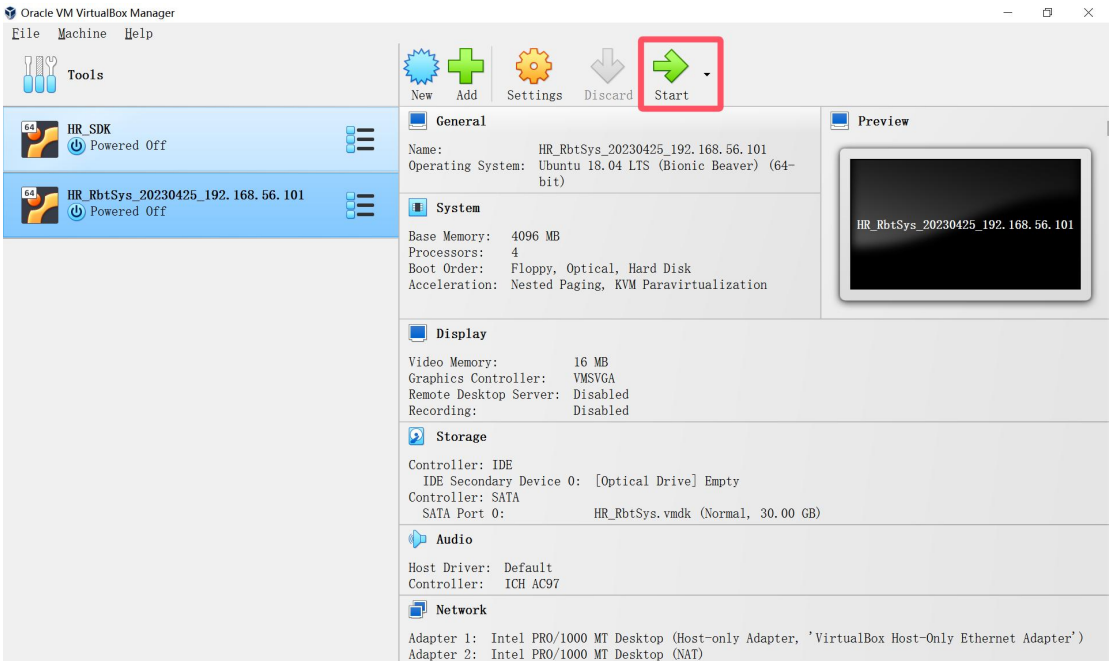
The HR_RbtSys_src.7z provided by our company and place it in a suitable location. Since the total size after decompression is about 30 G, it is recommended not to place it in the C drive. The following figure shows the decompressed file. Select HR_RbtSys.vbox, right-click and select Open.



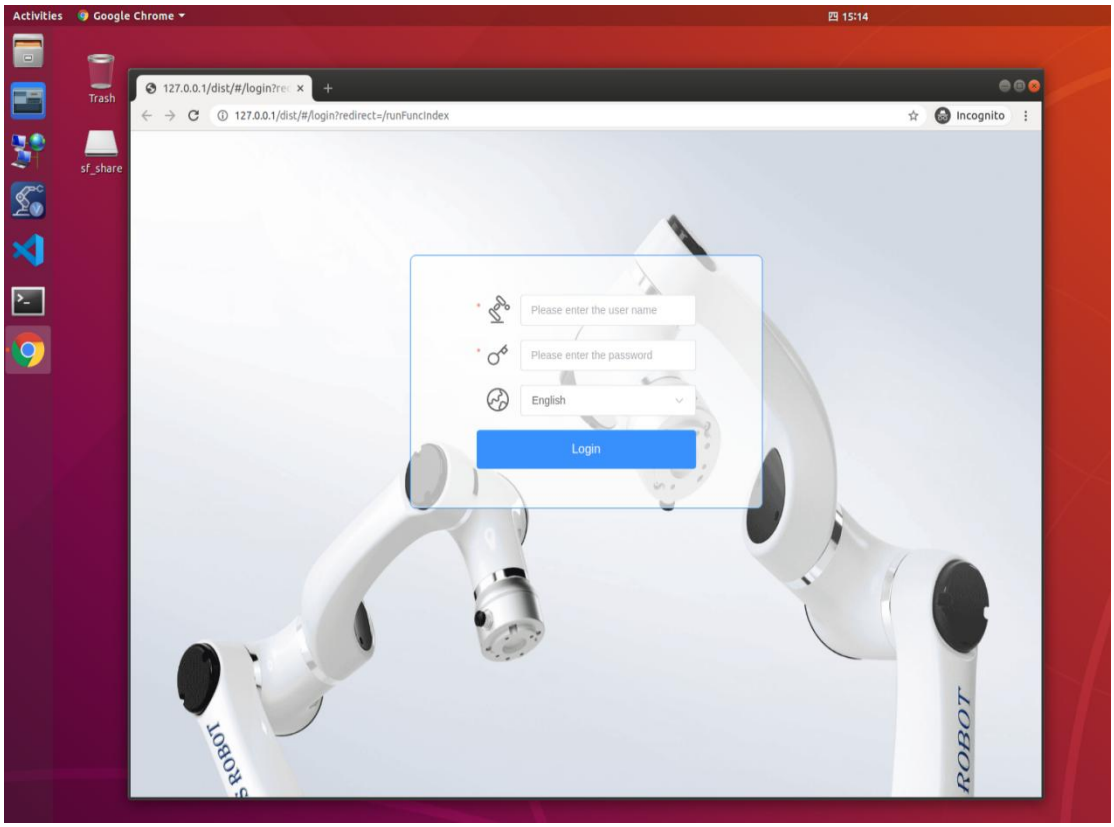
1. Open using VirtualBox Manager;



2. Click Start to start the virtual machine;



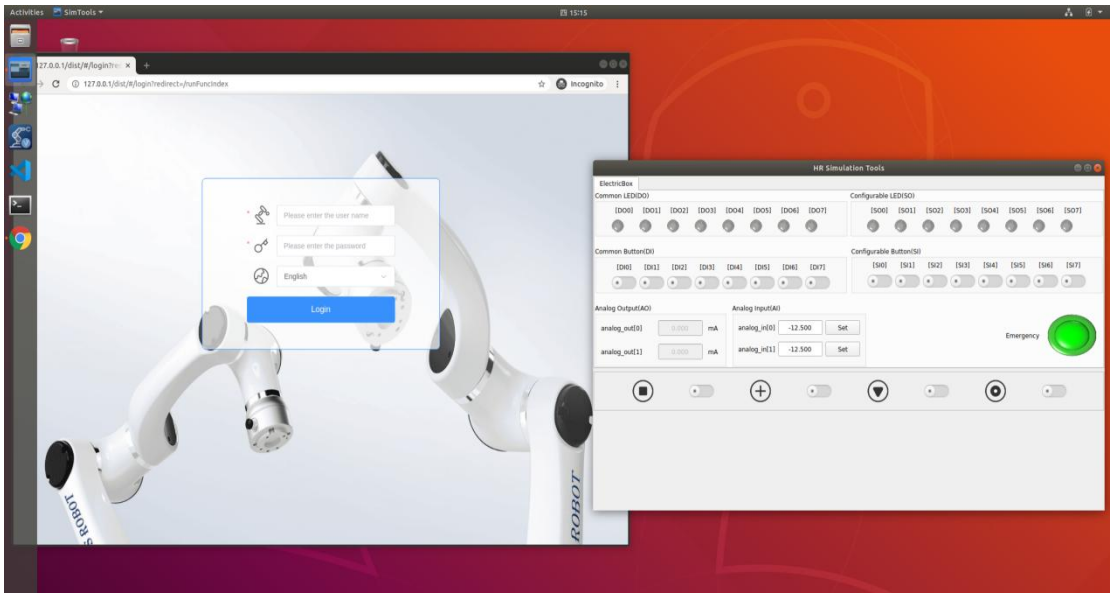
15.5. **Run the Virtual Machine**



Enter the user account and password, and you can operate the controller after logging in; please ask the supplier or technical support personnel for the administrator's account and password information.

15.6. Simulation Electric Box

Press the Win key on the keyboard to pop up the system menu. Click the Sim Tools icon to open the simulated electrical box, as shown below:



In this interface, you can debug the input port and observe the output port. On the right, you can turn on and off the robot power.